STAFF REPORT TO COUNCIL

Report Prepared By: Jake Belobaba, Director of Development Services
Report Reviewed By: **Erin Anderson, Acting CAO**
Meeting Date: May 19, 2020
File No: 6740-20
RE: **Waterfront Area Plan Implementation: Uplands Remediation Process**

RECOMMENDATION:
That Council:

1. Receive the Regulatory Path to Closure and Stage 1 Preliminary Site Investigation prepared by Golder Associates provided in Appendix A of the staff report from the Director of Development Services dated May 19, 2020;

2. Direct Staff to amend the 2020-2024 Financial Plan to include up to $400,000 from general surplus and development reserves to cover the cost of a detailed site investigation over the course of 2020-2021; and

3. Direct staff to:
    a. submit an application on behalf of the Town to the Federation of Canadian Municipalities to obtain funding for a detailed site investigation of the uplands and to obtain the services of a consultant, if required, to prepare the grant application; and
    b. seek the services of a qualified environmental engineering firm to complete a detailed site investigation of the uplands.

EXECUTIVE SUMMARY:
This report provides a summary of the updated findings of Golder Associates related to the nature of contamination on the "uplands" and a preliminary roadmap and funding strategy for remediation and provincial approval. Staff are recommending expediting further assessments and remediation and pursuing grant funding to cover further assessment costs.

PREVIOUS COUNCIL DIRECTION

Resolution Number	Date	Resolution
CS 2013-103	04/02/2013	It was moved, seconded and carried that the following comments be provided to the Province of British Columbia regarding Crown land referrals 1413402 and 1413408:

- The Town has an interest in the clean-up of the Ladysmith Harbour to its original natural state.

- In November 2012, the Province and the Town released a report on the environmental conditions of the Ladysmith waterfront outlining the extent of the contamination and a range of costs for alternatives to address remediation. These costs are significant.

- Consideration of a Crown grant or Crown lease should be subject to the applicant's financial commitment to appropriately addressing the environmental condition of the land consistent with Ladysmith's land use vision.

…

INTRODUCTION/BACKGROUND:

Town-owned parcels of land comprising areas referred to as the "uplands" in the Waterfront Area Plan are contaminated (Figure 1). In 2011, Golder Associates was hired to undertake a Stage 1 site investigation of the waterfront lands. In addition to identifying contamination on the Crown-owned water lots and foreshore, the study (available here https://www.ladysmith.ca/city-hall/reports-publications) revealed a number of Areas of Environmental Concern (AEC's) and Areas of Potential Environmental Concern (APEC's) on the Town-owned uplands. In late 2019, Golder Associates was again hired to prepare an updated Stage I investigation and remediation cost estimate focusing entirely on the uplands. The second study was needed to reflect the newly adopted Waterfront Area Plan, changes to provincial remediation standards and changes in remediation costs.

Figure 1: Uplands with AEC's and APECS

The Golder Reports

The Golder reports, attached as Appendix A, include: 1) an updated Stage I investigation; and 2) a regulatory "path to closure" which includes a preliminary estimate of remediation costs. The site investigation has confirmed a number AEC's and APEC's still exist on or near the uplands. Golder's preliminary remediation costs are estimated at $480,000-$700,000, excluding taxes and certain provincial fees. Completing a detailed site investigation would account for approximately $200,000-$400,000 of the total remediation cost. The timeframe to complete a detailed site investigation is estimated to be 4-12 months and is influenced by factors such as soil conditions and provincial testing standards. Golder has also provided a number of "paths" to regulatory closure (i.e. provincial approvals), some of which may allow the Town to develop the land in phases—i.e. develop and sell uncontaminated or remediated portions of the uplands while contaminated

portions are remediated. All remediation paths require a detailed site investigation. At this point, it is not necessary to decide on a preferred path, as such a decision is largely dependent on the results of a detailed site investigation and is ultimately approved by the Province.

RECOMMENDATION

Detailed Site Investigation
Staff are recommending that the detailed site investigation recommended in the Golder Report proceed as soon as possible. A detailed site investigation is a prerequisite for some types of redevelopment, all remediation options and a final decision on which remediation option is the most practical. Once complete, the Town can develop a remediation plan for provincial approval which may allow the Town to begin generating revenues from the uplands (e.g. through subdivision and sale or other forms of development) to offset remediation costs. If a phasing option is not approved by the Province, the Town can completely remediate the site prior to development. In this case, the remediation costs are likely to be recovered from future land sales of land, however, the timeframe to realize the revenues may be longer.

Funding
The FCM site remediation/ risk management study program, reimburses up to 50% of costs for a site investigation study to a maximum of $175,000, potentially reducing the Town's $200,000-$400,000 detailed site investigation cost to approximately $100,000-$225,000. Staff are recommending $400,000 be budgeted to cover the full cost of the site investigation, but if FCM funding is received, not all of this money will be used.

Staff note that consulting fees for preparing the FCM grant application are eligible expenses under the grant program. FCM funding is only available for site investigation work that has not occurred, so using a consultant to expedite the grant application accelerates the start of the detailed site investigation process, which accelerates the remediation process, which accelerates the development process and realization of the Waterfront Area Plan.

LONG TERM CONSIDERATIONS

Remediation and Provincial Approvals
Once a detailed site investigation is complete, the Town can develop and implement a formal remediation plan and commence the Provincial approval process. As noted above, options may be available to remediate the uplands in phases, possibly allowing revenue-generating activities like subdivision and sale to occur while remediation is underway. All remediation options require provincial approval, which can be expected to include a "freeze" on some or all development, a binding remediation plan and schedule, and bonding. The Province makes these decisions based on the information gathered in the detailed site investigation and provincial standards. In all scenarios, the process starts with a detailed site investigation and ends with provincially certified remediation.

Revenue from the sale of uncontaminated or remediated land is expected to more than cover the cost of uplands remediation, consistent with the "pay as you go" approach in the Waterfront Area Plan.

DISCUSSION
The estimated cost of further site investigations and remediation of the uplands is significantly lower than costs to remediate the Crown owned water lots and foreshore. As the owner of the uplands, the Town

has the ability initiate the remediation process at any time and it is prudent to do so as soon as possible. To successfully implement the Waterfront Area Plan further development is required and, to obtain approvals for further development remediation is required, the next step of which is a detailed site investigation. Seeking funding for remediation work will reduce the Town's costs, leaving more funds available for other aspects of the Waterfront Area Plan.

ALTERNATIVES:
Council can choose to:

1. Not proceed with further assessment and remediation of the uplands at this time and address site contamination as the need arises.

FINANCIAL IMPLICATIONS:
The upfront cost of a detailed site investigation and later remediation, even with FCM funding is significant. However, the eventual return on this investment in the form of development revenue is likely to be significantly higher. At this juncture, remediating and developing the uplands is expected to be economically viable, with substantial future revenues for the Town. It is a sound financial decision to remediate the uplands.

The funding for up to $400,000 can come from prior year surplus and other reserves.

LEGAL IMPLICATIONS:
The Town has a number of obligations under the *Environmental Management Act* related to contamination on the uplands. The proposed course of action is not in conflict with these obligations.

CITIZEN/PUBLIC RELATIONS IMPLICATIONS:
The Waterfront Area Plan involved extensive stakeholder and public engagement. The development plans for the uplands are based on the results of the Waterfront Area Planning process and remediation is a necessary step to implementing the community's vision for the waterfront.

INTERDEPARTMENTAL INVOLVEMENT/IMPLICATIONS:
N/A

ALIGNMENT WITH SUSTAINABILITY VISIONING REPORT:

☒Complete Community Land Use
☐Green Buildings
☒Innovative Infrastructure
☒Healthy Community
☐ Not Applicable
☐ Low Impact Transportation
☒ Multi-Use Landscapes
☐ Local Food Systems
☒ Local, Diverse Economy

ALIGNMENT WITH STRATEGIC PRIORITIES:

☐Infrastructure
☐Community
☒Waterfront
☒ Economy
☐ Not Applicable

I approve the report and recommendation(s).

Guillermo Ferrero, Chief Administrative Officer

ATTACHMENT(S):
Appendix A: Golder Report

REPORT

Stage 1 Preliminary Site Investigation

Lots 1, 4 and 5 Adjacent to Ladysmith Harbour

Submitted to:

Town of Ladysmith
132C Roberts St
Mail PO Box 220
Ladysmith, BC V9G 1A2

Attention: Jake Belobaba, Director of Development Services

Submitted by:

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18109842-001-R-Rev0

28 February 2020

Distribution List

eCopy - Town of Ladysmith

eCopy - Golder Associates Ltd.

Executive Summary

Golder Associates Ltd. (“Golder”) was retained by the Town of Ladysmith (“TOL”) to conduct a Stage 1 Preliminary Site Investigation (“Stage 1 PSI”) of Lots 1, 4, and 5 adjacent to Ladysmith Harbour, in the Town of Ladysmith, on Vancouver Island, British Columbia (the “Site”).

The Site is currently owned by the Town of Ladysmith. The Site is predominantly vacant, but is occupied by several structures in the central area of Lot 4.

Environmental site assessments have been conducted at the Site since the early 1990s. In 2011, Golder was retained by the Crown Lands Opportunities and Restoration Branch (CLORB) of the Ministry of Forests, Lands and Natural Resource Operations (MFLNRO) to conduct a Stage 1 PSI, Detailed Site Investigation (DSI), sediment investigation and preliminary geotechnical investigation at Ladysmith Harbour. The primary objectives of the investigation activities were to refine remedial and geotechnical costs and options for the Site. This investigation included Lots 1 and 4 of the Site, as well as the adjacent Slack Point and two water lots within Ladysmith Harbour (which are considered off-Site for the current investigation).

The primary objective of this Stage 1 PSI was to update the historical finding for the Site and identify, insofar as possible based on readily available information and without an intrusive investigation, former or current practices at the Site that may represent issues of actual or potential environmental concern. The scope of work included review of available analytical data and rescreening to current regulatory standards. Golder understands that the TOL is considering development of the Site and requires information concerning management of contaminated land. The Stage 1 PSI will be used to review the regulatory path to closure and update the environmental liabilities associated with the site. These tasks are completed under separate cover.

Authorization to proceed with this investigation was received by email from Mr. Jake Belobaba, Director of Development Services, on behalf of the Town of Ladysmith on 22 November 2019. This Stage 1 PSI report has been prepared for the use of the Town of Ladysmith and may not be relied upon by others without written consent from Golder.

Summary of Issues of Environmental Concern

The issues of environmental concern identified in this Stage 1 ESA are outlined in the following table:

AEC or APEC	Summary	PCOC, COC
On-Site		
AEC 1 - Former fuel pump islands, ASTs and possible PCB Storage.	Owing to the presence of soil and groundwater contamination, this area is considered an AEC. The extent of the contamination in this area of the Site is generally shallow in nature, to depths up to approximately 2.4 m below ground surface, and is estimated to have an approximate volume of 1,000 m^3	Soil: BTEX, VPH, LEPH, HEPH Groundwater: BTEC, VPH, LEPH Soil Vapour: VPHv, xylene, naphthalene
AEC 2 - Former Maintenance Area and Current Boat Repair and Construction Operations.	This area is considered an AEC. The extent of soil contamination in this area is generally shallow, to depths up to 4 metres. The volume of contaminated soils in the area to the northwest of the former maintenance building is estimated to be 2,000 m^3	Soil: BTEX, VPH, LEPH, HEPH, styrene,
AEC 3 - Former Waste Oil Storage Area and Compressor Storage Area.	The area is confirmed to be an AEC as soil contamination has been identified. However, based on the soil sampling results in the shallow soils, and the absence of groundwater contamination, the area of soil contamination is likely limited in area. The volume of contaminated soils is estimated to be 20 m^3	Soil: LEPH
APEC 4 – Stockpile of material from around sewage treatment plant	Fill of unknown quality was placed on-site, removed from the ground at the Ladysmith sewage treatment plant to facilitate expansion of the plant. While the fill is not known to be contaminated, it has been retained as an APEC, and further investigation of the material is recommended.	Soil: LEPH, HEPH, PAH, metals
APEC 5 – Fill of unknown quality on Lot 5	Lot 5 is adjacent to Slack Point, where coal fill and surficial fill were previously identified with concentrations of LEPH, HEPH, metals, and PAHs above CSR standards. Since previous intrusive investigations did not include Lot 5, there is a potential for fill material of similar quality to Slack Point to be located on Lot 5. It has been retained as an APEC, and further investigation of the soil quality is recommended.	Soil: LEPH, HEPH, metals, PAHs

AEC or APEC	Summary	PCOC, COC
Off-Site		
APEC 6 – Fill Material at Block B and D of DL 2016 (former Location of the Shingle Mill / Sawmill)	No soil or groundwater contamination identified during previous site investigations, as such, this area is not considered an AEC. However, the sampling program was limited in area, and additional sample collection along the filled area may be warranted, therefore, remains an APEC.	Soil: LEPH/HEPH, PAH, and metals Groundwater: LEPH/HEPH, PAH, metals Soil Vapour: VPHv, BTEX, naphthalene
AEC 7 - Former Log Dump (Lot 17G).	Contamination was identified during previous site investigations, but appears to be limited to a particular range of depth and may be associated with the log dump or filling activities. The extent of the contamination in this area of the Site was observed to be between 2.5 and 5.5 m below ground surface. The contamination is estimated to have an approximate volume of 5,800 m^3	Soil: VPH, LEPH Groundwater: PAH Soil Vapour: VPHv

Based on all the information obtained as part of this Stage 1 PSI, the following other Special Attention Items were identified on the Site:

- asbestos
- PCBs
- Lead

These Special Attention Items are not considered to represent an issue of potential environmental concern provided they are managed in accordance with applicable environmental, health, and safety legislation.

To address the issue(s) of potential environmental concern identified, Golder recommends the following:

- Conducting a Detailed Site Investigation to further investigate soil, groundwater and soil vapour quality at identified AECs.

Study Limitations

This report (the “Report”) was prepared for the exclusive use of the Town of Ladysmith for the express purpose of providing advice with respect to the environmental condition of the Site. In evaluating the site, Golder Associates Ltd. has relied in good faith on information provided by others as noted in the Report. We have assumed that the information provided is factual and accurate. We accept no responsibility for any deficiency, misstatement or inaccuracy contained in this report as a result of omissions, misinterpretations or fraudulent acts of persons interviewed or contacted.

Any use which a third party makes of this Report, or any reliance on or decisions to be made based on it, are the sole responsibility of the third parties. If a third party require reliance on this Report, written authorization from Golder is required. Golder disclaims responsibility of consequential financial effects on transactions or property values, or requirements for follow-up actions and costs.

The scope and the period of Golder’s assessment are described in this Report, and are subject to restrictions, assumptions and limitations. Except as noted herein, the work was conducted in accordance with the scope of work and terms and conditions within Golder’s proposal. Golder did not perform a complete assessment of all possible conditions or circumstances that may exist at the site referenced in the Report. Conditions may therefore exist which were not detected given the limited nature of the assessment Golder was retained to undertake with respect to the Site and additional environmental studies and actions may be required. In addition, it is recognized that the passage of time affects the information provided in the Report. Golder’s opinions are based upon information considered at the time of the writing of the Report. It is understood that the services provided for in the scope of work allowed Golder to form no more than an opinion of the actual conditions at the Site at the time the site was visited, and cannot be used to assess the effect of any subsequent changes in any laws, regulations, the environmental quality of the site or its surroundings. Asbestos and mould surveys were not performed. If a service is not expressly indicated, do not assume it has been provided.

The results of an assessment of this nature should in no way be construed as a warranty that the Site is free from any and all contamination from past or current practices.

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1.0 INTRODUCTION

1.1 Background and Objective

Golder Associates Ltd. (“Golder”) was retained by the Town of Ladysmith (“TOL”) to conduct a Stage 1 Preliminary Site Investigation (“Stage 1 PSI”) of Lots 1, 4, and 5 adjacent to Ladysmith Harbour, in the Town of Ladysmith, on Vancouver Island, British Columbia (the “Site”). The location, surroundings, and layout of the Site are shown in Figure 1 and Figure 2.

The Site is currently owned by the Town of Ladysmith. The Site is predominantly vacant, but is occupied by several structures in the central area of Lot 4.

Environmental site assessments have been conducted at the Site since the early 1990s. In 2011, Golder was retained by the Crown Lands Opportunities and Restoration Branch (CLORB) of the Ministry of Forests, Lands and Natural Resource Operations (MFLNRO) to conduct a Stage 1 PSI, DSI, sediment investigation and preliminary geotechnical investigation at Ladysmith Harbour. The primary objectives of the investigation activities were to refine remedial and geotechnical costs and options for the Site. This investigation included Lots 1 and 4 of the Site, as well as the adjacent Slack Point and two water lots within Ladysmith Harbour (which are considered off-Site for the current investigation).

The primary objective of this Stage 1 PSI was to update the historical finding for the Site and identify, insofar as possible based on readily available information and without an intrusive investigation, former or current practices at the Site that may represent issues of actual or potential environmental concern. Golder understands that the TOL is considering development of the Site and requires information concerning management of contaminated land. The Stage 1 PSI will be used to review the regulatory path to closure and update the environmental liabilities associated with the site. These tasks are completed under separate cover.

Authorization to proceed with this investigation was received by email from Mr. Jake Belobaba, Director of Development Services, on behalf of the Town of Ladysmith on 22 November 2019. This Stage 1 PSI report has been prepared for the use of the Town of Ladysmith and may not be relied upon by others without written consent from Golder.

1.2 Site Uses and Structures

Based on the TOL zoning bylaw (shown in Figure 3), the Site is zoned as a mix of park, residential, commercial and industrial land. The Site is largely vacant and covered by gravel roads, low-lying vegetation, shrubs and grass. Some buildings are present on the Site (e.g., former railway repair building, washroom, car shop, cable splicing shed and storage sheds).

1.3 Scope of Work

Golder's assessment was carried out in accordance with the British Columbia Ministry of Environment ("BC ENV") Technical Guidance on Contaminated Sites #10, *Guidance for a Stage 1 Preliminary Site Investigation* (dated August 2016), as well as in general accordance with Canadian Standards Association ("CSA") Standard Z768-01, *Phase I Environmental Site Assessment* (reaffirmed 2016), and involved the following scope of work:

- Reviewing readily available records to collect data on past and present activities on the Site.
- Visiting the Site to observe current Site conditions and operations and further assess any potential environmental concerns identified in the records review.
- Interviewing knowledgeable individual(s), to corroborate or augment the information gathered from the records review and Site visit.
- Collecting soil samples inside the former railway repair building. Excavations were open at the time of the Site visit for installation of new footings in the building.
- Evaluating the information from the records review, Site visit, and interviews.
- Rescreening of available historical analytical data.
- Preparing a Stage 1 PSI report.

For the purposes of this Stage 1 PSI, the assessment area included the Site and surrounding properties within 500 m of the Site.

In preparing this Stage 1 PSI, Golder has applied professional judgement in considering readily available information and has relied in good faith on information provided by others. This level of effort is a method of risk reduction rather than risk elimination. This assessment included a cursory overview of the neighbouring land uses and does not constitute a complete assessment of neighbouring land uses. Further reductions in risk can be achieved through a program of intrusive testing at the Site, including sample collection and analysis.

2.0 SITE DESCRIPTION

2.1 Site Location and Setting

The Site consists of a rectangular parcel of land approximately 10.5 hectares in area that is located at Ladysmith Harbour, Ladysmith, BC (Figure 1). The Site comprises several legal lots, which are identified with other relevant property information in Table 1 below.

Table 1: Property Information Summary

Civic Addresses	Civic addresses are available for the following buildings located at the Site: ■ The main repair shop (610 Oyster Bay Drive) ■ The washroom building (612 Oyster Bay Drive) ■ The roundhouse (614 Oyster Bay Drive) ■ The car shop (616 Oyster Bay Drive) ■ The cable splicing shed (840 Oyster Bay Drive) (building locations are shown on Figures 4a and 4b)
Legal District	Oyster District, Ladysmith.
Legal Description	■ Lot 1 District Lots 24 and 56, Oyster District Plan VIP64405 ■ Lot 4 District Lots (DL) 8G, 11G, 24 and 56, Oyster District, Plan 45800, except part in plans VIP64405, VIP71943 and VIP72131 ■ Lot 5 District Lots 24 and 56, Oyster District Plan 45800
Parcel Identifier	■ Lot 1: 023-652-926 ■ Lot 4: 010-208-828 ■ Lot 5: 010-208-861
Latitude/Longitude	48º 59’ 44” North, 123º 48’ 54” West (approximate centre of the Site).

Each lot comprising the Site is illustrated on Figure 2.

2.2 Topographic, Geologic, and Hydrogeologic Setting

The topography of the Site is typically flat with terraces separated from the shoreline areas by steep rock bluffs along the northeast. The nearest surface water body is Ladysmith Harbour, located adjacent to the east of the Site.

Surficial Geology of Nanaimo, British Columbia Map 27-1963, Sheet 92 G4 and 92 F-1 East (1:63,360) indicates that the surficial geology of the Site is composed of marine deposits of gravel, sand and mainly marine veneer commonly less than 1.5 metres thick and overlain by ground moraine deposits of till, lenses of gravel, sand and silt. Bedrock composed of volcanic and sedimentary rock outcrops along the foreshore.

Materials encountered at the Site during previous investigations (Hardy BBT 1990b; Levelton 2000b, Golder 2011) generally consisted of fill soils of variable thickness overlying native silt, sand and gravel overlying sedimentary bedrock. Historical industrial activities took place on the Site for many years. During these activities, fill soils of variable thickness and composition were placed throughout the area. The fill materials generally consisted of silt, sand and gravel, cobbles and boulders. In addition, organic material (roots and woody debris), isolated coal waste, and anthropogenic materials including concrete and metal, were encountered in the fill layers.

During the Golder 2011 field investigation, it was found that the thickness of the fill ranges overall from about 0.1 m to 3.4 m, and on average is about 1.3 m. In the area south of the bluff and along Oyster Bay Drive, the fill layer typically ranged in thickness from about 0.1 m to 1.6 m, and on average was about 0.7 m. The composition of the fill material encountered in this area consisted of silty sand with minor gravel to sand and gravel. Also included in this layer is cobbles, boulders and concrete, metal and wood debris.

Underlying the fill materials, native deposits of silts, sands and gravels were encountered overlying, in some locations, glacial till-like deposits. The depths to which these deposits extended ranged broadly from about 0.4 m to 5.9 metres below ground surface (m bgs), but were generally in the range of about 1 m to 4 m bgs.

Underlying the native soil deposits, sedimentary bedrock was encountered in approximately half of the test pits and monitoring wells/boreholes drilled on-site (Golder 2011). Where encountered, the depth to bedrock ranged widely among the boreholes and test pits, from 0.6 m to 8.1 m bgs; however, at most locations it was generally between 1 m and 4 m bgs. The rock consisted of sedimentary sandstone and mudstone (argillite).

Regional groundwater flow in the underlying aquifers is typically to the northeast towards Ladysmith Harbour. In previous field investigations, it was found that groundwater flow is directed toward the shoreline, despite some tidal influences (Golder 2011). Buried utilities, underground structures, and septic systems can affect local (shallow) groundwater flow conditions.

An online search of the iMapBC database was performed on 6 December 2019, to identify groundwater usage and water wells at the Site and in the immediate surrounding areas. According to iMapBC, the aquifer underlying the Ladysmith area, including the Site, is labelled Aquifer 168. Aquifer 168 is classified as IIIA under the BC Aquifer Classification System, indicating low demand relative to the aquifer's low productivity and high vulnerability of the aquifer to contamination from surface sources. The aquifer material is listed as bedrock.

The online search of the iMapBC database identified the nearest well was located approximately 475 metres (m) to the west of the Site, on the west side of the Island Highway. The detailed well record indicates that the well was constructed in 2007 and is 120 metres deep. A figure showing the results of the water well search of iMapBC can be found in Appendix A. In addition to current water use, the BC ENV requires that future drinking water use is considered when assessing potential threats to groundwater quality

3.0 PROPERTY USE INFORMATION REVIEW

3.1 Aerial Photos

Aerial photos of the Site and vicinity for the years 1952, 1957, 1962, 1968, 1975, 1984, 1988, 1993, 1998, and 2007, as well as Google Earth Images from 2017, were reviewed by Golder. The presence and absence of structures on the Site and on neighbouring properties were noted (Table 2).

Table 2: Aerial Photos

Date	Site Description	Surrounding Area
1952	A log dump is observed at Lot 4 DL11G. Several roads and rail spurs are present on the Site and along the foreshore. Buildings, in the configuration as they are currently known, are present at the Site, including the main repair shop, car shop, roundhouse, washroom and cable splicing shed. To the northeast of the main repair building, a clearing is observed, and a structure is visible in the approximate area of Lot 1.	Logging activities are present through-out the water adjacent to the northeast of the Site, including several log booms and sorting pockets. There appear to be several roadways through the area and a building appears on the western portion of Slack Point, approximately 100m east of Site. Several small structures appear on the north side of Slack Point. A parking area is noted on the east side of Slack Point. The current day government wharf (to the northwest of the Site) is not visible. The land adjoining the wharf, Blocks B and D of DL 2016 have not yet been filled. The shingle mill referred to in previous reports is visible (on a wharf) in the area of Block B. The area further to the north west of the Site has been filled and appears to have a hook-like pattern similar to the 1902 map of Ladysmith (Golder 2011; Appendix I). To the southwest of the Site, the railway and major highway are visible and several residential housing plots are observed occupying the Town of Ladysmith. To the southeast of the Site, the Loading Wharf and Transfer Wharf, as described on the 1902 map of Ladysmith (Golder 2011; Appendix I) are also visible.
1957	The Site appears similar to the 1952 aerial photograph; however, the photograph quality and scale of photograph prevent identification of specific Site features.	The Site appears similar to the 1952 aerial photograph; however, the photograph quality and scale of photograph prevent identification of specific Site features.
1962	The building structure southeast of the main repair building appears to have been removed and a new structure is observed in the area inferred to be the location of the former pump islands.	To the northwest of the Site the Government wharf has been constructed. A square shaped filled area has been constructed immediately adjacent to Blocks B and D, which is connected to the Government wharf. The shingle mill has been removed. The area to the southwest of the Site appears similar to 1952, with the exception of increased housing development to the southwest of the Site.The Loading Wharf to the southeast of the Site is no longer evident The northern tip of Slack Point (northeast of Site) has been expanded slightly and includes the distinctive hook-shaped point. There appears to be less evidence of logging activities; at least half of the sorting pockets have been removed in the water area.

Date	Site Description	Surrounding Area
1968	The Site appears generally unchanged.	There appears to have been more filling to the northwest of the Site and the former hook-like pattern observed in the 1952 photograph appears to have been completely filled in. A long breakwater has been constructed extending from Block B and D of DL 2016 into Ladysmith Harbour, and more fill appears to have been added in the area of Block B and D. The Transfer Wharf to the southeast of the Site is no longer evident. It appears that more fill has been added to the northern portion of Slack Point (northeast of the Site) and fewer logs, machinery and buildings are visible. Some areas of Slack Point are observed to be vegetated (trees are observed).
1975	The Site appears generally unchanged.	More filling has occurred northwest of the Site. Two large areas appear to have been paved and are utilized for industrial activity. Log sorting is observed along the water adjacent these areas. A wide road extends through Slack Point (northeast of Site) from Lot 4.
1984	The Site appears generally unchanged.	To the northwest of the Site, the two paved areas identified in 1975 were expanded and more fill added to this area. A hydraulic crane used for loading barges is visible in water lot DL 651. A large stockpile of material is present in the central portion of Slack Point. The road through Slack Point observed in 1975 is less prominent in the 1984 aerial photograph.
1988	The Site appears generally unchanged.	In the harbour adjacent to the Site, the majority of the logging operations have ceased, and a small marina has been constructed in the harbour. A few scattered log booms remain, and all of the sorting pockets have been removed. To the northwest of the Site extensive infilling is again observed and appears to connect the two paved areas observed in the 1975 aerial photograph. The area around the base of the Government Wharf and including Blocks B and D of DL 2016 has been further infilled and appears similar to the current configuration; it appears to be paved and utilized as a storage area. Slack Point is further vegetated and the building in the western area is no longer visible. The northern tip of Slack Point has been further expanded and appears slightly rounded.
1993	The Site appears generally unchanged.	The large stockpile of material on Slack Point remains present and is now covered in vegetation.
1998	The Site appears generally unchanged. The cable splicing shed is not visible, likely owing to tree cover.	The small marina in the harbour adjacent to the Site has been expanded and includes a narrow breakwater area and several boats are visible. The large stockpile of material on Slack Point is no longer visible.

Date	Site Description	Surrounding Area
2007	A road from the main highway leading into the area southwest of Slack Point has been constructed (Transfer Beach Boulevard). The structure to the southeast of the main repair building (inferred to be the location of the former pump islands) has been removed.	The marina has expanded to include at least two docks; the breakwater has been removed. Park-like features are observed on the southeast portion of Slack Point, including a baseball field and amphitheatre. Immediately east of the Site there is another residential development (what is now 63B Avenue). The hydraulic crane piles are no longer visible on DL 651; however, many boats and other structures are visible in the water northwest of Slack Point.
2017 (Google Earth Image)	Similar to the 2007 photograph.	Similar to the 2007 photograph, except a breakwater has been constructed to the North of the Site, adjacent to the Government Wharf, and Slack Point (northeast of site) is increasingly vegetated.

3.2 Fire Insurance Records

SCM Risk Management Services Inc (SCM) was contacted on 4 September 2009 for fire insurance information pertaining to the Site. Although a 1954 fire insurance map was available for the Town of Ladysmith, the map did not include the Site itself. Fire insurance maps were therefore not obtained during this investigation.

The 1999 Revelop report contained a 1911 fire insurance map. The map also did not have any details for the site. The map showed that the railway was present along the shoreline of Ladysmith in 1911. The map also includes information on the former Tyee Copper Smelter northwest of and cross gradient to the Site.

3.3 City Directories

On 4 September 2009, Golder contacted Vancouver Public Library's (VPL) Information and Research Centre to request city directories for the Town of Ladysmith. A representative from VPL indicated that there were no city directories with street indexes for Ladysmith. The response is included in Appendix E.

Revelop (1999) conducted a search of city directories for the Site; they reported that the Comox Logging Company and Railway Co. was listed at the Site on the city directories from 1950 to 1957 (not available for review by Golder). In addition, Revelop obtained an early 1900 microfiche business directory information (not reviewed by Golder) that included a description of Ladysmith Harbour in the 1902-1910 directories. The description was as follows:

> *"Ladysmith in the Nanaimo district.... is the shipping point for the Wellington Collieries where the largest ships afloat can come alongside and tie up to the capacious wharves. Here you will find the latest and most up-to-date improved machinery on the Pacific Coast for the quick dispatch of colliers. The wharves are substantially built on copper covered pile. Here also is the transfer wharf for the Canadian Pacific Railway. ..."* (Revelop 1999).

It was reported by Revelop that the directory indicated that the wharves may have been built on copper covered piles, potentially affecting the quality of sediments in the vicinity.

3.4 Land Title Information

Golder performed a current and historic land title search of the Site using the Land Title and Survey Authority (LTSA) of British Columbia. Historical searches and land lease agreements from previous reports were also reviewed.

Currently, the Site (Lots 1, 4, and 5) is owned by the Town of Ladysmith. For Lot 1, TOL has held the title since at least November 1996. For Lot 4, TOL has held the title since at least March 2001. Lot 5 has been owned by TOL since April 2019. Prior to this, several companies held the title for Lot 5 between March 1988 and April 2019. Notable owners include logging companies (TimberWest Forest Ltd., Elk Falls Wood Products Ltd., Crown Forest Industries Ltd.) and Sea Vision Resorts Development Ltd. Prior to March 1988, TOL held the title of Lot 5.

Copies of the land titles are included in Appendix B.

3.5 Agreement of Purchase and Sale

Golder was not provided with an agreement of purchase and sale.

3.6 Previous Environmental Reports

The following environmental reports related to the Site were obtained by Golder. Environmental site assessments have been conducted at the Site and in the surrounding area since the early 1990s. Reports are ordered from oldest to most recent. Golder consulted these reports to develop an understanding of issues previously identified for the Site and surrounding properties.

- Norecol Environmental Consultants Ltd. 1989. “Habitat Compensation Plan Ladysmith Waterfront Development Plan”. Dated August 1989.
- Dames & Moore. 1990a. “Fletcher Challenge Canada Environmental Inspection Report on Leases 101501, 101502 & 101503, Ladysmith Harbour”. Dated 24 August 1990.
- Dames & Moore. 1990b. “Fletcher Challenge Canada Decommissioning Investigations – Leases 101501, 101502 & 101503 Ladysmith Harbour”. Dated December 1990.
- Hardy BBT Limited. 1990a. “Preliminary Geotechnical Assessment for Ladysmith Waterfront Development Ladysmith, BC”. Dated October 1990.
- Hardy BBT Limited. 1990b. “Environmental Review for Town of Ladysmith Waterfront Development Slack Point Area, Ladysmith, BC”. Dated October 1990.
- Subsea Enterprises Inc. 1993. “Summary Report of an Underwater Video Survey of Fletcher Challenge Lease (Lot 651) in Ladysmith Harbour, BC”. Dated 30 September 1993.
- EBA Environmental Ltd. 1994a. “Elk Falls Forest Industries Limited Phase II Environmental Assessment Ladysmith Harbour Leases Project Ladysmith, BC”. Dated January 1994.
- EBA Environmental Ltd. 1994b. “Elk Falls Forest Industries Limited Supplemental Information Ladysmith Harbour Leases Project Ladysmith, BC”. Dated 4 February 1994.

- Triton Environmental Consultants Ltd., “Slack Point Biophysical Inventory and Fish Habitat Compensation Plan”. Dated May 1996.
- Town of Ladysmith, “Waterfront Area Plan”, January 1997.
- New Pacific Ventures, “Environmental Assessment of Proposed Developments at Burleith Log Sort Ladysmith Harbour”. Dated May 1998.
- The Revelop Group. 1999. “Stage 1 Preliminary Site Investigation Ladysmith Waterfront Development Ladysmith, BC”. Dated 10 August 1999.
- Phoenix Environmental Services Ltd. 1999. “Environmental Impact Assessment Report Proposed Ladysmith Marina and Waterfront Development Project Ladysmith, BC.” Dated December 1999.
- Levelton Engineering Ltd. 2000a. “Preliminary Site Investigation Stage I and Stage II Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District. Ladysmith, BC”. Dated 25 August 2000.
- Levelton Engineering Ltd. 2000b. “Detailed Site Investigation Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District. Ladysmith, BC”. Dated 25 October 2000.
- Levelton Engineering Ltd. 2000c. “Remediation Plan Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District Ladysmith, BC”. Dated 1 December 2000.
- Westmar Consultants Inc. 2001. “Town of Ladysmith, Report for Waterfront Redevelopment”. Dated June 2001.
- Baker & Osland Appraisals Ltd. 2002. “Property Appraisal Report, Private and Crown Lands Front and Within Ladysmith Harbour Ladysmith, British Columbia”. Dated 4 April 2002.
- Phoenix Environmental Services Ltd. 2002. “Sediment Chemistry Investigation Report Ladysmith Harbour Proposed Waterfront Re-Development Ladysmith, BC”. Dated September 2002.
- W.R. Colclough & Associates Ltd. and D.F. Brown. 2004. “Review of Existing Environmental Reports Relating to Ladysmith Harbour and Uplands Proposed Waterfront Re-Development Ladysmith, BC”. Dated November 2004.
- EBA Engineering Consultants Ltd. “Stage 1 Preliminary Site Investigation Lot 5, Plan 45800, DL 24 and 56, Ladysmith, BC”. Dated February 2005.
- G3 Consulting Ltd. 2005. “Report on Ladysmith Harbour Foreshore Fill Sampling & Analysis”. Dated March 2005.
- Golder Associates Ltd. 2005. “Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site Investigation Lot 16G Ladysmith Harbour Ladysmith, BC”. Dated 19 July 2005.
- Golder Associates Ltd. 2011. “Supplemental Stage 1 Preliminary Site Investigation and Detailed Site Investigation, Ladysmith Harbour Ladysmith BC”. Dated 21 December 2011.

While technical peer reviews of the reports were not completed, noteworthy findings are summarized below. Note that the current site under investigation is sometimes referred to as the “Uplands Area” in previous reports, to distinguish from surrounding properties including Slack Point that were previously under investigation.

3.6.1 Norecol Environmental Consultants Ltd. 1989 Habitat Compensation Plan

In 1989, Norecol Environmental Consultants Ltd. (Norecol) prepared a habitat compensation plan for a waterfront development plan. Details were documented in the report entitled “*Habitat Compensation Plan Ladysmith Waterfront Development Plan*” (Norecol 1989). As part of its environmental liability assessment in 2005, Golder was provided the sediment composition section of the report by the Town of Ladysmith. The section indicated that three sediment samples were collected from three locations within the Site. Sampling location details were not provided. Of the samples, one contained a mercury concentration exceeding both the BC *Contaminated Sites Regulation* (CSR) sediment criteria and ocean disposal limits applicable at the time of the assessment, and cadmium exceeding the ocean disposal limits only. The report stated that the only indication from the Site history that would suggest a possible source of mercury and cadmium was the previous discharge of sewage into Ladysmith Harbour.

3.6.2 Dames & Moore 1990 Environmental Inspection

In 1990, Dames & Moore (D&M) conducted an environmental inspection of three leased properties including 1) Slack Point, 2) the former Log Dump at Lot 17G, and 3) DL 651 (water lot) (Leases 101501, 101502 and 101503 respectively). Details were documented in the report entitled “*Fletcher Challenge Canada Environmental Inspection Report on Leases 101501, 101502 & 101503, Ladysmith Harbour”,* dated 24 August 1990 (D&M 1990a). The environmental inspection consisted of a field inspection and a review of available information. The purpose of the investigation was to determine the potential environmental impacts related to the occupancy of Fletcher Challenge Canada prior to the surrender of the leases back to the Crown.

D&M’s review of historical information indicated that coal mining and forestry operations had been the primary industrial activity in Ladysmith Harbour over the past century. A coal smelter, ore smelter, shingle mill and iron foundry had been established in the port by 1899. According to D&M’s report, the fill used to create Slack Point was composed of coal, which was derived from the washing of coal mine and coal smelter wastes.

During the global economic depression in the 1930s, the coal industry began to decline, and the logging industry began its rise. Between 1935 and 1953, the leased area was occupied by the Comox Logging and Railway Company. From 1953 to 1988, Crown Zellerbach Building Materials Ltd. (Crown Zellerbach) occupied the area. Fletcher Challenge assumed the lease in 1988. During logging operations, the railroad was extended out onto Slack Point. Logging activities in the area included the off-loading of logs from the railway to Ladysmith Harbour, where log dumping, sorting, storing, and shipping took place. In 1979, a hydraulic crane was installed on piles and was used to load barges.

The report stated that no treatment of wood reportedly occurred at the Site. D&M indicated that pressure treated piles were used for the dolphins associated with the wharfs on Slack Point. According to a former employee of Fletcher Challenge, interviewed by D&M, waste bark materials and other debris from the operations were burned on Slack Point.

Diesel engines replaced steam in the leased area around 1958. Diesel, gasoline and bunker fuels were reported to have been used over the years for various purposes related to the logging and railway operations in Ladysmith Harbour. Fuel was initially delivered to the tanks by barge, until the 1960s when fuel was delivered by trucks. Fuelling operations occurred on the Uplands portion of the Site, including operations near the former rail yard, where fuels from above-ground tanks were used to fuel trains and boats. Boats at the shore were fuelled by a gravity-fed pipeline. The pipeline system was in place at the time of D&M's inspection; however, their report did not indicate if the pipelines were above or below ground surface. Trains bringing logs to Site were fuelled at the main fuelling facility (presumably by the former pump islands shown on Figure 5a).

In 1981, three diesel fuel tanks were installed in a concrete containment, and were described as being located "on the upslope side of the lower track". For the purpose of this current PSI, the location has been interpreted as being in the vicinity of the present-day Ladysmith Maritime Society Marina parking lot (located at the base of the Ladysmith Maritime Society Marina and Wharf, shown on Figure 4). Pipelines from these tanks transferred fuel to a fuelling dock for the boats. Interviews conducted by D&M indicated that a leak was historically from one of the pipes from the upper aboveground storage tanks (ASTs), and the soil around a lower track ballast was subsequently removed. The area was replaced with clean fill, and the leaking pipe replaced. The time and precise location of the spill was not provided in the report.

No other historical spills were reported during the interview conducted by D&M. Small gasoline tanks (above ground) were used at various times and locations (including Slack Point) for fuelling gasoline-powered equipment. The report by D&M does not provide specific details about the usage of fuel or fuel storage locations at Slack Point.

During their occupancy (from approximately 1935), Comox Logging and Railway Company established a small boat yard in the west corner of Slack Point where wooden and steel vessels were built (shown on Figure 4). In the 1970s, the boat yard was used as a boat repair and service shop until it burned down in 1987.

According to D&M, an untreated sewage pipe extended to the inner harbour area through a short outfall immediately north of the Government Wharf (off-Site) between 1905 and 1965. After 1965, the pipe became a storm water outfall, although the outfall acted as an overflow for sewage until 1985 when all sewage was directed to Holland point. The report indicated that the sewage outfall may be of considerable concern for the sediments in the area with regard to heavy metals and organic compounds that may have originated from industrial and domestic sources in Ladysmith.

D&M commented that the harbour was previously known for its shellfish but water quality concerns, including sewage discharge, recreational boaters and the forest industry, were indicated as likely major sources of the disappearance of a viable shellfish market. It was thought that the diversion of the raw sewage outfall (between 1965 and 1985) may have increased the potential for shellfish in the vicinity of Slack Point.

D&M did not identify any potential for residual chemicals on the lease areas originating from Fletcher Challenge Canada's activities. The report states that the marine log sort area would have deposited bark and other debris; however, D&M reported that there was no requirement to remove the debris from the lease area. The report did indicate, however, that given the extended past use of the area for coal washing and smelting and a raw sewage and storm water outfall into the harbour (occurring before occupancy by Fletcher), the quality of the sediments in the area would be a concern.

3.6.3 Dames & Moore 1990 Proposed Decommissioning

Following the report by D&M in 1990, the Ministry of Crown Lands requested that Fletcher Challenge Canada's leased area be "environmentally clean and suitable for residential development" prior to the surrender of the land back to the Crown. In a letter regarding "*Your Files: 1403254, 140255, 1403256 Lease # 101501, 101502, 101503 – Ladysmith*" dated 5 September 1990 (Fletcher Challenge 1990), Fletcher Challenge Canada responded to the ministry and outlined a proposed clean up of the area, as described in the document entitled "*Fletcher Challenge Canada Decommissioning Investigations – Leases 101501, 101502 & 101503 Ladysmith Harbour*", dated December 1990 (D&M 1990b).

Proposed activities included the removal of the following: steel cables and coils along the waterfront; three oil tanks, bases, piping and pump house; loose logs, floats, and sunken buildings; miscellaneous machinery, equipment and metal objects; railroad ties and tracks on Slack Point; boat ways and a burned boat haul; miscellaneous garbage; and boom shack, floats, stairways and piers.

3.6.4 Hardy BBT Limited 1990 Geotechnical and Environmental Assessments

In 1990, Hardy BBT Limited (Hardy BBT) conducted a geotechnical assessment and an environmental review of the Site and Slack Point for the Town of Ladysmith. Details were documented in the reports entitled "*Preliminary Geotechnical Assessment for Ladysmith Waterfront Development Ladysmith, BC*", dated October 1990 (Hardy BBT 1990a), and "*Environmental Review for Town of Ladysmith Waterfront Development Slack Point Area, Ladysmith, BC*", dated October 1990 (Hardy BBT 1990b).

The Town of Ladysmith requested the work to assess the possibility of future development of the area. The objective of the environmental assessment was to determine if significant soil contamination was present in the proposed development area. The purpose of the geotechnical investigation was to identify any significant foundation problems for the potential future development of residential and/or commercial buildings.

Hardy BBT's reports indicated that historical activities in Ladysmith Harbour included the shipping and handling of coal from rail cars onto ships. The waste materials from coal washing operations were discharged into the harbour to form Slack Point. Hardy BBT indicated that after the coal mining industry ceased, the area was used by forest product companies, and Slack Point was generally used as a dry land sort for logs and as a storage/disposal area for waste material and other debris from the harbour bottom.

On-site, Hardy BBT observed the following:

- One 500-gallon (approximately 1,900 L) underground storage tank (UST) in the vicinity of the washroom building that was reported to have been installed in the mid 1960s and was reported to have contained both heating oil and diesel fuel[1].
- Three above-ground fuel tanks (ASTs) (size was not reported) described as being located, "adjacent to a disused railway spur line along the shoreline downslope of the Railway Museum" (interpreted as being in the vicinity of the present-day Maritime Society Marina parking lot).

[1] During the 2011 Stage 1 PSI, Golder concluded that the UST was installed as a component of a sewage pump-out facility connected to the Town of Ladysmith's municipal sewage system, and was not used for fuel storage. Therefore, it is not considered an area of potential environmental concern

- A pump island located approximately 100 m to the southeast of the main repair building, supplied by nearby ASTs.
- Storage of electrical transformers, likely containing polychlorinated biphenyls (PCBs) approximately 100 m southeast of the main repair building.
- A waste oil storage area adjacent to the southeast of the main repair shop.

The inferred locations of historical features are shown on Figures 4a and 4b.

At Slack Point, adjacent to the east of the Site, Hardy BBT observed the following:

- An approximately 4 m high stockpile of wood debris (with an estimated volume of 4,200 m^3) was observed in the central portion of Slack Point (shown on Figure 4), near the end of a former railroad track. Communications with the Site representative at the time of the Hardy BBT report suggested that the material was dredged from the log dump area and transported via rail.
- The presence of a timber bulkhead and log-hauling ramp at the north end of Slack Point.
- The burned remains of the former boat maintenance facility at the west corner of Slack Point.
- A drainage ditch at the south eastern perimeter of Slack Point.
- Midway along the northern side of Slack Point, in the intertidal zone, the remains of a pile of dredged coal waste from the harbour.
- A small landfill associated with the former logging activities was also identified on Slack Point; however, the exact location was not described.

The inferred locations of historical features on Slack Point are shown on Figure 4.

A total of 25 test pits were advanced during Hardy BBT's geotechnical and environmental assessments. Fifteen of the test pits were installed for geotechnical assessment purposes; ten were installed for environmental assessment purposes. Samples from select locations during the environmental assessment were analysed for potential contaminants of concern, including metals and petroleum hydrocarbons.

The test pits were excavated to total depths ranging from 1.4 m and 4.0 m bgs (depth was limited due to a high-water table and sloughing conditions). The report indicated that test pits installed for environmental assessment purposes were advanced in the vicinity of the pump islands, the UST, the railway siding used for railcar and engine maintenance, the scrap metal storage area (inferred to be located to the northwest of the main repair building in the rail yard area), the concrete bunker containing the three ASTs, the main repair building in the rail yard area, the former cable slicing shed, the stockpiled area of unknown fill on Slack Point, and the small landfill identified at Slack Point.

Soil samples showed exceedances of standards applicable at the time of the assessment. Exceedances of various metals standards (arsenic, nickel, zinc, and copper) corresponded to test pits installed at Slack Point, in the area that was understood to be the abandoned landfill. Exceedances of petroleum hydrocarbons, including LEPH, xylenes, ethylbenzene, and toluene, corresponded to test pits installed on-site near the former fuel pump island and the main repair building, as well as off-site on Slack Point.

3.6.5 EBA Environmental Ltd. 1994 Phase II ESA and Supplemental Investigation

3.6.5.1 Overview and Purpose

In 1994, a Phase II ESA was conducted at the Site by EBA Environmental Ltd. (EBA) on behalf of Elk Falls Forest Industries Limited (Elk Falls). A subsequent supplemental investigation was conducted to obtain additional information as requested of BC ENV. The investigations were conducted over the entire Site, as well as Slack Point and the adjacent water lots. Elk Falls was the lease holder of the investigated properties at the time of the investigations. The purpose of the investigations was to further characterize soil and groundwater such that the land leases could be returned to the Crown. Details of the investigations were documented in the reports entitled "*Elk Falls Forest Industries Limited Phase II Environmental Assessment Ladysmith Harbour Leases Project Ladysmith, BC*, dated January 1994 (EBA 1994a), and "*Elk Falls Forest Industries Limited Supplemental Information Ladysmith Harbour Leases Project Ladysmith, BC*", dated 4 February 1994 (EBA 1994b).

3.6.5.2 Historical Information

According to EBA, the area was used for log handing and sorting between 1935 and 1987. The operations involved transferring logs into the harbour. The boat repair facility located on the western portion of Slack Point burned down in the late 1980s and the burned debris was removed in 1992. The three ASTs located at the shoreline near DL 17G (shown on Figure 4) had been removed by 1994, and there was no evidence of surficial staining or distressed vegetation. The log dump area had been removed and backfilled and the railway tracks had been removed.

According to EBA, extensive clean-up activities were reported to have been completed in 1992 and included the removal of "piles, dolphins, wharfs and piers, floats, boom shacks, buildings, equipment, cable, burned boat-repair ways, miscellaneous dumped garbage, and other scattered debris" (EBA 1994a). No further documentation of the decommissioning activities at the Site was available for Golder's review.

3.6.5.3 Scope of Investigation

EBA's intrusive investigation consisted of the excavation of seventeen test pits, to a maximum depth of approximately 4.5 m bgs. Thirteen test pits were advanced off-Site around Slack Point; specifically, five were advanced near the former boar repair area, two were advanced in the vicinity of the landfill, and the remainder were advanced through-out Slack Point. Four additional test pits were hand excavated using a shovel on-site.

3.6.5.4 Observations

During the investigation, shallow groundwater was observed in some of the test pits at 1.5 m to 2 m below grade. Subsurface stratigraphy encountered at Slack Point was similar to that observed by Hardy BBT, which consisted of up to 3.5 m of wood waste, with other debris in some locations, underlain by coal waste. No visual evidence of contaminants or buried hazardous materials was noted at the former boat repair area, and no significant garbage or other debris was uncovered from the test pits near the inferred location of the former landfill (shown on Figure 4). EBA inferred that the area was potentially associated with unauthorized dumping.

3.6.5.5 Analytical Results

Chemical analysis of select soil samples collected from the test pits included mineral oil and grease (MOG), sulphur, metals (two samples only), and chlorinated phenols (one sample only). Results of the analysis were compared to CSR standards applicable at the time of the assessment, for commercial land-use (CL), industrial land-use (IL), residential land-use (RL) and park land-use (PL). Soil samples showed exceedances of MOG and sulfur, based on the BC ENV standards applicable at the time of the assessment.

EBA also collected groundwater from four of the test pits. These samples were reported to contain high amounts of suspended coal sediment. Chemical analysis of the groundwater samples included MOG, salinity, total metals, chloride, sulphate, and sulphur. Groundwater samples contained concentrations of several metals (copper, chromium, barium, mercury, nickel, and lead) and MOG exceeding the BC ENV standards applicable at the time of assessment.

3.6.5.6 Ministry of Environment Correspondence

Appended to EBA's supplemental investigation report (EBA 1994b) is a letter of correspondence from BC ENV (referred to as the Ministry of Environment Lands and Parks at the time of the report) to EBA providing a summary of the reports reviewed by BC ENV. One document identified in the letter was a proposal for an environmental assessment of Leases 101501, 101502, & 101503 (Norecol Environmental Consultants Ltd., titled "*Proposed Methodology for Environmental Site Assessment Leases 101501, 101502, & 101503, Ladysmith Harbour, BC*.", dated 12 August 1991). This document was not available to Golder for review. Additional information provided in the letter indicates that smelter waste (slag), originating from the former Tyee Copper Company in Ladysmith, may also have also been used as (deep) fill material at Slack Point. BC ENV concluded that Fletcher Challenge's responsibility for clean-up at the Site is limited to the wood waste and dredged materials dumped on the Site (estimated to have a total volume of 20,000 m^3), and to any soil and groundwater impacts related to the boat repair operations.

3.6.6 Town of Ladysmith 1997 Waterfront Area Plan

In 1997, a waterfront area plan (WAP) was developed by the Town of Ladysmith (Town of Ladysmith 1997) to provide guidance for future land use of the Ladysmith waterfront. The Site was planned for multi-family residential, mixed-use residential/commercial land use. The WAP illustrated the planned use of Slack Point as a mixture of park and residential/commercial properties.

The WAP did not identify areas or activities of environmental concern or potential environmental concern.

3.6.7 New Pacific Ventures 1998 Environmental Assessment at Burleith Log Sort

In 1998, New Pacific Ventures (New Pacific) conducted an Environmental Assessment at the Burleith Log Sort on Ladysmith Harbour, located north of the Site and across Ladysmith Harbour (off-Site), for the Burleith Log Sort. Details were documented in "*Environmental Assessment of Proposed Developments at Burleith Log Sort Ladysmith Harbour*", dated May 1998 (New Pacific 1998). The report indicated that the Burleith Log Sort area has been substantially altered from the natural condition by logs and wood debris associated with the log sorting activity, as well as the ongoing activity of boom boats. A thick layer of wood debris was observed in the intertidal and subtidal regions of the area.

3.6.8 The Revelop Group 1999 Stage 1 PSI

A Stage 1 PSI of the Site, Slack Point, and surrounding areas was conducted by the Revelop Group (Revelop) on behalf of Concept Bank Corporation and SVR Acquisitions Limited, who were reported to be the lease holder at the time. Details were documented in the report, "*Stage 1 Preliminary Site Investigation Ladysmith Waterfront Development Ladysmith, BC*", dated 10 August 1999 (ReVelop 1999).

The investigation included the Site, Slack Point, and the water lots adjacent to the Site. The general description of the investigation area was described as extending towards the Harbour shoreline from the rail lines to the waterfront shoreline, and from Transfer Beach Park to the Government Wharf. Information contained in the Stage 1 PSI conducted by Revelop (i.e., historical fire insurance plans, city directories, site inspection details, etc.) is referenced, where applicable, in the current investigation.

The report identified 15 Areas of Potential Environmental Concern (APECs) including:

- The abandoned fuel pump island southeast of main repair shop (location of historical Site features is shown on Figures 4a and 4b).
- Former electrical transformer storage area near pump island.
- Railway siding formerly used for railcar and engine maintenance.
- Former waste oil disposal area at southeast end of the main repair shop.
- Underground storage tank behind the washroom building (inferred by Golder to be the sewage pump-out facility connected to the Town of Ladysmith's municipal sewage system).
- Cable splicing shed.
- Coal waste material at Slack Point.
- Former landfill at Slack Point.
- Former logging building on Slack Point.
- Former log dump area (DL 17G).
- Former above ground storage tank at the shoreline.
- Sewage disposal outfalls in the harbour area.
- Off-site copper smelter.
- Areas surrounding railway buildings formerly used for maintenance.
- Railway buildings including:
  - The main repair building used for machining, welding and major mechanical repair of locomotives. The building was also leased by a small boat building companies and a wood pellet company.
  - The roundhouse that was used for short term maintenance on the locomotives; this building contained below-ground maintenance pits, which were reported to have been filled in.
  - Car shop located west of the roundhouse.

3.6.9 Phoenix Environmental 1999 Environmental Impact Assessment and 2002 Sediment Chemistry Investigation

In 1999, Phoenix Environmental Services Ltd. (Phoenix) was retained by the Town of Ladysmith to conduct an environmental impact assessment of the proposed marina and waterfront development project. Details were documented in the report, "*Environmental Impact Assessment Report Proposed Ladysmith Marina and Waterfront Development Project Ladysmith, BC*", dated December 1999 (Phoenix 1999). The project area assessed in this report included the Site, as well as the filled foreshore areas DL 2016 Blocks B, C and D. For the proposed development, an approximate area of 3.8 ha of existing intertidal and subtidal mudflats was proposed to be dredged to a minimum depth of 2 m bgs. Dredged material was anticipated to be deposited offshore at Porlier Pass, in accordance with Environment Canada's Ocean Disposal Limits (ODL).

Historical information provided in the report indicated that the Township of Ladysmith discharged untreated sewage through an outfall into the water lots in the vicinity of the Log Dump in DL 17G. A small sawmill was reported to have been historically located in the northwest corner of the Site near the boat launch adjacent the government wharf[2]. During their investigation, a timber cribbing retaining wall was observed along the shoreline of DL 11 and a small spit was observed "near the north edge at the former location of one of the former mills". The location of the spit was not clearly described in the report. Discarded batteries and other refuse in the water were visible from the small marina located in the centre of the Site. In addition, it was reported that onboard repairs, painting, and other marina related activities could have resulted in sediment contamination around the marina.

As part of their assessment, sediment samples were collected and analysed for metals, organic carbon and AVS/SEM ratio. The purpose of the sampling was to determine sediment quality with respect to the aquatic habitat and to determine the suitability of the sediments for ocean or upland disposal. A total of six samples were collected using a Ponar dredge. Results of the analysis indicated that none of the six samples had concentrations of metals above the standards applicable at the time of the assessment. However, Phoenix indicated that concentrations of cadmium were higher than the ODLs in sediment samples from four locations.

In 2002, additional sediment sampling was conducted by Phoenix on behalf of Water and Land British Columbia Inc. Details were documented in the report entitled, "*Sediment Chemistry Investigation Report Ladysmith Harbour Proposed Waterfront Re-Development Ladysmith, BC*", dated September 2002 (Phoenix 2002). During the sediment investigation, Phoenix identified five areas of potential environmental concern including:

- Former log booming, storage and barge loading areas.
- Former sawmill near to the northwest corner of the Site (Golder infers this area to comprise the former shingle mill on Block B and D of DL 2016 shown on Figure 4).
- Former coal trans-shipment area (Slack Point).
- Former sewage effluent pipe (south of Ladysmith Maritime Society Marina and Wharf).
- Existing small boat moorage.

[2] There has been no other reference to or evidence of a sawmill at this location, and Golder infers this area to be the location of the former shingle mill.

The sediment investigation included the collection of 58 surface sediment samples (40 grab samples, 4 core samples, plus duplicate samples) at 42 locations, which were analyzed for trace metals (55 samples), polycyclic aromatic hydrocarbons (PAHs) (22 samples), PCBs (7 samples), particle size (47 samples), and/or total organic carbon (TOC; 55 samples).

Of the 55 samples analysed for metals in sediment, none had concentrations above the CSR sediment standards applicable at the time of the assessment. Twenty of the samples contained concentrations of cadmium that were greater than the ODL for cadmium.

Of the 22 samples analysed for PAHs, 19 contained concentrations of total PAHs that were greater than the ODL for total PAHs, and 18 contained concentrations of one or more PAHs that were above the CSR sediment standards applicable at the time of the assessment.

Based on the results of the investigation, Phoenix concluded that ocean disposal appeared to be an unlikely candidate for disposal because cadmium concentrations were greater than the applicable criteria, with some concentrations that were more than double the ODLs. In addition, PAH contamination was widespread, with concentrations up to five times the ODLs in some locations. Phoenix also concluded that the source of the metals and/or PAHs in sediment could be attributed to past coal transhipment, log handling activities or sewage disposal, as well as small boat moorage uses.

3.6.10 Levelton Engineering 2000 Stage I and II PSI, DSI, and Remediation Plan

3.6.10.1 Overview

Levelton Engineering Ltd. (Levelton) was retained by the Town of Ladysmith in 2000 to carry out a Stage I and II PSI, a DSI, and a Remediation Plan for the Site. The purpose of the work was to identify potential environmental liabilities and prepare the area for the potential sale of the property. Details were documented in the following reports: "*Preliminary Site Investigation Stage I and Stage II Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District. Ladysmith, BC*", dated 25 August 2000 (Levelton 2000a); "*Detailed Site Investigation Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District. Ladysmith, BC*", dated 25 October 2000 (Levelton 2000b); and, "*Remediation Plan Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District Ladysmith, BC*", dated 1 December 2000 (Levelton 2000c). Note that these reports did not include Lot 5, which is considered as part of the Site for the current investigation.

3.6.10.2 Areas of Potential Environmental Concern

The Stage I identified both on-site and off-site areas of potential environmental concern. Additional information contained in the Stage I PSI is referenced, where applicable, in the current investigation. On-site areas of environmental concern included:

- Fill from copper smelting operations, located at the northwest edge of site.
- Cable splicing shed (locations of historical Site features are shown on Figures 4a and 4b).
- Railway maintenance area.
- Transformer and shop area.

- Former waste oil area.
- Former AST and fuel island area.
- Former scale pit, where polychlorinated biphenyls (PCB) containing transformers were suspected to exist.
- Waterfront area above the high tide mark.

Levelton also used ground penetrating radar to identify a UST by the washroom building (Levelton 2000c); however, Golder infers this to be the sewage pump out facility connected to the Town of Ladysmith's municipal sewage system.

Off-site areas of environmental concern included two gas service stations within 50 metres of the property boundary along the west side of Esplanade Avenue (currently a Petro Canada located at 435 Esplanade Avenue, and a Shell Canada located at 728 Esplanade Avenue).

Levelton also reported that a truck repair and salvage business was located northwest of the Site; however, the business was not considered an area of potential concern.

3.6.10.3 Stage II Test Pit Investigation

The Stage II PSI included the excavation of eleven test pits to a maximum depth of 4 m in areas of suspected contamination around the Site. Soil samples were analysed for metals, benzene, toluene, ethylbenzene and xylene (BTEX), EPH10-19, and EPH19-32, and PCBs. Two sediment samples were also collected using a hand auger and analysed for metals.

One soil sample had a concentration of EPH10-19 that exceeded the CSR CL, RL, and PL standards applicable at the time of the assessment. This test pit was located near the maintenance railway buildings (see Figure 4).

One sediment sample had concentrations of copper and lead that exceeded the CSR AW Level I standard applicable at the time of the assessment. This sample was taken at the base of the Ladysmith Maritime Society Marina and Wharf.

3.6.10.4 DSI Sampling Program

The DSI included the drilling of fourteen boreholes; five of the boreholes were completed as monitoring wells. The DSI also included the collection of four foreshore sediment samples, the collection of four groundwater samples, and the collection of one surface soil sample.

Two soil samples had concentrations of EPH10-19 or EPH19-32 that exceeded the CSR standards applicable at the time of the assessment. These samples were taken from the former waste oil disposal area located at the southeast end of the main shops building.

One groundwater sample, also located near the former waste oil disposal area at the southeast end of the main shops building, had a concentration of EPH10-19 that exceeded the CSR AW standards applicable at the time of the assessment.

3.6.10.5 Remediation Plan

Details of Levelton's Remediation Plan (Levelton 2000c) were not considered relevant to the current investigation.

3.6.11 Westmar Consultants Inc. 2001 Cost Estimate for Redevelopment

In 2001, Westmar Consultants Inc. (Westmar) was retained by the Town of Ladysmith to prepare an order-of-magnitude cost estimate for proposed redevelopment of the Ladysmith harbour. Proposed works included dredging, filing, slope protection, removal of wood waste, soil improvement at Slack Point, and a provision for basic site infrastructure. Details were documented in the report, *"Town of Ladysmith Report for: Waterfront Redevelopment,"* dated June 2001.

The report did not identify areas or activities of environmental concern or potential environmental concern.

3.6.12 Baker & Osland 2002 Property Appraisal

In 2002, Baker & Osland Appraisals Ltd. (Baker & Osland) prepared an appraisal report for the Site, including Slack Point, and the water lots adjacent to Site (not investigated in this report), to determine a total market value for the Site. Details were documented in the report entitled, "P*roperty Appraisal Report, Private and Crown Lands Front and Within Ladysmith Harbour Ladysmith, British Columbia*", dated 4 April 2002.

The report contained photographs, building and area descriptions, sizes and construction details as of 2002 (the Site in the 2002 photographs appeared similar to the current Site configuration). In addition, the report supplied civic addresses for the buildings on Lot 4 including: the main repair shop (610 Oyster Bay Drive), the washroom building (612 Oyster Bay Drive), the roundhouse (614 Oyster Bay Drive), the car shop (616 Oyster Bay Drive) and the cable splicing shed (840 Oyster Bay Drive).

The report did not identify areas or activities of environmental concern or potential environmental concern.

3.6.13 W.R. Colclough & Associated Ltd. 2004 Report Review

In 2004, W.R. Colclough & Associated Ltd. (Colclough) was retained by Land and Water British Columbia Inc. (now Integrated Land Management Bureau) to review existing environment reports relating to Ladysmith Harbour, and to identify areas where additional investigation was required. Details were documented in the report entitled, *"Review of Existing Environmental Reports Relating to Ladysmith Harbour and Uplands Proposed Waterfront Re-Development Ladysmith, British Columbia,"* dated November 2004.

The report reviewed 16 reports that were also reviewed by Golder. In addition, the Colclough review also included a review of a 1990 report entitled *"Yearly Mine Reports for Ore and Coal Processing,"* prepared for by the Ministry of Energy, Mines and Petroleum Resources, which provided information relating to the Tyee copper smelter. The report indicated that the smelter operated from 1902 to 1913 and that copper ore was received from various mines operating on the west coast and as far away as Mexico.

Based on their review, Colclough summarised that:

- The water lots adjacent to the Site contain coal and wood wastes on the bottom.

- Contamination levels vary across the harbour.
- Some samples indicated levels of cadmium exceeding the ODLs.
- Some samples indicated levels of total PAHs exceeding the ODLs.
- Cadmium and PAH contamination may not be bioavailable.
- Lot 16G and Lot 17G (Slack Point; considered off-Site for the current investigation) are contaminated.

3.6.14 EBA Engineering Consultants Ltd. 2005 Stage 1 PSI

In 2005, EBA conducted a Stage 1 PSI for Lot 5, Plan 45800, DL 24 and 56, of Ladysmith, BC on behalf of Land and Water British Columbia (now Integrated Land Management Bureau). Details of the investigation were documented in "Stage 1 Preliminary Site Investigation Lot 5, Plan 45800, DL 24 and 56, Ladysmith, BC", dated February 2005.

The investigation identified two areas of potential environmental concern on adjacent properties as follows:

- The former pump island and railway maintenance building.
- Coal fill on Slack Point.

No areas of potential environmental concern were identified on Lot 5.

3.6.15 Golder Associates Ltd. 2005 Stage 1 PSI and DSI

In 2005, Golder was retained by the Crown Contaminated Sites Branch to conduct a Supplemental PSI and a DSI on Slack Point. Details were documented in the report entitled, *"Supplemental Stage 1 Preliminary Site Investigation and Detailed Site Investigation Lot 16G Ladysmith Harbour Ladysmith, BC,"* dated 19 July 2005. This report only pertains to Slack Point, which is considered off-site for this investigation.

The DSI investigated six APECs on Slack Point identified during the Supplemental PSI. The APECs included:

- Coal Fill.
- Surficial wood waste fill.
- Buried refuse and possible landfills.
- Former boat repair shop.
- Former wood waste and dredgeate stockpile (that was present on the site between 1984 and 1994).
- A stockpile of imported material of unknown quality observed during the 2005 site reconnaissance (not observed during the Site reconnaissance associated with this current report).

The investigation consisted of a geophysical survey, test pitting, borehole drilling and subsequent groundwater monitoring well installation, soil and groundwater sampling, seepage water sampling, and soil vapour sampling.

Following the investigation of the soil, groundwater, and seepage water at Slack Point, three of the six APECs were considered Areas of Environmental Concern (AECs). These included:

- **Coal Fill** – Hydrogen sulphide odours were observed in the coal fill at depths below 12 metres at Slack Point, well below the surface of the groundwater table (generally 2 m to 3 m below ground surface). Light Extractable Petroleum Hydrocarbons (LEPH), Heavy Extractable Petroleum Hydrocarbons (HEPH), and PAH in soils were present at concentrations greater than the CSR RL/PL standards (applicable at the time of the assessment) throughout Slack Point. Elevated concentrations of sulphur were present in the coal fill. Groundwater samples collected during the DSI did not contain concentrations of potential contaminants of concern (PCOCs) at concentrations greater than the CSR AW standards. Of the non-regulated parameters, elevated concentrations of dissolved calcium, iron, manganese, magnesium and sodium were detected at several monitoring wells at the Site. These concentrations were suspected to be caused by leachate or saltwater intrusion (i.e., sea water). No other indications of leachate from the coal (i.e., low pH, elevated metal concentrations) were measured on Slack Point. A seepage water sample collected from the western foreshore area (near the former boat yard) indicated concentrations of LEPHw and a numerous PAHs that were greater than one-tenth of the CSR AW standards. It was suspected that the elevated concentrations of these parameters were a result of suspended coal particles (sediment) in the water, suggesting that PAHs associated with the coal may be transported via surface water and other modes of particulate transport.
- **Surficial wood waste fill** – LEPH, HEPH and metals were detected at concentrations greater than CSR RL/PL standards, and concentrations of zinc were greater than the CSR CL/IL standard. Of the hydrocarbons, HEPH was thought to be naturally occurring in wood that may have been entrained in the samples. The volume of the surface fill materials at Slack Point was estimated to be between about 34,000 m^3 and 67,000 m^3. The contamination identified appeared to be related to the quality of materials used as surface fill at the Site. Some of these materials may have included sediments historically dredged from Ladysmith Harbour.
- Buried **refuse and possible land filling** – The investigation identified areas containing metal debris and refuse along the northwest portion of the Slack Point. Concentrations of PCOCs in groundwater were below the CSR AW standards in this area; however, the area contained concentrations of LEPH, HEPH and metals above the CSR CL/IL and/or the CSR RL/PL soil standards applicable at the time of assessment.

The other APECs (including the former boat repair shop) identified in the supplemental PSI were investigated and no concentrations of PCOCs were found that exceeded the applicable standards. These APECS were not considered AECs for Slack Point and are not considered AECs for this current report.

During the investigation, two surface water samples were collected and analysed from a drainage ditch located on the eastern perimeter of Slack Point. It is inferred that the ditch carries storm water to the Harbour and that the water in the ditch is not associated with Slack Point. As such, the chemistry results for water samples collected from the ditch in 2005 have not been included in the current investigation.

3.6.16 Golder Associates Ltd. 2011 Supplemental Stage 1 PSI and DSI

Golder was retained by the Crown Lands Opportunities and Restoration Branch (CLORB) of the Ministry of Forests, Lands and Natural Resource Operations (MFLNRO) to conduct a Stage 1 PSI, DSI, sediment investigation and preliminary geotechnical investigation at Ladysmith Harbour. The primary objectives of the investigation activities were to refine remedial and geotechnical costs and options for the Site. This report presented the results of the Stage 1 PSI and DSI. Results of the sediment and preliminary geotechnical investigations were presented under separate covers. This report considered the current Site, as well as Slack Point and two water lots within Ladysmith Harbour (considered off-Site for the current investigation).

3.6.16.1 Stage 1 PSI

The Stage 1 PSI scope of work included a review of available historical reports. During the Stage 1 PSI, 21 on-site and 4 off-site likely APECs or AECs were identified, including the following:

Table 3: Summary of APECs and AECs identified as part of the 2011 Golder Stage 1 PSI

Information Obtained by Historic Review and Stage 1 PSI
Slack Point
Coal Fill at Slack Point.
Surficial Fill from Non-Coal sources at Slack Point.
Former Wood Waste and Dredgeate Stockpile.
Former Boat Repair Shop on Slack Point.
Buried Refuse and Possible Abandoned Landfill.
Stockpiles of Imported Sand and Gravel.
Former Buildings Associated with Logging Activities.
Upland Areas
Fill Material in the Uplands (Lot 1 and Lot 4).
Former Scale Pit and Possible PCB Storage (Lot 4).
Former Pump Islands, ASTs, Pump House and Possible PCB Storage (Lot 1).
Historic Fuel Pipelines from Uplands to the Harbour (Lot 1 and Lot 5).
Former Maintenance Area and Current Boat Repair and Construction Operations (Lot 4).
Former Waste Oil Storage Area and Compressor Storage Location (Lot 4).
Former Location of Oil Drum, Scrap Metal Storage, and Stockpiles of Unknown Quality (Lot 4).
Former Cable Splicing Shed (Lot 4).
Suspect UST Adjacent the Washroom Building (Lot 4).

Information Obtained by Historic Review and Stage 1 PSI
Filled Foreshore
Fill Material at Block B and D of DL 2016 (former Location of the Shingle Mill).
Small Sawmill.
Former Log Dump (DL17G).
Former Location of ASTs at the Foreshore.
Sediments
Foreshore Sediments – Quality of Sediment, Marina Activities, Sewage Outfall and Pressure Treated Piles.
Off-Site
Off-Site Service Stations at 435 and 728 Esplanade.
Copper Smelter.
Burleith Log Sort Facility.
Iron Foundry.

At the conclusion of the Stage 1 PSI, some of the APECs were eliminated based on historical information available, and as such, were not retained for further investigation. A total of eight AECs and nine APECS were retained for further investigation and or delineation. A scope of work was developed for the DSI to investigate and delineate the retained APECs and/or AECs. Nine of the APECs/AECs identified in this investigation are located in the Uplands area, which corresponds to the Site currently under investigation. Furthermore, several APECs/AECs are located adjacent to the current Site, and have the potential to impact the Site. Slack Point and the Former Log Dump (Filled Foreshore) are located adjacent to Lot 5; APECs/AECs identified in those areas could potentially cause impacts on Lot 5. Other areas in the Filled Foreshore (such as the former shingle mill and sawmill) are adjacent to Lot 4. Similarly, APECs/AECs identified in the Filled Foreshore areas could impact the Site currently under investigation.

3.6.16.2 DSI

The objectives for the DSI were to:

i) Assess soil, soil vapour, and groundwater quality in each of the APECs identified at the Site.

ii) Determine if the APECs should be considered areas of environmental concern (AEC).

iii) Characterize and delineate the extent of soil and/or groundwater contamination at the Site associated with the AECs.

iv) Provide supporting information for a risk assessment and remedial plan for the Site. The objective of the sediment investigation was to obtain data to refine remedial costs and options, including an assessment of whether ocean disposal is a viable option, for potential developers of the Site.

The remedial costs and options were presented under separate cover.

The DSI activities were conducted between November 2009 and February 2011 and included surface soil sampling, test pitting, drilling of boreholes and installation of monitoring wells, and soil and groundwater sampling.

Following the completion of the DSI, the following APECs/AEC, and their respective contamination of concern, were identified. Note that only the “Uplands” is located on the current Site, and the remaining areas are located adjacent to Site.

Table 4: Summary of AECs and APECs identified as part of the 2011 Golder DSI

AEC or APEC	Summary	Contaminant of Concern (COC)
Slack Point		
Coal Fill.	Coal fill material contains concentrations of PAHs, LEPH, and HEPH at concentrations above PL/RL standards.	Soil: naphthalene LEPH and HEPH Soil vapour: naphthalene
Surface Fill.	Samples collected form surface fill (non-coal) contained concentrations of LEPH, HEPH and metals exceeding the CSR PL/RL standards and zinc concentrations exceeding the CL/IL standards.	Soil: LEPH, HEPH, metals
Buried Refuse.	Samples collected from the area of the buried refuse contained concentrations of LEPH, HEPH and metals exceeding the CSR CL/IL standard and/or the CSR RL/PL standards.	Soil: LEPH, HEPH, metals
Uplands		
Former fuel pump islands, ASTs and possible PCB Storage.	Volatile Petroleum Hydrocarbon (VPH) contamination exceeding the CSR CL/IL standard was identified in the southern areas of AEC 10. Xylene and naphthalene were detected in soils but were below standards. In the northern area of AEC 10, surficial soil samples contained HEPH concentrations over the CSR RL/PL or CL/IL standards. The HEPH appears to be limited to a surficial area adjacent to a concrete slab; and is not adjoining the VPH contamination located in the southern area of AEC 10. One groundwater sample, in the southern area of AEC 10, contained VPH concentrations exceeding the CSR standards. While the groundwater contamination has not been delineated horizontally or vertically, groundwater contamination is expected to be limited to the area of soil contamination.	Soil: VPH, HEPH Groundwater: VPH Soil Vapour: VPHv, xylene, naphthalene
Former Maintenance Area and Current Boat Repair and Construction Operations.	LEPH contamination was observed in the area of AEC 12. Contamination appeared to be patchy, and not associated with one contiguous source/activity. In addition, styrene was observed at concentrations exceeding the PL/RL standards. LEPH contamination was historically observed in groundwater; however, subsequent sampling did not indicate the presence of groundwater contamination.	Soil: LEPH, styrene
Former Waste Oil Storage Area and Compressor Storage Area.	Near-surface soil contamination of LEPH and HEPH concentrations exceeding the CSR PL/RL standards. LEPH contamination was historically observed in the groundwater; however, subsequent sampling did not indicate the presence of groundwater contamination.	Soil: LEPH

AEC or APEC	Summary	Contaminant of Concern (COC)
Filled Foreshore		
Fill Material at Block B and D of DL 2016 (former Location of the Shingle Mill), Small Sawmill, Tyee Copper Smelter, and Iron Foundry	No soil or groundwater contamination identified during the DSI, as such, this area is not considered an APEC. However, the sampling program was limited in area, and additional sample collection along the filled area may be warranted, therefore, remains an APEC.	Potential COC: Soil: LEPH/HEPH, PAH, and metals Groundwater: LEPH/HEPH, PAH, metals Soil Vapour: VPHv, BTEX, naphthalene
Former Log Dump (Lot 17G).	Presence of NAPL was observed during drilling boreholes along the shoreline in the vicinity of the former log dump. VPH, LEPH and HEPH concentrations exceeding CSR CL/IL standards in soil, and PAH constituents exceeding. NAPL was not observed in monitoring wells during monitoring events. Contamination appears to be limited to a particular range of depth and may be associated with the log dump or filling activities. Contamination in this area does not appear to be related to migration of contamination from areas where railway maintenance activities occurred.	Soil: VPH, LEPH Groundwater: PAH Soil Vapour: VPHv
Sediments		
– Foreshore sediments.	Reported under separate cover.	

The COCs identified in these tables are based on the standards applicable at the time of the 2011 assessments. Historical data associated with the Site has been rescreened to current CSR standards, and is summarized in Section 8.0.

4.0 REGULATORY INFORMATION REVIEW

4.1 BC ENV Contaminated Site Registry

An on-line search of the BC ENV Site Registry for registered properties within 500 m of the centre of the Site was conducted on 10 December 2019 by Golder. Twelve Site IDs were found within 500 m from the Site boundaries and are listed in Table 5 below. The results of the search are presented in Appendix C.

Table 5: BC Site Registry Summary

Property ID#	Site Address	Description	Location Relative to the Site	Status
1700	Slack Point	Last updated: 13 October 2005. The details listed the entire Site as well as Slack Point (not assessed during this investigation).	Includes the Site area and areas adjacent to the East of the Site, and downgradient of Site.	Details indicated that the Site was Active and Under Assessment and included multiple suspected land uses: coal, dry docks, landfill, petroleum storage, bulk freight handling, and logging activities.
3034	209 Symonds Street	Last updated 7 October 1997. Appears to be a residential property on Google Street View.	360m southwest of Site, across the highway, up-to cross-gradient of Site.	Details indicate that the site was used for petroleum production, produce water storage, and had an aboveground or underground tank. On 11 July 1995, the site was remediated to commercial/industrial levels.
5352	Foot of Oyster Cove Road	Last updated: 14 March 2001.	Unclear: point on map places it adjacent to the northwest. Address shows the property is approximately 850 m southeast of and cross gradient to the Site.	Details suggested that a Phase I and 2 Environmental Site Assessment was performed on a harbour or port in 1997 and 1999 respectively, initiated by Transport Canada. It appears that the reports were submitted for external review in 1998. Currently registered inactive and no further action is required.
5775	840 First Avenue	Last updated: 17 March 2000 (currently occupied by a retail business adjacent a Big O Tire Shop).	Approximately 200 m southwest and upgradient of the Site.	The details did not indicate previous activities at the Site or the reason for the Site profile. A notice of independent remediation was filed in 1999. Currently registered inactive and no further action is required.
6836	728 Esplanade Avenue	Last updated: 10 August 2015 (currently occupied by a Shell Service Station).	Approximately 50 m southwest and upgradient of the Site.	A notice of independent remediation was filed in 2000. The most recent update lists that a Site Risk Classification Report is required. The Site is listed as active and under remediation.
9349	Forward Road (at the intersection of Dogwood Drive)	Last updated: 2 June 2005. Currently occupied by an automotive repair shop “Dalby’s Service”	235 m southwest and upgradient of Site, across the highway	A spill was reported on 20 January 2005. No other updates are available.

Property ID#	Site Address	Description	Location Relative to the Site	Status
11634	1111 First Avenue	Last updated: 23 April 2015. Currently occupied by a Save On Gas petrol station	230 m west of Site, across the highway, up-to cross-gradient of Site.	On 21 August 2009, a notice of independent remediation completion was submitted.
12891	1030 Oyster Bay Drive	Last updated: 3 March 2017. Currently occupied by Ladysmith Motorsports	260 m northwest and cross-gradient of Site	The most recent notations on file date to November 2011, when an independent remediation was conducted as part of a development permit application.
21424	422 First Avenue	Last updated: 15 March 2018. Based on an August 2014 Google Earth image, it was occupied by the Travellers Hotel, and for sale at the time.	250m southwest and upgradient of Site, across the highway	A notice of independent remediation was completed on 4 March 2018, and the site was classified as non-high risk. No information regarding the site use is available.
21774	941 Oyster Bay Drive and 930 Ludlow Road	Last updated: 24 June 2019. 930 Ludlow Road is currently occupied by "Wash me on Ludlow". 941 Oyster Bay Drive is occupied by multiple businesses including an overhead crane business and bottle depot.	Adjacent to the northwest of Site, cross-gradient from Site.	Notations on the file indicate the site was classified as non-high risk in June 2018. A release letter was granted July 2018 for development; however, conditions are noted such that an annual statement must be submitted by an approved professional, and that a remediation must be completed within five years.
22253	205 Bayview Avenue	Last updated: 22 March 2019. Based on a 2014 Google Earth Image, the site was previously used as a martial arts school.	300m south of Site, up-to cross-gradient from Site.	A notice of independent remediation was complete in March 2019. A notation on the file indicates it was for a commercial underground storage tank.
3687	610 First Avenue	Last updated: 30 December 2003 (currently occupied by a commercial business including a Canada Post Outlet).	Approximately 200 m southwest of and upgradient from the Site.	Described as "Tombstone data only for site registry" – Federal. Land. Currently registered inactive and no further action was required.

Site ID 1700 included the site and much of the surrounding area. Based on the review of the location of the registered sites, two properties were located within 100 metres of the Site and may have the potential to impact site soil or groundwater. The remainder of the 9 registered sites were located at distances greater than 200 metres and not considered to be of potential environmental concern. The three registered sites (ID 1700, 6836, 21774) are further discussed below.

4.1.1 Property ID 1700

Property ID #1700 is located at Slack Point, adjacent to the east of the Site. The property status remains active and under assessment. Suspected land uses listed in the details were coal, dry docks, landfill, petroleum storage, bulk freight handling, and logging practices. These indicate several CSR Schedule 2 activities. It is not known what triggered the Site Profile for the property; however, after the logging activities had ceased in the area, several investigations were performed, the likely basis for the site profile.

4.1.2 Property ID 6836

Property ID #6836 is currently a Shell Service Station. This property was listed as active and under remediation. The property is located approximately 50 metres upgradient from the Site, and therefore may have the potential to impact the quality of the soil and/or groundwater at the Site.

4.1.3 Property ID 21774

Property ID #21774 is listed with two different addresses, for 930 Ludlow Road, and 941 Oyster Bay Drive. 930 Ludlow Road is currently occupied by “Wash me on Ludlow”. 941 Oyster Bay Drive is occupied by multiple businesses including an overhead crane business and bottle depot. Property uses are listed as vehicle repair/salvage/wrecking, and construction/demolition of material including concrete and asphalt. Notations on the file indicate the property was classified as non-high risk in June 2018. A release letter was granted July 2018 for development; however, conditions are noted such that an annual statement must be submitted by an approved professional, and that a remediation must be completed within five years. The registry lists contaminated fill an issue for this property. The property is located adjacent to the northwest of the Site, and therefore may have the potential to impact the quality of the soil and/or groundwater at the Site.

4.2 Federal Contaminated Sites Inventory

An on-line search of the Federal Contaminated Sites Inventory (FCSI), on the Site and surrounding area, was conducted on 10 December 2019 by Golder. A summary of the search results is provided in Appendix D.

Three Federal Real Property (FRP) locations were found adjacent to the Site, with six associated FSCI Identifier number (or “Contaminated Sites”). The Identifier numbers indicated a Site Name, Contaminant Type, and Media Type. The details provided are summarised in Table 8, below.

Table 8: Details from FSCI Database for the surrounding area

FSCI Identifier	Site Name	Contaminant Type	Media Type	Site Status
00020503	Ladysmith Boat Basin No. 1	N/A	N/A	Closed: Historical review not required.
00021349	Ladysmith (Waste Oil ASTs)	PAHs, PHCs (petroleum hydrocarbons)	Sediment	Active: Initial testing completed. Detailed testing underway.

FSCI Identifier	Site Name	Contaminant Type	Media Type	Site Status
00021350	Ladysmith (Nearshore Sediment and surface water)	PAHs, PHCs	Sediment	Suspected: Historical review completed. Initial testing underway.
00021352	Ladysmith (Boat Grid)	Metal metalloid organometallic PAHs PHCs	Surface water	Closed: Initial testing completed. No Further action required.
00021353	Ladysmith (Fill Material)	Metal metalloid organometallic	Soil	Closed: Initial testing completed. No Further action required.
00021354	Ladysmith (Stormwater Discharge)	Metal metalloid organometallic	Sediment	Closed: Initial testing completed. No Further action required.

The inventory indicated several CSR Schedule 2 activities associated with these sites. Exact locations of the activities were not identified on the FCSI; however, the majority of the activities have been identified during the previously described historical investigations.

4.3 Town of Ladysmith

4.3.1 Historic Town of Ladysmith Map and Photographs

A copy of a 1902 Town of Ladysmith map was obtained from the archives (Ladysmith Archives 1902) and is included in Appendix E. The map illustrates the Site as of 1902.

On-site, a washer and scales are observed, which appear to be associated with the operations off-site on Slack Point. Railway tracks and a depot are also observed. In the northwestern portion of the Site, a shingle mill extends off-Site near the present-day Government wharf.

Off-Site, it appears that Slack Point does not extend as far into the Harbour as in current conditions. There are three wharfs visible in the area of Slack Point including: the loading wharf, the transfer wharf, and a T-shaped wharf. The area below the T-shaped wharf is labelled “Dirt Dump". To the northwest of the shingle mill is the Tyee Copper Company smelter and foundry, located off-Site and cross gradient to the Site.

4.3.2 Placement of Material from Sewage Treatment Plant

Following a kick-off call for this project, Jake Belobaba from the Town of Ladysmith raised a question regarding potential disposal of material from the sewage treatment plant on Lot 4.

Geoff Goodall, the Director of Infrastructure Service for the Town of Ladysmith, was contacted and he explained that the material taken to the site was native material that was removed from the ground at the sewage treatment plant site, to facilitate expansion of the plant. The material was not known to be contaminated, and he attached a figure showing the locations where the material was placed.

A copy of this communication is provided in Appendix E.

5.0 INTERVIEW AND SITE VISIT

5.1 Interview

Golder conducted an interview and a Site visit with Mr. Quentin Goodbody, the President of the Ladysmith and District Historical Society, on 16 December 2019. Mr. Goodbody has volunteered at the Site for two years. During the interview, Golder obtained information regarding the past land use at, and near, the Site. The following information presented in Table 6 was obtained from the interview.

Table 6: Past Uses of the Site and Neighbouring Properties

Reported Use	Details
Dry cleaner	None reported on-site or surrounding properties.
Industrial metal finishing, including painting or electroplating	Metal finishing and electroplating was not reported to have occurred on-site or surrounding properties; however, the Site Representative reported that some painting of trains in the main repair building was likely to have occurred.
Other industry	The Site Representative reported that logging activities and a railway yard were present at the Site. Tyee Copper Smelter was historically located approximately 400 m to the north of the Site.
Fuel storage	The Site Representative reported that two ASTs were historically present to the southeast of the of the main repair building. Fuel was gravity-fed from the ASTs downgradient to the southeast to the area of the former log dump for fuelling activities. The approximately location of these ASTs is shown on Figure 4.
Retail fuel outlet or vehicle service garage	Retail fuel outlets or service garages were not present on-site; however, fuelling activities took place on-site as reported above and industrial vehicle and locomotive repair and maintenance took place on-site in the main repair building. Retail fuel outlets are currently located off-site and within 100 m of the Site.
Landfilling or placement of fill	The Site Representative noted that fill material was placed along the west boundary of the Site in the 1980s during upgrades to the highway.
Wastewater impoundments	None reported.
Solid or liquid waste storage or disposal	None reported.
Environmental sampling, wells, or evidence of drilling	Previous environmental investigations have been conducted at the Site, as described in Section 3.6 and presented in Section 8.0.
Other activities that may have affected the environmental condition of the Site or neighbouring properties	None reported.

5.2 Site Visit

During the Site visit, the Site Assessor walked through and observed accessible areas of the interior and exterior of the Site, observed neighbouring properties, interviewed the Site Representative, and photographed representative Site features (Appendix F). The following table provides an overview of the Site visit (Table 7).

Table 7: Details of Site Visit

Item	Information
Weather	Overcast, raining
Restrictions on access or photography	Golder did not enter the first aid building and washroom building in the railway yard area on-site.

5.2.1 Present Site Uses

The Site includes Lot 1, Lot 4, and Lot 5 near the Ladysmith Harbour. There are currently six buildings on-site including the former railway repair building, washroom, car shop, roundhouse, first aid shed, and cable splicing shed. The locations of the buildings are shown on Figure 4.

The former railway repair building is approximately 62 m long by 23 m wide and 12 m high, with galvanized iron reinforced walls. This building was historically used for logging truck maintenance at the southeast end of the building and steam and diesel locomotive repair at the northwest end. The Site representative noted that during the time the building was used for logging truck and steam/diesel locomotive repair, there were pits present in the floor of the building for accessing the underside of trucks and locomotives. The front (east) portion of the building was partitioned into several compartments and storage. Until late 2019, the building was fronted by the Maritime Society office building, Liquid Gold Art Studios, and the Arts Council of Ladysmith and District. The rear of the building (or the northwest end) was occupied by two tenant operations including Atlantis Kayaks and Southwood Products. During the Site Visit, the building was vacant, and the concrete floor of the building was cut to excavate and install new footings for the building.

According to the Site representatives, a compressor house was historically located immediately adjacent to the southwest side of the former railway repair building. The location of the former compressor house is shown on Figure 4. The compressor was used for the air lines associated with the locomotives. During Golder's Site Visit for the 2011 Supplemental Stage 1 PSI, the Site representatives commented that the compressor leaked on a steady basis; they were unsure when the compressor had been removed. In addition to the compressor, a waste oil tank car was stored in this area. Oil from locomotive and truck maintenance was generally stored in the waste oil tank car and used by surrounding municipalities to irrigate the roads to keep the dust down. During the 2011 Site visit, the Site representatives mentioned that oil was frequently discarded onto the ground as opposed to disposal in the waste oil tank car. This area also contained buckets of pyrene used for washing metal parts. The use of the waste oil area likely ceased when the logging operations halted in the mid 1980s. Currently this area is paved with asphalt and concrete stairs are located on the slope to the west.

According to the Site representatives interviewed by Golder in 2011, the washroom building was historically used as the electrical shop for the logging activities and railway (Golder 2011). The building currently consists of two washrooms, showers and a changing area. The building was not entered during the Site visit.

The roundhouse building is approximately 23 m by 8 m and is similar in construction to the main repair shop. It is currently used by the Ladysmith Maritime Society for restoration of a locomotive. Several 4 litre cans of paint and 20 litre pails of grease were stored on workbenches inside the building. According to the Site representative, locomotive maintenance was historically performed in this area (i.e., oil changes, small engine repair). The area was historically used by the logging industry to store the locomotives. A rail spur was visible in the building. During the 2011 Site Visit conducted by Golder for the Supplemental Stage 1 PSI, the Site representative noted that there was a maintenance pit located under plywood between the rail spur and the pit appeared to have been infilled and covered over.

The former car shop is approximately 25 m by 12 m and is located southwest of the roundhouse. It is currently used by the Ladysmith Maritime Society for vessel restoration products and storage. Historically, the car shop was used as a repair facility for train wheels and axles.

The first aid building is located north of the roundhouse and is generally used as a storage shed.

Two smaller sheds were observed in the vicinity of the car shops and were used for storage.

The former cable splicing shed is located in the northwest portion of Lot 4. According to the Site representatives, the building has been used for wood and galvanized steel siding construction, wood manufacturing, storage of explosives and streetlights, and used for cable splicing. The building is not currently in use.

A railyard area is located adjacent to the north of the Site buildings, it is currently used as a display area for several railcars and historical equipment. The Site representative noted that the area to the west of the rail spurs in the railyard area was historically used as a boneyard for derelict equipment. This boneyard is identified on Figure 4.

5.2.2 Site Buildings and Equipment

At the time of the Site visit, the Site was developed with six buildings as described in Section 5.2.1.

Table 8: Site Building(s) and Equipment

Topic	Findings and Assessor Comments
Below-ground floors and uses	None.
Frequently occupied rooms in contact with the ground	The former railway repair building, car shop and roundhouse are frequently occupied, and the floor of the building is in contact with the ground. Each of these three buildings has a concrete floor.
Number of tenant units	The Ladysmith Maritime Society is currently occupying the Site buildings.
Building exterior	Former railway repair building: corrugated steel siding, glass windows with wooden frames, roof material unknown. Car Shop: painted wood and corrugated steel siding, glass windows with wooden frames, roof material unknown. Roundhouse and Washroom: painted wood siding, glass windows with wooden frames, roof material unknown. First aid shed: painted wood siding, glass windows with wooden frames, metal roof. Cable splicing shed: corrugated steel siding, metal roof.

Topic	Findings and Assessor Comments
Building interior	Former railway repair building: concrete floor, wood and drywall interior walls, wood ceiling. Car Shop, washroom, first aid shed, cable splicing shed: did not enter interior of these buildings. Roundhouse: wood walls and ceiling, concrete floor.
Heating/Cooling system	At the time of the Site visit there were no heating or cooling systems for the on-site buildings.
Backup power supply	At the time of the Site visit there was no backup power supply on-site.
Potable water supply	Municipal water.
Other services	The Site is serviced with electricity from the municipal grid. The wastewater is pumped on-site into the municipal sewer line.
Hydraulic lift equipment	No hydraulic lift equipment was noted on-site.
Other mechanical equipment	No other mechanical equipment was noted during the Site visit.
X-ray equipment	None

5.2.3 Pesticides and Fertilizers

Generally, the use of herbicides and pesticides is a practise around railways and in railway yards. During the Site visit a representative from the Town of Ladysmith working on the footing installations in the former railway repair building noted that soil underneath the office rooms on the east side of the building had been treated with pesticides for termites.

5.2.4 Storage Tanks

At the time of the Site reconnaissance, USTs and ASTs were not observed on the Site. Previous reports indicated that trains and boats were fuelled using a pump island located approximately 100 m to the southeast of the former railway repair building. The pump island was reported to be connected to above-ground tanks. Fuel was gravity fed down to the shoreline via pipes and used to fuel boats, locomotives and trucks. A structure was observed in this area on aerial photographs, between 1962 and 1998. According to the Site representatives' interview in Golder 2011, a pump house/shed cantilevered over the cliff to the northeast and was held up by stilts. In Golder 2011, it was noted that the Site representatives said that PCB storage may have been located in this area; however, the contents, locations and size were unknown. This area is currently vacant and covered by gravel.

5.2.5 Chemical Storage

At the time of the Site reconnaissance, chemical products were observed in the roundhouse building. Chemical storage was generally limited to several 4 litre cans of paint and 20 L plastic buckets of grease. The containers were all closed and stored on shelves or on the concrete floor of the building. Staining or cracks in the concrete floor were not observed in the roundhouse during the Site visit; however, much of the floor was not visible beneath the locomotive being restored in the building.

5.2.6 Waste Generation and Handling

One garbage bin was observed adjacent to the former railway repair building for construction work within the building. No other garbage bins were located on-site. According to Golder 2011, municipal garbage collection is not available at the Ladysmith Harbour and when the Site buildings were occupied by other tenants, bins were present on-site for collection of garbage which were picked up on an as-needed basis.

Coal sorting bins were historically stored along the side of the road southeast of the main repair shop. The number of bins was unknown, and these bins were not observed during the Site reconnaissance (Golder 2011).

5.2.7 Surrounding Properties

During the Site reconnaissance, surrounding properties were observed to be commercial, industrial and parkland. The following summarizes land use in the area of the Site:

Southeast (cross-gradient of the Site): Southeast of the Site is Transfer Beach Park. The park includes a playground, basketball and beach volleyball courts, a washroom building, and a building occupied by a kayak rental company.

Southwest (upgradient of the Site): Immediately southwest of the Site is the Trans Canada Highway, followed by a mixture of commercial businesses, residential homes and parkland. Three service stations (a Save On Gas at 1111 1st Avenue, a Petro Canada at 434 Esplanade Avenue, and a Shell at 728 Esplanade Avenue) were observed to be southwest of the Site.

North/northwest (cross-gradient of the Site): To the northwest are industrial activities, including the Government wharf to the north and the Western Forest Products sawmill and sorting facility beyond that, followed by the Town of Ladysmith. A Co-Op Cardlock service station and the “Wash me on Ludlow” carwash is located adjacent to the northwest of and cross gradient to the Site.

Northeast: The area to the northeast generally comprises open water passage and Ladysmith Harbour followed by Burleith Arm.

6.0 SPECIAL ATTENTION ITEMS

Information about Special Attention Items was collected during the Site visit and the interview with the Site Representative. The TOL noted that in 2019 a hazardous building material assessment was conducted for the former railway repair building and that lead paint was noted near stairs within the building and no other hazardous building materials (HBMs) were identified. Golder did not review the report as part of this PSI and other HBMs could be present in other buildings on the Site. The following section provide an overview of potential special attention items that may be present on-site.

6.1 Polychlorinated Biphenyls

Polychlorinated biphenyls (“PCBs”) are a group of organic chemicals that were widely used in caulking and electrical equipment manufactured between the 1950s and 1980. Caulking potentially containing PCBs can found in any building constructed during this period. Electrical equipment potentially containing PCBs is restricted to transformers, capacitors, heat transfer equipment, hydraulic equipment, electromagnets and vapour diffusion pumps manufactured prior to September 1977 and in lamp ballasts manufactured prior to July 1980. The PCB regulations of the Canadian Environmental Protection Act (SOR/2008-273) require that equipment containing more than 50 mg/kg PCBs should be decommissioned, with PCB use to be eliminated by 2025.

Based on the reported age of the Site buildings (built in approximately 1935 [Golder 2011]), PCB-containing electrical caulking and equipment may be present on-site. Fluorescent lights were observed in the roundhouse building; however, lamp ballasts for these fluorescent light fixtures were not inspected. The Site Representative was unaware whether the lamp ballasts contained PCBs. In addition, it was reported that PCB may have been stored near the storage tanks, refer to Section 5.2.4.

6.2 Asbestos-Containing Materials

The use of asbestos-containing building materials (“ACMs”), including both friable and non-friable asbestos, in building construction materials significant declined on a voluntarily basis in the mid-1970s. The use of materials containing friable asbestos in Canada was effectively discontinued by 1986 as a result of strict provincial regulation. Typical examples of friable ACMs include thermal, fire-proofing or acoustical insulating materials and can include deteriorated materials containing non-friable ACM (e.g., Transite™ pipe). Typical examples of non-friable ACMs can include packings, gaskets, sealants, resilient flooring, asphalt roofing, mastics, drywall joint compounds, stuccos, cementitious and Transite™ materials (including drains and downspouts), and Transite™ shingles. Buildings constructed prior to 1986 potentially contain both friable and non-friable ACMs. Buildings constructed after 1986 potentially contain non-friable ACMs; however, as a practical matter, the condition of some non-friable ACMs can deteriorate, releasing asbestos fibres if disturbed.

Based on the age of the Site buildings (built in approximately 1935 [Golder 2011]), building materials may contain ACMs. Building materials were observed to be present that potentially contain ACMs including window putty, and drywall joint compound. The TOL noted that in 2019 a hazardous building material assessment was conducted for the former railway repair building and that asbestos containing material was not present.

6.3 Lead and Lead-Containing Surface Coatings

Paints manufactured prior to 1960 commonly contained significant lead. In 1976, the Canadian Hazardous Products Act restricted the lead content of paints and other surface coatings on furniture, household products, children's products, and exterior and interior surfaces and since that time lead content of paints has continued to decline. Lead-containing surface coatings in good condition are not typically associated with health risks to building occupants; however, unacceptable lead exposures can occur during building renovations, modifications or demolition activities. Other potential sources of lead in buildings include soldered plumbing joints installed prior to 1986 and lead plumbing pipe (used up until 1975). Lead is present in leaded glass and other type of radiation shielding that are used where radiation sources are present (e.g., medical and dental clinics).

Based on the reported age of the Site buildings (built in approximately 1935 [Golder 2011]), it is likely that interior or exterior surface coatings contain significant lead concentrations. The observed painted surfaces of the Site building were generally noted to be in good condition. The TOL noted that in 2019 a hazardous building material assessment was conducted for the former railway repair building and that lead paint was noted near stairs within the building.

6.4 Ozone-Depleting Substances

Refrigeration and air conditioning equipment in service prior to 1998 may contain chlorofluorocarbon refrigerants that are designated as ozone-depleting substances (ODSs). Non-ODS refrigerants have been developed and are available to replace these materials in newer equipment. Other ODSs include halons, methyl chloroform and carbon tetrachloride. Under the Ozone Depleting Substances Regulations 1998 of the Canadian Environmental Protection Act (SOR/99-07), all ODSs are being phased out of use in Canada.

ODS-containing equipment were not observed or reported to be present at the Site.

6.5 Urea Formaldehyde Foam Insulation

Urea formaldehyde foam insulation (“UFFI”) is low-density foam that was used as an insulating material in the 1970s until it was banned from use in Canada in 1980. UFFI was commonly injected through walls by drilling injection holes, typically in walls, roof structures, ceilings and overhangs.

The Site Representative was not aware of UFFI having been used at the Site.

6.6 Radon and Radioactive Substances

Radon is a radioactive gas formed by the decay of naturally occurring uranium. In 2012 Health Canada released a major study demonstrating that radon concentrations in 7% of Canadian homes exceed the recommended guideline of 200 Becquerels/m^3. It is likely that similar proportion of commercial and industrial buildings are also impacted by radon at concentrations exceeding this recommended guideline. Although radon concentrations vary significantly across Canada, no geographic area is radon-free, and Health Canada recommends the completion of long-term radon testing to determine radon concentrations within a building.

Radioactive sources are found within a wide range of testing equipment including lasers, x-ray sources, imaging and radiography equipment, industrial gauges (including density gauges and other materials testing equipment), and smoke detectors. Radioactive products include any uranium containing material and medical isotopes.

The Site Representative was unaware if radon testing was conducted at the Site. No radioactive sources were observed or reported to be present at the Site.

6.7 Mercury

Mercury may be present in both mechanical and electrical equipment including thermometers, thermostats, switch gears, barometers, vacuum gauges, gas pressure regulators, electrical switches/relays, batteries and electrolytic manufacturing processes. Small amounts of mercury are present in some fluorescent lights, including mercury vapour, metal halide, and sodium vapour lamps.

During the Site visit, Golder did not observe equipment or products potentially containing mercury.

6.8 Mould or Water Damage

Mould can grow on damp building materials such as ceiling tiles, drywall, carpeting, and areas. Mould growth is commonly associated with water leakage.

The Site Representative stated that mould has not been of concern at the Site. Areas of moisture, water ingress and/or suspected mould growth were not observed or reported to be present at the time of the Site reconnaissance.

6.9 Noise and Vibration

No major or persistent sources of noise and vibration were observed or reported to be present other than that typical of vehicular traffic on the adjacent roadways.

6.10 Non-ionizing Electromagnetic Radiation

Non-ionizing types of electromagnetic radiation include radiofrequency and microwave radiation, which can be associated with tissue damage through heating. Radiofrequency radiation is produced by radio and TV transmitters, induction heaters, and dielectric sealers. Microwave radiation is produced by microwave ovens, parabolic antennas, radar devices, and diathermy applicators. No human health risks are known to be associated with the low-energy electromagnetic radiation from commercial electronics and power transmission lines.

No sources of non-ionizing electromagnetic radiation were observed or reported to be present at the time of the Site reconnaissance.

7.0 SURFICIAL SOIL SAMPLING

During the Site visit on 16 December 2019, construction work was taking place inside the former railway repair building to install new footings. As the former railway repair building was retained as an AEC as part of the DSI undertaken by Golder in 2011, there was a potential for contamination under the building. While the footing excavations were open, Golder collected soil samples from the soil beneath the floor of the former railway repair building. Three soil samples and one duplicate soil sample were collected within the former railway repair building. One sample was collected outside the building to the south, as the TOL representative on-site noted that material would also need to be removed from this area during construction activities. The locations of the soil samples collected are presented on Figure 8.

7.1 Field Methodology

Prior to the collection of the soil samples, tools used to collect the samples were washed with laboratory grade detergent and rinsed with distilled water. Nitrile gloves were worn when handling sampling equipment and samples and were changed between sample locations.

The soil samples were collected using a shovel. The soil conditions were recorded and logged by Golder staff, including soil descriptions, headspace measurements using a photoionization detector (PID), and observations of potential contamination (if encountered). The soil samples collected within the former railway repair building were taken between 0.3 to 0.6 m below the bottom of the concrete floor and the sample collected on the bank to the south of the building was collected at 0.3 to 0.6 m below the ground surface.

Field-screening of soil samples was completed using the dry headspace method, where plastic headspace bags were filled two-thirds full of soil, then sealed, shaken, and left to stand for several minutes. The headspace over the soil was then monitored for the presence of organic vapours using a MiniRAE 2000 photoionization detector (PID) containing a 10.6 eV ultraviolet lamp calibrated to 100 parts per million (“ppm”) isobutylene.

The soil samples were placed in pre-cleaned 125 millilitre (mL) glass jars with Teflon®-lined lids supplied by the laboratory. The samples were submitted for the analysis of metals; extractable petroleum hydrocarbons (EPH); polycyclic aromatic hydrocarbons (PAHs); benzene, toluene, ethylbenzene, and xylenes (BTEX); and methyl t-butyl ether (MTBE). The samples were labelled, registered on a chain-of-custody form, and placed in a cooler with ice packs for transport to the laboratory within the recommended temperatures and hold times.

7.2 Applicable Regulatory Standards

In British Columbia, environmental matters pertaining to contaminated sites generally fall under the jurisdiction of the Ministry of Environment & Climate Change Strategy (ENV), pursuant to the Environmental Management Act (EMA, SBC 2003, Chapter 53 assented to 23 October 2003, updated to 16 May 2019. The key regulation under the EMA that relates to the assessment and remediation of contaminated sites is the Contaminated Sites Regulation (CSR; BC Reg. 375/96, O.C. 1480/96 and M271/2004, as updated [includes amendments up to BC Reg. 13/2019, 24 January 2019]).

The CSR provides numerical standards for the evaluation of soil, groundwater, sediment and soil vapour quality. The CSR numerical standards applied to analytical soil data were based on current and potential future land use including urban park (PL), commercial (CL), and industrial (IL) land use. Referenced standards include

groundwater flow to surface used by marine aquatic life (AW-M), intake of contaminated soil (I), toxicity to invertebrates and plants (T), drinking water (DW), ecological health protection (EH), and human health protection (HH).

7.3 Analytical Results

The soil under the concrete floor of the former railway repair building and on the bank to the south of the building generally consisted of silty sand or silty clay. Wood debris was observed in the soil collected from locations HS19-01 and HS19-05. The organic vapour headspace measurements collected in the field for the soil samples were less than or equal to 0.1 ppm.

ALS of Burnaby, BC performed the chemical analyses of the soil samples. Copies of the analytical reports and the corresponding chain-of-custody form are provided in Appendix G.

The results of the soil sample analyses were compared to applicable BC CSR standards for PL, CL and IL land use and are presented in Table H-1 and H-2 in Appendix H. The concentration of parameters of concern were less than the applicable BC CSR standards in the samples analyzed with the exception of the concentration of copper in sample HS19-04 collected on the bank to the south of the former railway repair building which was greater than the AW-M standard for PL, CL, and IL land use.

7.4 Quality Assurance and Quality Control

The target for QA/QC for a field investigation was a minimum of one replicate sample for every ten samples analyzed for a particular compound (i.e., 10 percent field replicates).

The relative percent difference (RPD), the absolute difference between the two values, divided by the mean of replicate analysis was used to evaluate the sample result variability. Where the concentration of a given parameter is less than five times the method detection limit (MDL), the laboratory results are considered to be less precise and the RPD was not calculated. For parameters with concentrations less than five times the MDL, the difference factor (DF) between the sample and its replicate is calculated. For soils, data quality objectives (DQOs) are an RPD value of less than 30% for metals and less than 50% for organic compounds.

A review of the laboratory quality control reports indicated that the laboratory appears to have met their own standards and internal targets with the exception of a laboratory duplicate result outside the laboratory DQO due to sample heterogeneity. Copies of the laboratory analytical reports including laboratory quality control data are provided in Appendix G.

The soil field QA/QC program consisted of one duplicate soil sample pair collected at location HS19-02. The results of the QA/QC program indicated that the calculated RPDs and DFs were within Golder's DQOs.

Full results of the soil QA/QC program are presented in Table H-3 in Appendix H.

8.0 HISTORICAL DATA RESCREENING

The BC CSR and associated guidance have undergone significant revision since the analytical data was last reviewed at the Site as part of Golder 2011. In order to define the work required at the Site going forward, it is important to consider the Site under the current regulatory framework. Based on Golder's understanding of the Project, Golder has rescreened the available data for Lots 1 and 4 and sampling locations located near to and adjacent to Lot 5 to the current applicable CSR standards. The results of the data rescreening are summarized in Table 9 below. Results are included in Appendix I, and are illustrated on Figures 6a, 6b, 7a, and 7b.

Table 9: Summary of Historical Data Rescreening

AEC or APEC	Summary	COC
On-Site		
Former fuel pump islands, ASTs and possible PCB Storage.	In the southern area, ■ Historic soil concentrations at CPT 1 (Hardy BBT 1999) exceed current CSR standards for benzene, toluene, and xylene. ■ At TP09-02 and TP09-03 (Golder 2011) VPH concentrations in soil exceed the current CSR RL/PL or CL/IL standards. Naphthalene in soil also exceeds CSR RL/PL standards at TP09-03. ■ One groundwater sample (MW09-1; Golder 2011) in the southern part had concentrations of benzene, toluene, and MTBE that exceeded CSR-DW standards. This sample also had VPH concentrations exceeding CSR AW-M standards. In the northern area, ■ Surficial soil samples (SA6 and SA7; Golder 2011) had HEPH concentrations that exceed current CSR RL/PL or CL/IL standards. ■ One groundwater sample and its duplicate (MW09-8; Golder 2011) had concentrations of pyrene, benzo(a)pyrene, and benzo(b)fluoranthene that exceeded CSR standards in 2009. These concentrations decreased after resampling in 2011.	Soil: VPH, HEPH, naphthalene, BTEX Groundwater: VPH, BTEX Soil Vapour: VPH, xylene, naphthalene
Former Maintenance Area and Current Boat Repair and Construction Operations.	Rescreened historical data showed the following: ■ Historic soil concentrations at CPT 2 and 3 (Hardy BBT 1999) exceed current CSR standards for LEPH and VPH. ■ Soil at 3 m bgs in TP6 (Levelton 2002) exceeded the CSR PL/RL and CL/IL standards for barium. ■ LEPH concentrations in soil above CSR CL/IL standards were measured at BH00-08 (0.2 m) and at TP4 (2.0 m and 4.0 m) (Levelton 2002). ■ Shallow soil samples in two test pits (TP09-6 and TP11-07; Golder 2011) had LEPH concentrations exceeding CSR PL/RL and CL/IL standards, as well as HEPH concentrations exceeding CSR PL/RL standards. Exceedances of volatiles in soil including ethylbenzene, xylene, benzene, and styrene of CSR PL/RL or CL/IL standards were also found in the area.	Soil: LEPH, HEPH, BTEX, styrene

AEC or APEC	Summary	COC
	■ Soil samples in the area also had concentrations of zinc (MW09-11) and lead (TP09-05) exceeding CSR PL/RL or CL/IL standards (Golder 2011). ■ In groundwater, MW09-9 and MW09-11 (Golder 2011) had concentrations of manganese that exceeded CSR-DW standards. ■ Downgradient location MW11-02 (Golder 2011) had concentrations of chloroform and arsenic that exceeded CSR standards.	
Former Waste Oil Storage Area and Compressor Storage Area.	■ Historic shallow soil concentrations at BH00-03 (Levelton 2002) were rescreened showing exceedances of CSR PL/RL standards for LEPH and HEPH at 0.2 m bgs. A deeper sample at 0.5m bgs was below CSR PL/RL standards. ■ New soil samples taken as part of the 2011 DSI showed no exceedance of LEPH/HEPH standards. ■ In groundwater, MW11-02 located downgradient had concentrations of arsenic and chloroform exceeding CSR standards.	Soil: LEPH, HEPH
Off-Site		
Fill Material at Block B and D of DL 2016 (former Location of the Shingle Mill), Small Sawmill; Tyee Copper Smelter, and Iron Foundry	■ One soil sample at MW09-05 (Golder 2011) showed concentrations of benzene that exceed the current CSR PL/CL standard. ■ No other soil or groundwater contamination identified during the 2011 Golder DSI.	Potential COC: Soil: LEPH/HEPH, PAH, and metals Groundwater: LEPH/HEPH, PAH, metals Soil Vapour: VPHv, BTEX, naphthalene
Former Log Dump (Lot 17G) located downgradient and adjacent to Lot 5.	In soil, ■ Historic investigations found concentrations of Mineral Oil and Grease (MOG) inferred to be above current CSR RL/PL and CL/IL standards for LEPH, at EBA's Test Pit 14 (EBA 1994). ■ At Test Pit 11 (EBA 1994), concentrations of MOG are inferred to be above current CSR RL/PL standards for LEPH. ■ MW-09-8 (Golder 2011) showed concentrations of VPH, LEPH, and HEPH in exceedance of current CSR RL/PL and CL/IL standards at a depth of 2.9-3.0 m bgs. Concentrations of these parameters were below the standards at a depth of 3.9 – 5.0 m bgs. ■ Concentrations of LEPH exceeded the CL/IL standard and concentrations of HEPH exceeded the CSR RL/PL in BH09-13, BH09-14, and BH09-15 (Golder 2011).	Soil: VPH, LEPH, HEPH, BTEX Groundwater: PAH Soil Vapour: VPHv

AEC or APEC	Summary	COC
	■ Concentrations of benzene also exceeded the CSR RL/PL and CL standards at BH09-13 (Golder 2011). In groundwater, ■ Concentrations of PAH's at MW11-01 (Golder 2011) exceeded CSR standards, for benzo(a)pyrene (CSR AW-M), benzo(b)fluoranthene (CSR DW), and pyrene (CSR AW-M) ■ One groundwater sample and its duplicate (MW09-8; Golder 2011) had concentrations of pyrene, benzo(a)pyrene, and benzo(b)fluoranthene that exceeded CSR standards in 2009. These concentrations decreased after resampling in 2011. ■ In groundwater, MW11-02 located upgradient had concentrations of arsenic and chloroform exceeding CSR standards.	

9.0 ISSUES OF ENVIRONMENTAL CONCERN

Based on all the information obtained as part of this Stage 1 PSI, the following issues of potential environmental concern were identified:

9.1 AEC 1 – Former Fuel Pump Islands, ASTs and Possible PCB Storage

A former pump island and possible associated ASTs containing bunker fuel, gasoline and/or diesel fuel (D&M 1990a) were historically located approximately 100 m southeast of the main repair building (on Lot 1). Historical investigations identified concentrations of hydrocarbon constituents in soil and groundwater that exceeded the standards applicable at the time of the investigation. Historical data suggested that storage of PCBs may also have taken place in this area.

In the southern area, historic soil concentrations at CPT 1 (Hardy BBT 1999) exceed current CSR standards for benzene, toluene, and xylene. As part of the DSI undertaken by Golder in 2011, this AEC was retained, and investigation to delineate soil and confirm groundwater quality was completed.

In the southern area,

- At TP09-02 and TP09-03 (Golder 2011) VPH concentrations in soil exceed the current CSR RL/PL or CL/IL standards. Naphthalene in soil also exceeds CSR RL/PL standards at TP09-03.
- One groundwater sample (MW09-1; Golder 2011) in the southern part of the AEC had concentrations of benzene, toluene, and MTBE that exceeded CSR-DW standards. This sample also had VPH concentrations exceeding CSR AW-M standards.

In the northern area,

- Surficial soil samples (SA6 and SA7; Golder 2011) had HEPH concentrations that exceed current CSR RL/PL or CL/IL standards.
- One groundwater sample and its duplicate (MW09-8; Golder 2011) had concentrations of pyrene that exceeded CSR AW-M standards in 2009. These concentrations decreased after resampling in 2011.

The HEPH appears to be limited to a surficial area adjacent to a concrete slab; PCBs were not detected. Based on the results of the soil sampling and field observation, the soil contamination has been delineated. While the groundwater contamination has not been delineated horizontally or vertically, groundwater contamination is expected to be limited to the area of soil contamination.

Owing to the presence of soil and groundwater contamination, this area is considered an AEC. The extent of the contamination in this area of the Site is generally shallow in nature, to depths up to approximately 2.4 m below ground surface, and is estimated to have an approximate volume of 1,000 m^3.

9.2 AEC 2 – Former Maintenance Area and Current Boat Repair and Construction Operations

During historic logging operations, the former railcar and engine maintenance area (rail yard) contained several sub-areas for train and truck repair, including: the main repair shop, car shop and roundhouse. The shops historically used underground pits for maintenance. The current operations in the area include kayak manufacturing, boat construction and repair; specialty wood product construction occurs in the main repair shop and roundhouse. As part of the 2011 Golder site investigation, a chemical storage area containing drums of polyester composites, resins and acetone were present between the buildings. Staining was observed on the floors of the buildings and in the outside general storage area (Golder 2011).

Historical investigations in this area show exceedances of current CSR standards:

- Historic soil concentrations at CPT 2 and 3 (Hardy BBT 1999) exceed current CSR standards for LEPH and VPH.
- Soil at 3 m bgs in TP6 (Levelton 2002) exceeded the CSR PL/RL and CL/IL standards for barium.
- LEPH concentrations in soil above CSR CL/IL standards were measured at BH00-08 (0.2 m) and at TP4 (2.0 m and 4.0 m) (Levelton 2002).

As part of the DSI undertaken by Golder in 2011, this area was retained as an AEC, and investigation to delineate soil and confirm groundwater quality was completed. During this investigation:

- Shallow soil samples in two test pits (TP09-6 and TP11-07; Golder 2011) had LEPH concentrations exceeding CSR PL/RL and CL/IL standards, as well as HEPH concentrations exceeding CSR PL/RL standards. Exceedances of volatiles in soil including ethylbenzene, xylene, benzene, and styrene of CSR PL/RL or CL/IL standards were also found in the area.
- Soil samples in the area also had concentrations of zinc (MW09-11), and lead (TP09-05) exceeding CSR PL/RL or CL/IL standards (Golder 2011).
- In groundwater, MW09-9 and MW09-11 (Golder 2011) had concentrations of manganese that exceeded CSR-DW standards.
- Downgradient at location MW11-02 (Golder 2011) had concentrations of chloroform and arsenic that exceeded CSR standards.

Concentrations in exceedances of standards were found on the northwest side of the former maintenance building. In addition, the contamination appeared to be patchy, and not associated with one contiguous source in the former railway maintenance area. A series of test pits delineated contamination to the northwest and southeast, and a downgradient monitoring well delineates the area of contamination to the northeast. Soil contamination is not delineated towards the west (upgradient); however, based on the review of the data and the heterogeneous nature of the contamination in the soil, the contamination is not expected to be widespread and would be limited in area and depth.

This area is considered an AEC. The extent of soil contamination in this area is generally shallow, to depths up to 4 metres. The volume of contaminated soils in the area to the northwest of the former maintenance building is estimated to be 2,000 m^3.

9.3 AEC 3 – Former Waste Oil Storage Area and Compressor Storage Area

Previous reports indicated that the former waste oil area was located on the southeast side of the main repair shop (currently used as a waste bin storage area). In addition, Site representatives indicated that this area formerly comprised a large compressor that was prone to leaks. Historic LEPH concentrations in a soil sample at BH00-03 at 0.2 m bgs (Levelton 2002) from this area are greater than the current CSR PL/RL standard. Although a deeper sample at 0.5 m bgs was collected and found to be below CSR PL/RL standards, the hydrocarbon contamination in soil was not delineated laterally, or sampled at the water table.

As part of the DSI undertaken by Golder in 2011, the extent of hydrocarbon contamination in groundwater was delineated to confirm the extent of the contamination. As part of the DSI, an additional monitoring well was installed downgradient to the northeast. Additionally, a well was installed to investigate off-Site APECs to the southeast.

New soil samples taken as part of the 2011 DSI showed no exceedance of LEPH/HEPH standards.

In groundwater, MW11-02 located downgradient had concentrations of arsenic and chloroform exceeding CSR standards. Groundwater sampling results indicated that hydrocarbon concentrations met the CSR standards in new and existing wells in the area.

The area is confirmed to be an AEC as soil contamination has been identified. However, based on the soil sampling results in the shallow soils, and the absence of groundwater contamination, the area of soil contamination is likely limited in area. The volume of contaminated soils is estimated to be 20 m³.

9.4 APEC 4 – Stockpile of Material from Sewage Treatment Plant

During communication with representatives from the Town of Ladysmith, it was reported that there was disposal of material from the sewage treatment plant on a portion of Lot 4. The TOL representatives confirmed that the material was removed from the ground at the Ladysmith sewage treatment plant to facilitate expansion of the plant. To their knowledge there was no contamination in this material, and that it was a very sandy material. The client provided a sketch showing the general location of the fill placement.

Since this material was placed on-site, and because the exact nature of the material is unknown, it has been retained as an APEC. Further investigation of the material is recommended.

9.5 APEC 5 – Fill of Unknown Quality on Lot 5

Previous investigations on-site have not included Lot 5, which is located between Lot 4 and Slack Point. In Golder's previous work (Golder 2005 and Golder 2011), several AEC's were in identified in the area of Slack Point, including the following:

- **Coal fill at Slack Point**: Coal fill material contains concentrations of PAHs, LEPH, and HEPH at concentrations above CSR PL/RL standards applicable at the time of assessment. The observed depth of the coal fill at Slack Point ranged from 6.5 m to 10 m below surface level in the most southwestern (upland) portion of Slack Point, to 16.6 metres below surface in the remainder of Slack Point. It is estimated that the

total volume of coal fill present at Slack Point is approximately 725,000 m^3 (roughly one million tons), and that an additional 100,000 to 225,000 m^3 (140,000 to 300,000 tons) of coal fill likely extends into the harbour (based on an angle of repose of 15° to 30°).

- **Surficial Fill from Non-Coal Sources at Slack Point**: Surface fill materials (not including coal fill) were observed as discontinuous units across Slack Point and ranged in thickness up to a maximum of 3 m. Little to no surface fill materials were observed above the coal fill unit in the south corner of Slack Point. Samples collected from surface fill (non-coal) contained concentrations of LEPH, HEPH and metals exceeding the CSR PL/RL standards and zinc concentrations exceeding the CSR CL/IL standards applicable at the time of assessment. The extent of the surface fill materials at Slack Point, including the concentrated regions of buried metal debris and refuse (summarized below), are estimated to have an approximate volume between 34,000 m^3 and 67,000 m^3.
- **Buried Refuse and Possible Abandoned Landfill**: Regions of concentrated buried metal debris and refuse are present on Slack Point, but no underground storage tanks were identified. Areas investigated were found to have metal debris, car parts, bricks, wire cable, oil and paint cans, and wood waste. Samples collected from the area of the buried refuse contained concentrations of LEPH, HEPH and metals exceeding the CSR CL/IL standard and/or the CSR RL/PL standards.

Since previous intrusive investigations did not include Lot 5, there is a potential for fill material of similar quality to Slack Point to be located on Lot 5. It has been retained as an APEC, and further investigation of the soil quality is recommended.

9.6 APEC 6 – Former Shingle Mill/Sawmill (Off-Site)

Blocks B and D of DL 2016 adjacent to Site were created by infilling the water lot, in the area previously occupied by the shingle mill wharf. Based on the aerial photograph review, filling of this area appeared to take place between 1962 and 1988. The investigation by Phoenix in 1999 indicated that a small sawmill was located near to the northwest corner of the Site on Block B and D of DL 2016, adjacent to the government wharf (there has been no other reference to or evidence of a sawmill at this location). Golder infers this area to be the location of the former shingle mill.

No previous intrusive investigations occurred at Blocks B and D, other than the DSI undertaken by Golder in 2011. The DSI included soil and groundwater sampling at one location (MW09-16), with no soil or groundwater contamination identified. Results of the investigation indicated that the fill soil and groundwater did not contain any other metals or hydrocarbon concentrations exceeding the CSR standards. One sampling location, MW09-05 located on the northeast area of the Site contained benzene exceeding the CSR standard, and may be attributable to this off-Site APEC. It is noted that the sampling program was limited, and additional sample collection along the filled area may be advisable to provide greater certainty. As such, this area is not considered an AEC; however, retained as an off-site APEC.

9.7 AEC 7 – Former Log Dump (Lot 17G; Off-Site)

The former log dump on Lot 17G was established in the 1930s and used to transport logs from railcars to the Harbour for bundling and shipping. The area is an extension of the natural foreshore and therefore contains fill materials of unknown quality. The log dump was reported to have been removed in 1992 (EBA 1994).

Historic investigations found concentrations of Mineral Oil and Grease (MOG) inferred to be above current CSR RL/PL and CL/IL standards for LEPH, at EBA's Test Pit 14 (EBA 1994). At Test Pit 11, concentrations of MOG are inferred to be above current CSR RL/PL standards for LEPH.

As part of the DSI undertaken by Golder in 2011, observations of Site soil conditions indicated hydrocarbon like odours, sheet and product droplets, likely indicative of contamination. As such, several boreholes, and subsequent monitoring wells, were completed as step outs along the shoreline to the northwest. Soil and groundwater contamination were confirmed at several of these locations. Based on the results of the DSI and historical investigation activities, the contamination has been delineated along the shoreline. In addition, as part of the sediment sampling program in Ladysmith Harbour, several sediment samples were collected near to the shoreline, adjacent to the log dump area. No indication of hydrocarbon contamination was observed during the sediment sampling program (Golder 2011c), and the contamination is inferred to be limited to the near shore filled area. However, there was uncertainty as to whether this contamination was from former log dump, filling activities, or subsurface migration to the area from the former railway yard activities.

To address this issue, a deep groundwater well was completed in the bedrock, on the bedrock terrace located up gradient from the former log dump, and down gradient of the former maintenance building (MW11-02; Golder 2011). This well was installed at a depth intersecting the elevation of the contamination observed at the log dump. This well had concentrations of arsenic and chloroform exceeding CSR standards, but no hydrocarbon exceedances. Consequently, it was concluded that the contamination at the former log dump appears to be associated with log dump or filling activities in the filled foreshore area, and is not related to the former railway maintenance area. As such, the area is generally delineated to the southwest; however, it is not known whether contamination extends onto Lot 5. The former log dump area has been retained as an AEC.

The extent of the contamination in this area of the Site was observed to be between 2.5 and 5.5 m below ground surface. The contamination is estimated to have an approximate volume of 5,800 m^3.

9.8 Summary of Issues of Environmental Concern

The issues of environmental concern identified in this Stage 1 ESA are outlined in the following table:

AEC or APEC	Summary	PCOC, COC
On-Site		
AEC 1 – Former fuel pump islands, ASTs and possible PCB Storage.	Owing to the presence of soil and groundwater contamination, this area is considered an AEC. The extent of the contamination in this area of the Site is generally shallow in nature, to depths up to approximately 2.4 m below ground surface, and is estimated to have an approximate volume of 1,000 m^3	Soil: BTEX, VPH, LEPH, HEPH Groundwater: BTEC, VPH, LEPH Soil Vapour: VPHv, xylene, naphthalene
AEC 2 – Former Maintenance Area and Current Boat Repair and Construction Operations.	This area is considered an AEC. The extent of soil contamination in this area is generally shallow, to depths up to 4 metres. The volume of contaminated soils in the area to the northwest of the former maintenance building is estimated to be 2,000 m^3	Soil: BTEX, VPH, LEPH, HEPH, styrene,
AEC 3 – Former Waste Oil Storage Area and Compressor Storage Area.	The area is confirmed to be an AEC as soil contamination has been identified. However, based on the soil sampling results in the shallow soils, and the absence of groundwater contamination, the area of soil contamination is likely limited in area. The volume of contaminated soils is estimated to be 20 m^3	Soil: LEPH
APEC 4 – Stockpile of material from around sewage treatment plant	Fill of unknown quality was placed on-site, removed from the ground at the Ladysmith sewage treatment plant to facilitate expansion of the plant. While the fill is not known to be contaminated, it has been retained as an APEC, and further investigation of the material is recommended.	Soil: LEPH, HEPH, PAH, metals
APEC 5 – Fill of unknown quality on Lot 5	Lot 5 is adjacent to Slack Point, where coal fill and surficial fill were previously identified with concentrations of LEPH, HEPH, metals, and PAHs above CSR standards. Since previous intrusive investigations did not include Lot 5, there is a potential for fill material of similar quality to Slack Point to be located on Lot 5. It has been retained as an APEC, and further investigation of the soil quality is recommended.	Soil: LEPH, HEPH, metals, PAHs
Off-Site		
APEC 6 – Fill Material at Block B and D of DL 2016 (former Location of the Shingle Mill / Sawmill)	No soil or groundwater contamination identified during previous site investigations, as such, this area is not considered an AEC. However, the sampling program was limited in area, and additional sample collection along the filled area may be warranted, therefore, remains an APEC.	Soil: LEPH/HEPH, PAH, and metals Groundwater: LEPH/HEPH, PAH, metals Soil Vapour: VPHv, BTEX, naphthalene
AEC 7 – Former Log Dump (Lot 17G).	Contamination was identified during previous site investigations, but appears to be limited to a particular range of depth and may be associated with the log dump or filling activities. The extent of the contamination in this area of the Site was observed to be between 2.5 and 5.5 m below ground surface. The contamination is estimated to have an approximate volume of 5,800 m^3	Soil: VPH, LEPH Groundwater: PAH Soil Vapour: VPHv

Based on all the information obtained as part of this Stage 1 PSI, the following other Special Attention Items were or are likely identified on the Site:

- asbestos
- PCBs
- Lead

These Special Attention Items are not considered to represent an issue of potential environmental concern provided they are managed in accordance with applicable environmental, health, and safety legislation.

10.0 RECOMMENDATIONS

To address the issue(s) of potential environmental concern identified, Golder recommends the following:

- Conducting a Detailed Site Investigation to further investigate soil, groundwater and soil vapour quality at identified AECs.

11.0 QUALIFICATIONS OF SITE ASSESSOR

Pursuant to Section 63 of the Contaminated Sites Regulation (CSR), Golder confirms that this report has been prepared in accordance with the applicable sections of the CSR (Part 14) and related BC ENV technical guidance (TG) documents.

This report was authored by Robyn Chatwin-Davies, MASc, EIT, and by Dawn Flotten, PEng, CSAP. Robyn Chatwin-Davies is an environmental engineer-in-training with 2 years experience, and Dawn Flotten is a senior environmental engineer with over 23 years of experience in the assessment of contaminated sites, and is familiar with the assessments carried out at the Ladysmith Harbour.

12.0 CLOSURE

We trust that this report meets your immediate requirements. If you have any questions regarding the content of this report, please do not hesitate to contact this office.

Golder Associates Ltd.



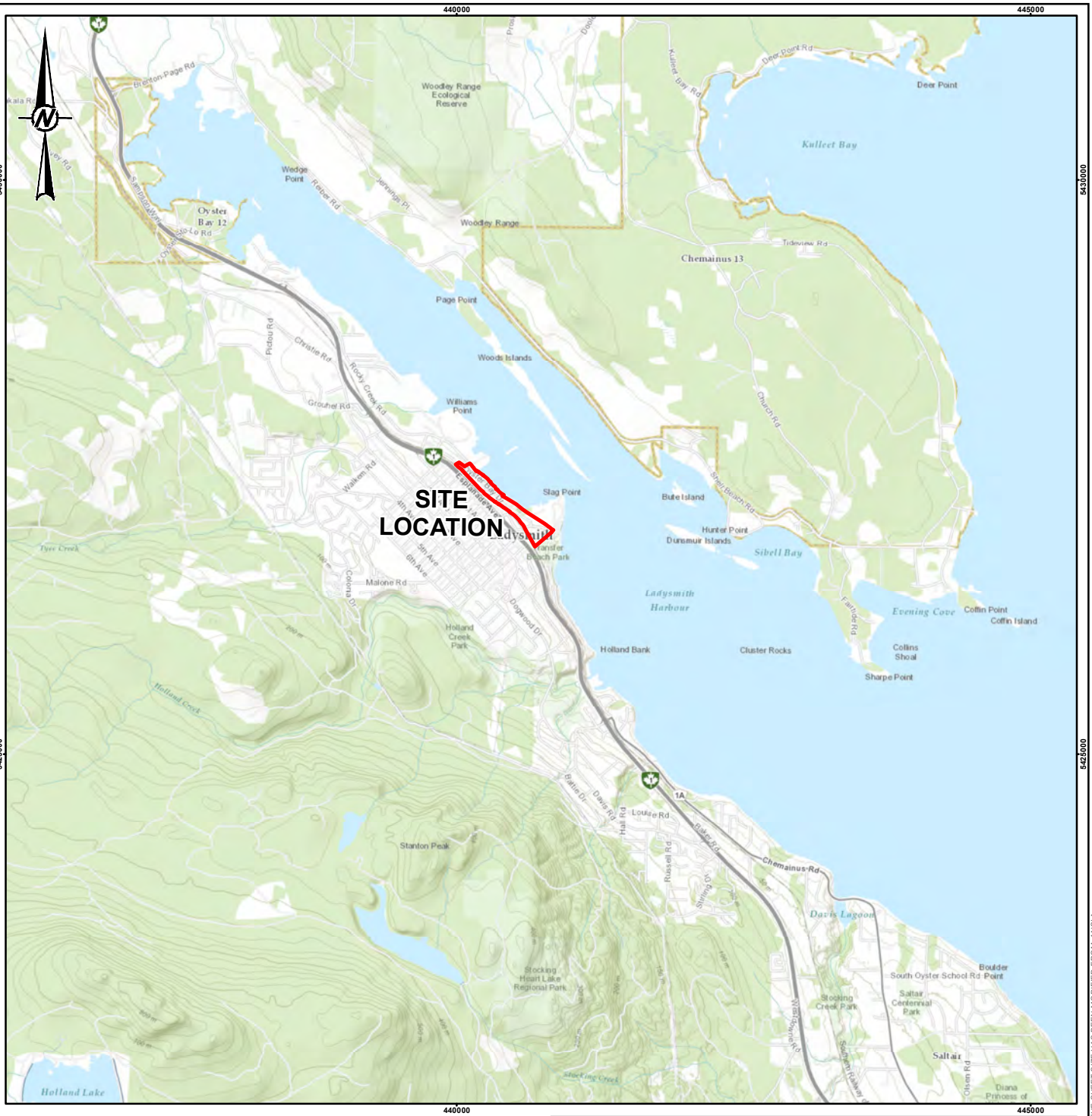
Robyn Chatwin-Davies, MASc, EIT
Environmental Engineer

Dawn Flotten, PEng, CSAP
Principal, Senior Environmental Engineer

RCD/DF/asd

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LEGEND

SITE BOUNDARY

NOT FOR CONSTRUCTION

0 1,000 2,000

1:50,000 METRES

NOTE(S)

1. LOCATIONS ARE APPROXIMATE.

REFERENCE(S)

1. TOPOGRAPHIC MAP © ESRI AND ITS LICENSORS. USED UNDER LICENSE, ALL RIGHTS RESERVED.

COORDINATE SYSTEM: NAD 1983 UTM ZONE 10N

CLIENT
TOWN OF LADYSMITH

PROJECT
LOTS 1, 4 AND 5 ADJACENT TO LADYSMITH HARBOUR

TITLE
KEY PLAN

CONSULTANT
GOLDER

YYYY-MM-DD	2020-02-27
DESIGNED	MN
PREPARED	MH/CDAB
REVIEWED	MN
APPROVED	DF

PROJECT NO.	CONTROL	REV.	FIGURE
18108342	1000	A	1

PATH: N:\Client\Town of Ladysmith\Ladysmith Harbour\99_PROJECTS\18108342\1000\02_PRODUCTION\MXD\Report\18108342-1001-FIG-1_Key_Plan.mxd PRINTED ON: 2020-02-27 AT: 10:46:07 AM

IF THIS MEASUREMENT DOES NOT MATCH WHAT IS SHOWN, THE SHEET SIZE HAS BEEN MODIFIED FROM: ANSI A

25mm

0



LEGEND

- SITE BOUNDARY
- Filled Foreshore
- Uplands
- Water Lot

NOT FOR CONSTRUCTION

0 125 250

1:6,000 METRES

NOTE(S)

1. LOCATIONS ARE APPROXIMATE.

REFERENCE(S)

1. IMAGERY COPYRIGHT © 20191219 ESRI AND ITS LICENSORS. SOURCE: ESRI. USED UNDER LICENSE, ALL RIGHTS RESERVED.
2. OTHER BASEDATA OBTAINED FROM WMS IN THE PROVINCE OF BRITISH COLUMBIA. COORDINATE SYSTEM: NAD 1983 UTM ZONE 10N

CLIENT
TOWN OF LADYSMITH

PROJECT
LOTS 1, 4 AND 5 ADJACENT TO LADYSMITH HARBOUR

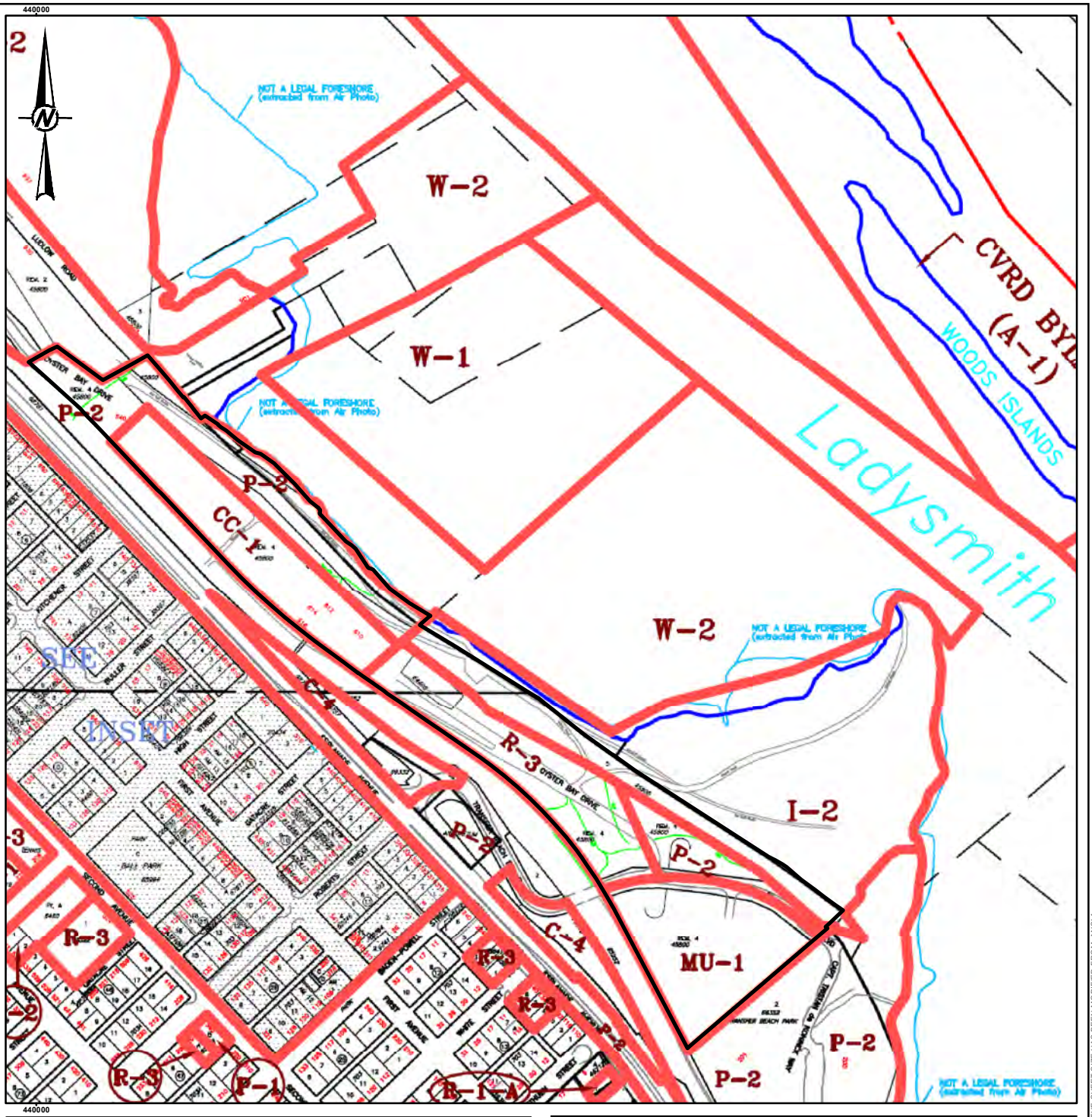
TITLE
SITE PLAN

CONSULTANT
GOLDER

YYYY-MM-DD	2020-02-27
DESIGNED	MN
PREPARED	MH/CDAB
REVIEWED	MN
APPROVED	DF

PROJECT NO.	CONTROL	REV.	FIGURE
1810842	1000	A	2

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25mm
0



LEGEND

SITE BOUNDARY

NOT FOR CONSTRUCTION

0 125 250

1:6,000 METRES

NOTE(S)

1. LOCATIONS ARE APPROXIMATE.

REFERENCE(S)

1. IMAGE OBTAINED FROM THE TOWN OF LADYSMITH ZONING BYLAW.

COORDINATE SYSTEM: NAD 1983 UTM ZONE 10N

CLIENT
TOWN OF LADYSMITH

PROJECT
LOTS 1, 4 AND 5 ADJACENT TO LADYSMITH HARBOUR

TITLE
SURROUNDING LAND USE

CONSULTANT

GOLDER

YYYY-MM-DD	2020-02-27
DESIGNED	MN
PREPARED	MH/CDAB
REVIEWED	MN
APPROVED	DF

PROJECT NO.	CONTROL	REV.	FIGURE
1810842	1000	A	3

PATH: N:\Client\Town of Ladysmith\Ladysmith Harbour\99_PROJECTS\1810842\101002_PRODUCTION\MXD\Report\1810842_1001_FIG 3_Site_Location_Surrounding_Landuse.mxd PRINTED ON: 2020-02-27 AT: 10:45:52 AM

IF THIS MEASUREMENT DOES NOT MATCH WHAT IS SHOWN, THE SHEET SIZE HAS BEEN MODIFIED FROM: ANSI A
25mm
0



LEGEND

- SITE BOUNDARY
- LOT BOUNDARY
- HISTORIC FEATURE

NOTES

1. HISTORIC LOCATIONS ARE APPROXIMATE.

REFERENCES

AERIAL IMAGE © 2019 GOOGLE OBTAINED FROM GOOGLE EARTH PRO, USED UNDER LICENSE.
IMAGE DATE: 8/18/2016. GOOGLE EARTH IMAGE IS NOT TO SCALE.

LOT BOUNDARIES OBTAINED FROM BC LAND TITLE AND SURVEY ON 2019-12-18

DATUM: NAD83, PROJECTION: UTM ZONE 10

0 25 50
1:1,000 METRES

0 50 100
1:2,000 METRES

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CLIENT
TOWN OF LADYSMITH

PROJECT
LOTS 1, 4 AND 5 ADJACENT TO LADYSMITH HARBOUR
LADYSMITH HARBOUR, LADYSMITH, B.C.

TITLE
SITE PLAN WITH HISTORICAL FEATURES

CONSULTANT
GOLDER

YYYY-MM-DD	2020-02-26
DESIGNED	MN
PREPARED	RTJ
REVIEWED	GG
APPROVED	DF

PROJECT NO.	PHASE	REV.	FIGURE
18109842	1000	0	4

IF THIS MEASUREMENT DOES NOT MATCH WHAT IS SHOWN, THE SHEET SIZE HAS BEEN MODIFIED FROM: ANSI D
0 25 mm

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LEGEND

- SITE BOUNDARY
- LOT BOUNDARY
- MONITORING WELL LOCATION
- BOREHOLE LOCATION
- TEST PIT LOCATION
- SURFACE SAMPLE LOCATION

NOTES

1. SAMPLING LOCATIONS ARE APPROXIMATE.

REFERENCES

AERIAL IMAGE © 2019 GOOGLE OBTAINED FROM GOOGLE EARTH PRO, USED UNDER LICENSE.
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LOT BOUNDARIES OBTAINED FROM BC LAND TITLE AND SURVEY ON 2019-12-18

DATUM: NAD83, PROJECTION: UTM ZONE 10

0 25 50
1:1,000 METRES

0 50 100
1:2,000 METRES

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CLIENT
TOWN OF LADYSMITH

PROJECT
LOTS 1, 4 AND 5 ADJACENT TO LADYSMITH HARBOUR
LADYSMITH HARBOUR, LADYSMITH, B.C.

TITLE
SITE PLAN SHOWING INVESTIGATION LOCATIONS

CONSULTANT
GOLDER

YYYY-MM-DD	2020-02-26
DESIGNED	MN
PREPARED	RTJ
REVIEWED	GG
APPROVED	DF

PROJECT NO.	PHASE	REV.	FIGURE
18109842	1000	0	5

IF THIS MEASUREMENT DOES NOT MATCH WHAT IS SHOWN, THE SHEET SIZE HAS BEEN MODIFIED FROM: ANSI D

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Location	MW09-1
Date Sampled	11/10/2009
Depth of Sample (m)	0.6-0.76
Moisture (%)	18.5
pH	5.7
Metals	<

Location	MW09-3
Date Sampled	11/10/2009
Depth of Sample (m)	0.6-0.76
Moisture (%)	8.4
pH	5.7
Metals	<

Location	TP09-01
Date Sampled	15/12/2009
Depth of Sample (m)	0.6-0.8
Moisture (%)	-
pH	5.9
Metals	<

Location	TP09-04	
Date Sampled	15/12/2009	
Depth of Sample (m)	1-1.2	
Moisture (%)	17.0	
pH	7.2	
Arsenic	17	
Metals	<	

Location	MW09-5	
Date Sampled	11/11/2009	
Depth of Sample (m)	0.6-0.8	1.4-1.6
Moisture (%)	16.8	-
pH	5.9	6.2
Metals	<	<

Location	MW09-16		
Date Sampled	14/11/2009		
Depth of Sample (m)	1.2-1.37	3.35-3.5	3.35-3.5
Moisture (%)	10.3	18.7	18.1
pH	6.3	7.3	7.5
Metals	<	<	<

Location	MW09-9
Date Sampled	13/11/2009
Depth of Sample (m)	1.5-1.7
Moisture (%)	12.6
pH	8.1
Metals	<

Location	MW09-10	
Date Sampled	13/11/2009	
Depth of Sample (m)	0.6-0.76	1.5-1.7
Moisture (%)	8.50	15.40
pH	7.6	7.7
Metals	<	<

Location	MW09-11
Date Sampled	13/11/2009
Depth of Sample (m)	0.3-0.4
Moisture (%)	21.20
pH	6.6
Metals	<
Zinc	736

Location	KE SA1
Date Sampled	25/11/2009
Depth of Sample (m)	0.3-0.4
Moisture (%)	9.0
pH	6.2
Metals	<

Location	KE SA2
Date Sampled	25/11/2009
Depth of Sample (m)	0.2-0.3
Moisture (%)	10.8
pH	6
Metals	<

Location	KE SA4
Date Sampled	25/11/2009
Depth of Sample (m)	0.2-0.3
Moisture (%)	6.90
pH	6.1
Metals	<

Location	TP09-05
Date Sampled	15/12/2009
Depth of Sample (m)	0.3-0.5
Moisture (%)	-
pH	7.9
Lead	736
Metals	<

Location	MW09-06
Date Sampled	11/12/2009
Depth of Sample (m)	2.0-2.2
Moisture (%)	-
pH	6.8
Metals	<

Location	TP09-09	
Date Sampled	15/12/2009	
Depth of Sample (m)	0.4-0.6	1.0-1.2
Moisture (%)	15.9	16.3
pH	7.2	7.4
Metals	<	<

Location	BH09-13
Date Sampled	14/11/2009
Depth of Sample (m)	3.8-4.0
Moisture (%)	19.3
pH	7.4
Arsenic	4
Metals	<

Location	BH09-14
Date Sampled	14/11/2009
Depth of Sample (m)	4.1-4.3
Moisture (%)	9.5
pH	8.1
Metals	<

Location	MW09-7
Date Sampled	11/12/2009
Depth of Sample (m)	0.5-0.7
Moisture (%)	-
pH	6.3
Metals	<

Location	MW09-8	
Date Sampled	13/11/2009	
Depth of Sample (m)	2.9-3.0	
Moisture (%)	20.0	24.3
pH	8	7.9
Metals	<	<

Location	TP09-10	
Date Sampled	15/12/2009	
Depth of Sample (m)	0.3-0.6	1.6-1.8
Moisture (%)	9.0	7.2
pH	7.5	6.9
Metals	<	<

Location	TP09-11
Date Sampled	15/12/2009
Depth of Sample (m)	0.2-0.4
Moisture (%)	13.2
pH	6.7
Metals	<

Location	TP09-12		
Date Sampled	15/12/2009		
Depth of Sample (m)	0.1-0.3	0.1-0.3	1.3-1.5
Moisture (%)	15.0	18.4	12.4
pH	6.8	6.8	7.3
Metals	<	<	<

Location	CPT 1
Date Sampled	18/09/1990
Depth of Sample (m)	1.55
pH	4.87
Metals	<

Location	CPT 4
Date Sampled	18/09/1990
Depth of Sample (m)	2.5
pH	7.2
Metals	<

Location	CPT 5
Date Sampled	18/09/1990
Depth of Sample (m)	0.7
pH	6.62
Metals	<

Location	TP 1
Date Sampled	18/07/2000
Depth of Sample (m)	1
pH	6.2
Metals	<

Location	TP 2
Date Sampled	18/07/2000
Depth of Sample (m)	1
pH	6.5
Metals	<

Location	TP 3
Date Sampled	18/07/2000
Depth of Sample (m)	1
pH	6.5
Metals	<

Location	TP 4	
Date Sampled	18/07/2000	
Depth of Sample (m)	2.00	4.00
pH	6.8	6.3
Metals	<	<

Location	TP 8
Date Sampled	18/07/2000
Depth of Sample (m)	3
pH	8
Barium	1,000
Metals	<

Location	TP 12
Date Sampled	18/07/2000
Depth of Sample (m)	3.00
pH	-
Metals	<

Location	LOT 11G
Date Sampled	21/08/2000
Depth of Sample (m)	0.5
pH	6.4
Chromium	100
Metals	<

LEGEND

- SITE BOUNDARY
- LOT BOUNDARY
- MONITORING WELL LOCATION
- BOREHOLE LOCATION
- TEST PIT LOCATION
- SURFACE SAMPLE LOCATION
- SOIL PARAMETER CONCENTRATION MEETS APPLICABLE STANDARDS
- SOIL PARAMETER CONCENTRATION IS GREATER THAN APPLICABLE CSR IL, CSR CL AND CSR PL STANDARDS

TABLE

10	CONCENTRATION IN SOIL EXCEEDS APPLICABLE CSR PL STANDARD
10	CONCENTRATION IN SOIL EXCEEDS APPLICABLE CSR CL STANDARD
10	CONCENTRATION IN SOIL EXCEEDS APPLICABLE CSR IL STANDARD

ABBREVIATIONS

IL	INDUSTRIAL LAND USE
CL	COMMERCIAL LAND USE
PL	PARK LAND USE
FD	FIELD DUPLICATE
FDA	FIELD DUPLICATE AVAILABLE

NOTES

1. ALL VALUES ARE IN MICROGRAMS PER LITRE (µg/g), UNLESS OTHERWISE NOTED.
2. SAMPLING LOCATIONS ARE APPROXIMATE.

REFERENCES

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LOT BOUNDARIES OBTAINED FROM BC LAND TITLE AND SURVEY ON 2019-12-18

DATUM: NAD83, PROJECTION: UTM ZONE 10

0 25 50 — 1:1,000 METRES

0 50 100 — 1:2,000 METRES

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CLIENT
TOWN OF LADYSMITH

CONSULTANT
GOLDER

PROJECT
LOTS 1, 4 AND 5 ADJACENT TO LADYSMITH HARBOUR
LADYSMITH HARBOUR, LADYSMITH, B.C.

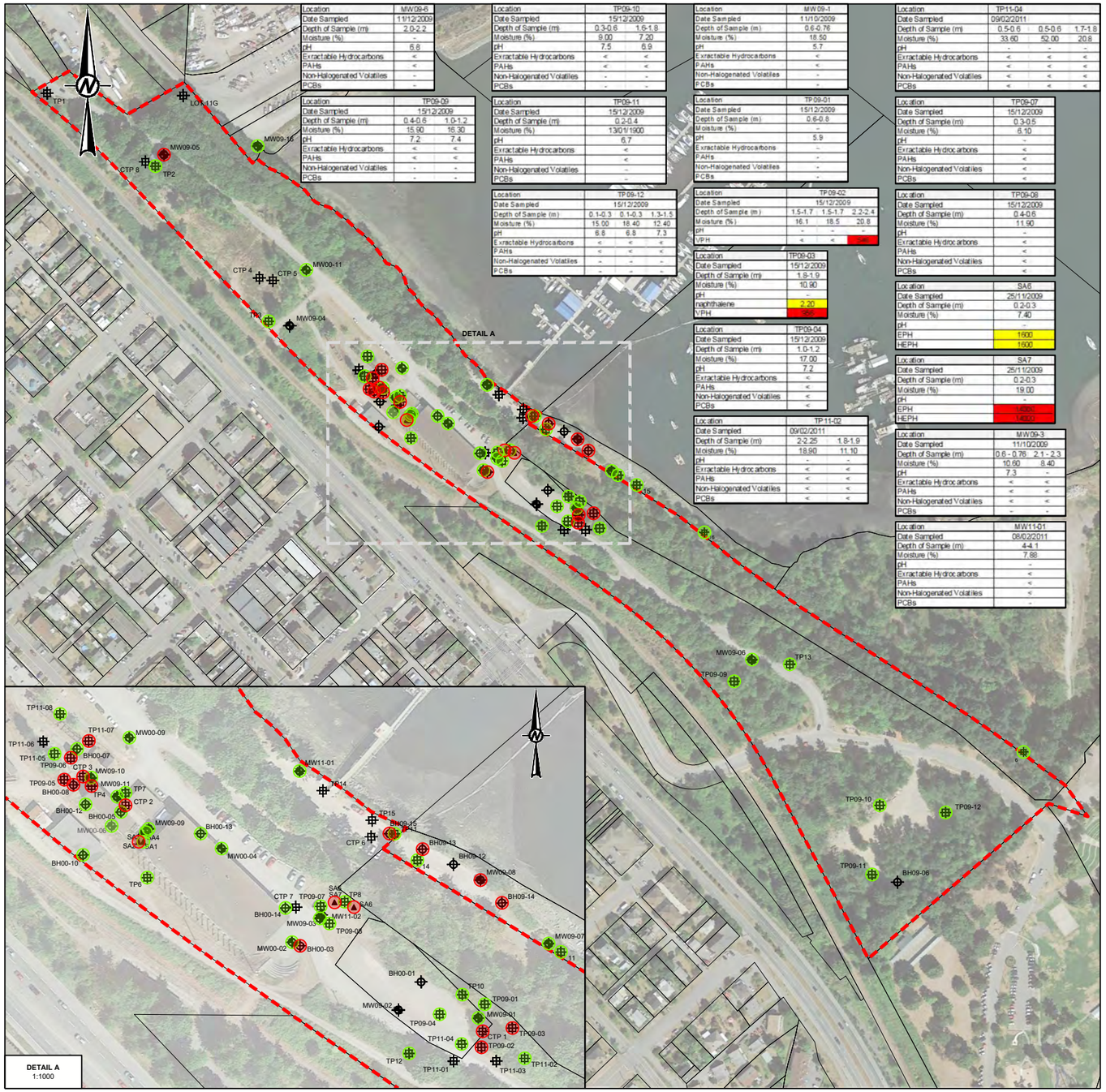
TITLE
SOIL ANALYTICAL RESULTS - METALS

YYYY-MM-DD	2020-02-26
DESIGNED	MN
PREPARED	RTJ
REVIEWED	GG
APPROVED	DF

PROJECT NO.	PHASE	REV.	FIGURE
18109842	1000	0	6A

Path: \\golder.gds\gal\Kelowna\CAD-GIS\Client\TOWN OF LADYSMITH\Ladysmith Harbour\09_PROJECTS\18109842_Enviro1000\[Task]\18109842_1000\02_PRODUCTION\DWG\ | File Name: 18109842_1000 - Results.dwg | Last Edited By: rylames Date: 2020-02-28 Time:11:00:19 AM | Printed By: RYJames Date: 2020-02-28 Time:11:03:19 AM

0 25mm IF THIS MEASUREMENT DOES NOT MATCH WHAT IS SHOWN, THE SHEET SIZE HAS BEEN MODIFIED FROM: ANSI D



Location	MW09-6
Date Sampled	11/12/2009
Depth of Sample (m)	2.0-2.2
Moisture (%)	-
pH	6.8
Extractable Hydrocarbons	<
PAHs	<
Non-Halogenated Volatiles	-
PCBs	-

Location	TP09-09	
Date Sampled	15/12/2009	
Depth of Sample (m)	0.4-0.6	1.0-1.2
Moisture (%)	15.90	16.30
pH	7.2	7.4
Extractable Hydrocarbons	<	<
PAHs	<	<
Non-Halogenated Volatiles	-	-
PCBs	-	-

Location	TP09-10	
Date Sampled	15/12/2009	
Depth of Sample (m)	0.3-0.6	1.6-1.8
Moisture (%)	9.00	7.20
pH	7.5	6.9
Extractable Hydrocarbons	<	<
PAHs	<	<
Non-Halogenated Volatiles	-	-
PCBs	-	-

Location	TP09-11	
Date Sampled	15/12/2009	
Depth of Sample (m)	0.2-0.4	
Moisture (%)	13/01/1900	
pH	6.7	
Extractable Hydrocarbons	<	
PAHs	<	
Non-Halogenated Volatiles	-	
PCBs	-	

Location	TP09-12		
Date Sampled	15/12/2009		
Depth of Sample (m)	0.1-0.3	0.1-0.3	1.3-1.5
Moisture (%)	15.00	16.40	12.40
pH	6.8	6.8	7.3
Extractable Hydrocarbons	<	<	<
PAHs	<	<	<
Non-Halogenated Volatiles	-	-	-
PCBs	-	-	-

Location	MW09-1
Date Sampled	11/10/2009
Depth of Sample (m)	0.6-0.76
Moisture (%)	18.50
pH	5.7
Extractable Hydrocarbons	<
PAHs	<
Non-Halogenated Volatiles	-
PCBs	-

Location	TP09-01
Date Sampled	15/12/2009
Depth of Sample (m)	0.6-0.8
Moisture (%)	-
pH	5.9
Extractable Hydrocarbons	-
PAHs	-
Non-Halogenated Volatiles	-
PCBs	-

Location	TP09-02		
Date Sampled	15/12/2009		
Depth of Sample (m)	1.5-1.7	1.5-1.7	2.2-2.4
Moisture (%)	16.1	18.5	20.8
pH	-	-	-
VPH	<	<	340
PAHs	<	<	<
Non-Halogenated Volatiles	-	-	-
PCBs	-	-	-

Location	TP09-03
Date Sampled	15/12/2009
Depth of Sample (m)	1.8-1.9
Moisture (%)	10.90
pH	-
naphthalene	2.20
VPH	350

Location	TP09-04
Date Sampled	15/12/2009
Depth of Sample (m)	1.0-1.2
Moisture (%)	17.00
pH	7.2
Extractable Hydrocarbons	<
PAHs	<
Non-Halogenated Volatiles	<
PCBs	-

Location	TP11-02	
Date Sampled	09/02/2011	
Depth of Sample (m)	2-2.25	1.8-1.9
Moisture (%)	18.90	11.10
pH	-	-
Extractable Hydrocarbons	<	<
PAHs	<	<
Non-Halogenated Volatiles	<	<
PCBs	<	<

Location	TP11-04		
Date Sampled	09/02/2011		
Depth of Sample (m)	0.5-0.6	0.5-0.6	1.7-1.8
Moisture (%)	33.60	52.00	20.8
pH	-	-	-
Extractable Hydrocarbons	<	<	<
PAHs	<	<	<
Non-Halogenated Volatiles	<	<	<
PCBs	<	<	<

Location	TP09-07
Date Sampled	15/12/2009
Depth of Sample (m)	0.3-0.5
Moisture (%)	6.10
pH	-
Extractable Hydrocarbons	<
PAHs	<
Non-Halogenated Volatiles	<
PCBs	<

Location	TP09-08
Date Sampled	15/12/2009
Depth of Sample (m)	0.4-0.6
Moisture (%)	11.90
pH	-
Extractable Hydrocarbons	<
PAHs	<
Non-Halogenated Volatiles	<
PCBs	<

Location	SA8
Date Sampled	25/11/2009
Depth of Sample (m)	0.2-0.3
Moisture (%)	7.40
pH	-
EPH	1600
HEPH	1600

Location	SA7
Date Sampled	25/11/2009
Depth of Sample (m)	0.2-0.3
Moisture (%)	19.00
pH	-
EPH	14000
HEPH	14000

Location	MW09-3	
Date Sampled	11/10/2009	
Depth of Sample (m)	0.6 - 0.76	2.1 - 2.3
Moisture (%)	10.60	8.40
pH	7.3	-
Extractable Hydrocarbons	<	<
PAHs	<	<
Non-Halogenated Volatiles	<	<
PCBs	-	-

Location	MW11-01
Date Sampled	08/02/2011
Depth of Sample (m)	4-4.1
Moisture (%)	7.88
pH	-
Extractable Hydrocarbons	<
PAHs	<
Non-Halogenated Volatiles	<
PCBs	-

Location	MW09-16		
Date Sampled	14/11/2009		
Depth of Sample (m)	1.2 - 1.37	3.35 - 3.50	3.35 - 3.50
Moisture (%)	10.30	18.70	18.1
pH	8.3	7.3	7.5
Extractable Hydrocarbons	<	<	<
PAHs	<	<	<
Non-Halogenated Volatiles	-	-	-
PCBs	-	-	-

Location	MW09-5	
Date Sampled	11/11/2009	
Depth of Sample (m)	1.4 - 1.6	7.1-7.3
Moisture (%)	-	12.90
pH	6.2	-
Benzene	<	0.12

Location	MW09-9
Date Sampled	13/11/2009
Depth of Sample (m)	1.5-1.7
Moisture (%)	12.60
pH	8.1
Extractable Hydrocarbons	<
PAHs	<
Non-Halogenated Volatiles	-
PCBs	-

Location	MW09-10	
Date Sampled	13/11/2009	
Depth of Sample (m)	0.6 - 0.76	1.5 - 1.7
Moisture (%)	8.50	15.40
pH	7.6	7.7
Extractable Hydrocarbons	<	<
PAHs	<	<
Non-Halogenated Volatiles	-	-
PCBs	-	-

Location	MW09-11	
Date Sampled	13/11/2009	
Depth of Sample (m)	0.3-0.4	
Moisture (%)	21.20	
pH	6.6	
Extractable Hydrocarbons	<	
PAHs	<	
Non-Halogenated Volatiles	-	
PCBs	-	

Location	SA1
Date Sampled	25/11/2009
Depth of Sample (m)	0.2-0.3
Moisture (%)	9.00
pH	6.2
Extractable Hydrocarbons	<
PAHs	<
Non-Halogenated Volatiles	<
PCBs	-

Location	SA2
Date Sampled	25/11/2009
Depth of Sample (m)	0.2-0.3
Moisture (%)	10.80
pH	6
benzene	0.08
styrene	29

Location	SA4
Date Sampled	25/11/2009
Depth of Sample (m)	0.2-0.3
Moisture (%)	6.90
pH	6.1
Extractable Hydrocarbons	<
PAHs	<
Non-Halogenated Volatiles	<
PCBs	-

Location	TP09-05		
Date Sampled	15/12/2009		
Depth of Sample (m)	0.3-0.5	0.7-0.9	1.6-1.8
Moisture (%)	-	22.40	17.9
pH	7.9	-	-
ethylbenzene	42	<	<
Total xylene	96	<	<

Location	TP09-05		
Date Sampled	15/12/2009		
Depth of Sample (m)	0.9-1.1	0.9-1.1	1.7-1.9
Moisture (%)	19.50	19.80	19.2
pH	-	-	-
EPH10-19	1000	1000	<
EPH19-32	1500	2000	<
LEPH	1000	1000	<
HEPH	1500	2000	<

Location	TP11-05
Date Sampled	11/02/2009
Depth of Sample (m)	0.3-0.4
Moisture (%)	15.50
pH	-
Extractable Hydrocarbons	<
PAHs	<
Non-Halogenated Volatiles	<
PCBs	<

Location	TP11-07		
Date Sampled	16/12/2009		
Depth of Sample (m)	1.5-1.6	2.5-3.0	2.8-3.0
Moisture (%)	22.80	8.71	10.5
pH	-	-	-
EPH10-19	1000	<	<
EPH19-32	1000	<	<
LEPH	1000	<	<
HEPH	1000	<	<

Location	TP11-08		
Date Sampled	11/02/2009		
Depth of Sample (m)	1.4-1.5		
Moisture (%)	16.30		
pH	-		
Extractable Hydrocarbons	<		
PAHs	<		
Non-Halogenated Volatiles	<		
PCBs	-		

Location	MW09-8		
Date Sampled	13/11/2009		
Depth of Sample (m)	2.9 - 3.0	2.9 - 3.0	3.9 - 5.0
Moisture (%)	20.00	24.30	11.4
pH	8	7.9	-
EPH10-19	1900	1900	<
EPH19-32	2000	1700	<
LEPH	1900	1800	<
HEPH	2000	1700	<
VPH	1900	<	<

Location	BH09-13
Date Sampled	14/11/2009
Depth of Sample (m)	3.8-4.0
Moisture (%)	19.30
pH	7.4
EPH10-19	1900
EPH19-32	1800
LEPH	1900
HEPH	1800

Location	BH09-14
Date Sampled	14/11/2009
Depth of Sample (m)	4.1-4.3
Moisture (%)	9.50
pH	8.1
EPH10-19	1900
EPH19-32	1800
LEPH	1900
HEPH	1800

Location	BH09-15	
Date Sampled	14/11/2009	
Depth of Sample (m)	3.65-3.8	
Moisture (%)	14.50	11.70
pH	-	-
EPH10-19	1900	1900
EPH19-32	3100	3300
LEPH	3100	3300
HEPH	3100	3300

Location	CPT-1
Date Sampled	18/09/1990
Depth of Sample (m)	1.56
pH	4.97
Benzene	1.02
Toluene	34
Total Xylene	104.7

Location	CPT-2
Date Sampled	18/09/1990
Depth of Sample (m)	1.80
pH	-
Total Extractable Hydrocarbons	3800

Location	CPT-3
Date Sampled	18/09/1990
Depth of Sample (m)	1 1.8
pH	- -
Oil and Grease	2150 2150
Total Extractable Hydrocarbons	- 2450

Location	TP-2
Date Sampled	18/07/2000
Depth of Sample (m)	1
pH	6.5
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	TP-3
Date Sampled	18/07/2000
Depth of Sample (m)	3
pH	-
Non-Halogenated Volatiles	<
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	TP-4	
Date Sampled	18/07/2000	
Depth of Sample (m)	2	4
pH	6.8	6.3
EPH10-19	4300	3200

Location	TP-6
Date Sampled	18/07/2000
Depth of Sample (m)	3
pH	8
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	TP-7
Date Sampled	18/07/2000
Depth of Sample (m)	3
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	TP-8
Date Sampled	18/07/2000
Depth of Sample (m)	-
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	-
General	-

Location	TP-10	
Date Sampled	18/07/2000	
Depth of Sample (m)	2	2
pH	-	-
Non-Halogenated Volatiles	<	<
PAHs	-	-
Extractable Hydrocarbons	<	<
General	-	-

Location	TP-11	
Date Sampled	18/07/2000	
Depth of Sample (m)	1	1
pH	-	-
Non-Halogenated Volatiles	-	-
PAHs	-	-
Extractable Hydrocarbons	<	<
General	-	-

Location	TP-12
Date Sampled	18/07/2000
Depth of Sample (m)	3
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	TP-13
Date Sampled	18/07/2000
Depth of Sample (m)	1
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	MW00-02
Date Sampled	20/09/2000
Depth of Sample (m)	1.4
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	BH00-03	
Date Sampled	20/09/2000	
Depth of Sample (m)	0.2	0.5
pH	-	-
EPH10-19	1350	875
EPH19-32	1450	680

Location	MW00-04
Date Sampled	20/09/2000
Depth of Sample (m)	1.4
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	BH00-05
Date Sampled	20/09/2000
Depth of Sample (m)	2.6
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	MW00-06
Date Sampled	20/09/2000
Depth of Sample (m)	1.5
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	BH00-07	
Date Sampled	20/09/2000	
Depth of Sample (m)	1.5	2.9
pH	-	
Non-Halogenated Volatiles	-	-
PAHs	-	-
Extractable Hydrocarbons	<	<
General	-	-

Location	BH00-08		
Date Sampled	20/09/2000		
Depth of Sample (m)	0.8	1.8	2.9
pH	-	-	-
EPH10-19	<	2300	<

Location	MW00-09	
Date Sampled	20/09/2000	
Depth of Sample (m)	2.4	2.4
pH	-	-
Non-Halogenated Volatiles	-	-
PAHs	-	-
Extractable Hydrocarbons	<	<
General	-	-

Location	BH00-10
Date Sampled	21/09/2000
Depth of Sample (m)	1.4 3.5
pH	- -
Non-Halogenated Volatiles	- -
PAHs	- -
Extractable Hydrocarbons	< <
General	- -

Location	MW00-11
Date Sampled	21/09/2000
Depth of Sample (m)	4.6
pH	-
Non-Halogenated Volatiles	<
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	BH00-12
Date Sampled	21/09/2000
Depth of Sample (m)	2.1
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	BH00-13
Date Sampled	21/09/2000
Depth of Sample (m)	1.2
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	BH00-14
Date Sampled	21/09/2000
Depth of Sample (m)	1.2
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	11
Date Sampled	27/09/1993
Depth of Sample (m)	0.80
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	14
Date Sampled	16/07/1993
Depth of Sample (m)	0.6
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	15
Date Sampled	10/08/1993
Depth of Sample (m)	0.6
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	-
General	<

Location	16
Date Sampled	10/08/1993
Depth of Sample (m)	0.6
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	-
General	<

Location	6
Date Sampled	07/05/1993
Depth of Sample (m)	4.5
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

Location	1
Date Sampled	07/05/1993
Depth of Sample (m)	3
pH	-
Non-Halogenated Volatiles	-
PAHs	-
Extractable Hydrocarbons	<
General	-

LEGEND

- SITE BOUNDARY
- LOT BOUNDARY
- MONITORING WELL LOCATION
- BOREHOLE LOCATION
- TEST PIT LOCATION
- SURFACE SAMPLE LOCATION
- SOIL PARAMETER CONCENTRATION MEETS APPLICABLE STANDARDS
- SOIL PARAMETER CONCENTRATION IS GREATER THAN APPLICABLE CSR IL, CSR CL AND CSR PL STANDARDS

TABLE

- CONCENTRATION IN SOIL EXCEEDS APPLICABLE CSR PL STANDARD
- 10 CONCENTRATION IN SOIL EXCEEDS APPLICABLE CSR CL STANDARD
- 10 CONCENTRATION IN SOIL EXCEEDS APPLICABLE CSR IL STANDARD

ABBREVIATIONS

IL	INDUSTRIAL LAND USE
CL	COMMERCIAL LAND USE
PL	PARK LAND USE
FD	FIELD DUPLICATE
FDA	FIELD DUPLICATE AVAILABLE

NOTES

1. ALL VALUES ARE IN MICROGRAMS PER LITRE (µg/g), UNLESS OTHERWISE NOTED.
2. SAMPLING LOCATIONS ARE APPROXIMATE.

REFERENCES

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DATUM: NAD83, PROJECTION: UTM ZONE 10

0 25 50 1:1,000 METRES

0 50 100 1:2,000 METRES

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CONSULTANT
GOLDER

PROJECT
LOTS 1, 4 AND 5 ADJACENT TO LADYSMITH HARBOUR
LADYSMITH HARBOUR, LADYSMITH, B.C.

TITLE
SOIL ANALYTICAL RESULTS - HYDROCARBONS

YYYY-MM-DD	2020-02-26
DESIGNED	MN
PREPARED	RTJ
REVIEWED	GG
APPROVED	DF

PROJECT NO.	PHASE	REV.	FIGURE
18109842	1000	0	6B

IF THIS MEASUREMENT DOES NOT MATCH WHAT IS SHOWN, THE SHEET SIZE HAS BEEN MODIFIED FROM: ANSI D

Path: \\golder.gds\gal\kelowna\CAD-GIS\Clerit\TOWN OF LADYSMITH\Ladysmith Harbour\99_PROJECTS\18109842_Enviro\1000\[Task\]\18109842_1000-002_PRODUCTION\DWG\ | File Name: 18109842_1000 - Results.dwg | Last Edited By: rylames Date: 2020-02-28 Time:11:00:19 AM | Printed By: RYJames Date: 2020-02-28 Time:11:04:53 AM



Location	MW00-2
Date Sampled	18/11/2009
Hardness (mg/L)	125
pH	7.55
Metals	<

Location	MW00-4
Date Sampled	11/12/2009
Hardness (mg/L)	101
pH	6.66
Zinc	0.19
Metals	<

Location	MW00-09
Date Sampled	11/12/2009
Hardness (mg/L)	56
pH	6.33
Metals	<

Location	MW00-11
Date Sampled	11/12/2009
Hardness (mg/L)	198
pH	7.33
Metals	<

Location	MW09-1
Date Sampled	18/11/2009
Hardness (mg/L)	106
pH	7.97
Metals	<

Location	MW09-2
Date Sampled	18/11/2009
Hardness (mg/L)	151
pH	7.36
Manganese	2.37
Metals	<

Location	MW09-3
Date Sampled	18/11/2009
Hardness (mg/L)	144
pH	7.69
Metals	<

Location	MW09-4
Date Sampled	19/11/2009
Hardness (mg/L)	206
pH	7.64
Metals	<

Location	MW09-5
Date Sampled	19/11/2009
Hardness (mg/L)	175
pH	7.6
Metals	<

Location	MW09-9	
Date Sampled	25/11/2009	
Hardness (mg/L)	149.00	149.00
pH	6.75	6.75
Manganese	2.45	2.47
Metals	<	<

Location	MW09-10
Date Sampled	19/11/2009
Hardness (mg/L)	149
pH	8.13
Metals	<

Location	MW09-11
Date Sampled	19/11/2009
Hardness (mg/L)	145
pH	7.88
Manganese	3.77
Metals	<

Location	MW09-16
Date Sampled	20/11/2009
Hardness (mg/L)	32
pH	7.62
Metals	<

Location	MW11-01	
Date Sampled	16/02/2011	22/02/2011
Hardness (mg/L)	425	4330
pH	7.69	6.76
Dissolved Chloride	2000	12800
Dissolved Sulfate (SO4)	<	1790
Sodium	869	7520
Strontium	<	5.37
Metals	<	<

Location	MW11-02	
Date Sampled	16/02/2011	22/02/2011
Hardness (mg/L)	83.50	73.20
pH	8.05	7.87
Arsenic	0.0109	0.0304
Metals	<	<

Location	MW09-07		
Date Sampled	16/02/2011	20/11/2009	22/02/2011
Hardness (mg/L)	185	102	1010
pH	6.34	7.22	6.49
Dissolved Chloride	384	-	2170
Iron	<	<	7.68
Sodium	287	<	1210
Metals	<	<	<

Location	MW09-08				
Date Sampled	16/02/2011	16/02/2011	22/02/2011	20/11/2009	20/11/2009
Hardness (mg/L)	694	684	3650	1460	1890
pH	6.99	6.99	6.68	6.92	6.92
Dissolved Chloride	2340	2150	10800	5410	6480
Dissolved Sulfate (SO4)	<	<	1510	686	843
Arsenic	<	<	<	0.011	0.014
Lithium	<	<	<	0.035	0.039
Sodium	1270	1250	6310	2800	3450
Strontium	<	<	4.79	2.8	3.43
Metals	<	<	<	<	<

LEGEND

- SITE BOUNDARY
- LOT BOUNDARY
- MONITORING WELL LOCATION
- BOREHOLE LOCATION
- TEST PIT LOCATION
- SURFACE SAMPLE LOCATION
- GROUNDWATER PARAMETER CONCENTRATION MEETS APPLICABLE STANDARDS
- GROUNDWATER PARAMETER CONCENTRATION IS GREATER THAN APPLICABLE STANDARDS

TABLE

100	CONCENTRATION IN GROUNDWATER EXCEEDS APPLICABLE CSR DW STANDARD
100	CONCENTRATION IN GROUNDWATER EXCEEDS APPLICABLE CSR AW-F STANDARD
100	CONCENTRATION IN GROUNDWATER EXCEEDS APPLICABLE CSR DW AND CSR AW-F STANDARDS

ABBREVIATIONS

DW	DRINKING WATER
AW-F	AQUATIC LIFE, MARINE
FD	FIELD DUPLICATE
FDA	FIELD DUPLICATE AVAILABLE

NOTES

1. ALL VALUES ARE IN MICROGRAMS PER LITRE (µg/L), UNLESS OTHERWISE NOTED.
2. SAMPLING LOCATIONS ARE APPROXIMATE.

REFERENCES

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0 50 100 — 1:2,000 METRES

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TOWN OF LADYSMITH

CONSULTANT
GOLDER

YYYY-MM-DD	2020-02-26
DESIGNED	MN
PREPARED	RTJ
REVIEWED	GG
APPROVED	DF

PROJECT
LOTS 1, 4 AND 5 ADJACENT TO LADYSMITH HARBOUR
LADYSMITH HARBOUR, LADYSMITH, B.C.

TITLE
GROUNDWATER ANALYTICAL RESULTS - METALS

PROJECT NO.	PHASE	REV.	FIGURE
18109842	1000	0	7A

Path: \\golder.gds\gal\Kelowna\CAD-GIS\Client\TOWN OF LADYSMITH\Ladysmith Harbour\99_PROJECTS\18109842_Enviro1000\Task\18109842_1000\02_PRODUCTION\DWG\ | File Name: 18109842_1000- Results.dwg | Last Edited By: rylames Date: 2020-02-28 Time:11:00:19 AM | Printed By: RYJames Date: 2020-02-28 Time:11:03:49 AM
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0 25 mm



Location	MW00-9
Date Sampled	10/12/2009
Hardness (mg/L)	56
pH	6.33
Monoaromatic Hydrocarbons	<
PAHs	<
Other Hydrocarbons	<

Location	MW00-11
Date Sampled	10/12/2009
Hardness (mg/L)	198
pH	7.33
Monoaromatic Hydrocarbons	<
PAHs	<
Other Hydrocarbons	<

Location	MW09-1
Date Sampled	18/11/2009
Hardness (mg/L)	106
pH	7.97
Benzene	56
Toluene	198
Total Xylene	1200
VPH	3800

Location	MW09-2
Date Sampled	18/11/2009
Hardness (mg/L)	151
pH	7.36
Monoaromatic Hydrocarbons	<
PAHs	<
Other Hydrocarbons	<

Location	MW09-3
Date Sampled	18/11/2009
Hardness (mg/L)	144
pH	7.69
Monoaromatic Hydrocarbons	<
PAHs	<
Other Hydrocarbons	<

Location	MW09-4
Date Sampled	19/11/2009
Hardness (mg/L)	206
pH	7.64
Monoaromatic Hydrocarbons	<
PAHs	<
Other Hydrocarbons	<

Location	MW09-5
Date Sampled	19/11/2009
Hardness (mg/L)	175
pH	7.6
Monoaromatic Hydrocarbons	<
PAHs	<
Other Hydrocarbons	<

Location	MW09-6
Date Sampled	19/11/2009
Hardness (mg/L)	-
pH	6.76
Monoaromatic Hydrocarbons	<
PAHs	<
Other Hydrocarbons	<

Location	MW09-7		
Date Sampled	16/02/2011	22/02/2011	20/11/2009
Hardness (mg/L)	185	-	102
pH	6.34	6.49	7.22
Monoaromatic Hydrocarbons	<	<	<
PAHs	<	<	<
Other Hydrocarbons	<	<	<

Location	MW09-8				
Date Sampled	16/02/2011	16/02/2011	20/11/2009	20/11/2009	22/02/2011
Hardness (mg/L)	694	684	1480	1890	-
pH	6.99	6.99	6.92	6.92	6.66
benzo(a)pyrene	<	<	0.03	0.03	<
benzo(b)fluoranthene	<	<	0.09	<	<
pyrene	<	<	0.23	0.23	<

Location	MW09-9	
Date Sampled	25-Nov-09	25-Nov-09
Hardness (mg/L)	149	149
pH	6.75	6.75
Monoaromatic Hydrocarbons	<	<
PAHs	<	<
Other Hydrocarbons	<	<

Location	MW09-10
Date Sampled	19/11/2009
Hardness (mg/L)	149
pH	8.13
Monoaromatic Hydrocarbons	<
PAHs	<
Other Hydrocarbons	<

Location	MW09-11
Date Sampled	19/11/2009
Hardness (mg/L)	145
pH	7.88
Monoaromatic Hydrocarbons	<
PAHs	<
Other Hydrocarbons	<

Location	MW09-16
Date Sampled	20/11/2009
Hardness (mg/L)	32
pH	7.62
Monoaromatic Hydrocarbons	<
PAHs	<
Other Hydrocarbons	<

Location	MW11-01	
Date Sampled	16/02/2011	22/02/2011
Hardness (mg/L)	425	-
pH	7.69	6.76
benzo(a)pyrene	0.399	<
benzo(b)fluoranthene	0.467	<
pyrene	0.854	<

Location	MW11-02	
Date Sampled	16/02/2011	22/02/2011
Hardness (mg/L)	83.50	-
pH	8.05	7.87
Monoaromatic Hydrocarbons	<	<
PAHs	<	<
Other Hydrocarbons	<	<

Location	MW00-02	
Date Sampled	25/09/2000	18/11/2009
Hardness (mg/L)	-	125.00
pH	-	7.55
Monoaromatic Hydrocarbons	-	<
PAHs	-	<
Other Hydrocarbons	<	<

Location	MW00-04	
Date Sampled	25/09/2000	10/12/2009
Hardness (mg/L)	-	101.00
pH	-	6.66
Monoaromatic Hydrocarbons	-	<
PAHs	-	<
Other Hydrocarbons	<	<

Location	MW00-06
Date Sampled	25/09/2000
Hardness (mg/L)	-
pH	-
Monoaromatic Hydrocarbons	-
PAHs	-
Other Hydrocarbons	<

Location	MW00-09
Date Sampled	25/09/2000
Hardness (mg/L)	-
pH	-
Monoaromatic Hydrocarbons	-
PAHs	-
Other Hydrocarbons	<

LEGEND

- SITE BOUNDARY
- LOT BOUNDARY
- MONITORING WELL LOCATION
- BOREHOLE LOCATION
- TEST PIT LOCATION
- SURFACE SAMPLE LOCATION
- GROUNDWATER PARAMETER CONCENTRATION MEETS APPLICABLE STANDARDS
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ABBREVIATIONS

DW	DRINKING WATER
AW-F	AQUATIC LIFE, MARINE
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FDA	FIELD DUPLICATE AVAILABLE

NOTES

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2. SAMPLING LOCATIONS ARE APPROXIMATE.

REFERENCES

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0 25 50 METRES 1:1,000

0 50 100 METRES 1:2,000

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CONSULTANT
GOLDER

YYYY-MM-DD	2020-02-26
DESIGNED	MN
PREPARED	RTJ
REVIEWED	GG
APPROVED	DF

PROJECT
LOTS 1, 4 AND 5 ADJACENT TO LADYSMITH HARBOUR
LADYSMITH HARBOUR, LADYSMITH, B.C.

TITLE
GROUNDWATER ANALYTICAL RESULTS - HYDROCARBONS

PROJECT NO.	PHASE	REV.	FIGURE
18109842	1000	0	7B

IF THIS MEASUREMENT DOES NOT MATCH WHAT IS SHOWN, THE SHEET SIZE HAS BEEN MODIFIED FROM: ANSI D

0 25 mm

Path: \\golder.gds\gal\Kelowna\CAD-GIS\Client\TOWN OF LADYSMITH\Ladysmith Harbour\99_PROJECTS\18109842_Enviro\1000\[Task]\18109842_1000\02_PRODUCTION\DWG\ | File Name: 18109842_1000 - Results.dwg | Last Edited By: rylames Date: 2020-02-28 Time:11:00:19 AM | Printed By: RYJames Date: 2020-02-28 Time:11:04:17 AM



LEGEND

- ⊕ HAND AUGER SAMPLE
- LEGAL BOUNDARY
- SITE BOUNDARY

NOT FOR CONSTRUCTION

0 20 40

1:1,000 METRES

NOTE(S)

1. LOCATIONS ARE APPROXIMATE.

REFERENCE(S)

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CLIENT

TOWN OF LADYSMITH

PROJECT

LOTS 1, 4 AND 5 ADJACENT TO LADYSMITH HARBOUR

TITLE

DECEMBER 2019 SOIL SAMPLE LOCATIONS

CONSULTANT

GOLDER

YYYY-MM-DD	2020-02-27
DESIGNED	AU
PREPARED	CDAB
REVIEWED	AU
APPROVED	DF

PROJECT NO.	CONTROL	REV.	FIGURE
18109842	1000	A	8

PATH: N:\Client\Town_of_Ladysmith\Ladysmith Harbour\99_PROJECTS\18109842-1000\02_PRODUCTION\MXD\Report\18109842_1000_FIG_8_December_2019_Soil_Sample_Locations.mxd PRINTED ON: 2020-02-27 AT: 10:46:39 AM

IF THIS MEASUREMENT DOES NOT MATCH WHAT IS SHOWN, THE SHEET SIZE HAS BEEN MODIFIED FROM: ANSI B

0 25mm



LEGEND

- SITE BOUNDARY
- LOT BOUNDARY
- APPROXIMATE APEC/AEC BOUNDARY

APEC/AEC	DESCRIPTION
AEC 1	FORMER FUEL PUMP ISLANDS, ASTs AND POSSIBLE PCB STORAGE
AEC 2	FORMER RAILWAY MAINTENANCE AREAS AND CURRENT BOAT REPAIR AND CONSTRUCTION OPERATIONS
AEC 3	FORMER WASTE OIL STORAGE AREA AND COMPRESSOR STORAGE AREA
APEC 4	STOCKPILE OF MATERIAL FROM AROUND SEWAGE TREATMENT PLANT
APEC 5	FILL OF UNKNOWN QUALITY ON LOT 5
APEC 6	FILL MATERIAL AT BLOCK B AND D OF DL 2016 (FORMER LOCATION OF THE SHINGLE MILL/SAWMILL
AEC 7	FORMER LOG DUMP (LOT 17G)

NOTES

1. SAMPLING LOCATIONS ARE APPROXIMATE.
2. AEC = AREA OF ENVIRONMENTAL CONCERN
   APEC = AREA OF POTENTIAL ENVIRONMENTAL CONCERN

REFERENCES

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TOWN OF LADYSMITH

PROJECT
LOTS 1, 4 AND 5 ADJACENT TO LADYSMITH HARBOUR
LADYSMITH HARBOUR, LADYSMITH, B.C.

TITLE
AREAS OF ENVIRONMENTAL CONCERN AND AREAS OF POTENTIAL ENVIRONMENTAL CONCERN

CONSULTANT

GOLDER

YYYY-MM-DD	2020-02-26
DESIGNED	MN
PREPARED	RTJ
REVIEWED	GG
APPROVED	DF

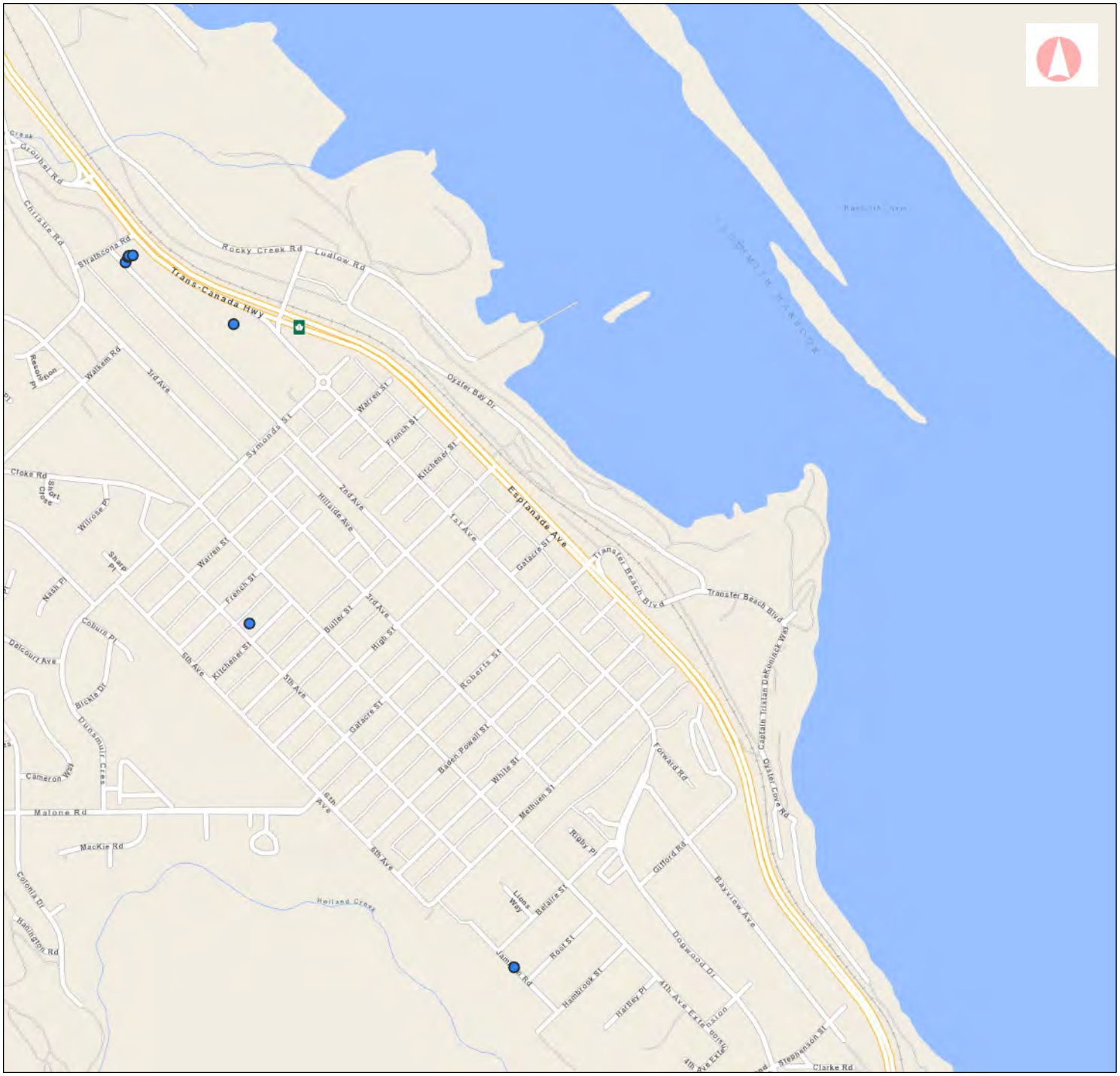
PROJECT NO.	PHASE	REV.	FIGURE
18109842	1000	0	9

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APPENDIX A

iMapBC



Water Wells near Site

Legend

- Water Wells - All

0 0.41 0.81 km

1: 20,000

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Datum: NAD83

Projection: WGS_1984_Web_Mercator_Auxiliary_Sphere

Key Map of British Columbia



APPENDIX B

Land Title Search

TITLE SEARCH PRINT 2019-12-06, 14:47:48

File Reference: 181098421000 Requestor: Greg Krewski

Declared Value $292000

CURRENT INFORMATION ONLY - NO CANCELLED INFORMATION SHOWN

Land Title District VICTORIA
Land Title Office VICTORIA

Title Number CA7424608
From Title Number ET76069

Application Received 2019-04-01

Application Entered 2019-04-01

Registered Owner in Fee Simple
Registered Owner/Mailing Address: TOWN OF LADYSMITH
410 ESPLANADE, PO BOX 220
LADYSMITH, BC
V9G 1A2

Taxation Authority Ladysmith, Town of

Description of Land
Parcel Identifier: 010-208-861
Legal Description:
LOT 5, DISTRICT LOTS 24 AND 56, OYSTER DISTRICT, PLAN 45800

Legal Notations
HERETO IS ANNEXED EASEMENT EB15415 OVER DISTRICTS LOTS 138, 142 AND 6G OYSTER DISTRICT AND LOT 4, PLAN 45800
SERVIENT TENEMENT CANCELLED AS TO DISTRICT LOT 6G, OYSTER DISTRICT, BY EP107093, 2000-12-20

Charges, Liens and Interests
Nature: UNDERSURFACE RIGHTS
Registration Number: D23415
Registration Date and Time: 1975-01-27 08:10
Registered Owner: HER MAJESTY THE QUEEN IN RIGHT OF THE PROVINCE OF BRITISH COLUMBIA
Remarks: INTER ALIA
ASSIGNMENT OF 84658G AND 195198G (SEE 74848G, 379783G, 325302G, 19225N AND 325301G)

Duplicate Indefeasible Title NONE OUTSTANDING

Transfers	NONE
Pending Applications	NONE

CURRENT INFORMATION ONLY - NO CANCELLED INFORMATION SHOWN

Land Title District VICTORIA
Land Title Office VICTORIA

Title Number ET76069
From Title Number EM113875

Application Received 2002-07-04

Application Entered 2002-07-11

Title Cancelled **2019-04-01**

Registered Owner in Fee Simple
Registered Owner/Mailing Address: INVET HOLDINGS CORP., INC.NO. 48357-A
801 - 1039 - 17TH AVENUE S.W.
CALGARY, AB
T2T 0B2

Taxation Authority Ladysmith, Town of

Description of Land
Parcel Identifier: 010-208-861
Legal Description:
LOT 5, DISTRICT LOTS 24 AND 56, OYSTER DISTRICT, PLAN 45800

Legal Notations
EXPROPRIATION ACT NOTICE, SEE CA7163118, 2018-10-31
DEALINGS RESTRICTED

HERETO IS ANNEXED EASEMENT EB15415 OVER DISTRICTS LOTS 138, 142 AND 6G
OYSTER DISTRICT AND LOT 4, PLAN 45800
SERVIENT TENEMENT CANCELLED AS TO DISTRICT LOT 6G, OYSTER DISTRICT,
BY EP107093, 2000-12-20

NOTICE OF INTEREST, BUILDERS LIEN ACT (S.3(2)), SEE ET83785
FILED 2002-07-22

Charges, Liens and Interests

Nature:	UNDERSURFACE RIGHTS
Registration Number:	D23415
Registration Date and Time:	1975-01-27 08:10
Registered Owner:	HER MAJESTY THE QUEEN IN RIGHT OF THE PROVINCE OF BRITISH COLUMBIA
Remarks:	INTER ALIA ASSIGNMENT OF 84658G AND 195198G (SEE 74848G, 379783G, 325302G, 19225N AND 325301G)

Duplicate Indefeasible Title NONE OUTSTANDING

Transfers

Registration Date:	2019-04-01
Description:	ALL EPROPRATION ACT CA7424608

TITLE SEARCH PRINT 2019-12-06, 14:47:48

File Reference: 181098421000 Requestor: Greg Krewski

Declared Value $ 605000

CURRENT INFORMATION ONLY - NO CANCELLED INFORMATION SHOWN

Land Title District	VICTORIA
Land Title Office	VICTORIA
Title Number	EM113875
From Title Number	EK27544
Application Received	1998-11-27
Application Entered	1998-12-02
Title Cancelled	**2002-07-11**

Registered Owner in Fee Simple

Registered Owner/Mailing Address: S.V.R. ACQUISITIONS LTD., INC.NO. 48357-A
801 - 1039 - 17TH AVENUE S.W.
CALGARY, AB
T2T 0B2

Taxation Authority Ladysmith, Town of

Description of Land

Parcel Identifier: 010-208-861

Legal Description:

LOT 5, DISTRICT LOTS 24 AND 56, OYSTER DISTRICT, PLAN 45800

Legal Notations

HERETO IS ANNEXED EASEMENT EB15415 OVER DISTRICTS LOTS 138, 142 AND 6G OYSTER DISTRICT AND LOT 4, PLAN 45800
SERVIENT TENEMENT CANCELLED AS TO DISTRICT LOT 6G, OYSTER DISTRICT, BY EP107093, 2000-12-20

Charges, Liens and Interests

Nature:	UNDERSURFACE RIGHTS
Registration Number:	D23415
Registration Date and Time:	1975-01-27 08:10
Registered Owner:	HER MAJESTY THE QUEEN IN RIGHT OF THE PROVINCE OF BRITISH COLUMBIA
Remarks:	ASSIGNMENT OF 84658G AND 195198G (SEE 74848G, 379783G, 325302G, 19225N AND 325301G)

Nature:	OPTION TO PURCHASE
Registration Number:	EP40977
Registration Date and Time:	2000-05-26 14:56
Registered Owner:	GRANBY HARBOUR DEVELOPMENT CORPORATON INCORPORATION NO. 593903

Nature:	MORTGAGE
Registration Number:	EP83606
Registration Date and Time:	2000-10-04 09:40
Registered Owner:	MONTREAL TRUST COMPANY OF CANADA INCORPORATION NO. 34,811A
Remarks:	OF EP40977

Duplicate Indefeasible Title NONE OUTSTANDING

Transfers

Registration Date:	2002-07-11
Description:	ALL ET76069

TITLE SEARCH PRINT 2019-12-06, 14:47:49

File Reference: 181098421000 Requestor: Greg Krewski

Declared Value $ 605000

CURRENT INFORMATION ONLY - NO CANCELLED INFORMATION SHOWN

Land Title District	VICTORIA
Land Title Office	VICTORIA
Title Number	EK27544
From Title Number	EH148391
Application Received	1996-03-15
Application Entered	1996-03-26
Title Cancelled	**1998-12-02**

Registered Owner in Fee Simple

Registered Owner/Mailing Address: SEA VISION RESORTS DEVELOPMENT LTD., INC.NO. A42285,
19 GATACRE STREET,
P.O. BOX 1589,
LADYSMITH, BC
V0R 2E0.

Taxation Authority Ladysmith, Town of

Description of Land

Parcel Identifier: 010-208-861

Legal Description:

LOT 5, DISTRICT LOTS 24 AND 56, OYSTER DISTRICT, PLAN 45800

Legal Notations

HERETO INTER ALIA IS ANNEXED RESTRICTIVE COVENANT EB15414 OVER DISTRICT LOT 6G, OYSTER DISTRICT.
DOMINANT TENEMENT CANCELLED AS TO ALL EXCEPT LOT 5, PLAN 45800, SEE EK27539/41, 15-03-1996, K. JACQUES PER DBC.

HERETO IS ANNEXED EASEMENT EB15415 OVER DISTRICTS LOTS 138, 142 AND 6G OYSTER DISTRICT AND LOT 4, PLAN 45800

Charges, Liens and Interests

Nature:	UNDERSURFACE RIGHTS
Registration Number:	D23415
Registration Date and Time:	1975-01-27 08:10
Registered Owner:	HER MAJESTY THE QUEEN IN RIGHT OF THE PROVINCE OF BRITISH COLUMBIA
Remarks:	ASSIGNMENT OF 84658G AND 195198G (SEE 74848G, 379783G, 325302G, 19225N AND 325301G)

Duplicate Indefeasible Title NONE OUTSTANDING

Transfers

Registration Date:	1998-12-02
Description:	ALL EM113875

TITLE SEARCH PRINT 2019-12-06, 14:47:49

File Reference: 181098421000 Requestor: Greg Krewski

Declared Value $NOMINAL

CURRENT INFORMATION ONLY - NO CANCELLED INFORMATION SHOWN

Land Title District	VICTORIA
Land Title Office	VICTORIA
Title Number	EH148391
From Title Number	EG17932
Application Received	1994-11-10
Application Entered	1994-11-29
Title Cancelled	**1996-03-26**

Registered Owner in Fee Simple

Registered Owner/Mailing Address: TIMBERWEST FOREST I LIMITED, INC.NO. 440252
P.O. BOX 10058, PACIFIC CENTRE
VANCOUVER, BC
V7V 1J7

Taxation Authority Ladysmith, Town of

Description of Land

Parcel Identifier: 010-208-861

Legal Description:

LOT 5, DISTRICT LOTS 24 AND 56, OYSTER DISTRICT, PLAN 45800

Legal Notations

HERETO INTER ALIA IS ANNEXED RESTRICTIVE COVENANT EB15414 OVER DISTRICT LOT 6G, OYSTER DISTRICT.
DOMINANT TENEMENT CANCELLED AS TO ALL EXCEPT LOT 5, PLAN 45800, SEE EK27539/41, 15-03-1996, K. JACQUES PER DBC.

HERETO IS ANNEXED EASEMENT EB15415 OVER DISTRICTS LOTS 138, 142 AND 6G OYSTER DISTRICT AND LOT 4, PLAN 45800

Charges, Liens and Interests

Nature:	UNDERSURFACE RIGHTS
Registration Number:	D23415
Registration Date and Time:	1975-01-27 08:10
Registered Owner:	HER MAJESTY THE QUEEN IN RIGHT OF THE PROVINCE OF BRITISH COLUMBIA
Remarks:	ASSIGNMENT OF 84658G AND 195198G (SEE 74848G, 379783G, 325302G, 19225N AND 325301G)

Duplicate Indefeasible Title NONE OUTSTANDING

Transfers

Registration Date:	1996-03-26
Description:	ALL EK27544

TITLE SEARCH PRINT 2019-12-06, 14:47:49

File Reference: 181098421000 Requestor: Greg Krewski

Declared Value $43,700.00

CURRENT INFORMATION ONLY - NO CANCELLED INFORMATION SHOWN

Land Title District VICTORIA
Land Title Office VICTORIA

Title Number EG17932
From Title Number EG10509

Application Received 1993-02-15

Application Entered 1993-04-08

Title Cancelled **1994-11-29**

Registered Owner in Fee Simple
Registered Owner/Mailing Address: ELK FALLS WOOD PRODUCTS LIMITED, INC.NO. 440252
9TH FLOOR, 700 WEST GEORGIA STREET
VANCOUVER, BC
V7Y 1J7

Taxation Authority Ladysmith, Town of

Description of Land
Parcel Identifier: 010-208-861
Legal Description:
LOT 5, DISTRICT LOTS 24 AND 56, OYSTER DISTRICT, PLAN 45800

Legal Notations
HERETO INTER ALIA IS ANNEXED RESTRICTIVE COVENANT EB15414 OVER DISTRICT LOT 6G, OYSTER DISTRICT

HERETO INTER ALIA IS ANNEXED EASEMENT EB15415 OVER DISTRICTS LOTS 138, 142 AND 6G, OYSTER DISTRICT AND OVER LOT 4, PLAN 45800

Charges, Liens and Interests
Nature: UNDERSURFACE RIGHTS
Registration Number: D23415
Registration Date and Time: 1975-01-27 08:10
Registered Owner: HER MAJESTY THE QUEEN IN RIGHT OF THE PROVINCE OF BRITISH COLUMBIA
Remarks: ASSIGNMENT OF 84658G AND 195198G (SEE 74848G, 379783G, 325302G, 19225N AND 325301G)

Nature:	RIGHT OF FIRST REFUSAL
Registration Number:	EB15410
Registration Date and Time:	1988-02-24 12:19
Registered Owner:	TOWN OF LADYSMITH

Duplicate Indefeasible Title NONE OUTSTANDING

Transfers

Registration Date:	1994-11-29
Description:	ALL EH148391

PARCEL IDENTIFIER (PID): 010-208-828

SHORT LEGAL DESCRIPTION:S/45800/////4
MARG:* REM.
TAXATION AUTHORITY:
1 Ladysmith, Town of

FULL LEGAL DESCRIPTION: CURRENT
LOT 4, DISTRICT LOTS 8G, 11G, 24 AND 56, OYSTER DISTRICT, PLAN 45800,
EXCEPT PART IN PLANS VIP64405, VIP71943 AND VIP72131

MISCELLANEOUS NOTES:
SRW PLAN 45801

ASSOCIATED PLAN NUMBERS:
PLAN VIP45800
STATUTORY RIGHT OF WAY PLAN VIP45801
SUBDIVISION PLAN VIP64405
PLAN VIP71943
SUBDIVISION PLAN VIP72131

AFB/IFB: MN: Y PE: 3 SL: 1 TI: 1

PARCEL IDENTIFIER (PID): 010-208-861

SHORT LEGAL DESCRIPTION:S/45800/////5
MARG:
TAXATION AUTHORITY:
1 Ladysmith, Town of

FULL LEGAL DESCRIPTION: CURRENT
LOT 5, DISTRICT LOTS 24 AND 56, OYSTER DISTRICT, PLAN 45800

MISCELLANEOUS NOTES:

ASSOCIATED PLAN NUMBERS:
PLAN VIP45800

AFB/IFB: MN: N PE: 0 SL: 1 TI: 1

PARCEL IDENTIFIER (PID): 023-652-926

SHORT LEGAL DESCRIPTION:S/VIP64405/////1
MARG:
TAXATION AUTHORITY:
1 Ladysmith, Town of

FULL LEGAL DESCRIPTION: CURRENT
LOT 1 DISTRICT LOTS 24 AND 56 OYSTER DISTRICT PLAN VIP64405

MISCELLANEOUS NOTES:
SRW PLAN 45801

ASSOCIATED PLAN NUMBERS:
PLAN VIP45800
STATUTORY RIGHT OF WAY PLAN VIP45801
SUBDIVISION PLAN VIP64405

AFB/IFB: MN: Y PE: 0 SL: 1 TI: 1

TITLE SEARCH PRINT 2019-12-06, 14:47:49

File Reference: 181098421000 Requestor: Greg Krewski

Declared Value $NOMINAL

CURRENT INFORMATION ONLY - NO CANCELLED INFORMATION SHOWN

Title Issued Under SECTION 98 LAND TITLE ACT

Land Title District VICTORIA
Land Title Office VICTORIA

Title Number EB15407
From Title Number S110892
19225N

Application Received 1988-02-24

Application Entered 1988-03-07

Title Cancelled **1988-03-07**

Registered Owner in Fee Simple
Registered Owner/Mailing Address: TOWN OF LADYSMITH,
410 ESPLANADE
LADYSMITH, BC
V0R 2E0

Taxation Authority Ladysmith, Town of

Description of Land
Parcel Identifier: 010-208-861
Legal Description:
LOT 5, DISTRICT LOTS 24 AND 56, OYSTER DISTRICT, PLAN 45800

Legal Notations NONE

Charges, Liens and Interests
Nature: UNDERSURFACE RIGHTS
Registration Number: D23415
Registration Date and Time: 1975-01-27 08:10
Registered Owner: HER MAJESTY THE QUEEN IN RIGHT OF THE PROVINCE OF BRITISH COLUMBIA
Remarks: ASSIGNMENT OF 84658G AND 195198G (SEE 74848G, 379783G, 325302G, 19225N AND 325301G)

Duplicate Indefeasible Title NONE OUTSTANDING

Transfers

Registration Date:	1988-03-07
Description:	ALL EB15409

TITLE SEARCH PRINT 2019-12-06, 14:47:49

File Reference: 181098421000 Requestor: Greg Krewski

Declared Value $40,000

CURRENT INFORMATION ONLY - NO CANCELLED INFORMATION SHOWN

Land Title District VICTORIA
Land Title Office VICTORIA

Title Number EB15409
From Title Number EB15407

Application Received 1988-02-24

Application Entered 1988-03-07

Title Cancelled **1993-02-08**

Registered Owner in Fee Simple
Registered Owner/Mailing Address: CROWN FOREST INDUSTRIES LIMITED, INC.NO. 338,579
600 - 815 WEST HASTINGS STREET
VANCOUVER, BC
V6C 2Y4

Taxation Authority Ladysmith, Town of

Description of Land
Parcel Identifier: 010-208-861
Legal Description:
LOT 5, DISTRICT LOTS 24 AND 56, OYSTER DISTRICT, PLAN 45800

Legal Notations
HERETO INTER ALIA IS ANNEXED RESTRICTIVE COVENANT EB15414 OVER DISTRICT LOT 6G, OYSTER DISTRICT

HERETO INTER ALIA IS ANNEXED EASEMENT EB15415 OVER DISTRICTS LOTS 138, 142 AND 6G, OYSTER DISTRICT AND OVER LOT 4, PLAN 45800

Charges, Liens and Interests
Nature: UNDERSURFACE RIGHTS
Registration Number: D23415
Registration Date and Time: 1975-01-27 08:10
Registered Owner: HER MAJESTY THE QUEEN IN RIGHT OF THE PROVINCE OF BRITISH COLUMBIA
Remarks: ASSIGNMENT OF 84658G AND 195198G (SEE 74848G, 379783G, 325302G, 19225N AND 325301G)

Nature:	RIGHT OF FIRST REFUSAL
Registration Number:	EB15410
Registration Date and Time:	1988-02-24 12:19
Registered Owner:	TOWN OF LADYSMITH

Duplicate Indefeasible Title NONE OUTSTANDING

Transfers

Registration Date:	1993-02-08
Description:	ALL EG10509

TITLE SEARCH PRINT 2019-12-06, 14:49:10

File Reference: 181098421000 Requestor: Greg Krewski

CURRENT INFORMATION ONLY - NO CANCELLED INFORMATION SHOWN

Title Issued Under SECTION 98 LAND TITLE ACT

Land Title District VICTORIA
Land Title Office VICTORIA

Title Number EK132565
From Title Number EB15406

Application Received 1996-11-25

Application Entered 1997-01-21

Registered Owner in Fee Simple
Registered Owner/Mailing Address: TOWN OF LADYSMITH
P.O. BOX 220
410 ESPLANDE
LADYSMITH, BC
V0R 2E0

Taxation Authority Ladysmith, Town of

Description of Land
Parcel Identifier: 023-652-926
Legal Description:
LOT 1 DISTRICT LOTS 24 AND 56 OYSTER DISTRICT PLAN VIP64405

Legal Notations
HERETO IS ANNEXED EASEMENT EL8116 OVER LOT 4, PLAN 45800 EXCEPT PART IN PLAN VIP64405

Charges, Liens and Interests
Nature: EXCEPTIONS AND RESERVATIONS
Registration Number: M76300
Registered Owner: ESQUIMALT AND NANAIMO RAILWAY COMPANY
Remarks: INTER ALIA
A.F.B. 9.693.7434A, SECTION 172(3)
FOR ACTUAL DATE AND TIME OF REGISTRATION SEE ORIGINAL GRANT FROM E AND N RAILWAY COMPANY AS TO PART FORMERLY PARCEL B (DD 10119N), DISTRICT LOT 24, OYSTER DISTRICT
FOR ACTUAL DATE AND TIME OF REGISTRATION SEE ORIGINAL GRANT FROM E & N RAILWAY COMPANY

Nature:	UNDERSURFACE RIGHTS
Registration Number:	D23415
Registration Date and Time:	1975-01-27 08:10
Registered Owner:	HER MAJESTY THE QUEEN IN RIGHT OF THE PROVINCE OF BRITISH COLUMBIA
Remarks:	INTER ALIA AS TO PART FORMERLY LOT 1, INCLUDING PARCEL "A" OF PLAN 4224 AND TO THAT PART FORMERLY DISTRICT LOT 56, OYSTER DISTRICT AND TO THAT PART IN DISTRICT LOT 24 OYSTER DISTRICT SHOWN COLOURED PINK ON PLAN 2030 ASSIGNMENT OF 195198G AND 84658G (SEE 74848G, 379783G, 325301G, 325302G, 19225N AND 325303G)

Nature:	EASEMENT
Registration Number:	EB15415
Registration Date and Time:	1988-02-24 12:21
Remarks:	INTER ALIA, APPURTENANT TO LOT 5, PLAN 45800 AND TO LEASE EB15413 REGISTERED AGAINST DISTRICT LOT 16G, OYSTER DISTRICT AND TO LEASE EB15411 REGISTERED AGAINST DISTRICT LOT 17G, OYSTER DISTRICT AND TO LEASE EB15412 REGISTERED AGAINST DISTRICT LOT 651, COWICHAN DISTRICT DOMINANT TENEMENT RE: LEASE EB15413 CANCELLED BY EK27539, 15-03-1996, K. JACQUES PER DBC DOMINANT TENEMENT RE. LEASE EB15411 CANCELLED BY EK27540 - 15.03.1996 - K JACQUES PER DBC. DOMINANT TENEMENT RE. LEASE EB15412 CANCELLED BY EK27541 - 15.03.1996 - K JACQUES PER DBC.

Duplicate Indefeasible Title NONE OUTSTANDING

Transfers NONE

Pending Applications NONE

CURRENT AND CANCELLED INFORMATION SHOWN

Land Title District VICTORIA
Land Title Office VICTORIA

Title Number ES24122
From Title Number ES11109

Application Received 2001-03-30

Application Entered 2001-04-03

Registered Owner in Fee Simple
Registered Owner/Mailing Address: TOWN OF LADYSMITH
P.O. BOX 220
410 ESPLANDE
LADYSMITH, BC
V0R 2E0

Taxation Authority Ladysmith, Town of

Description of Land
Parcel Identifier: 010-208-828
Legal Description:
LOT 4, DISTRICT LOTS 8G, 11G, 24 AND 56, OYSTER DISTRICT, PLAN 45800, EXCEPT PART IN PLANS VIP64405, VIP71943 AND VIP72131

Legal Notations
NOTICE OF INTEREST BUILDERS LIEN ACT, (S.3(2)) SEE ET83784 2002 07 22 REGARDING EB15408

Charges, Liens and Interests
Nature: EXCEPTIONS AND RESERVATIONS
Registration Number: M76300
Registered Owner: ESQUIMALT AND NANAIMO RAILWAY COMPANY
Remarks: INTER ALIA
A.F.B. 9.693.7434A, SECTION 172(3)
FOR ACTUAL DATE AND TIME OF REGISTRATION SEE ORIGINAL GRANT FROM E AND N RAILWAY COMPANY AS TO PART FORMERLY PARCEL B (DD 10119N), DISTRICT LOT 24, OYSTER DISTRICT
FOR ACTUAL DATE AND TIME OF REGISTRATION SEE ORIGINAL GRANT FROM E & N RAILWAY COMPANY

Nature:	UNDERSURFACE RIGHTS
Registration Number:	D23415
Registration Date and Time:	1975-01-27 08:10
Registered Owner:	HER MAJESTY THE QUEEN IN RIGHT OF THE PROVINCE OF BRITISH COLUMBIA
Remarks:	INTER ALIA AS TO PART FORMERLY LOT 1, INCLUDING PARCEL "A" OF PLAN 4224 AND TO THAT PART FORMERLY DISTRICT LOT 56, OYSTER DISTRICT AND TO THAT PART IN DISTRICT LOT 24 OYSTER DISTRICT SHOWN COLOURED PINK ON PLAN 2030 ASSIGNMENT OF 195198G AND 84658G (SEE 74848G, 379783G, 325301G, 325302G, 19225N AND 325303G)

Nature:	UNDERSURFACE AND OTHER EXC & RES
Registration Number:	S6166
Registration Date and Time:	1987-01-27 14:18
Registered Owner:	HER MAJESTY THE QUEEN IN RIGHT OF THE PROVINCE OF BRITISH COLUMBIA
Remarks:	SECTION 47, LAND ACT, DD S6163 AS TO PART IN DISTRICT LOT 11G, OYSTER DISTRICT

Nature:	UNDERSURFACE AND OTHER EXC & RES
Registration Number:	S6170
Registration Date and Time:	1987-01-27 14:26
Registered Owner:	HER MAJESTY THE QUEEN IN RIGHT OF THE PROVINCE OF BRITISH COLUMBIA
Remarks:	SECTION 47, LAND ACT, DD S6167 AS TO PART IN DISTRICT LOT 8G, OYSTER DISTRICT

Nature:	STATUTORY RIGHT OF WAY
Registration Number:	EB15408
Registration Date and Time:	1988-02-22 12:18
Registered Owner:	CROWN FOREST INDUSTRIES LIMITED, (INC. NO. 338,579)
Transfer Number:	EB15408 TRANSFERRED TO EJ69244
Registered Owner:	CROWN FOREST INDUSTRIES LIMITED INCORPORATION NO. A37357
Transfer Number:	EJ69244 TRANSFERRED TO EK27546
Registered Owner:	SEA VISION RESORTS DEVELOPMENT LTD.
Transfer Number:	EK27546 TRANSFERRED TO EM113867
Registered Owner:	S.V.R. ACQUISITIONS LTD. INCORPORATION NO. 48357-A
Transfer Number:	EM113867 TRANSFERRED TO ET76070
Registered Owner:	INVET HOLDING CORP. INCORPORATION NO. A48357
Transfer Number:	ET76070
Remarks:	PART IN PLAN 45801

Nature:	COVENANT
Registration Number:	EB15402
Registration Date and Time:	1988-02-24 12:17
Registered Owner:	HER MAJESTY THE QUEEN IN RIGHT OF THE PROVINCE OF BRITISH COLUMBIA
Remarks:	PURSUANT TO SECTION 215 LAND TITLE ACT INCLUDES INDEMNITY PURSUANT TO SECTION 215(2)(A) LAND TITLE ACT, AS TO PARTS IN DISTRICT LOT 8G AND 11G, OYSTER DISTRICT

Nature:	EASEMENT
Registration Number:	EB15415
Registration Date and Time:	1988-02-24 12:21
Remarks:	INTER ALIA, APPURTENANT TO LOT 5, PLAN 45800 AND TO LEASE EB15413 REGISTERED AGAINST DISTRICT LOT 16G, OYSTER DISTRICT AND TO LEASE EB15411 REGISTERED AGAINST DISTRICT LOT 17G, OYSTER DISTRICT AND TO LEASE EB15412 REGISTERED AGAINST DISTRICT LOT 651, COWICHAN DISTRICT DOMINANT TENEMENT RE: LEASE EB15413 CANCELLED BY EK27539, 15-03-1996, K. JACQUES PER DBC DOMINANT TENEMENT RE. LEASE EB15411 CANCELLED BY EK27540 - 15.03.1996 - K JACQUES PER DBC. DOMINANT TENEMENT RE. LEASE EB15412 CANCELLED BY EK27541 - 15.03.1996 - K JACQUES PER DBC.

Nature:	EASEMENT
Registration Number:	EL8116
Registration Date and Time:	1997-01-20 11:02
Remarks:	INTER ALIA APPURTENANT TO LOT 1, PLAN VIP64405

Nature:	OPTION TO PURCHASE
Registration Number:	ES2683
Registration Date and Time:	2001-01-15 09:30
Registered Owner:	GRANBY HARBOUR DEVELOPMENT CORP. INCORPORATION NO. 593903
Remarks:	INTER ALIA
Cancelled By:	**EV58154**
Cancelled Date:	**2003-06-04**

Nature:	CAVEAT
Registration Number:	EV63640
Registration Date and Time:	2003-06-17 14:39
Registered Owner:	GRANBY HARBOUR DEVELOPMENT CORP.
Remarks:	INTER ALIA
Cancelled By:	**EV98275**
Cancelled Date:	**2003-08-27**

Duplicate Indefeasible Title NONE OUTSTANDING

Transfers NONE

Pending Applications

Parcel Identifier: 010-208-828
Application Number/Type: CA7808088 TO CA7808090 CHARGE RELEASE

Corrections

ET62424A CHARGE OWNER NAME CORRECTED M76300 2002-06-05 09:30:00

APPENDIX C

BC ENV Contaminated Site Registry



BC MoE Site Registry

Legend

- Environmental Remediation

0 0.20 0.41 km

1: 10,000

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Datum: NAD83

Projection: WGS_1984_Web_Mercator_Auxiliary_Sphere

Key Map of British Columbia

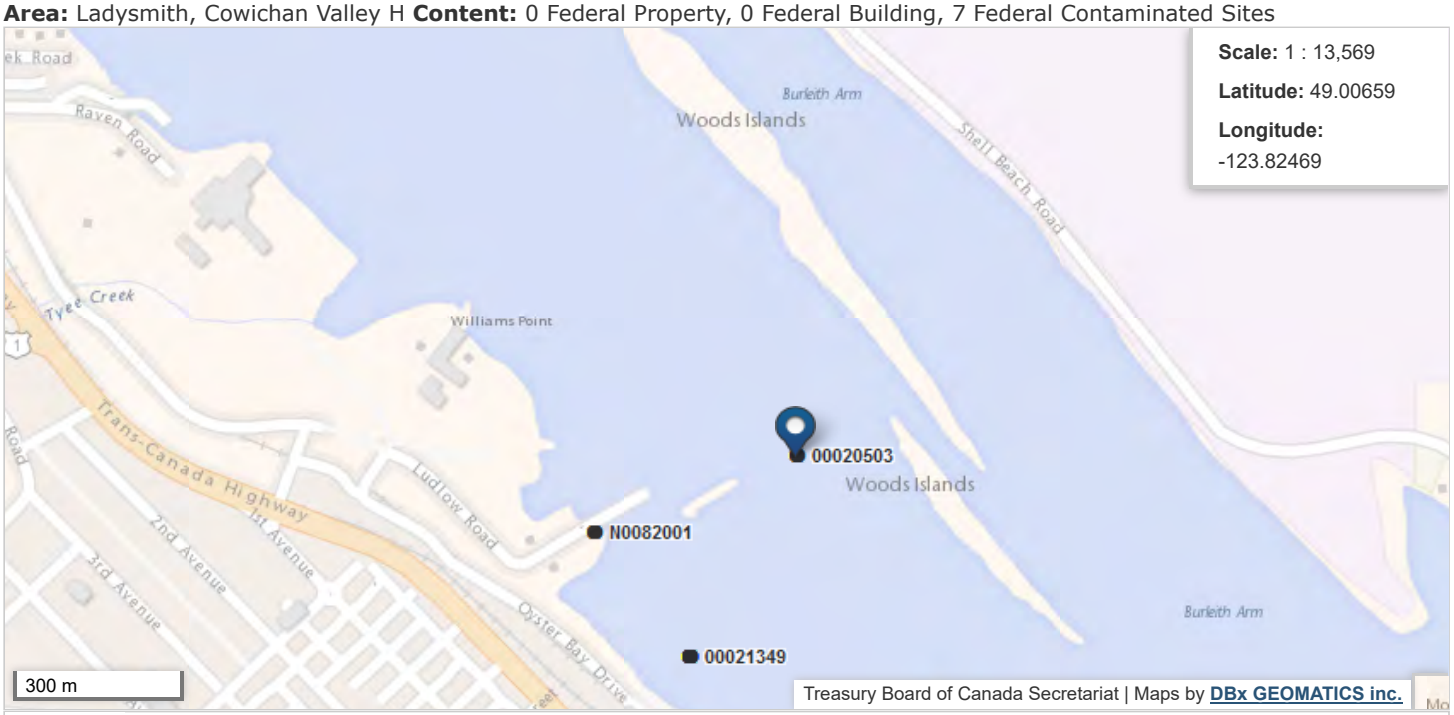


APPENDIX D

Federal Contaminated Site Inventory

DFRP/FCSI - Map Navigator

Area: Ladysmith, Cowichan Valley H **Content:** 0 Federal Property, 0 Federal Building, 7 Federal Contaminated Sites



Layers

- Federal Properties
- Federal Buildings
- Federal Contaminated Sites
- Economic Region
- Census Divisions
- Census Subdivisions
- Metropolitan Areas
- Federal Electoral Districts
- Treaty Areas

[1] This layer is visible only when the map scale is smaller than 1:3,000,000.

[2] Suspected Active Closed

[3] Google base maps are only available when the map scale is smaller than 1:60,000.

The tables content is currently representating the actual map.

UPDATE TABLES

Federal Properties (0) / Parcels (0) | **Federal Buildings (0)** | **Federal Contaminated Sites (7)**

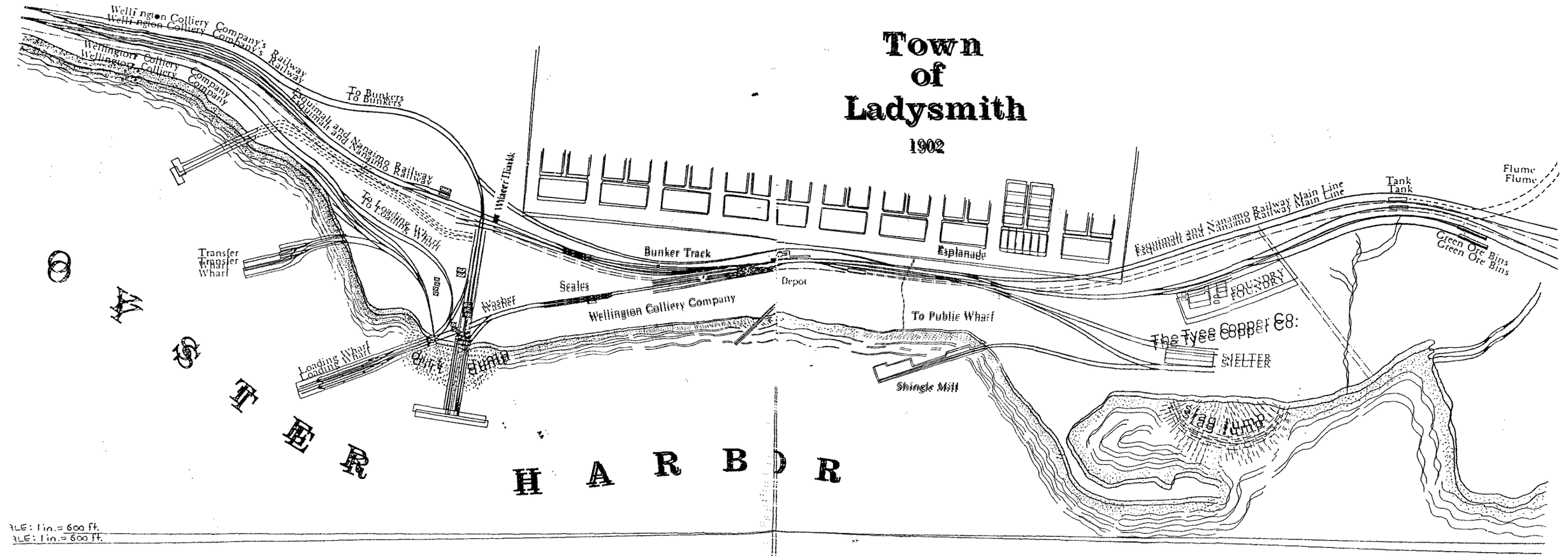
Number of rows per page: 25 Page(s): 1 / 1

Site Number	Reporting Organization or Branch	Internal Identifier	Site Name
00020503	Fisheries and Oceans Canada	P K 01227	Ladysmith Boat Basin No. 1 - Unassessed
00021349	Fisheries and Oceans Canada	P S 06086	Ladysmith (Waste Oil ASTs)
00021350	Fisheries and Oceans Canada	P S 06086	Ladysmith (Nearshore Sediment and surface water)
00021352	Fisheries and Oceans Canada	P S 06086	Ladysmith (Boat Grid)
00021353	Fisheries and Oceans Canada	P S 06086	Ladysmith (Fill Material)
00021354	Fisheries and Oceans Canada	P S 06086	Ladysmith (Stormwater Discharge)
N0082001	Transport Canada	N0082001	Waterlot

Number of rows per page: 25 Page(s): 1 / 1

APPENDIX E

Local and Regional Government Responses


Town
of
Ladysmith
1902
Wellington Colliery Company's Railway
Wellington Colliery Company's Railway
Wellington Colliery Company
Wellington Colliery Company
To Bunkers
To Bunkers
Esquimalt and Nanaimo Railway
Esquimalt and Nanaimo Railway
To Loading Wharf
To Loading Wharf
Water Tank
Transfer Wharf
Transfer Wharf
Bunker Track
Esplanade
Depot
Scales
Washer
Wellington Colliery Company
To Public Wharf
Loading Wharf
Loading Wharf
Shingle Mill
Esquimalt and Nanaimo Railway Main Line
Esquimalt and Nanaimo Railway Main Line
Tank
Tank
Flume
Flume
Green Ore Bins
Green Ore Bins
FOUNDRY
FOUNDRY
SMELTER
SMELTER
Stag Jump
OYSTER HARBOR

From: User - Infoaction [infoaction@vpl.ca]
Sent: September 4, 2009 4:43 PM
To: Daniels, Amber
Subject: RE: Directory Search

Hi Amber,

I'm terribly sorry but there are no city directories with street indexes for Ladysmith, so we will not be able to carry out this search for you.

Regrets,

Claudia

InfoAction
Vancouver Public Library's Information & Research Centre
Level 5, 350 West Georgia St.
Vancouver, BC V6B 6B1
t. 604-331-3612
f. 604-331-3611
e. infoaction@vpl.ca

http://www.infoaction.ca

From: Daniels, Amber [Amber_Daniels@golder.com]
Sent: Friday, September 04, 2009 4:01 PM
To: User - Infoaction
Subject: Directory Search

Hello! I was hoping someone could perform an historical directory search for Ladysmith: Please go back as far as possible, if you will go over 100 or so dollars please give me a call. I am specifically looking for these streets:

All of: Oyster Bay Drive
Captain de Konnick Way
Transfer Beach Blvd
Oyster Cove Rd
Ludlow Rd

Blocks 100-600 for First and Second Avenues
Blocks 0-200 for Buller St, High St, Gatacre St, Roberts St. Baden-Powell St, White Street, and Methuen Street

Thanks so much, PO # 09-1436-5008

Amber Daniels

Amber Daniels (B.Sc., GradTech.) | Environmental Scientist | **Golder Associates Ltd.**
500 - 4260 Still Creek Drive, Burnaby, British Columbia, Canada V5C 6C6
T: +1 (604) 296 4200 | **D:** +1 (604) 296-4308 | **F:** +1 (604) 298 5253 | **E:** Amber_Daniels@golder.com |
www.golder.com

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Please consider the environment before printing this email.



LEGEND

- Site Boundary
- Filled Foreshore
- Uplands
- Water Lot

* Lot sizes derived from Baker & Osland Appraisals Ltd. (2002)

DRAFT

REFERENCE

Image © 2009 TerraMetrics. Captured from Google Earth Pro © 2009.
Other base data obtained from WMS in the Province of British Columbia.
Projection: UTM Zone 10 Datum: NAD 83

PROJECT	CROWN CONTAMINATED SITES PROGRAM, CROWN LAND OPPORTUNITIES AND RESTORATION BRANCH B.C. MINISTRY OF FORESTS, LANDS AND NATURAL RESOURCE OPERATIONS LADYSMITH HARBOUR, LADYSMITH, B.C.
TITLE	**SITE PLAN SHOWING LEGAL LOTS**

Golder Associates
Greater Vancouver Office, B.C.

PROJECT No. 09-1436-5008			PHASE No. 3000	
DESIGN	LK	02SEP10	SCALE AS SHOWN	REV. 1
GIS	MH	02SEP10	**FIGURE 2**	
CHECK				
REVIEW				

N:\Bur-Graphics\Projects\2009\1436\09-1436-5008\GIS\projects\3000\figure-02_siteplan.mxd

From:	Geoff Goodall
To:	Newman, Mark; Jake Belobaba; Flotten, Dawn
Cc:	Chatwin-Davies, Robyn
Subject:	RE: Stage 1: Uplands
Date:	December 4, 2019 8:37:05 AM
Attachments:	image006.jpg image008.png image004.png image005.jpg 20191204084605826.pdf

There may be some confusion here, the material that was taken to the site was native material that was removed from the ground at the sewage treatment plant site to facilitate expansion of the plant. To our knowledge there was no contamination in this material. Apparently it was a very sandy material. I have enclosed a map that indicates the approximate area where the material was placed.

Geoff Goodall
Director
Infrastructure Services
250-245-6440
330 6th Avenue MAIL PO Box 220 Ladysmith, BC V9G 1A2
Working together to build our future

From: Newman, Mark [mailto:Mark_Newman@golder.com]
Sent: December 3, 2019 5:33 PM
To: Jake Belobaba; Flotten, Dawn
Cc: Geoff Goodall; Chatwin-Davies, Robyn
Subject: RE: Stage 1: Uplands

Hi Jake,

Thanks for this. If Geoff is able to provide some more details on where and when the byproducts were deposited, that would be useful (I have attached a figure showing the outlines of the lots for reference).

If you have any information to suggest chemicals (flocculants for example) were disposed of there, that would be useful to know also.

Just to give you a brief update – I have not yet had time to reach out to Kelly or Clayton yet regarding the site visit. I'm hoping to line that up for next week, depending on their schedule.

Thanks,
Mark

Mark Newman (MSc, PGeo)
Environmental Geoscientist

Golder Associates Ltd.
2nd floor, 3795 Carey Road, Victoria, British Columbia, Canada V8Z 6T8
D: +1 250 419 4930 | **T:** +1 250 881 7372 | **C:** +1 604 354 4539 | **golder.com**
LinkedIn | Facebook | Twitter

Work Safe, Home Safe

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Please consider the environment before printing this email.

From: Jake Belobaba <jbelobaba@ladysmith.ca>
Sent: December-02-19 3:41 PM
To: Flotten, Dawn <Dawn_Flotten@golder.com>; Newman, Mark <Mark_Newman@golder.com>
Cc: Geoff Goodall <ggoodall@ladysmith.ca>
Subject: RE: Stage 1: Uplands

EXTERNAL EMAIL

Hi Dawn and Mark

One more thing I forgot to mention, in the past there had been some disposal of byproducts from the sewage treatment plant on the "jewel" portion of Lot 4. I've cc'd Geoff who can provide you with more information about this. We believe this was occurring pre-2012, but will need to do a bit of digging to be sure.

Cheers

From: Jake Belobaba
Sent: November 27, 2019 4:25 PM
To: Flotten, Dawn <Dawn_Flotten@golder.com>; 'Newman, Mark' <Mark_Newman@golder.com>
Cc: Kelly Giesbrecht (kgiesbrecht@ladysmith.ca) <kgiesbrecht@ladysmith.ca>; Clayton Postings (cpostings@ladysmith.ca) <cpostings@ladysmith.ca>
Subject: Stage 1: Uplands

Hi Dawn and Mark

Thanks for the call today. I've attached Kelly and Clayton's contact info. Please contact them to set up site visits and interviews. Clayton should be able to provide you with some contacts of folks from the historical society that worked in the rail facility (if necessary). As I mentioned, there have been some excavations on site that may be useful for the site visit (Kelly can provide more detail). There's been no industrial use of the site since the original Golder Report in 2012

As always, feel free to contact me if you need any additional info.

Cheers

Jake Belobaba, RPP, MCIP
Director of Development Services
Development Services Department
Phone: 250.245.6405
Cell: 250.616.3755
132C Roberts St MAIL PO Box 220 Ladysmith, BC V9G 1A2

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APPENDIX F

Site Photos



Photo 1: Former Railway Repair Building (looking northwest)



Photo 2: Bank on the south side of Former Railway Repair Building (looking northwest)



Photo 3: Interior of Former Railway Repair Building (looking northwest)



Photo 4: Excavation for footing installation inside Former Railway Repair Building (soil sample location HS19-01)



Photo 5: Railyard area on Site (looking southeast)



Photo 6: Septic access (round metal plate) at the north end of the Washroom Building (looking southwest)



Photo 7: Car Shop Building on Site (looking southeast)



Photo 8: Stockpiled soil removed from footing installation excavations inside the Former Railway Repair Building located southeast of the Former Railway Repair Building (looking southeast)



Photo 9: Historical location of Former Compressor House, oil drum storage area and ASTs (looking southeast)



Photo 10: Former Cable Splicing Shed (looking west)



Photo 11: Co-Op Cardlock and "Wash me on Ludlow" car wash located adjacent to the north of the Site (looking southeast)



Photo 12: Slack Point Park located adjacent to the south east of the southeast of the Site (looking north)



Photo 13: Shell service station located at 728 Esplanade Avenue located southwest of the Site across the Trans Canada Highway (looking west)

APPENDIX G

Laboratory Certificate of Analysis

GOLDER ASSOCIATES LTD.
ATTN: Alanna Umphrey
3795 Carey Road, Second Floor
Victoria BC V8Z 6T8

Date Received: 18-DEC-19
Report Date: 27-DEC-19 14:55 (MT)
Version: FINAL

Client Phone: 250-881-7372

Certificate of Analysis

Lab Work Order #: L2398084

Project P.O. #: NOT SUBMITTED
Job Reference: 18109842
C of C Numbers: 07043
Legal Site Desc:

Amber Springer, B.Sc
Account Manager

[This report shall not be reproduced except in full without the written authority of the Laboratory.]

ALS ENVIRONMENTAL ANALYTICAL REPORT

Sample ID	L2398084-1	L2398084-2	L2398084-3	L2398084-4	L2398084-5
Description	SO	SO	SO	SO	SO
Sampled Date	16-DEC-19	16-DEC-19	16-DEC-19	16-DEC-19	16-DEC-19
Sampled Time					
Client ID	07043-01	07043-02	07043-03	07043-04	07043-05
Grouping / **Analyte**					
SOIL					
Physical Tests Moisture (%)	19.0	22.8	18.3	21.4	34.0
pH (1:2 soil:water) (pH)	6.86	8.15	8.04	7.32	5.89
Metals Aluminum (Al) (mg/kg)	17300	20900	21000	25400	21600
Antimony (Sb) (mg/kg)	0.12	0.14	0.15	0.20	2.78
Arsenic (As) (mg/kg)	6.12	7.01	8.43	7.33	7.46
Barium (Ba) (mg/kg)	67.1	73.7	83.1	81.1	325
Beryllium (Be) (mg/kg)	0.31	0.28	0.28	0.35	0.45
Bismuth (Bi) (mg/kg)	<0.20	<0.20	<0.20	<0.20	0.29
Boron (B) (mg/kg)	<5.0	<5.0	<5.0	<5.0	10.3
Cadmium (Cd) (mg/kg)	0.093	0.031	0.041	0.113	0.655
Calcium (Ca) (mg/kg)	4500	9760	10200	5540	7380
Chromium (Cr) (mg/kg)	29.3	38.6	36.5	37.0	36.0
Cobalt (Co) (mg/kg)	8.35	12.3	12.1	12.3	11.4
Copper (Cu) (mg/kg)	23.4	49.4	49.6	31.4	118
Iron (Fe) (mg/kg)	22500	30600	31800	35300	23200
Lead (Pb) (mg/kg)	4.29	2.57	2.80	5.88	59.4
Lithium (Li) (mg/kg)	17.5	13.7	13.1	13.0	17.9
Magnesium (Mg) (mg/kg)	4640	6660	6570	5540	5560
Manganese (Mn) (mg/kg)	205	532	553	918	301
Mercury (Hg) (mg/kg)	0.055	0.063	0.051	0.051	0.175
Molybdenum (Mo) (mg/kg)	0.78	0.21	0.25	1.98	1.17
Nickel (Ni) (mg/kg)	16.1	21.9	21.1	20.8	32.4
Phosphorus (P) (mg/kg)	194	557	529	253	473
Potassium (K) (mg/kg)	350	590	610	440	590
Selenium (Se) (mg/kg)	<0.20	<0.20	<0.20	0.27	0.83
Silver (Ag) (mg/kg)	<0.10	<0.10	<0.10	<0.10	0.21
Sodium (Na) (mg/kg)	246	604	668	314	332
Strontium (Sr) (mg/kg)	32.0	67.8	72.8	59.5	68.6
Sulfur (S) (mg/kg)	<1000	<1000	<1000	<1000	<1000
Thallium (Tl) (mg/kg)	<0.050	<0.050	<0.050	<0.050	0.063
Tin (Sn) (mg/kg)	<2.0	<2.0	<2.0	<2.0	5.6
Titanium (Ti) (mg/kg)	1350	1520	1760	1740	1280
Tungsten (W) (mg/kg)	<0.50	<0.50	<0.50	<0.50	<0.50
Uranium (U) (mg/kg)	0.533	0.258	0.271	0.404	0.515
Vanadium (V) (mg/kg)	78.8	82.5	91.5	116	82.6
Zinc (Zn) (mg/kg)	30.8	34.0	33.8	33.5	132
Zirconium (Zr) (mg/kg)	3.9	5.5	6.1	3.2	4.4

* Please refer to the Reference Information section for an explanation of any qualifiers detected.

ALS ENVIRONMENTAL ANALYTICAL REPORT

	Sample ID	L2398084-1	L2398084-2	L2398084-3	L2398084-4	L2398084-5
	Description	SO	SO	SO	SO	SO
	Sampled Date	16-DEC-19	16-DEC-19	16-DEC-19	16-DEC-19	16-DEC-19
	Sampled Time					
	Client ID	07043-01	07043-02	07043-03	07043-04	07043-05
Grouping	**Analyte**					
SOIL						
Volatile Organic Compounds	VOC Sample Container	Field MeOH	Field MeOH	Field MeOH	Field MeOH	Field MeOH
	Benzene (mg/kg)	<0.0050	<0.0050	<0.0050	<0.0050	<0.0050
	Ethylbenzene (mg/kg)	<0.015	<0.015	<0.015	<0.015	<0.015
	Methyl t-butyl ether (MTBE) (mg/kg)	<0.20	<0.20	<0.20	<0.20	<0.20
	Styrene (mg/kg)	<0.050	<0.050	<0.050	<0.050	<0.050
	Toluene (mg/kg)	<0.050	<0.050	<0.050	<0.050	<0.050
	ortho-Xylene (mg/kg)	<0.050	<0.050	<0.050	<0.050	<0.050
	meta- & para-Xylene (mg/kg)	<0.050	<0.050	<0.050	<0.050	<0.050
	Xylenes (mg/kg)	<0.075	<0.075	<0.075	<0.075	<0.075
	Surrogate: 4-Bromofluorobenzene (SS) (%)	95.7	96.0	91.3	89.3	87.5
	Surrogate: 1,4-Difluorobenzene (SS) (%)	87.8	88.3	85.2	84.0	80.8
Hydrocarbons	EPH10-19 (mg/kg)	<200	<200	<200	<200	<200
	EPH19-32 (mg/kg)	<200	<200	<200	<200	<200
	LEPH (mg/kg)	<200	<200	<200	<200	<200
	HEPH (mg/kg)	<200	<200	<200	<200	<200
	Surrogate: 2-Bromobenzotrifluoride (%)	99.4	103.3	94.9	98.1	103.0
Polycyclic Aromatic Hydrocarbons	Acenaphthene (mg/kg)	<0.0050	<0.0050	<0.0050	<0.0050	<0.040 DLCI
	Acenaphthylene (mg/kg)	<0.0050	<0.0050	<0.0050	<0.0050	<0.030 DLCI
	Anthracene (mg/kg)	<0.0040	<0.0040	<0.0040	<0.0040	<0.090 DLCI
	Benz(a)anthracene (mg/kg)	<0.010	<0.010	<0.010	<0.010	0.113
	Benzo(a)pyrene (mg/kg)	<0.010	<0.010	<0.010	<0.010	0.072
	Benzo(b&j)fluoranthene (mg/kg)	<0.010	<0.010	<0.010	<0.010	0.135
	Benzo(b+j+k)fluoranthene (mg/kg)	<0.015	<0.015	<0.015	<0.015	0.177
	Benzo(g,h,i)perylene (mg/kg)	<0.010	<0.010	<0.010	<0.010	0.073
	Benzo(k)fluoranthene (mg/kg)	<0.010	<0.010	<0.010	<0.010	0.042
	Chrysene (mg/kg)	<0.010	<0.010	<0.010	<0.010	0.125
	Dibenz(a,h)anthracene (mg/kg)	<0.0050	<0.0050	<0.0050	<0.0050	0.0137
	Fluoranthene (mg/kg)	<0.010	<0.010	<0.010	<0.010	0.196
	Fluorene (mg/kg)	<0.010	<0.010	<0.010	<0.010	0.066
	Indeno(1,2,3-c,d)pyrene (mg/kg)	<0.010	<0.010	<0.010	<0.010	0.056
	1-Methylnaphthalene (mg/kg)	<0.050	<0.050	<0.050	<0.050	0.499
	2-Methylnaphthalene (mg/kg)	<0.010	<0.010	<0.010	<0.010	0.521
	Naphthalene (mg/kg)	<0.010	<0.010	<0.010	<0.010	0.183
	Phenanthrene (mg/kg)	<0.010	<0.010	<0.010	<0.010	0.706
	Pyrene (mg/kg)	<0.010	<0.010	<0.010	<0.010	0.171
	Quinoline (mg/kg)	<0.050	<0.050	<0.050	<0.050	<0.050

* Please refer to the Reference Information section for an explanation of any qualifiers detected.

ALS ENVIRONMENTAL ANALYTICAL REPORT

Grouping	Analyte	L2398084-1 SO 16-DEC-19 07043-01	L2398084-2 SO 16-DEC-19 07043-02	L2398084-3 SO 16-DEC-19 07043-03	L2398084-4 SO 16-DEC-19 07043-04	L2398084-5 SO 16-DEC-19 07043-05
Sample ID / Description / Sampled Date / Sampled Time / Client ID						
SOIL						
Polycyclic Aromatic Hydrocarbons	Surrogate: Chrysene d12 (%)	102.2	98.6	101.8	96.8	96.9
	Surrogate: Naphthalene d8 (%)	97.7	95.0	98.7	95.0	97.6
	Surrogate: Phenanthrene d10 (%)	104.3	111.8	125.6	117.9	116.1
	B(a)P Total Potency Equivalent (mg/kg)	<0.020	<0.020	<0.020	<0.020	0.122
	IACR (CCME)	<0.15	<0.15	<0.15	<0.15	1.79

* Please refer to the Reference Information section for an explanation of any qualifiers detected.

Reference Information

QC Samples with Qualifiers & Comments:

QC Type Description	Parameter	Qualifier	Applies to Sample Number(s)
Duplicate	Cadmium (Cd)	DUP-H	L2398084-1, -2, -3, -4, -5
Duplicate	Manganese (Mn)	DUP-H	L2398084-1, -2, -3, -4, -5

Qualifiers for Individual Parameters Listed:

Qualifier	Description
DLCI	Detection Limit Raised: Chromatographic Interference due to co-elution.
DUP-H	Duplicate results outside ALS DQO, due to sample heterogeneity.

Test Method References:

ALS Test Code	Matrix	Test Description	Method Reference**
EPH-TUMB-FID-VA	Soil	EPH in Solids by Tumbler and GCFID	BC MOE EPH GCFID

Analysis is in accordance with BC MOE Lab Manual method "Extractable Petroleum Hydrocarbons in Solids by GC/FID", v2.1, July 1999. Soil samples are extracted with a 1:1 mixture of hexane and acetone using a rotary extraction technique modified from EPA 3570 prior to gas chromatography with flame ionization detection (GC-FID). EPH results include Polycyclic Aromatic Hydrocarbons (PAH) and are therefore not equivalent to Light and Heavy Extractable Petroleum Hydrocarbons (LEPH/HEPH).

HG-200.2-CVAF-VA	Soil	Mercury in Soil by CVAAS	EPA 200.2/1631E (mod)

Soil samples are digested with hot nitric and hydrochloric acids, followed by CVAAS analysis. This method is fully compliant with the BC SALM strong acid leachable metals digestion method.

LEPH/HEPH-CALC-VA	Soil	LEPHs and HEPHs	BC MOE LEPH/HEPH

LEPHs and HEPHs are measures of Light and Heavy Extractable Petroleum Hydrocarbons in soil. Results are calculated by subtraction of applicable PAH concentrations from EPH10-19 and EPH19-32, as per the BC Lab Manual LEPH/HEPH calculation procedure.

LEPHs = EPH10-19 minus Naphthalene and Phenanthrene.

HEPHs = EPH19-32 minus Benz(a)anthracene, Benzo(a)pyrene, Benzo(b)fluoranthene, Benzo(k)fluoranthene, Dibenz(a,h)anthracene, indeno(1,2,3-c,d)pyrene, and Pyrene.

MET-200.2-CCMS-VA	Soil	Metals in Soil by CRC ICPMS	EPA 200.2/6020A (mod)

Soil/sediment is dried, disaggregated, and sieved (2 mm). Strong Acid Leachable Metals in the <2mm fraction are solubilized by heated digestion with nitric and hydrochloric acids. Instrumental analysis is by Collision / Reaction Cell ICPMS.

Limitations: This method is intended to liberate environmentally available metals. Silicate minerals are not solubilized. Some metals may be only partially recovered (matrix dependent), including Al, Ba, Be, Cr, S, Sr, Ti, Tl, V, W, and Zr. Elemental Sulfur may be poorly recovered by this method. Volatile forms of sulfur (e.g. sulfide, H2S) may be excluded if lost during sampling, storage, or digestion.

MOISTURE-VA	Soil	Moisture content	CCME PHC in Soil - Tier 1 (mod)

This analysis is carried out gravimetrically by drying the sample at 105 C for a minimum of two hours.

PAH-TMB-H/A-MS-VA	Soil	PAH - Rotary Extraction (Hexane/Acetone)	EPA 3570/8270

This analysis is carried out using procedures adapted from "Test Methods for Evaluating Solid Waste" SW-846, Methods 3570 & 8270, published by the United States Environmental Protection Agency (EPA). The procedure uses a mechanical shaking technique to extract a subsample of the sediment/soil with a 1:1 mixture of hexane and acetone. The extract is then solvent exchanged to toluene. The final extract is analysed by capillary column gas chromatography with mass spectrometric detection (GC/MS). Surrogate recoveries may not be reported in cases where interferences from the sample matrix prevent accurate quantitation. Because the two isomers cannot be readily chromatographically separated, benzo(j)fluoranthene is reported as part of the benzo(b)fluoranthene parameter.

Benzo(a)pyrene Total Potency Equivalents [B(a)P TPE] represents the sum of estimated cancer potency relative to B(a)P for all potentially carcinogenic unsubstituted PAHs, and is calculated as per the CCME PAH Soil Quality Guidelines reference document (2010).

PH-1:2-VA	Soil	pH in Soil (1:2 Soil:Water Extraction)	BC WLAP METHOD: PH, ELECTROMETRIC, SOIL

This analysis is carried out in accordance with procedures described in "pH, Electrometric in Soil and Sediment - Prescriptive Method", Rev. 2005, Section B Physical, Inorganic and Misc. Constituents, BC Environmental Laboratory Manual. The procedure involves mixing the dried (at <60°C) and sieved (No. 10 / 2mm) sample with deionized/distilled water at a 1:2 ratio of sediment to water. The pH of the solution is then measured using a standard pH probe.

VOC7-L-HSMS-VA	Soil	VOCs in soil by Headspace GCMS	EPA 5035A/5021A/8260C

The soil methanol extract is added to water and reagents, then heated in a sealed vial to equilibrium. The headspace from the vial is transferred into a gas chromatograph. Target compound concentrations are measured using mass spectrometry detection.

VOC7/VOC-SURR-MS-VA	Soil	VOC7 and/or VOC Surrogates for Soils	EPA 5035A/5021A/8260C
XYLENES-CALC-VA	Soil	Sum of Xylene Isomer Concentrations	EPA 8260B & 524.2

Calculation of Total Xylenes

Total Xylenes is the sum of the concentrations of the ortho, meta, and para Xylene isomers. Results below detection limit (DL) are treated as zero. The DL for Total Xylenes is set to a value no less than the square root of the sum of the squares of the DLs of the individual Xylenes.

Reference Information

** ALS test methods may incorporate modifications from specified reference methods to improve performance.

The last two letters of the above test code(s) indicate the laboratory that performed analytical analysis for that test. Refer to the list below:

Laboratory Definition Code	Laboratory Location
VA	ALS ENVIRONMENTAL - VANCOUVER, BRITISH COLUMBIA, CANADA

Chain of Custody Numbers:

07043

GLOSSARY OF REPORT TERMS

Surrogate - A compound that is similar in behaviour to target analyte(s), but that does not occur naturally in environmental samples. For applicable tests, surrogates are added to samples prior to analysis as a check on recovery.
mg/kg - milligrams per kilogram based on dry weight of sample.
mg/kg wwt - milligrams per kilogram based on wet weight of sample.
mg/kg lwt - milligrams per kilogram based on lipid-adjusted weight of sample.
mg/L - milligrams per litre.
< - Less than.
D.L. - The reported Detection Limit, also known as the Limit of Reporting (LOR).
N/A - Result not available. Refer to qualifier code and definition for explanation.

Test results reported relate only to the samples as received by the laboratory.
UNLESS OTHERWISE STATED, ALL SAMPLES WERE RECEIVED IN ACCEPTABLE CONDITION.
Analytical results in unsigned test reports with the DRAFT watermark are subject to change, pending final QC review.

Quality Control Report

Workorder: L2398084 Report Date: 27-DEC-19

Client: GOLDER ASSOCIATES LTD.
3795 Carey Road, Second Floor
Victoria BC V8Z 6T8
Contact: Alanna Umphrey

Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
EPH-TUMB-FID-VA	**Soil**							
Batch R4945676								
WG3245924-3 DUP		**L2398084-4**						
EPH10-19		<200	<200	RPD-NA	mg/kg	N/A	40	20-DEC-19
EPH19-32		<200	<200	RPD-NA	mg/kg	N/A	40	20-DEC-19
WG3245924-4 IRM		**ALS PHC RM3**						
EPH10-19			104.7		%		70-130	20-DEC-19
EPH19-32			111.1		%		70-130	20-DEC-19
WG3245924-2 LCS								
EPH10-19			107.3		%		70-130	20-DEC-19
EPH19-32			102.7		%		70-130	20-DEC-19
WG3245924-1 MB								
EPH10-19			<200		mg/kg		200	20-DEC-19
EPH19-32			<200		mg/kg		200	20-DEC-19
Surrogate: 2-Bromobenzotrifluoride			101.9		%		60-140	20-DEC-19
HG-200.2-CVAF-VA	**Soil**							
Batch R4949181								
WG3245929-4 CRM		**VA-CANMET-TILL2**						
Mercury (Hg)			106.4		%		70-130	20-DEC-19
WG3245929-2 DUP		**L2398084-4**						
Mercury (Hg)		0.051	0.059		mg/kg	16	40	20-DEC-19
WG3245929-3 LCS								
Mercury (Hg)			105.7		%		80-120	20-DEC-19
WG3245929-1 MB								
Mercury (Hg)			<0.0050		mg/kg		0.005	20-DEC-19
MET-200.2-CCMS-VA	**Soil**							
Batch R4947508								
WG3245929-4 CRM		**VA-CANMET-TILL2**						
Aluminum (Al)			98.1		%		70-130	19-DEC-19
Antimony (Sb)			101.6		%		70-130	19-DEC-19
Arsenic (As)			102.1		%		70-130	19-DEC-19
Barium (Ba)			101.4		%		70-130	19-DEC-19
Beryllium (Be)			98.4		%		70-130	19-DEC-19
Bismuth (Bi)			96.9		%		70-130	19-DEC-19
Cadmium (Cd)			102.1		%		70-130	19-DEC-19
Calcium (Ca)			105.5		%		70-130	19-DEC-19
Copper (Cu)			98.1		%		70-130	19-DEC-19
Iron (Fe)			106.8		%		70-130	19-DEC-19
Lead (Pb)			93.9		%		70-130	19-DEC-19

Quality Control Report

Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
MET-200.2-CCMS-VA	**Soil**							
Batch R4947508								
WG3245929-4 CRM		**VA-CANMET-TILL2**						
Lithium (Li)			108.7		%		70-130	19-DEC-19
Magnesium (Mg)			101.0		%		70-130	19-DEC-19
Manganese (Mn)			99.1		%		70-130	19-DEC-19
Molybdenum (Mo)			98.9		%		70-130	19-DEC-19
Nickel (Ni)			102.4		%		70-130	19-DEC-19
Phosphorus (P)			94.3		%		70-130	19-DEC-19
Potassium (K)			103.7		%		70-130	19-DEC-19
Selenium (Se)			0.40		mg/kg		0.15-0.55	19-DEC-19
Silver (Ag)			0.26		mg/kg		0.16-0.36	19-DEC-19
Sodium (Na)			103.0		%		70-130	19-DEC-19
Strontium (Sr)			100.3		%		70-130	19-DEC-19
Thallium (Tl)			95.9		%		70-130	19-DEC-19
Tin (Sn)			2.4		mg/kg		0.2-4.2	19-DEC-19
Titanium (Ti)			102.8		%		70-130	19-DEC-19
Tungsten (W)			1.27		mg/kg		1-2	19-DEC-19
Uranium (U)			96.4		%		70-130	19-DEC-19
Vanadium (V)			104.4		%		70-130	19-DEC-19
Zinc (Zn)			96.2		%		70-130	19-DEC-19
WG3245929-3 LCS								
Aluminum (Al)			104.0		%		80-120	19-DEC-19
Antimony (Sb)			91.3		%		80-120	19-DEC-19
Arsenic (As)			97.5		%		80-120	19-DEC-19
Barium (Ba)			106.5		%		80-120	19-DEC-19
Beryllium (Be)			93.7		%		80-120	19-DEC-19
Bismuth (Bi)			88.1		%		80-120	19-DEC-19
Boron (B)			94.0		%		80-120	19-DEC-19
Cadmium (Cd)			97.4		%		80-120	19-DEC-19
Calcium (Ca)			99.1		%		80-120	19-DEC-19
Chromium (Cr)			104.3		%		80-120	19-DEC-19
Cobalt (Co)			99.7		%		80-120	19-DEC-19
Copper (Cu)			94.5		%		80-120	19-DEC-19
Iron (Fe)			110.9		%		80-120	19-DEC-19
Lead (Pb)			92.2		%		80-120	19-DEC-19
Lithium (Li)			93.7		%		80-120	19-DEC-19

Quality Control Report

Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
MET-200.2-CCMS-VA	**Soil**							
Batch R4947508								
WG3245929-3 LCS								
Magnesium (Mg)			105.6		%		80-120	19-DEC-19
Manganese (Mn)			104.4		%		80-120	19-DEC-19
Molybdenum (Mo)			95.7		%		80-120	19-DEC-19
Nickel (Ni)			98.5		%		80-120	19-DEC-19
Phosphorus (P)			100.3		%		80-120	19-DEC-19
Potassium (K)			103.8		%		80-120	19-DEC-19
Selenium (Se)			97.9		%		80-120	19-DEC-19
Silver (Ag)			94.5		%		80-120	19-DEC-19
Sodium (Na)			99.9		%		80-120	19-DEC-19
Strontium (Sr)			92.7		%		80-120	19-DEC-19
Sulfur (S)			94.5		%		80-120	19-DEC-19
Thallium (Tl)			87.8		%		80-120	19-DEC-19
Tin (Sn)			92.5		%		80-120	19-DEC-19
Titanium (Ti)			99.5		%		80-120	19-DEC-19
Tungsten (W)			96.3		%		80-120	19-DEC-19
Uranium (U)			97.7		%		80-120	19-DEC-19
Vanadium (V)			103.4		%		80-120	19-DEC-19
Zinc (Zn)			94.9		%		80-120	19-DEC-19
Zirconium (Zr)			99.1		%		70-130	19-DEC-19
WG3245929-1 MB								
Aluminum (Al)			<50		mg/kg		50	19-DEC-19
Antimony (Sb)			<0.10		mg/kg		0.1	19-DEC-19
Arsenic (As)			<0.10		mg/kg		0.1	19-DEC-19
Barium (Ba)			<0.50		mg/kg		0.5	19-DEC-19
Beryllium (Be)			<0.10		mg/kg		0.1	19-DEC-19
Bismuth (Bi)			<0.20		mg/kg		0.2	19-DEC-19
Boron (B)			<5.0		mg/kg		5	19-DEC-19
Cadmium (Cd)			<0.020		mg/kg		0.02	19-DEC-19
Calcium (Ca)			<50		mg/kg		50	19-DEC-19
Chromium (Cr)			<0.50		mg/kg		0.5	19-DEC-19
Cobalt (Co)			<0.10		mg/kg		0.1	19-DEC-19
Copper (Cu)			<0.50		mg/kg		0.5	19-DEC-19
Iron (Fe)			<50		mg/kg		50	19-DEC-19
Lead (Pb)			<0.50		mg/kg		0.5	19-DEC-19

Quality Control Report

Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
MET-200.2-CCMS-VA	**Soil**							
Batch R4947508								
WG3245929-1 MB								
Lithium (Li)			<2.0		mg/kg		2	19-DEC-19
Magnesium (Mg)			<20		mg/kg		20	19-DEC-19
Manganese (Mn)			<1.0		mg/kg		1	19-DEC-19
Molybdenum (Mo)			<0.10		mg/kg		0.1	19-DEC-19
Nickel (Ni)			<0.50		mg/kg		0.5	19-DEC-19
Phosphorus (P)			<50		mg/kg		50	19-DEC-19
Potassium (K)			<100		mg/kg		100	19-DEC-19
Selenium (Se)			<0.20		mg/kg		0.2	19-DEC-19
Silver (Ag)			<0.10		mg/kg		0.1	19-DEC-19
Sodium (Na)			<50		mg/kg		50	19-DEC-19
Strontium (Sr)			<0.50		mg/kg		0.5	19-DEC-19
Sulfur (S)			<1000		mg/kg		1000	19-DEC-19
Thallium (Tl)			<0.050		mg/kg		0.05	19-DEC-19
Tin (Sn)			<2.0		mg/kg		2	19-DEC-19
Titanium (Ti)			<1.0		mg/kg		1	19-DEC-19
Tungsten (W)			<0.50		mg/kg		0.5	19-DEC-19
Uranium (U)			<0.050		mg/kg		0.05	19-DEC-19
Vanadium (V)			<0.20		mg/kg		0.2	19-DEC-19
Zinc (Zn)			<2.0		mg/kg		2	19-DEC-19
Zirconium (Zr)			<1.0		mg/kg		1	19-DEC-19
Batch R4954687								
WG3245929-2 DUP		**L2398084-4**						
Aluminum (Al)		25400	24700		mg/kg	2.7	40	23-DEC-19
Antimony (Sb)		0.20	0.26		mg/kg	25	30	23-DEC-19
Arsenic (As)		7.33	8.65		mg/kg	17	30	23-DEC-19
Barium (Ba)		81.1	94.6		mg/kg	15	40	23-DEC-19
Beryllium (Be)		0.35	0.37		mg/kg	6.2	30	23-DEC-19
Bismuth (Bi)		<0.20	<0.20	RPD-NA	mg/kg	N/A	30	23-DEC-19
Boron (B)		<5.0	<5.0	RPD-NA	mg/kg	N/A	30	23-DEC-19
Cadmium (Cd)		0.113	0.179	DUP-H	mg/kg	45	30	23-DEC-19
Calcium (Ca)		5540	5640		mg/kg	1.9	30	23-DEC-19
Chromium (Cr)		37.0	35.8		mg/kg	3.2	30	23-DEC-19
Cobalt (Co)		12.3	13.6		mg/kg	10	30	23-DEC-19
Copper (Cu)		31.4	33.8		mg/kg	7.6	30	23-DEC-19

Quality Control Report

Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
MET-200.2-CCMS-VA	**Soil**							
Batch R4954687								
WG3245929-2 DUP		**L2398084-4**						
Iron (Fe)		35300	33500		mg/kg	5.2	30	23-DEC-19
Lead (Pb)		5.88	6.21		mg/kg	5.4	40	23-DEC-19
Lithium (Li)		13.0	12.0		mg/kg	7.6	30	23-DEC-19
Magnesium (Mg)		5540	5200		mg/kg	6.4	30	23-DEC-19
Manganese (Mn)		918	2830	DUP-H	mg/kg	102	30	23-DEC-19
Molybdenum (Mo)		1.98	2.40		mg/kg	19	40	23-DEC-19
Nickel (Ni)		20.8	22.6		mg/kg	8.2	30	23-DEC-19
Phosphorus (P)		253	261		mg/kg	3.2	30	23-DEC-19
Potassium (K)		440	460		mg/kg	5.0	40	23-DEC-19
Selenium (Se)		0.27	0.23		mg/kg	17	30	23-DEC-19
Silver (Ag)		<0.10	<0.10	RPD-NA	mg/kg	N/A	40	23-DEC-19
Sodium (Na)		314	298		mg/kg	5.1	40	23-DEC-19
Strontium (Sr)		59.5	55.9		mg/kg	6.2	40	23-DEC-19
Sulfur (S)		<1000	<1000	RPD-NA	mg/kg	N/A	30	23-DEC-19
Thallium (Tl)		<0.050	0.061	RPD-NA	mg/kg	N/A	30	23-DEC-19
Tin (Sn)		<2.0	<2.0	RPD-NA	mg/kg	N/A	40	23-DEC-19
Titanium (Ti)		1740	1710		mg/kg	2.1	40	23-DEC-19
Tungsten (W)		<0.50	<0.50	RPD-NA	mg/kg	N/A	30	23-DEC-19
Uranium (U)		0.404	0.402		mg/kg	0.6	30	23-DEC-19
Vanadium (V)		116	117		mg/kg	0.8	30	23-DEC-19
Zinc (Zn)		33.5	34.0		mg/kg	1.3	30	23-DEC-19
Zirconium (Zr)		3.2	3.0		mg/kg	4.7	30	23-DEC-19
MOISTURE-VA	**Soil**							
Batch R4945927								
WG3245930-3 DUP		**L2398084-2**						
Moisture		22.8	21.1		%	7.4	20	18-DEC-19
WG3245930-2 LCS								
Moisture			100.5		%		90-110	18-DEC-19
WG3245930-1 MB								
Moisture			<0.25		%		0.25	18-DEC-19
PAH-TMB-H/A-MS-VA	**Soil**							
Batch R4946245								
WG3245924-3 DUP		**L2398084-4**						
Acenaphthene		<0.0050	<0.0050	RPD-NA	mg/kg	N/A	50	22-DEC-19
Acenaphthylene		<0.0050	<0.0050	RPD-NA	mg/kg	N/A	50	22-DEC-19

Quality Control Report

Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
PAH-TMB-H/A-MS-VA	**Soil**							
Batch R4946245								
WG3245924-3 DUP		**L2398084-4**						
Anthracene		<0.0040	<0.0040	RPD-NA	mg/kg	N/A	50	22-DEC-19
Benz(a)anthracene		<0.010	<0.010	RPD-NA	mg/kg	N/A	50	22-DEC-19
Benzo(a)pyrene		<0.010	<0.010	RPD-NA	mg/kg	N/A	50	22-DEC-19
Benzo(b&j)fluoranthene		<0.010	<0.010	RPD-NA	mg/kg	N/A	50	22-DEC-19
Benzo(g,h,i)perylene		<0.010	<0.010	RPD-NA	mg/kg	N/A	50	22-DEC-19
Benzo(k)fluoranthene		<0.010	<0.010	RPD-NA	mg/kg	N/A	50	22-DEC-19
Chrysene		<0.010	<0.010	RPD-NA	mg/kg	N/A	50	22-DEC-19
Dibenz(a,h)anthracene		<0.0050	<0.0050	RPD-NA	mg/kg	N/A	50	22-DEC-19
Fluoranthene		<0.010	<0.010	RPD-NA	mg/kg	N/A	50	22-DEC-19
Fluorene		<0.010	<0.010	RPD-NA	mg/kg	N/A	50	22-DEC-19
Indeno(1,2,3-c,d)pyrene		<0.010	<0.010	RPD-NA	mg/kg	N/A	50	22-DEC-19
1-Methylnaphthalene		<0.050	<0.050	RPD-NA	mg/kg	N/A	50	22-DEC-19
2-Methylnaphthalene		<0.010	<0.010	RPD-NA	mg/kg	N/A	50	22-DEC-19
Naphthalene		<0.010	<0.010	RPD-NA	mg/kg	N/A	50	22-DEC-19
Phenanthrene		<0.010	<0.010	RPD-NA	mg/kg	N/A	50	22-DEC-19
Pyrene		<0.010	<0.010	RPD-NA	mg/kg	N/A	50	22-DEC-19
Quinoline		<0.050	<0.050	RPD-NA	mg/kg	N/A	50	22-DEC-19
WG3245924-5 IRM		**ALS PAH RM2**						
Acenaphthene			98.9		%		60-130	22-DEC-19
Acenaphthylene			111.4		%		60-130	22-DEC-19
Anthracene			116.3		%		60-130	22-DEC-19
Benz(a)anthracene			92.7		%		60-130	22-DEC-19
Benzo(a)pyrene			89.6		%		60-130	22-DEC-19
Benzo(b&j)fluoranthene			98.1		%		60-130	22-DEC-19
Benzo(g,h,i)perylene			94.0		%		60-130	22-DEC-19
Benzo(k)fluoranthene			99.4		%		60-130	22-DEC-19
Chrysene			86.2		%		60-130	22-DEC-19
Dibenz(a,h)anthracene			98.7		%		60-130	22-DEC-19
Fluoranthene			96.0		%		60-130	22-DEC-19
Fluorene			95.5		%		60-130	22-DEC-19
Indeno(1,2,3-c,d)pyrene			93.8		%		60-130	22-DEC-19
1-Methylnaphthalene			93.3		%		60-130	22-DEC-19
2-Methylnaphthalene			91.1		%		60-130	22-DEC-19
Naphthalene			92.7		%		50-130	22-DEC-19

Quality Control Report

Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
PAH-TMB-H/A-MS-VA	**Soil**							
Batch R4946245								
WG3245924-5 IRM		**ALS PAH RM2**						
Phenanthrene			94.1		%		60-130	22-DEC-19
Pyrene			98.0		%		60-130	22-DEC-19
WG3245924-2 LCS								
Acenaphthene			94.5		%		60-130	22-DEC-19
Acenaphthylene			93.1		%		60-130	22-DEC-19
Anthracene			96.4		%		60-130	22-DEC-19
Benz(a)anthracene			92.5		%		60-130	22-DEC-19
Benzo(a)pyrene			92.9		%		60-130	22-DEC-19
Benzo(b&j)fluoranthene			94.1		%		60-130	22-DEC-19
Benzo(g,h,i)perylene			80.8		%		60-130	22-DEC-19
Benzo(k)fluoranthene			92.2		%		60-130	22-DEC-19
Chrysene			85.1		%		60-130	22-DEC-19
Dibenz(a,h)anthracene			85.1		%		60-130	22-DEC-19
Fluoranthene			91.9		%		60-130	22-DEC-19
Fluorene			93.8		%		60-130	22-DEC-19
Indeno(1,2,3-c,d)pyrene			88.3		%		60-130	22-DEC-19
1-Methylnaphthalene			91.9		%		60-130	22-DEC-19
2-Methylnaphthalene			94.3		%		60-130	22-DEC-19
Naphthalene			93.0		%		50-130	22-DEC-19
Phenanthrene			92.4		%		60-130	22-DEC-19
Pyrene			92.3		%		60-130	22-DEC-19
Quinoline			94.1		%		60-130	22-DEC-19
WG3245924-1 MB								
Acenaphthene			<0.0050		mg/kg		0.005	22-DEC-19
Acenaphthylene			<0.0050		mg/kg		0.005	22-DEC-19
Anthracene			<0.0040		mg/kg		0.004	22-DEC-19
Benz(a)anthracene			<0.010		mg/kg		0.01	22-DEC-19
Benzo(a)pyrene			<0.010		mg/kg		0.01	22-DEC-19
Benzo(b&j)fluoranthene			<0.010		mg/kg		0.01	22-DEC-19
Benzo(g,h,i)perylene			<0.010		mg/kg		0.01	22-DEC-19
Benzo(k)fluoranthene			<0.010		mg/kg		0.01	22-DEC-19
Chrysene			<0.010		mg/kg		0.01	22-DEC-19
Dibenz(a,h)anthracene			<0.0050		mg/kg		0.005	22-DEC-19
Fluoranthene			<0.010		mg/kg		0.01	22-DEC-19
Fluorene			<0.010		mg/kg		0.01	22-DEC-19

Quality Control Report

Workorder: L2398084 Report Date: 27-DEC-19

Test	Matrix	Reference	Result	Qualifier	Units	RPD	Limit	Analyzed
PAH-TMB-H/A-MS-VA	**Soil**							
Batch R4946245								
WG3245924-1 MB								
Indeno(1,2,3-c,d)pyrene			<0.010		mg/kg		0.01	22-DEC-19
1-Methylnaphthalene			<0.050		mg/kg		0.05	22-DEC-19
2-Methylnaphthalene			<0.010		mg/kg		0.01	22-DEC-19
Naphthalene			<0.010		mg/kg		0.01	22-DEC-19
Phenanthrene			<0.010		mg/kg		0.01	22-DEC-19
Pyrene			<0.010		mg/kg		0.01	22-DEC-19
Quinoline			<0.050		mg/kg		0.05	22-DEC-19
Surrogate: Naphthalene d8			92.3		%		50-130	22-DEC-19
Surrogate: Phenanthrene d10			97.3		%		60-130	22-DEC-19
Surrogate: Chrysene d12			94.6		%		60-130	22-DEC-19
PH-1:2-VA	**Soil**							
Batch R4947830								
WG3245929-2 DUP		**L2398084-4**						
pH (1:2 soil:water)		7.32	7.30	J	pH	0.02	0.2	19-DEC-19
VOC7-L-HSMS-VA	**Soil**							
Batch R4933986								
WG3246251-2 LCS								
Benzene			103.0		%		70-130	19-DEC-19
Ethylbenzene			123.2		%		70-130	19-DEC-19
Methyl t-butyl ether (MTBE)			105.4		%		70-130	19-DEC-19
Styrene			105.1		%		70-130	19-DEC-19
Toluene			96.6		%		70-130	19-DEC-19
meta- & para-Xylene			104.0		%		70-130	19-DEC-19
ortho-Xylene			99.6		%		70-130	19-DEC-19
WG3246251-1 MB								
Benzene			<0.0050		mg/kg		0.005	19-DEC-19
Ethylbenzene			<0.015		mg/kg		0.015	19-DEC-19
Methyl t-butyl ether (MTBE)			<0.20		mg/kg		0.2	19-DEC-19
Styrene			<0.050		mg/kg		0.05	19-DEC-19
Toluene			<0.050		mg/kg		0.05	19-DEC-19
meta- & para-Xylene			<0.050		mg/kg		0.05	19-DEC-19
ortho-Xylene			<0.050		mg/kg		0.05	19-DEC-19

Legend:

Limit	ALS Control Limit (Data Quality Objectives)
DUP	Duplicate
RPD	Relative Percent Difference
N/A	Not Available
LCS	Laboratory Control Sample
SRM	Standard Reference Material
MS	Matrix Spike
MSD	Matrix Spike Duplicate
ADE	Average Desorption Efficiency
MB	Method Blank
IRM	Internal Reference Material
CRM	Certified Reference Material
CCV	Continuing Calibration Verification
CVS	Calibration Verification Standard
LCSD	Laboratory Control Sample Duplicate

Sample Parameter Qualifier Definitions:

Qualifier	Description
DUP-H	Duplicate results outside ALS DQO, due to sample heterogeneity.
J	Duplicate results and limits are expressed in terms of absolute difference.
RPD-NA	Relative Percent Difference Not Available due to result(s) being less than detection limit.

Hold Time Exceedances:

All test results reported with this submission were conducted within ALS recommended hold times.

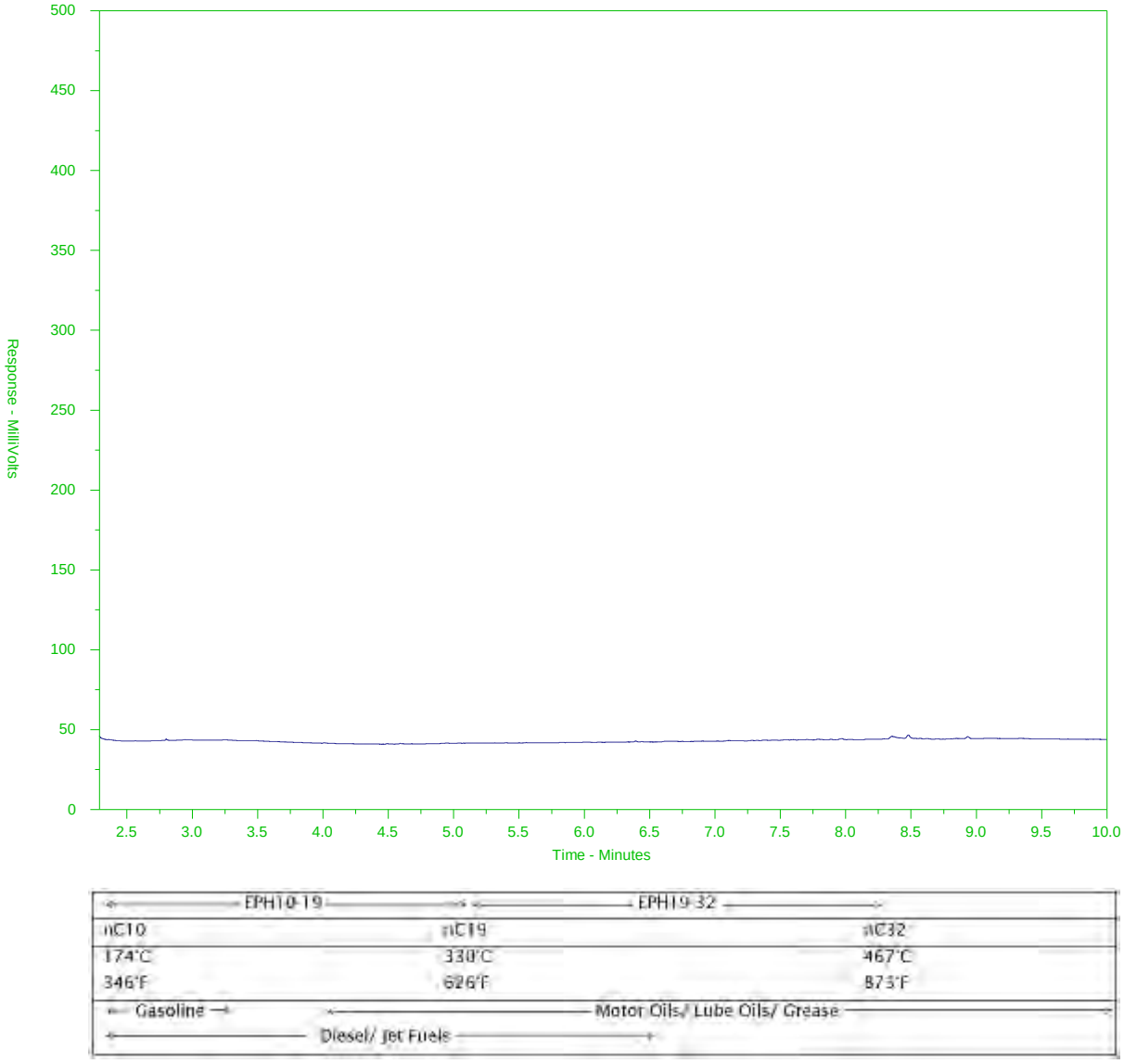
ALS recommended hold times may vary by province. They are assigned to meet known provincial and/or federal government requirements. In the absence of regulatory hold times, ALS establishes recommendations based on guidelines published by the US EPA, APHA Standard Methods, or Environment Canada (where available). For more information, please contact ALS.

The ALS Quality Control Report is provided to ALS clients upon request. ALS includes comprehensive QC checks with every analysis to ensure our high standards of quality are met. Each QC result has a known or expected target value, which is compared against pre-determined data quality objectives to provide confidence in the accuracy of associated test results.

Please note that this report may contain QC results from anonymous Sample Duplicates and Matrix Spikes that do not originate from this Work Order.

BC EPH HYDROCARBON DISTRIBUTION REPORT

ALS Sample ID: L2398084-1
Client Sample ID: 07043-01



The BC EPH Hydrocarbon Distribution Report (HDR) is intended to assist you in characterizing hydrocarbon products that may be present in your sample.

The scale at the bottom of the chromatogram indicates the approximate retention times of common petroleum products and three n-alkane hydrocarbon marker compounds. Retention times may vary between samples, but general patterns and distributions will remain similar.

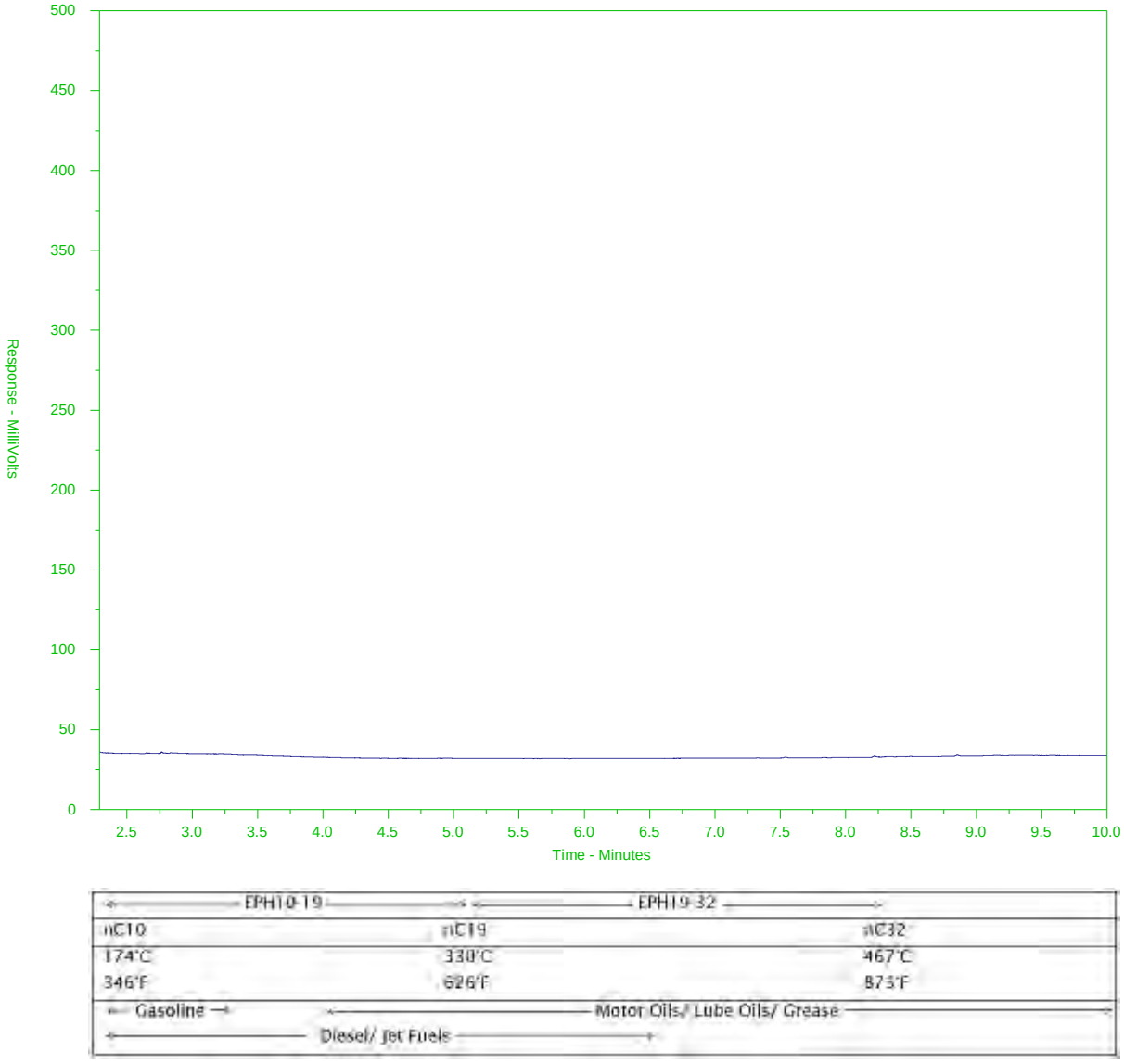
Peak heights in this report are a function of the sample concentration, the sample amount extracted, the sample dilution factor, and the scale at left.

A "-L-" in the sample ID denotes a low level sample. A "-S-" denotes a silica gel cleaned sample.

Note: This chromatogram was produced using GC conditions that are specific to the ALS Canada EPH method. Refer to the ALS Canada EPH Hydrocarbon Library for a collection of chromatograms from common reference samples (fuels, oils, etc.). The HDR library can be found at www.alsglobal.com.

BC EPH HYDROCARBON DISTRIBUTION REPORT

ALS Sample ID: L2398084-2
Client Sample ID: 07043-02



The BC EPH Hydrocarbon Distribution Report (HDR) is intended to assist you in characterizing hydrocarbon products that may be present in your sample.

The scale at the bottom of the chromatogram indicates the approximate retention times of common petroleum products and three n-alkane hydrocarbon marker compounds. Retention times may vary between samples, but general patterns and distributions will remain similar.

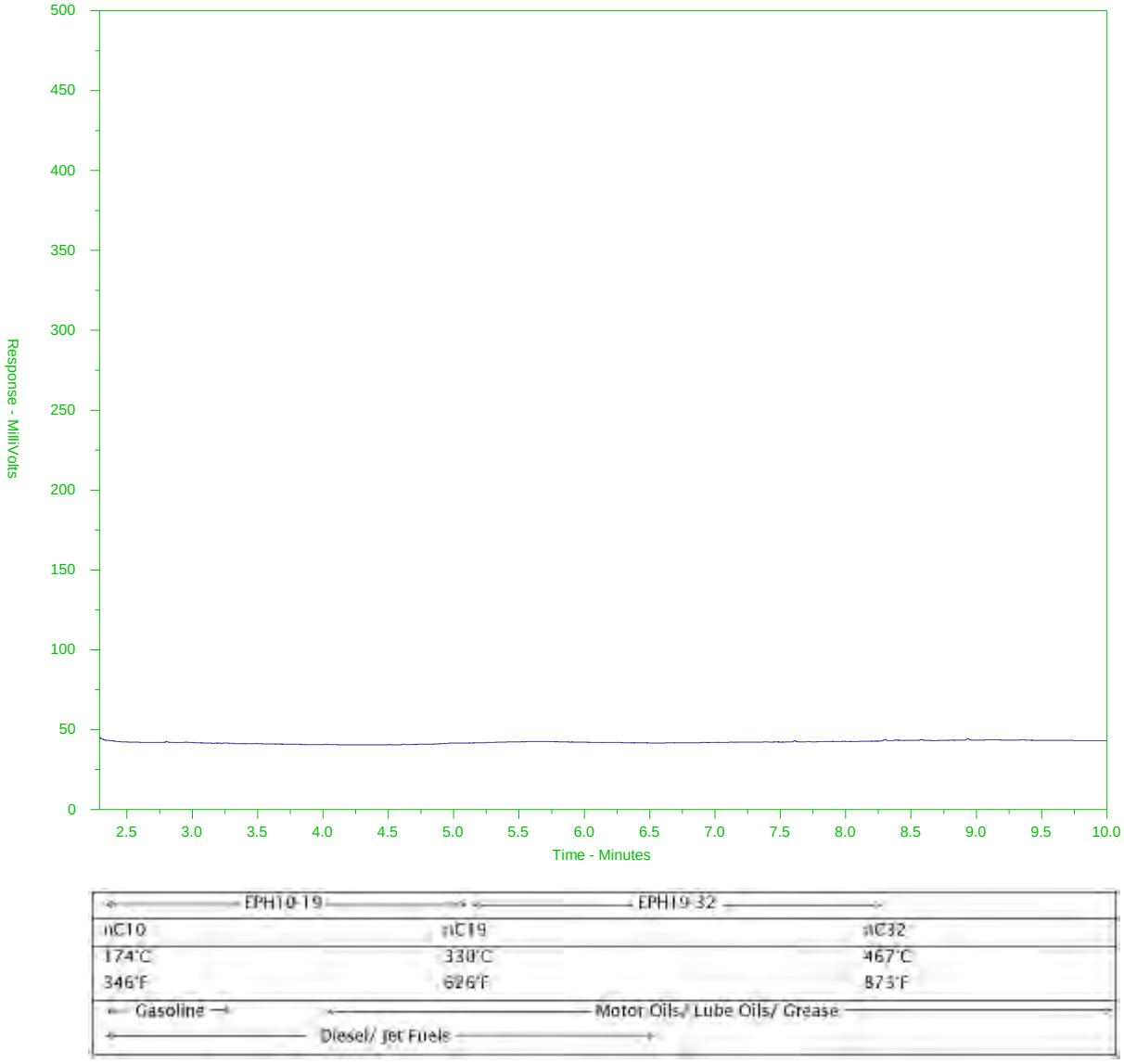
Peak heights in this report are a function of the sample concentration, the sample amount extracted, the sample dilution factor, and the scale at left.

A "-L-" in the sample ID denotes a low level sample. A "-S-" denotes a silica gel cleaned sample.

Note: This chromatogram was produced using GC conditions that are specific to the ALS Canada EPH method. Refer to the ALS Canada EPH Hydrocarbon Library for a collection of chromatograms from common reference samples (fuels, oils, etc.). The HDR library can be found at www.alsglobal.com.

BC EPH HYDROCARBON DISTRIBUTION REPORT

ALS Sample ID: L2398084-3
Client Sample ID: 07043-03



The BC EPH Hydrocarbon Distribution Report (HDR) is intended to assist you in characterizing hydrocarbon products that may be present in your sample.

The scale at the bottom of the chromatogram indicates the approximate retention times of common petroleum products and three n-alkane hydrocarbon marker compounds. Retention times may vary between samples, but general patterns and distributions will remain similar.

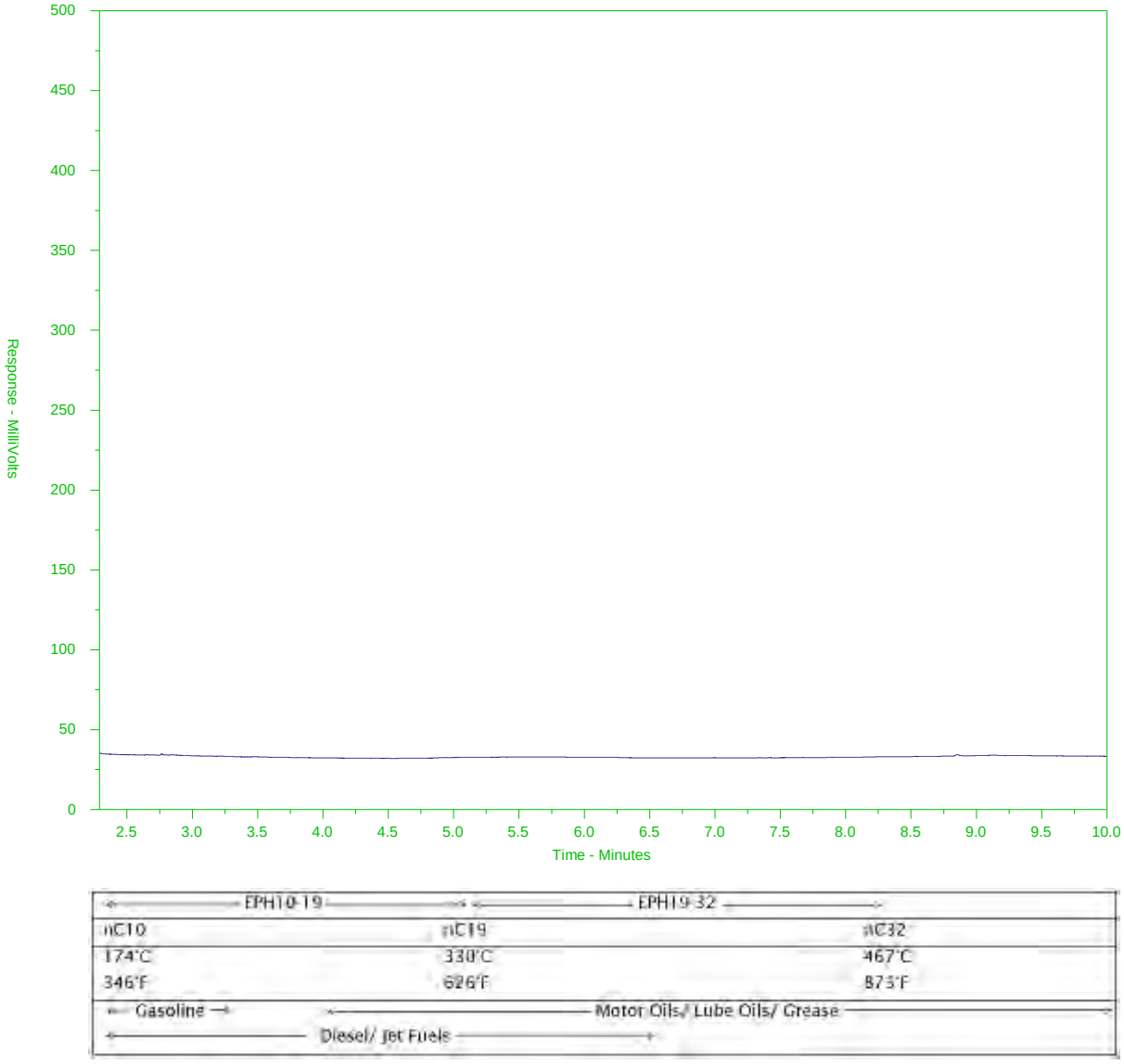
Peak heights in this report are a function of the sample concentration, the sample amount extracted, the sample dilution factor, and the scale at left.

A "-L-" in the sample ID denotes a low level sample. A "-S-" denotes a silica gel cleaned sample.

Note: This chromatogram was produced using GC conditions that are specific to the ALS Canada EPH method. Refer to the ALS Canada EPH Hydrocarbon Library for a collection of chromatograms from common reference samples (fuels, oils, etc.). The HDR library can be found at www.alsglobal.com.

BC EPH HYDROCARBON DISTRIBUTION REPORT

ALS Sample ID: L2398084-4
Client Sample ID: 07043-04



The BC EPH Hydrocarbon Distribution Report (HDR) is intended to assist you in characterizing hydrocarbon products that may be present in your sample.

The scale at the bottom of the chromatogram indicates the approximate retention times of common petroleum products and three n-alkane hydrocarbon marker compounds. Retention times may vary between samples, but general patterns and distributions will remain similar.

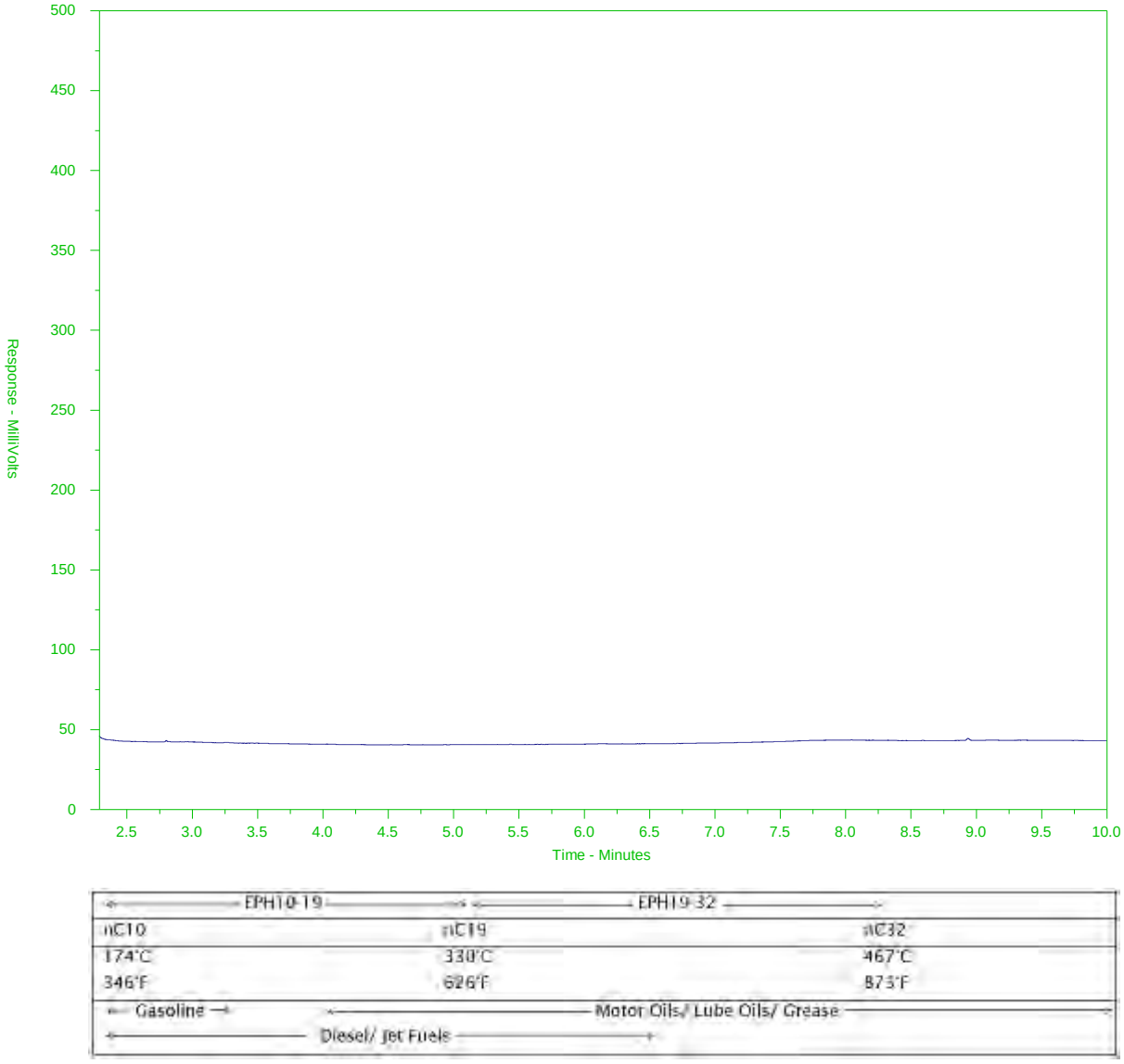
Peak heights in this report are a function of the sample concentration, the sample amount extracted, the sample dilution factor, and the scale at left.

A "-L-" in the sample ID denotes a low level sample. A "-S-" denotes a silica gel cleaned sample.

Note: This chromatogram was produced using GC conditions that are specific to the ALS Canada EPH method. Refer to the ALS Canada EPH Hydrocarbon Library for a collection of chromatograms from common reference samples (fuels, oils, etc.). The HDR library can be found at www.alsglobal.com.

BC EPH HYDROCARBON DISTRIBUTION REPORT

ALS Sample ID: WG3245924-3#L2398084-4
Client Sample ID: 07043-04



The BC EPH Hydrocarbon Distribution Report (HDR) is intended to assist you in characterizing hydrocarbon products that may be present in your sample.

The scale at the bottom of the chromatogram indicates the approximate retention times of common petroleum products and three n-alkane hydrocarbon marker compounds. Retention times may vary between samples, but general patterns and distributions will remain similar.

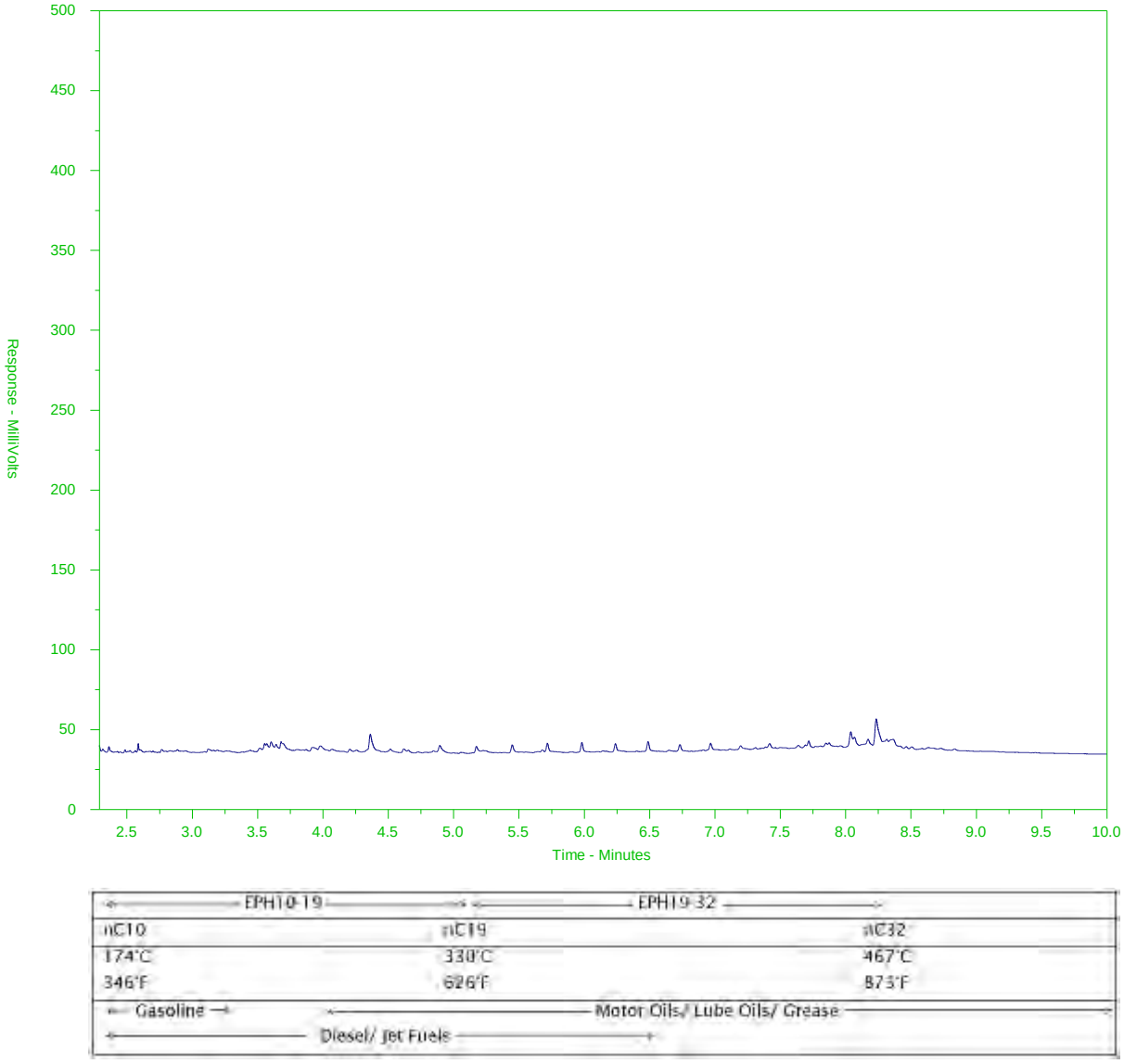
Peak heights in this report are a function of the sample concentration, the sample amount extracted, the sample dilution factor, and the scale at left.

A "-L-" in the sample ID denotes a low level sample. A "-S-" denotes a silica gel cleaned sample.

Note: This chromatogram was produced using GC conditions that are specific to the ALS Canada EPH method. Refer to the ALS Canada EPH Hydrocarbon Library for a collection of chromatograms from common reference samples (fuels, oils, etc.). The HDR library can be found at www.alsglobal.com.

BC EPH HYDROCARBON DISTRIBUTION REPORT

ALS Sample ID: L2398084-5
Client Sample ID: 07043-05



The BC EPH Hydrocarbon Distribution Report (HDR) is intended to assist you in characterizing hydrocarbon products that may be present in your sample.

The scale at the bottom of the chromatogram indicates the approximate retention times of common petroleum products and three n-alkane hydrocarbon marker compounds. Retention times may vary between samples, but general patterns and distributions will remain similar.

Peak heights in this report are a function of the sample concentration, the sample amount extracted, the sample dilution factor, and the scale at left.

A "-L-" in the sample ID denotes a low level sample. A "-S-" denotes a silica gel cleaned sample.

Note: This chromatogram was produced using GC conditions that are specific to the ALS Canada EPH method. Refer to the ALS Canada EPH Hydrocarbon Library for a collection of chromatograms from common reference samples (fuels, oils, etc.). The HDR library can be found at www.alsglobal.com.

CHAIN OF CUSTODY RECORD/ANALYSIS REQUEST

No. 07043 page 1 of 1

Golder Associates
200 – 2920 Virtual Way
Vancouver, British Columbia, Canada V5M 0C4
Telephone (604) 296-4200 Fax (604) 298-5253

Project Number: 18109842			Laboratory Name: ALS
Short Title: ToL	Golder Contact: Alanna Umphrey		Address: 8081 Lougheed Hwy Burnaby
Golder E-mail Address 1: AUmphrey@golder.com	Golder E-mail Address 2: MNewman @golder.com	Telephone/Fax: 604-253-4188	Contact: Amber Springer

Office Name: Victoria

EQuIS Facility Code: _____
EQuIS upload: ☐

Turnaround Time: ☐ 24 hr ☐ 48 hr ☐ 72 hr ☒ Regular (5 Days)
Criteria: ☒ CSR ☐ CCME ☐ BC Water Quality ☐ Other

Note: Final Reports to be issued by e-mail

Quote No.:

Analyses Required

Sample Control Number (SCN)	Sample Location	Sa. #	Sample Depth (m)	Sample Matrix (over)	Date Sampled (D / M / Y)	Time Sampled (HH:MM)	Sample Type (over)	QAQC Code (over)	Related SCN (over)	Number of Containers	BTEX	LEPH/HEPH/PAH	Metals	RUSH (Select TAT above)	Remarks (over)
07043 - 01	HS19-01	—	0.3-0.6	SO	16/12/19	—	GRAB	—	—	4	X	X	X		
07043 - 02	HS19-02	—	0.3-0.6	SO	16/12/19	—	GRAB	FDA	07043-03	4	X	X	X		
07043 - 03	HS19-02	—	0.3-0.6	SO	16/12/19	—	GRAB	FD	07043-02	4	X	X	X		
07043 - 04	HS19-03	—	0.3-0.6	SO	16/12/19	—	GRAB	—	—	4	X	X	X		
07043 - 05	HS19-04	—	0.3-0.6	SO	16/12/19	—	GRAB	—	—	4	X	X	X		
- 06															
- 07															
- 08															
- 09															
- 10															
- 11															
- 12															

L2398084-COFC

Sampler's Signature:	Relinquished by: Signature	Company: Golder	Date: 17 Dec 2019	Time: 14:45	Received by: Signature	Company
Comments: ON ICE	Method of Shipment: Purolator	Waybill No.:	Received for Lab by: JG	Date: 18 Dec 19	Time: 8:30AM	
	Shipped by:	Shipment Condition: Seal Intact:	Temp (°C)	Cooler opened by: SC	Date	Time

WHITE: Golder Copy YELLOW: Lab Copy

ESED

APPENDIX H

Analytical Tables

Table H-1
Stage 1 Update Surface Soil Samples
Results of Soil Analyses - Metals
Ladysmith Harbour, Ladysmith, BC

Location	CSR standards for PL	MCS	CSR standards for CL	MCS	CSR standards for IL	MCS	HS19-01	HS19-02	HS19-02	HS19-03	HS19-04
SCN							07043-01	07043-02	07043-03	07043-04	07043-05
Depth (m bgs)							0.3-0.6	0.3-0.6	0.3-0.6	0.3-0.6	0.3-0.6
Date Sampled (mm/dd/yyyy)							2019-12-16	2019-12-16	2019-12-16	2019-12-16	2019-12-16
QA/QC								FDA	FD		
Field Parameters											
Soil Vapours (ppm)							0.1	0.0	0.0	0.1	0.1
Physical Parameters											
moisture (%)							19.0	22.8	18.3	21.4	34.0
pH (pH units)							6.86	8.15	8.04	7.32	5.89
Total Metals											
*aluminum	55000	HH	250000	HH	250000	HH	17300	20900	21000	25400	21600
antimony	20	EH	40	EH	40	EH	0.12	0.14	0.15	0.20	2.78
arsenic	10	AW-M/DW	**10**	AW-M	10	AW-M/DW	6.12	7.01	8.43	7.33	7.46
barium	350	DW	350	DW	350	DW	67.1	73.7	83.1	81.1	325
beryllium[pH]	1-150	AW-M/I/DW	1-350	DW/T	1-350	DW/T	0.31	0.28	0.28	0.35	0.45
bismuth							<0.20	<0.20	<0.20	<0.20	0.29
boron (hot water soluble)							<5.0	<5.0	<5.0	<5.0	10.3
cadmium[pH]	1-30	AW-M/T	1-70	DW/AW-M	1-70	AW-M/DW	0.093	0.031	0.041	0.113	0.655
calcium							4500	9760	10200	5540	7380
*chromium, total	65	AW-M/DW	65	DW/AW-M	65	AW-M/DW	29.3	38.6	36.5	37.0	36.0
*cobalt	30	AW-M/DW	30	DW/AW-M	30	AW-M/DW	8.35	12.3	12.1	12.3	11.4
*copper[pH]	100-150	AW-M/T	**100-300**	AW-M/T	100-300	AW-M/T	23.4	49.4	49.6	31.4	**118**
*iron	70000	HH	150000	HH	150000	HH	22500	30600	31800	35300	23200
lead[pH]	120	I	**120-150**	I/DW	120-1000	DW/T	4.29	2.57	2.80	5.88	59.4
lithium	65	HH	**450**	HH	450	HH	17.5	13.7	13.1	13.0	17.9
magnesium							4640	6660	6570	5540	5560
*manganese	5000	DW/T	5000	DW/T	5000	DW/T	205	532	553	918	301
mercury (inorganic)	25	I	75	I/T	75	T	0.055	0.063	0.051	0.051	0.175
molybdenum	15	DW	15	DW	15	DW	0.78	0.21	0.25	1.98	1.17
nickel[pH]	70-150	AW-M/DW	70-250	AW-M/DW	70-250	AW-M/DW	16.1	21.9	21.1	20.8	32.4
phosphorus							194	557	529	253	473
potassium							350	590	610	440	590
*selenium	4	AW-M/DW	4	AW-M/DW	4	AW-M/DW	<0.20	<0.20	<0.20	0.27	0.83
silver	20	EH	40	EH	40	EH	<0.10	<0.10	<0.10	<0.10	0.21
sodium	200	T/ion	**1000**	T/ion	1000	T/ion	246	604	668	314	332
strontium (stable)	20000	HH	150000	HH	150000	HH	32.0	67.8	72.8	59.5	68.6
sulphur							<1000	<1000	<1000	<1000	<1000
thallium	9	EH	25	EH	25	EH	<0.050	<0.050	<0.050	<0.050	0.063
tin	50	EH	300	EH	300	EH	<2.0	<2.0	<2.0	<2.0	5.6
titanium							1350	1520	1760	1740	1280
tungsten	25	HH	200	HH	200	HH	<0.50	<0.50	<0.50	<0.50	<0.50
uranium	30	DW	30	DW	30	DW	0.533	0.258	0.271	0.404	0.515
*vanadium	200	DW	200	DW	200	DW	78.8	82.5	91.5	116	82.6
zinc[pH]	150-200	AW-M	**150-200**	AW-M	150-200	AW-M	30.8	34.0	33.8	33.5	132
zirconium							3.9	5.5	6.1	3.2	4.4

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 24 January 2019).
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial)
MCS: most conservative standard based on applicable site-specific standards
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; DW Drinking Water; AW = Groundwater Flow to Surface Water used by Aquatic Life, and M = Marine Aquatic Life.
EH = generic standard for the protection of ecological health
HH = generic standard for the protection of human health
pH = standard is pH dependant
FDA = field duplicate available, FD = field duplicate
< = less than laboratory reporting limit; - = not analyzed
QA/QC = quality assurance/quality control
V = Standard is valence dependant: III - trivalent chromium (Cr^{3+}); VI - hexa-valent chromium (Cr^{6+})
italics = Laboratory Method Detection Limit above applicable standards
*Background concentrations applied as detailed in Protocol 4 is applied to replace CSR Standards.

Table H-2
Stage 1 Update Surface Soil Samples
Results of Soil Analyses - Hydrocarbons and PCBs
Ladysmith Harbour, Ladysmith, BC

Location / **SCN** / **Depth (m bgs)** / **Date Sampled (mm/dd/yyyy)** / **QA/QC**	CSR Standards for PL	**MCS**	**CSR Standards for CL**	**MCS**	CSR Standards for IL	**MCS**	HS19-01 07043-01 0.3-0.6 2019-12-16	HS19-02 07043-02 0.3-0.6 2019-12-16 FDA	HS19-02 07043-03 0.3-0.6 2019-12-16 FD	HS19-03 07043-04 0.3-0.6 2019-12-16	HS19-04 07043-05 0.3-0.6 2019-12-16
Field Parameters											
Soil Vapours (ppm unless otherwise indicated)							0.1	0.0	0.0	0.1	0.1
Physical Parameters											
moisture (%)							19.0	22.8	18.3	21.4	34.0
pH (pH units)							6.86	8.15	8.04	7.32	5.89
Extractable Hydrocarbons											
EPH_{C10-19} [a]	*1000*		*2,000*		*2,000*		<200	<200	<200	<200	<200
EPH_{C19-32} [a]	*1000*		*5000*		*5,000*		<200	<200	<200	<200	<200
LEPH	1000	EH/HH	2,000	EH/HH	2,000	EH/HH	<200	<200	<200	<200	<200
HEPH	1000	EH/HH	5000	EH/HH	5,000	EH/HH	<200	<200	<200	<200	<200
Polycyclic Aromatic Hydrocarbons											
1-Methylnaphthalene	500	HH	1000	HH	1000	HH	<0.050	<0.050	<0.050	<0.050	0.499
2-methylnaphthalene	100	HH	950	HH	950	HH	<0.010	<0.010	<0.010	<0.010	0.521
acenaphthene	2000	HH	15000	HH	15000	HH	<0.0050	<0.0050	<0.0050	<0.0050	<0.040
acenaphthylene							<0.0050	<0.0050	<0.0050	<0.0050	<0.030
anthracene	2.5	T	30	T	30	T	<0.0040	<0.0040	<0.0040	<0.0040	<0.090
benzo(a)anthracene	1	EH	10	EH	10	EH	<0.010	<0.010	<0.010	<0.010	0.113
benzo(a)pyrene	10	I	30	I	50	I	<0.010	<0.010	<0.010	<0.010	0.072
Benzo(b&j)fluoranthene	1	EH	10	EH	10	EH	<0.010	<0.010	<0.010	<0.010	0.135
Benzo(b+j+k)fluoranthene							<0.015	<0.015	<0.015	<0.015	0.177
benzo(g,h,i)perylene							<0.010	<0.010	<0.010	<0.010	0.073
benzo(k)fluoranthene	1	EH	10	EH	10	EH	<0.010	<0.010	<0.010	<0.010	0.042
chrysene	400	HH	4500	HH	4500	HH	<0.010	<0.010	<0.010	<0.010	0.125
dibenz(a,h)anthracene	1	EH	10	EH	10	EH	<0.0050	<0.0050	<0.0050	<0.0050	0.0137
fluoranthene	50	T	200	T	200	T	<0.010	<0.010	<0.010	<0.010	0.196
fluorene	1000	HH	9500	HH	9500	HH	<0.010	<0.010	<0.010	<0.010	0.066
indeno(1,2,3-c,d)pyrene	1	EH	10	EH	10	EH	<0.010	<0.010	<0.010	<0.010	0.056
naphthalene	0.6	T	20	T	20	T	<0.010	<0.010	<0.010	<0.010	0.183
phenanthrene	5	EH	50	EH	50	EH	<0.010	<0.010	<0.010	<0.010	0.706
pyrene	10	EH	100	EH	100	EH	<0.010	<0.010	<0.010	<0.010	0.171
quinoline	4.5	HH	10	HH	10	HH	<0.050	<0.050	<0.050	<0.050	<0.050
B(a)P Total Potency Equivalent							<0.020	<0.020	<0.020	<0.020	0.122
IACR (CCME)							<0.15	<0.15	<0.15	<0.15	1.79
Non-Halogenated Volatiles											
benzene	0.035	DW	0.035	DW	0.035	DW	<0.0050	<0.0050	<0.0050	<0.0050	<0.0050
ethylbenzene	15	DW	15	DW	15	DW	<0.015	<0.015	<0.015	<0.015	<0.015
styrene	5	EH	50	EH	50	EH	<0.050	<0.050	<0.050	<0.050	<0.050
toluene	6	DW	6	DW	6	DW	<0.050	<0.050	<0.050	<0.050	<0.050
ortho-Xylene							<0.050	<0.050	<0.050	<0.050	<0.050
meta- & para-Xylene							<0.050	<0.050	<0.050	<0.050	<0.050
Total xylene	6.5	DW	6.5	DW	6.5	DW	<0.075	<0.075	<0.075	<0.075	<0.075
Methyl t-butyl ether (MTBE)	8000	HH	20000	HH	20000	HH	<0.20	<0.20	<0.20	<0.20	<0.20

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial).
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; AW = Groundwater Flow to Surface Water used by Aquatic Life, M = Marine; DW = Drinking Water; EH = generic standard for the protection of ecological health; HH = generic standard for the protection of human health
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 24 January 2019)
MCS = most conservative standard based on applicable site-specific standards
QA/QC = quality assurance/quality control
a. LEPH and HEPH criteria used as a conservative screen for EPH10-19 and EPH19-32, respectively
FDA = Field duplicate available; FD = Field duplicate.
italics = Laboratory Method Detection Limit above applicable standards
< = less than laboratory reporting limit; - = not analyzed
EPH_{C10-19} = extractable petroleum hydrocarbons, carbon range 10-19
EPH_{C19-32} = extractable petroleum hydrocarbons, carbon range 19-32
LEPH = light extractable petroleum hydrocarbons
HEPH = heavy extractable petroleum hydrocarbons
PAH = polycyclic aromatic hydrocarbon
ppm = parts per milion

Table H-3
Stage 1 Update Surface Soil Samples
Results of Soil Analyses - Quality Assurance/Quality Control
Ladysmith Harbour, Ladysmith, BC

	Location	HS19-02	HS19-02	Method Reporting Limit	Mean	Relative Percent Difference	Difference Factor (DF)
	SCN	07043-02	07043-03				
	Laboratory ID	L2398084-2	L2398084-3				
	Depth (m bgs)	0.3-0.6	0.3-0.6				
	Date Sampled	2019-12-16	2019-12-16				
	QA/QC	FDA	FD				
Total Metals							
*aluminum		20900	21000	50	20950	0.48%	NA
antimony		0.14	0.15	0.10	0.145	NA	0.10
arsenic		7.01	8.43	0.10	7.72	18%	NA
barium		73.7	83.1	0.50	78.4	12%	NA
beryllium[pH]		0.28	0.28	0.10	0.28	NA	0.00
bismuth		<0.20	<0.20	0.20	NC	NC	NC
boron (hot water soluble)		<5.0	<5.0	5.0	NC	NC	NC
cadmium[pH]		0.031	0.041	0.020	0.036	NA	0.50
calcium		9760	10200	50	9980	4.4%	NA
*chromium, total		38.6	36.5	0.50	37.55	5.6%	NA
*cobalt		12.3	12.1	0.10	12.2	1.6%	NA
*copper[pH]		49.4	49.6	0.50	49.5	0.40%	NA
*iron		30600	31800	50	31200	3.8%	NA
lead[pH]		2.57	2.80	0.50	2.685	8.6%	NA
lithium		13.7	13.1	2.0	13.4	4.5%	NA
magnesium		6660	6570	20	6615	1.4%	NA
*manganese		532	553	1.0	542.5	3.9%	NA
mercury (inorganic)		0.063	0.051	0.05	0.057	NA	0.24
molybdenum		0.21	0.25	0.10	0.23	NA	0.40
nickel[pH]		21.9	21.1	0.50	21.5	3.7%	NA
phosphorus		557	529	50	543	5.2%	NA
potassium		590	610	100	600	3.33%	NA
*selenium		<0.20	<0.20	0.20	NC	NC	NC
silver		<0.10	<0.10	0.10	NC	NC	NC
sodium		604	668	50	636	10%	NA
strontium (stable)		67.8	72.8	0.50	70.3	7.1%	NA
sulphur		<1000	<1000	1000	NC	NC	NC
thallium		<0.050	<0.050	0.050	NC	NC	NC
tin		<2.0	<2.0	2.0	NC	NC	NC
titanium		1520	1760	1.0	1640	15%	NA
tungsten		<0.50	<0.50	0.50	NC	NC	NC
uranium		0.258	0.271	0.050	0.2645	4.9%	NA
*vanadium		82.5	91.5	0.20	87	10%	NA
zinc[pH]		34.0	33.8	2.0	33.9	0.59%	NA
zirconium		5.5	6.1	1.0	5.8	10%	NA
Extractable Hydrocarbons					-	-	-
EPH_{C10-19} [a]		<200	<200	200	NC	NC	NC
EPH_{C19-32} [a]		<200	<200	200	NC	NC	NC
LEPH		<200	<200	200	NC	NC	NC
HEPH		<200	<200	200	NC	NC	NC
Polycyclic Aromatic Hydrocarbons							
1-Methylnaphthalene		<0.050	<0.050	0.050	NC	NC	NC
2-methylnaphthalene		<0.010	<0.010	0.010	NC	NC	NC
acenaphthene		<0.0050	<0.0050	0.0050	NC	NC	NC
acenaphthylene		<0.0050	<0.0050	0.0050	NC	NC	NC
anthracene		<0.0040	<0.0040	0.0040	NC	NC	NC
benzo(a)anthracene		<0.010	<0.010	0.010	NC	NC	NC
benzo(a)pyrene		<0.010	<0.010	0.010	NC	NC	NC
Benzo(b&j)fluoranthene		<0.010	<0.010	0.010	NC	NC	NC
Benzo(b+j+k)fluoranthene		<0.015	<0.015	0.015	NC	NC	NC
benzo(g,h,i)perylene		<0.010	<0.010	0.010	NC	NC	NC
benzo(k)fluoranthene		<0.010	<0.010	0.010	NC	NC	NC
chrysene		<0.010	<0.010	0.010	NC	NC	NC
dibenz(a,h)anthracene		<0.0050	<0.0050	0.0050	NC	NC	NC
fluoranthene		<0.010	<0.010	0.010	NC	NC	NC
fluorene		<0.010	<0.010	0.010	NC	NC	NC
indeno(1,2,3-c,d)pyrene		<0.010	<0.010	0.010	NC	NC	NC
naphthalene		<0.010	<0.010	0.010	NC	NC	NC
phenanthrene		<0.010	<0.010	0.010	NC	NC	NC
pyrene		<0.010	<0.010	0.010	NC	NC	NC
quinoline		<0.050	<0.050	0.050	NC	NC	NC
B(a)P Total Potency Equivalent		<0.020	<0.020	0.020	NC	NC	NC
IACR (CCME)		<0.15	<0.15	0.15	NC	NC	NC
Non-Halogenated Volatiles							
benzene		<0.0050	<0.0050	0.0050	NC	NC	NC
ethylbenzene		<0.015	<0.015	0.015	NC	NC	NC
styrene		<0.050	<0.050	0.050	NC	NC	NC
toluene		<0.050	<0.050	0.050	NC	NC	NC
ortho-Xylene		<0.050	<0.050	0.050	NC	NC	NC
meta- & para-Xylene		<0.050	<0.050	0.050	NC	NC	NC
Total xylene		<0.075	<0.075	0.075	NC	NC	NC
Methyl t-butyl ether (MTBE)		<0.20	<0.20	0.20	NC	NC	NC

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
SCN = sample control number
FDA = field duplicate available
FD = field duplicate
QA/QC = quality assurance/quality control

Method Reporting Limit indicates the minimum concentration that could be measured by laboratory instrumentation for a specific sample.

Mean indicates the mean or average value calculated of a field duplicate pair (the FDA and the FD).
Relative Percent Difference (RPD) is calculated when the mean value is greater than five times the method reporting limit; Golder's internal QA/QC target is less than 30% for metals and less than 50% for organic compounds.

Difference Factor (DF) is calculated when the mean value is less than five times the method reporting limit; Golder's internal QA/QC target is less than 2.

NA = not applicable; NC = not calculated
BOLD font indicates the parameter analyzed

APPENDIX I

Historical Data Rescreening

Table I-1a
Results of Soil Analyses - Metals
Ladysmith Harbour, Ladysmith, BC
Updated December 2019

APEC							AEC 10								
Location							MW09-1	MW09-3	MW09-3	TP09-01	TP09-02	TP09-02	TP09-02	TP09-03	TP09-04
SCN							21360-02	21360-07	21360-08	21381-01	21381-04	21381-05	21381-06	21381-09	21381-11
Depth (m bgs)							0.6 - 0.76	0.6 - 0.76	2.1 - 2.3	0.6-0.8	1.5-1.7	1.5-1.7	2.2-2.4	1.8-1.9	1.0-1.2
Date Sampled (mm/dd/yyyy)							2009-11-10	2009-11-10	2009-11-11	2009-12-15	2009-12-15	2009-12-15	2009-12-15	2009-12-15	2009-12-15
QA/QC											FDA	FD			
Field Parameters															
Soil Vapours (ppm)							0	10	23	20					22
Physical Parameters															
moisture (%)							18.5	10.6	8.4	-	16.1	18.5	20.8	10.9	17
pH (pH units)							5.7	7.3	-	5.9	-	-	-	-	7.2
Total Metals															
*aluminum	55000	HH	250000	HH	250000	HH	20900	11900	-	13600	-	-	-	-	19100
antimony	20	EH	40	EH	40	EH	< 10	< 10	-	< 10	-	-	-	-	< 10
arsenic	10	AW-M/DW	**10**	AW-M	10	AW-M/DW	< 10	< 10	-	< 10	-	-	-	-	**11**
barium	350	DW	350	DW	350	DW	135	42	-	116	-	-	-	-	153
beryllium^{pH}	1-150	AW-M/I/DW	1-350	DW/T	1-350	DW/T	< 1	< 1	-	< 1	-	-	-	-	< 1
boron (hot water soluble)							17	13	-	< 1	-	-	-	-	3
cadmium^{pH}	1-30	AW-M/T	1-70	DW/AW-M	1-70	AW-M/DW	0.8	< 0.5	-	< 0.5	-	-	-	-	1.8
calcium							4860	4470	-	3650	-	-	-	-	7800
*chromium, total	65	AW-M/DW	65	DW/AW-M	65	AW-M/DW	40	30	-	27	-	-	-	-	35
*cobalt	30	AW-M/DW	30	DW/AW-M	30	AW-M/DW	11	8	-	12	-	-	-	-	16
*copper^{pH}	100-150	AW-M/T	100-300	AW-M/T	100-300	AW-M/T	89	49	-	82	-	-	-	-	119
*iron	70000	HH	150000	HH	150000	HH	24900	18800	-	20500	-	-	-	-	28100
lead^{pH}	120	I	**120-150**	I/DW	120-1000	DW/T	89	23	-	75	-	-	-	-	52
magnesium							4220	5430	-	5190	-	-	-	-	3890
*manganese	5000	DW/T	5000	DW/T	5000	DW/T	558	274	-	493	-	-	-	-	1070
mercury (inorganic)	25	I	75	I/T	75	T	0.06	0.02	-	0.07	-	-	-	-	0.06
molybdenum	15	DW	15	DW	15	DW	< 4	< 4	-	< 4	-	-	-	-	< 4
nickel^{pH}	70-150	AW-M/DW	70-250	AW-M/DW	70-250	AW-M/DW	21	17	-	24	-	-	-	-	25
phosphorus							447	440	-	582	-	-	-	-	915
potassium							348	741	-	450	-	-	-	-	411
*selenium	4	AW-M/DW	4	AW-M/DW	4	AW-M/DW	*< 2*	*< 2*	-	*< 2*	-	-	-	-	*< 2*
silver	20	EH	40	EH	40	EH	< 2	< 2	-	< 2	-	-	-	-	< 2
sodium							246	408	-	172	-	-	-	-	168
strontium (stable)	20000	HH	150000	HH	150000	HH	38	19	-	25	-	-	-	-	106
tin	50	EH	300	EH	300	EH	< 5	< 5	-	< 5	-	-	-	-	< 5
titanium							953	939	-	964	-	-	-	-	1280
*vanadium	200	DW	200	DW	200	DW	102	42	-	67	-	-	-	-	92
zinc^{pH}	150-200	AW-M	**150-200**	AW-M	150-200	AW-M	71	49	-	81	-	-	-	-	143
zirconium							4	3	-	3	-	-	-	-	5

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019).
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial)
MCS: most conservative standard based on applicable site-specific standards
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; DW Drinking Water; AW = Groundwater Flow to Surface Water used by Aquatic Life, and M = Marine Aquatic Life.
EH = generic standard for the protection of ecological health
HH = generic standard for the protection of human health
pH = standard is pH dependant
FDA = field duplicate available, FD = field duplicate
< = less than laboratory reporting limit; - = not analyzed
QA/QC = quality assurance/quality control
V = Standard is valence dependant: III - trivalent chromium (Cr^{3+}); VI - hexa-valent chromium (Cr^{6+})
italics = Laboratory Method Detection Limit above applicable standards
*Background concentrations applied as detailed in Protocol 4 is applied to replace CSR Standards.

Table I-1a
Results of Soil Analyses - Metals
Ladysmith Harbour, Ladysmith, BC
Updated December 2019

APEC					APEC 15				APEC 17		
Location					MW09-5	MW09-5	BH09-15	BH09-15	MW09-16	MW09-16	MW09-16
SCN	CSR standards for PL	MCS	**CSR standards for CL**	MCS	21361-08	21361-09	21366-03	21366-04	21366-07	21366-09	21366-10
Depth (m bgs)					0.6 - 0.8	1.4 - 1.6	3.65 - 3.8	3.65 - 3.8	1.2 - 1.37	3.35 - 3.50	3.35 - 3.50
Date Sampled (mm/dd/yyyy)					2009-11-11	2009-11-11	11/14/2009	11/14/2009	11/14/2009	11/14/2009	11/14/2009
QA/QC							FDA	FD		FDA	FD
Field Parameters											
Soil Vapours (ppm)					30	15	340	340	16	26	26
Physical Parameters											
moisture (%)					16.8	-	14.5	11.7	10.3	18.7	18.1
pH (pH units)					5.9	6.2	-	-	6.3	7.3	7.5
Total Metals											
*aluminum	55000	HH	250000	HH	23800	19800	-	-	20500	11400	10900
antimony	20	EH	40	EH	< 10	< 10	-	-	< 10	< 10	< 10
arsenic	10	AW-M/DW	**10**	AW-M	< 10	< 10	-	-	< 10	< 10	< 10
barium	350	DW	350	DW	122	71	-	-	91	117	109
beryllium[pH]	1-150	AW-M/I/DW	1-350	DW/T	< 1	< 1	-	-	< 1	< 1	< 1
boron (hot water soluble)					18	21	-	-	18	14	14
cadmium[pH]	1-30	AW-M/T	1-70	DW/AW-M	< 0.5	< 0.5	-	-	< 0.5	< 0.5	< 0.5
calcium					3170	4420	-	-	3970	4940	4860
*chromium, total	65	AW-M/DW	65	DW/AW-M	33	25	-	-	29	15	15
*cobalt	30	AW-M/DW	30	DW/AW-M	12	12	-	-	10	7	7
*copper[pH]	100-150	AW-M/T	100-300	AW-M/T	44	52	-	-	39	29	29
*iron	70000	HH	150000	HH	24900	22000	-	-	22200	16600	16600
lead[pH]	120	I	**120-150**	I/DW	10	< 5	-	-	13	< 5	< 5
magnesium					6680	6580	-	-	5110	5140	5030
*manganese	5000	DW/T	5000	DW/T	337	323	-	-	418	242	235
mercury (inorganic)	25	I	75	I/T	0.05	0.04	-	-	0.04	0.02	0.03
molybdenum	15	DW	15	DW	< 4	< 4	-	-	< 4	< 4	< 4
nickel[pH]	70-150	AW-M/DW	70-250	AW-M/DW	21	17	-	-	23	16	15
phosphorus					493	470	-	-	538	440	432
potassium					401	607	-	-	552	630	593
*selenium	4	AW-M/DW	4	AW-M/DW	*< 2*	*< 2*	-	-	*< 2*	*< 2*	*< 2*
silver	20	EH	40	EH	< 2	< 2	-	-	< 2	< 2	< 2
sodium					162	364	-	-	176	557	525
strontium (stable)	20000	HH	150000	HH	25	28	-	-	21	29	26
tin	50	EH	300	EH	< 5	< 5	-	-	< 5	< 5	< 5
titanium					1300	1270	-	-	1230	913	902
*vanadium	200	DW	200	DW	82	67	-	-	65	45	46
zinc[pH]	150-200	AW-M	**150-200**	AW-M	51	33	-	-	47	29	28
zirconium					6	6	-	-	8	4	5

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Standards shown from the Contaminated Sites Regulation (“CSR”; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amenc
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial)
MCS: most conservative standard based on applicable site-specific standards
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; DW Drinking Wat
EH = generic standard for the protection of ecological health
HH = generic standard for the protection of human health
pH = standard is pH dependant
FDA = field duplicate available, FD = field duplicate
< = less than laboratory reporting limit; - = not analyzed
QA/QC = quality assurance/quality control
V = Standard is valence dependant: III - trivalent chromium (Cr^{3+}); VI - hexa-valent chromium (Cr^{6+})
italics = Laboratory Method Detection Limit above applicable standards
*Background concentrations applied as detailed in Protocol 4 is applied to replace CSR Standards.

Table I-1a
Results of Soil Analyses - Metals
Ladysmith Harbour, Ladysmith, BC
Updated December 2019

APEC					AEC 12								APEC 9		
Location					MW09-9	MW09-10	MW09-10	MW09-11	KE SA1	KE SA2	KE SA4	TP09-05	MW09-6	TP09-09	TP09-09
SCN	CSR standards for PL	**MCS**	**CSR standards for CL**	**MCS**	21364-02	21364-05	21364-07	21364-09	21370-01	21370-02	21370-04	21383-01	21362-06	21383-12	21581-01
Depth (m bgs)					1.5 - 1.7	0.6 - 0.76	1.5 - 1.7	0.3 - 0.4	0.2 - 0.3	0.2 - 0.3	0.2 - 0.3	0.3-0.5	2.0 - 2.2	0.4-0.6	1.0-1.2
Date Sampled (mm/dd/yyyy)					11/13/2009	11/13/2009	11/13/2009	11/13/2009	11/25/2009	11/25/2009	11/25/2009	2009-12-15	2009-12-11	2009-12-15	2009-12-15
QA/QC															
Field Parameters															
Soil Vapours (ppm)					16	10	22	5	-	-	-	20	10	10	15
Physical Parameters															
moisture (%)					12.6	8.5	15.4	21.2	9	10.8	6.9	-	-	15.9	16.3
pH (pH units)					8.1	7.6	7.7	6.6	6.2	6	6.1	7.9	6.8	7.2	7.4
Total Metals															
*aluminum	55000	HH	250000	HH	10400	16000	11700	21800	10800	11100	11300	12600	10100	11300	8200
antimony	20	EH	40	EH	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10
arsenic	10	AW-M/DW	**10**	AW-M	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10
barium	350	DW	350	DW	40	189	46	103	43	61	65	273	46	181	68
beryllium[pH]	1-150	AW-M/I/DW	1-350	DW/T	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1
boron (hot water soluble)					16	24	15	22	8	37	10	21	7	8	< 1
cadmium[pH]	1-30	AW-M/T	1-70	DW/AW-M	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	1.1	< 0.5	< 0.5	< 0.5
calcium					7070	7730	7290	4820	2920	3380	3000	11200	4430	6310	4030
*chromium, total	65	AW-M/DW	65	DW/AW-M	16	25	20	39	13	17	13	27	25	23	18
*cobalt	30	AW-M/DW	30	DW/AW-M	7	10	9	13	7	10	9	11	8	11	6
*copper[pH]	100-150	AW-M/T	100-300	AW-M/T	31	62	39	42	52	138	47	140	29	61	20
*iron	70000	HH	150000	HH	20100	20700	19200	26900	13800	16800	14900	20500	18600	18400	12400
lead[pH]	120	I	**120-150**	I/DW	< 5	45	< 5	19	15	80	12	**175**	< 5	20	< 5
magnesium					4500	5700	5510	5150	4460	5060	4680	5260	3880	5410	3340
*manganese	5000	DW/T	5000	DW/T	482	335	220	747	190	265	237	335	207	280	133
mercury (inorganic)	25	I	75	I/T	0.04	0.08	0.04	0.06	0.05	0.05	0.02	0.16	0.03	0.08	0.02
molybdenum	15	DW	15	DW	< 4	< 4	< 4	< 4	< 4	< 4	< 4	< 4	< 4	< 4	< 4
nickel[pH]	70-150	AW-M/DW	70-250	AW-M/DW	13	25	16	21	12	14	11	32	14	31	12
phosphorus					461	683	478	263	546	605	509	750	428	670	376
potassium					413	630	504	380	330	509	376	541	244	595	151
*selenium	4	AW-M/DW	4	AW-M/DW	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*
silver	20	EH	40	EH	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
sodium					504	402	524	208	201	205	204	289	210	237	144
strontium (stable)	20000	HH	150000	HH	41	111	35	47	16	18	20	166	45	91	32
tin	50	EH	300	EH	< 5	< 5	< 5	< 5	< 5	9	< 5	8	< 5	< 5	< 5
titanium					828	773	969	1160	396	478	478	786	952	458	1080
*vanadium	200	DW	200	DW	59	60	63	96	33	40	35	57	65	49	56
zinc[pH]	150-200	AW-M	**150-200**	AW-M	25	61	30	**210**	34	77	66	133	23	85	20
zirconium					4	6	5	8	3	2	3	4	4	3	1

Notes:
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V = Standard is valence dependant: III - trivalent chromium (Cr^{3+}); VI - hexa-valent chromium (Cr^{6+})
italics = Laboratory Method Detection Limit above applicable standards
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Table I-1a
Results of Soil Analyses - Metals
Ladysmith Harbour, Ladysmith, BC
Updated December 2019

APEC					APEC 20					APEC 8					
Location					BH09-13	BH09-14	MW09-7	MW09-8	MW09-8	TP09-10	TP09-10	TP09-11	TP09-12	TP09-12	TP09-12
SCN	CSR standards for PL	**MCS**	**CSR standards for CL**	**MCS**	21365-05	21365-11	21362-11	21363-07	21363-08	21581-03	21581-04	21581-06	21581-08	21581-09	21581-10
Depth (m bgs)					3.8 - 4.0	4.1 - 4.3	0.5 - 0.7	2.9 - 3.0	2.9 - 3.0	0.3-0.6	1.6-1.8	0.2-0.4	0.1-0.3	0.1-0.3	1.3-1.5
Date Sampled (mm/dd/yyyy)					11/14/2009	11/14/2009	2009-12-11	11/13/2009	11/13/2009	2009-12-15	2009-12-15	2009-12-15	2009-12-15	2009-12-15	2009-12-15
QA/QC								FDA	FD				FDA	FD	
Field Parameters															
Soil Vapours (ppm)					460	210	5	440	440	15	20	30	30	30	25
Physical Parameters															
moisture (%)					19.3	9.5	-	20	24.3	9	7.2	13.2	15	18.4	12.4
pH (pH units)					7.4	8.1	6.3	8	7.9	7.5	6.9	6.7	6.8	6.8	7.3
Total Metals															
*aluminum	55000	HH	250000	HH	23200	11600	14500	12100	11000	13200	12500	14300	14000	15400	19000
antimony	20	EH	40	EH	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10
arsenic	10	AW-M/DW	**10**	AW-M	**16**	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10	< 10
barium	350	DW	350	DW	95	55	45	55	52	94	108	91	130	131	87
beryllium[pH]	1-150	AW-M/I/DW	1-350	DW/T	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1	< 1
boron (hot water soluble)					38	19	8	19	18	< 1	< 1	< 1	1	< 1	< 1
cadmium[pH]	1-30	AW-M/T	1-70	DW/AW-M	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5
calcium					1920	9380	3810	14800	33000	5690	3940	5020	6280	5820	4840
*chromium, total	65	AW-M/DW	65	DW/AW-M	39	20	17	20	19	26	24	25	27	29	21
*cobalt	30	AW-M/DW	30	DW/AW-M	28	7	8	9	8	11	12	13	11	12	11
*copper[pH]	100-150	AW-M/T	100-300	AW-M/T	50	21	40	64	56	46	26	57	63	64	25
*iron	70000	HH	150000	HH	36300	18700	19500	18400	18200	22100	23800	21700	20800	22900	21800
lead[pH]	120	I	**120-150**	I/DW	30	19	5	36	54	16	5	23	58	60	6
magnesium					11500	6530	4900	6000	5890	5870	5560	5660	5390	5610	4270
*manganese	5000	DW/T	5000	DW/T	1190	329	236	425	427	510	734	694	464	608	505
mercury (inorganic)	25	I	75	I/T	0.11	0.07	0.05	0.2	0.26	0.04	0.02	0.05	0.09	0.09	0.05
molybdenum	15	DW	15	DW	7	< 4	< 4	5	< 4	< 4	< 4	< 4	< 4	< 4	< 4
nickel[pH]	70-150	AW-M/DW	70-250	AW-M/DW	54	17	13	22	18	22	19	22	21	22	18
phosphorus					441	373	570	492	469	635	1240	700	766	801	3010
potassium					3790	1420	412	815	796	647	347	384	422	442	318
*selenium	4	AW-M/DW	4	AW-M/DW	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*	*< 2*
silver	20	EH	40	EH	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2	< 2
sodium					3560	1760	293	1470	1720	298	149	197	366	406	249
strontium (stable)	20000	HH	150000	HH	34	70	20	117	266	45	19	36	65	56	49
tin	50	EH	300	EH	< 5	< 5	< 5	< 5	< 5	< 5	< 5	< 5	< 5	< 5	< 5
titanium					186	809	868	597	677	1010	617	1240	1060	1140	944
*vanadium	200	DW	200	DW	58	34	61	46	41	63	60	64	67	71	51
zinc[pH]	150-200	AW-M	**150-200**	AW-M	131	61	30	68	75	53	57	69	108	123	51
zirconium					5	3	4	3	3	3	2	3	3	3	4

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amenc
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial)
MCS: most conservative standard based on applicable site-specific standards
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; DW Drinking Wat
EH = generic standard for the protection of ecological health
HH = generic standard for the protection of human health
pH = standard is pH dependant
FDA = field duplicate available, FD = field duplicate
< = less than laboratory reporting limit; - = not analyzed
QA/QC = quality assurance/quality control
V = Standard is valence dependant: III - trivalent chromium (Cr^{3+}); VI - hexa-valent chromium (Cr^{6+})
italics = Laboratory Method Detection Limit above applicable standards
*Background concentrations applied as detailed in Protocol 4 is applied to replace CSR Standards.

Table I-1b
Results of Soil Analyses - Hydrocarbons and PCBs
Ladysmith Harbour, Ladysmith, BC

APEC							APEC 9			APEC 8							APEC 10													
Fill Samples									APEC 8								APEC 8								APEC 8	APEC 8	APEC 8	APEC 8	APEC 8
Location	CSR Standards for PL	MCS	CSR Standards for CL	MCS	CSR Standards for IL	MCS	MW09-6	TP09-09	TP09-09	TP09-10	TP09-10	TP09-11	TP09-12	TP09-12	TP09-12	MW09-1	TP09-01	TP09-02	TP09-02	TP09-02	TP09-03	TP09-04	TP11-02	TP11-02	TP11-03	TP11-04	TP11-04	TP11-04	TP09-07	TP09-08	ear TP-8) SA	ear TP-8) SA	ear TP-8) SA
SCN							21362-06	21383-12	21581-01	21581-03	21581-04	21581-06	21581-08	21581-09	21581-10	21360-02	21381-01	21381-04	21381-05	21381-06	21381-09	21381-11	0527-04	0527-05	0527-09	0528-02	0528-03	0528-04	21383-08	21383-11	21370-05	21370-06	21370-07
Depth (m bgs)							2.0 - 2.2	0.4-0.6	1.0-1.2	0.3-0.6	1.6-1.8	0.2-0.4	0.1-0.3	0.1-0.3	1.3-1.5	0.6 - 0.76	0.6-0.8	1.5-1.7	1.5-1.7	2.2-2.4	1.8-1.9	1.0-1.2	2-2.25	1.8-1.9	2.0-2.1	0.5-0.6	0.5-0.6	1.7-1.8	0.3-0.5	0.4-0.6	0.2 - 0.3	0.2 - 0.3	0.2 - 0.3
Date Sampled (mm/dd/yyyy)							11-12-09	15-12-09	15-12-09	15-12-09	15-12-09	15-12-09	15-12-09 FDA	15-12-09 FD	15-12-09	11-10-09	15-12-09	15-12-09 FDA	15-12-09 FD	15-12-09	15-12-09	15-12-09	09-02-11	09-02-11	09-02-11	09-02-11 FDA	09-02-11 FD	09-02-11	15-12-09	15-12-09	11/25/2009	11/25/2009	11/25/2009
QA/QC																																
Field Parameters																																
Soil Vapours (ppm unless otherwise indicated)							10	10	15	15	20	30	30	30	25	0	20	580	580	810	5% LEL	22	0	0	0	0.6	0.6	0	20	10	-	-	-
Physical Parameters																																
moisture (%)							-	15.9	16.3	9	7.2	13.2	15	18.4	12.4	18.5	-	16.1	18.5	20.8	10.9	17	18.9	11.1	21.4	33.6	52.0	20.8	6.1	11.9	20.2	7.4	19
pH (pH units)							6.8	7.2	7.4	7.5	6.9	6.7	6.8	6.8	7.3	5.7	5.9	-	-	-	-	7.2	-	-	-	-	-	-	-	-	-	-	-
Extractable Hydrocarbons																																
EPH_{C10-19} [a]	*1000*		***2,000***		*2,000*		< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	-	270	300	430	< 250	< 250	<200	<200	<200	<200	<200	<200	< 250	< 250	< 250	< 250	1,000
EPH_{C10-19} (sg)							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
EPH_{C19-32} [a]	*1000*		***5000***		*5,000*		< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	-	830	980	400	< 250	< 250	<200	<200	<200	<200	<200	<200	< 250	830	< 250	1,600	**14,000**
EPH_{C19-32} (sg)							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
LEPH	1000	EH/HH	**2,000**	EH/HH	2,000	EH/HH	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	-	270	300	430	< 250	< 250	<200	<200	<200	<200	<200	<200	< 250	< 250	< 250	< 250	1,000
HEPH	1000	EH/HH	**5000**	EH/HH	5,000	EH/HH	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	-	830	980	400	< 250	< 250	<200	<200	<200	<200	<200	<200	< 250	830	< 250	1,600	**14,000**
Polycyclic Aromatic Hydrocarbons																																
2-methylnaphthalene	100	HH	950	HH	950	HH	< 0.05	0.98	< 0.05	0.21	< 0.05	0.17	0.16	0.13	< 0.05	0.08	-	< 0.05	< 0.05	0.16	3.2	< 0.05	<0.050	0.250	<0.050	0.060	0.106	<0.050	< 0.05	0.06	< 0.05	0.08	0.29
acenaphthene	2000	HH	15000	HH	15000	HH	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	0.11	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
acenaphthylene							< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
anthracene	2.5	T	30	T	30	T	< 0.05	0.06	< 0.05	< 0.05	< 0.05	< 0.05	0.12	0.07	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
benzo(a)anthracene	1	EH	10	EH	10	EH	< 0.05	0.13	< 0.05	< 0.05	< 0.05	0.11	0.26	0.34	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.050	0.068	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
benzo(a)pyrene	10	I	30	I	50	I	< 0.05	0.09	< 0.05	0.05	< 0.05	0.13	0.33	0.39	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.050	<0.071	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	0.1
benzo(b)fluoranthene							< 0.05	0.1	< 0.05	0.06	< 0.05	0.14	0.38	0.46	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.050	0.095	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
benzo(g,h,i)perylene							< 0.05	0.07	< 0.05	< 0.05	< 0.05	0.07	0.22	0.26	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	0.47
benzo(k)fluoranthene	1	EH	10	EH	10	EH	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	0.08	0.19	0.16	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	0.08
chrysene	400	HH	4500	HH	4500	HH	< 0.05	0.15	< 0.05	0.05	< 0.05	0.12	0.32	0.45	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.050	0.087	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	0.11
dibenz(a,h)anthracene	1	EH	10	EH	10	EH	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	0.08	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
fluoranthene	50	T	200	T	200	T	< 0.05	0.09	< 0.05	0.05	< 0.05	0.16	0.52	0.44	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.050	0.103	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
fluorene	1000	HH	9500	HH	9500	HH	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	0.11	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
indeno(1,2,3-c,d)pyrene	1	EH	10	EH	10	EH	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	0.06	0.18	0.21	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	0.1	0.31
naphthalene	0.6	T	20	T	20	T	< 0.05	0.45	< 0.05	0.12	< 0.05	0.09	0.11	0.1	< 0.05	0.07	-	< 0.05	< 0.05	< 0.05	2.20	< 0.05	<0.050	0.136	<0.050	<0.050	0.060	<0.050	< 0.05	< 0.05	< 0.05	0.07	0.27
phenanthrene	5	EH	50	EH	50	EH	< 0.05	0.98	< 0.05	0.19	< 0.05	0.22	0.62	0.41	< 0.05	< 0.05	-	< 0.05	< 0.05	0.09	< 0.05	< 0.05	<0.050	0.144	<0.050	0.069	0.127	<0.050	< 0.05	0.06	< 0.05	0.13	0.32
pyrene	10	EH	100	EH	100	EH	< 0.05	0.15	< 0.05	0.07	< 0.05	0.18	0.56	0.58	< 0.05	< 0.05	-	< 0.05	0.08	< 0.05	< 0.05	< 0.05	<0.050	0.106	<0.050	<0.050	<0.050	<0.050	< 0.05	< 0.05	< 0.05	0.07	0.23
Non-Halogenated Volatiles																																
benzene	0.035	DW	**0.035**	DW	6.5	DW	-	-	-	-	-	-	-	-	-	*< 0.04*	-	*< 0.04*	-	*< 0.04*	*< 0.04*	*< 0.04*	*<0.040*	*<0.040*	*<0.040*	*<0.040*	*<0.040*	*<0.040*	-	-	-	-	-
ethylbenzene	15	DW	**15**	DW	15	DW	-	-	-	-	-	-	-	-	-	< 0.5	-	< 0.5	-	< 0.5	< 0.5	< 0.5	<0.050	<0.050	<0.050	<0.050	<0.050	<0.050	-	-	-	-	-
styrene	5	EH	50	EH	50	EH	-	-	-	-	-	-	-	-	-	< 0.1	-	< 0.5	-	< 0.1	< 0.1	< 0.1	<0.050	<0.050	<0.050	<0.050	<0.050	<0.050	-	-	-	-	-
toluene	6	DW	6	DW	6	DW	-	-	-	-	-	-	-	-	-	< 0.5	-	< 0.1	-	< 0.5	< 0.5	< 0.5	<0.050	<0.050	<0.050	<0.050	<0.050	<0.050	-	-	-	-	-
ortho-Xylene							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	<0.050	<0.050	<0.050	<0.050	<0.050	<0.050	-	-	-	-	-
meta- & para-Xylene							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	<0.050	0.060	<0.050	<0.050	<0.050	<0.050	-	-	-	-	-
Total xylene	6.5	DW	**6.5**	DW	6.5	DW	-	-	-	-	-	-	-	-	-	< 0.1	-	< 0.1	-	< 0.1	< 0.1	0.2	<0.071	<0.071	<0.071	<0.071	<0.071	<0.071	-	-	-	-	-
Methyl t-butyl ether (MTBE)	8000	HH	20000	HH	20000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	<0.20	<0.20	<0.20	<0.20	<0.20	<0.20	-	-	-	-	-
VPH	200	HH/EH	**200**	EH/HH	200	EH/HH	-	-	-	-	-	-	-	-	-	< 100	-	118	< 100	**546**	**956**	< 100	<100	<100	<100	<100	<100	<100	-	-	-	-	-
VH6-10							-	-	-	-	-	-	-	-	-	< 100	-	118	< 100	546	843	< 100	<100	<100	<100	<100	<100	<100	-	-	-	-	-
Polychlorinated Biphenyls																																
Arochlor 1242							-	< 0.03	< 0.03	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	< 0.03	< 0.03	< 0.03
Arochlor 1248							-	< 0.03	< 0.03	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	< 0.03	< 0.03	< 0.03
Arochlor 1254							-	< 0.03	< 0.03	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	< 0.03	< 0.03	< 0.03
Arochlor 1260							-	< 0.03	< 0.03	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	< 0.03	< 0.03	0.45
polychlorinated biphenyls (PCB-total)	1.5	T	35	I/T	35	I/T	-	< 0.03	< 0.03	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	< 0.03	< 0.03	0.45

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial).
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; AW = Groundwater Flow to Surface Water used by Aquatic Life, M = Marine; DW = Drinking Water
EH = generic standard for the protection of ecological health; HH = generic standard for the protection of human health
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including
amendments up to BC Reg. 13/2019, updated to 16 April 2019)
MCS = most conservative standard based on applicable site-specific standards
QA/QC = quality assurance/quality control
a. LEPH and HEPH criteria used as a conservative screen for EPH10-19 and EPH19-32, respectively
FDA = Field duplicate available; FD = Field duplicate.
italics = Laboratory Method Detection Limit above applicable standards
< = less than laboratory reporting limit; - = not analyzed
EPH_{C10-19} = extractable petroleum hydrocarbons, carbon range 10-19
EPH_{C19-32} = extractable petroleum hydrocarbons, carbon range 19-32
LEPH = light extractable petroleum hydrocarbons
HEPH = heavy extractable petroleum hydrocarbons
VPH = volatile petroleum hydrocarbons
VH (C6-C10) = volatile hydrocarbons, carbon range 6-10
PAH = polycyclic aromatic hydrocarbon
ppm = parts per milion

Table I-1b
Results of Soil Analyses - Hydrocarbons and PCBs
Ladysmith Harbour, Ladysmith, BC

APEC							AEC 13			APEC 17			APEC 15		AEC 12															
Fill Samples															APEC 8		APEC 8	APEC 8	APEC 8	APEC 8	APEC 8							APEC 8			
Location	CSR	MCS	CSR	MCS	CSR	MCS	MW09-3	MW09-3	MW11-01	MW09-16	MW09-16	MW09-16	MW09-5	MW09-5	MW09-9	MW09-10	MW09-10	MW09-11	KE SA1	KE SA2	KE SA4	TP09-05	TP09-05	TP09-06	TP09-06	TP09-06	TP11-05	TP11-07	TP11-07	TP11-07	TP11-08
SCN	Standards for		Standards		Standards for		21360-07	21360-08	0526-03	21366-07	21366-09	21366-10	21361-09	21362-02	21364-02	21364-05	21364-07	21364-09	21370-01	21370-02	21370-04	21383-01	21383-02	21383-03	21383-05	21383-06	21383-07	0528-10	0529-03	0529-07	0529-10	0529-12
Depth (m bgs)	PL		for CL		IL		0.6 - 0.76	2.1 - 2.3	4 - 4.1	1.2 - 1.37	3.35 - 3.50	3.35 - 3.50	1.4 - 1.6	7.1-7.3	1.5 - 1.7	0.6 - 0.76	1.5 - 1.7	0.3 - 0.4	0.2 - 0.3	0.2 - 0.3	0.2 - 0.3	0.3-0.5	0.7-0.9	1.6-1.8	0.9-1.1	0.9-1.1	1.7-1.9	0.3-0.4	1.5-1.6	2.8-3.0	2.8-3.0	1.4-1.5
Date Sampled (mm/dd/yyyy)							11-10-09	11-10-09	08-02-11	11/14/2009	11/14/2009	11/14/2009	11-11-09	11-11-09	11/13/2009	11/13/2009	11/13/2009	11/13/2009	11/25/2009	11/25/2009	11/25/2009	15-12-09	15-12-09	15-12-09	15-12-09	15-12-09	15-12-09	09-02-11	09-02-11	09-02-11	09-02-11	09-02-11
QA/QC											FDA	FD													FDA	FD			FDA	FD	
Field Parameters																															
Soil Vapours (ppm unless otherwise indicated)							10	23	0.4	16	26	26	15	23	16	10	22	5	-	-	-	20	260	35	410	410	-	0	165.2	35	0.1	0
Physical Parameters																															
moisture (%)							10.6	8.4	7.88	10.3	18.7	18.1		12.9	12.6	8.5	15.4	21.2	9	10.8	6.9	-	22.4	17.9	19.5	19.8	19.2	15.5	22.8	8.71	10.5	16.3
pH (pH units)							7.3	-	-	6.3	7.3	7.5	6.2	-	8.1	7.6	7.7	6.6	6.2	6	6.1	7.9	-	-	-	-	-	-	-	-	-	-
Extractable Hydrocarbons																															
EPH_{C10-19} [a]	*1000*		***2,000***		*2,000*		< 250	< 250	<200	< 250	< 250	< 250	-	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	-	< 250	< 250	***5,100***	***6,600***	1,000	<200	***4370***	<200	<200	<200
EPH_{C10-19} (sg)							-	-	<200	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
EPH_{C19-32} [a]	*1000*		***5000***		*5,000*		< 250	< 250	<200	< 250	< 250	< 250	-	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	-	< 250	< 250	*1500*	*2000*	370	<200	*1650*	<200	<200	650
EPH_{C19-32} (sg)							-	-	<200	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
LEPH	1000	EH/HH	**2,000**	EH/HH	2,000	EH/HH	< 250	< 250	<200	< 250	< 250	< 250	-	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	-	< 250	< 250	**5,100**	**6,600**	1000	<200	**4,370**	<200	<200	<200
HEPH	1000	EH/HH	**5000**	EH/HH	5,000	EH/HH	< 250	< 250	<200	< 250	< 250	< 250	-	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	-	< 250	< 250	1,500	2,000	370	<200	1,650	<200	<200	650
Polycyclic Aromatic Hydrocarbons																															
2-methylnaphthalene	100	HH	950	HH	950	HH	< 0.05	< 0.05	0.093	< 0.05	< 0.05	< 0.05	-	0.2	< 0.05	0.28	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	< 0.05	0.154	<0.050	<0.050	<0.050	<0.50
acenaphthene	2000	HH	15000	HH	15000	HH	< 0.05	< 0.05	<0.050	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	0.57	< 0.05	<0.050	<0.46	<0.050	<0.050	<0.50
acenaphthylene							< 0.05	< 0.05	<0.050	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	< 0.05	<0.050	<0.16	<0.050	<0.050	<0.50
anthracene	2.5	T	30	T	30	T	< 0.05	< 0.05	<0.050	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	< 0.05	<0.050	<0.13	<0.050	<0.050	<0.50
benzo(a)anthracene	1	EH	10	EH	10	EH	< 0.05	< 0.05	0.107	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.50
benzo(a)pyrene	10	I	30	I	50	I	< 0.05	< 0.05	0.107	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.50
benzo(b)fluoranthene							< 0.05	< 0.05	0.122	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.50
benzo(g,h,i)perylene							< 0.05	< 0.05	0.072	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	0.06	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.50
benzo(k)fluoranthene	1	EH	10	EH	10	EH	< 0.05	< 0.05	<0.050	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.50
chrysene	400	HH	4500	HH	4500	HH	< 0.05	< 0.05	0.116	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	< 0.05	<0.050	<0.056	<0.050	<0.050	<0.50
dibenz(a,h)anthracene	1	EH	10	EH	10	EH	< 0.05	< 0.05	<0.050	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.50
fluoranthene	50	T	200	T	200	T	< 0.05	< 0.05	0.283	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	0.06	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	0.19	<0.050	<0.073	<0.050	<0.050	<0.50
fluorene	1000	HH	9500	HH	9500	HH	< 0.05	< 0.05	<0.050	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	0.6	1.1	< 0.05	<0.050	0.898	<0.050	<0.050	<0.50
indeno(1,2,3-c,d)pyrene	1	EH	10	EH	10	EH	< 0.05	< 0.05	0.074	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	< 0.05	<0.050	<0.050	<0.050	<0.050	<0.50
naphthalene	0.6	T	20	T	20	T	< 0.05	< 0.05	0.051	< 0.05	< 0.05	< 0.05	-	0.1	< 0.05	0.14	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	0.08	0.084	<0.54	<0.050	<0.050	<0.50
phenanthrene	5	EH	50	EH	50	EH	< 0.05	0.05	0.156	< 0.05	< 0.05	< 0.05	-	0.11	< 0.05	0.3	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	0.15	0.090	<0.16	<0.050	<0.050	<0.50
pyrene	10	EH	100	EH	100	EH	< 0.05	< 0.05	0.263	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	0.07	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.5	< 0.5	0.15	<0.050	0.118	<0.050	<0.050	<0.50
Non-Halogenated Volatiles																															
benzene	0.035	DW	**0.035**	DW	6.5	DW	*< 0.04*	*< 0.04*	*<0.040*	-	-	-	-	**0.12**	< 0.03	-	< 0.03	-	< 0.03	**0.06**	< 0.03	*< 0.04*	*< 0.04*	-	*< 0.04*	-	-	*<0.040*	*<0.040*	*<0.040*	*<0.040*	*<0.040*
ethylbenzene	15	DW	**15**	DW	15	DW	< 0.5	< 0.5	<0.050	-	-	-	-	0.07	< 0.03	-	< 0.03	-	< 0.03	0.45	< 0.03	**17**	< 0.5	-	< 0.5	-	-	<0.050	<0.050	<0.050	<0.050	<0.050
styrene	5	EH	50	EH	50	EH	< 0.1	< 0.1	<0.050	-	-	-	-	< 0.03	< 0.03	-	< 0.03	-	0.07	29	0.24	< 0.1	< 0.1	-	< 0.1	-	-	<0.050	<0.050	<0.050	<0.050	<0.050
toluene	6	DW	6	DW	6	DW	< 0.5	< 0.5	<0.050	-	-	-	-	0.24	< 0.03	-	< 0.03	-	< 0.03	0.17	< 0.03	< 0.5	< 0.5	-	< 0.5	-	-	<0.050	<0.050	<0.050	<0.050	<0.050
ortho-Xylene							-	-	<0.050	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	<0.050	<0.050	<0.050	<0.050	<0.050
meta- & para-Xylene							-	-	<0.050	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	<0.050	<0.050	<0.050	<0.050	0.052
Total xylene	6.5	DW	**6.5**	DW	6.5	DW	< 0.1	< 0.1	<0.071	-	-	-	-	0.3	< 0.03	-	< 0.03	-	< 0.03	2.9	0.05	**96**	0.3	-	< 0.1	-	-	<0.071	<0.071	<0.071	<0.071	<0.071
Methyl t-butyl ether (MTBE)	8000	HH	20000	HH	20000	HH	-	-	<0.20	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	<0.20	<0.20	<0.20	<0.20	<0.20
VPH	200	HH/EH	**200**	EH/HH	200	EH/HH	< 100	< 100	<100	-	-	-	-	< 100	< 100	-	< 100	-	< 100	100	< 100	-	< 100	-	-	-	-	<100	<100	<100	<100	<100
VH6-10							< 100	< 100	<100	-	-	-	-	< 100	< 100	-	< 100	-	< 100	< 100	< 100	-	< 100	-	-	-	-	<100	<100	<100	<100	<100
Polychlorinated Biphenyls																															
Arochlor 1242							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Arochlor 1248							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Arochlor 1254							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Arochlor 1260							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
polychlorinated biphenyls (PCB-total)	1.5	T	35	I/T	35	I/T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial).
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; AW = Groundw
EH = generic standard for the protection of ecological health; HH = generic standard for the protection of human health
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019)
MCS = most conservative standard based on applicable site-specific standards
QA/QC = quality assurance/quality control
a. LEPH and HEPH criteria used as a conservative screen for EPH10-19 and EPH19-32, respectively
FDA = Field duplicate available; FD = Field duplicate.
italics = Laboratory Method Detection Limit above applicable standards
< = less than laboratory reporting limit; - = not analyzed
EPH_{C10-19} = extractable petroleum hydrocarbons, carbon range 10-19
EPH_{C19-32} = extractable petroleum hydrocarbons, carbon range 19-32
LEPH = light extractable petroleum hydrocarbons
HEPH = heavy extractable petroleum hydrocarbons
VPH = volatile petroleum hydrocarbons
VH (C6-C10) = volatile hydrocarbons, carbon range 6-10
PAH = polycyclic aromatic hydrocarbon
ppm = parts per milion

Table I-1b
Results of Soil Analyses - Hydrocarbons and PCBs
Ladysmith Harbour, Ladysmith, BC

APEC							APEC 15	APEC 19 and 20									
Fill Samples																	
Location							MW09-5	MW09-7	MW09-7	MW09-7	MW09-8	MW09-8	MW09-8	BH09-13	BH09-14	BH09-15	BH09-15
SCN	CSR Standards for PL	**MCS**	**CSR Standards for CL**	**MCS**	CSR Standards for IL	**MCS**	21361-08	21362-11	21363-02	21363-04	21363-07	21363-08	21363-10	21365-05	21365-11	21366-03	21366-04
Depth (m bgs)							0.6 - 0.8	0.5 - 0.7	2.1 - 2.3	3.9 - 4.1	2.9 - 3.0	2.9 - 3.0	3.9 - 5.0	3.8 - 4.0	4.1 - 4.3	3.65 - 3.8	3.65 - 3.8
Date Sampled (mm/dd/yyyy)							11-11-09	11-12-09	11-12-09	11-12-09	11/13/2009	11/13/2009	11/13/2009	11/14/2009	11/14/2009	11/14/2009	11/14/2009
QA/QC											FDA	FD				FDA	FD
Field Parameters																	
Soil Vapours (ppm unless otherwise indicated)							30	5	10	5	440	440	60	460	210	340	340
Physical Parameters																	
moisture (%)							16.8	-	16.3	15.7	20	24.3	11.4	19.3	9.5	14.5	11.7
pH (pH units)							5.9	6.3	-	-	8	7.9	-	7.4	8.1	-	-
Extractable Hydrocarbons																	
EPH_{C10-19} [a]	*1000*		***2,000***		*2,000*		< 250	-	< 250	< 250	*1,900*	***7,500***	< 250	***5,200***	***3,500***	***2,100***	***2,500***
EPH_{C10-19} (sg)							-	-	-	-	-	-	-	-	-	-	-
EPH_{C19-32} [a]	*1000*		***5000***		*5,000*		< 250	-	330	< 250	*2,000*	***7,700***	< 250	***9,800***	*1,800*	*3,100*	*3,300*
EPH_{C19-32} (sg)							-	-	-	-	-	-	-	-	-	-	-
LEPH	1000	EH/HH	**2,000**	EH/HH	2,000	EH/HH	< 250	-	< 250	< 250	1,900	**7,500**	< 250	**5,200**	**3,500**	**2,100**	**2,500**
HEPH	1000	EH/HH	**5000**	EH/HH	5,000	EH/HH	< 250	-	330	< 250	2,000	**7,700**	< 250	**9,800**	1,800	3,100	3,300
Polycyclic Aromatic Hydrocarbons																	
2-methylnaphthalene	100	HH	950	HH	950	HH	0.2	-	0.22	0.09	0.74	0.66	< 0.05	1.5	< 0.05	< 0.05	< 0.05
acenaphthene	2000	HH	15000	HH	15000	HH	0.82	-	< 0.05	< 0.05	3	1.7	< 0.05	1.7	< 0.05	< 0.05	< 0.05
acenaphthylene							< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
anthracene	2.5	T	30	T	30	T	1.5	-	0.06	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
benzo(a)anthracene	1	EH	10	EH	10	EH	0.4	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
benzo(a)pyrene	10	I	30	I	50	I	0.13	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
benzo(b)fluoranthene							0.19	-	0.1	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
benzo(g,h,i)perylene							< 0.05	-	0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
benzo(k)fluoranthene	1	EH	10	EH	10	EH	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
chrysene	400	HH	4500	HH	4500	HH	0.3	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
dibenz(a,h)anthracene	1	EH	10	EH	10	EH	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
fluoranthene	50	T	200	T	200	T	1.9	-	0.07	< 0.05	< 0.05	0.84	< 0.05	0.71	< 0.05	< 0.05	< 0.05
fluorene	1000	HH	9500	HH	9500	HH	0.87	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	1.3	< 0.05	< 0.05	< 0.05
indeno(1,2,3-c,d)pyrene	1	EH	10	EH	10	EH	< 0.05	-	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
naphthalene	0.6	T	20	T	20	T	0.42	-	0.15	0.09	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05
phenanthrene	5	EH	50	EH	50	EH	4	-	0.27	< 0.05	< 0.05	< 0.05	0.05	4.2	< 0.05	< 0.05	< 0.05
pyrene	10	EH	100	EH	100	EH	1.2	-	0.07	< 0.05	5.2	3	0.09	3	0.05	0.77	0.66
Non-Halogenated Volatiles																	
benzene	0.035	DW	**0.035**	DW	6.5	DW	< 0.03	-	*< 0.04*	*< 0.04*	*< 0.04*	*< 0.04*	*< 0.04*	**0.13**	*< 0.04*	*< 0.04*	*< 0.04*
ethylbenzene	15	DW	**15**	DW	15	DW	< 0.03	-	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5
styrene	5	EH	50	EH	50	EH	< 0.03	-	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
toluene	6	DW	6	DW	6	DW	< 0.03	-	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5
ortho-Xylene							-	-	-	-	-	-	-	-	-	-	-
meta- & para-Xylene							-	-	-	-	-	-	-	-	-	-	-
Total xylene	6.5	DW	**6.5**	DW	6.5	DW	< 0.03	-	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	0.3	< 0.1	< 0.1	< 0.1
Methyl t-butyl ether (MTBE)	8000	HH	20000	HH	20000	HH	-	-	-	-	-	-	-	-	-	-	-
VPH	200	HH/EH	**200**	EH/HH	200	EH/HH	< 100	-	< 100	< 100	**250**	140	< 100	< 100	< 100	< 100	< 100
VH6-10							< 100	-	< 100	< 100	250	140	< 100	< 100	< 100	< 100	< 100
Polychlorinated Biphenyls																	
Arochlor 1242							-	-	-	-	-	-	-	-	-	-	-
Arochlor 1248							-	-	-	-	-	-	-	-	-	-	-
Arochlor 1254							-	-	-	-	-	-	-	-	-	-	-
Arochlor 1260							-	-	-	-	-	-	-	-	-	-	-
polychlorinated biphenyls (PCB-total)	1.5	T	35	I/T	35	I/T	-	-	-	-	-	-	-	-	-	-	-

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a. LEPH and HEPH criteria used as a conservative screen for EPH10-19 and EPH19-32, respectively
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EPH_{C19-32} = extractable petroleum hydrocarbons, carbon range 19-32
LEPH = light extractable petroleum hydrocarbons
HEPH = heavy extractable petroleum hydrocarbons
VPH = volatile petroleum hydrocarbons
VH (C6-C10) = volatile hydrocarbons, carbon range 6-10
PAH = polycyclic aromatic hydrocarbon
ppm = parts per milion

Table I-1c
Results of Soil Analyses - Volatile Organic Compounds
Ladysmith Harbour, Ladysmith, BC

Location							MW09-5	MW09-5	MW09-9	MW09-10	KE SA1	KE SA2	KE SA4
SCN	CSR Standards for PL	MCS	**CSR Standards for CL**	MCS	CSR Standards for IL	MCS	21361-08	21362-02	21364-02	21364-07	21370-01	21370-02	21370-04
Depth (m bgs)							0.6 - 0.8	7.1-7.3	1.5 - 1.7	1.5 - 1.7	0.2 - 0.3	0.2 - 0.3	0.2 - 0.3
Date Sampled (mm/dd/yyyy)							2009-11-11	2009-11-11	11/13/2009	11/13/2009	11/25/2009	11/25/2009	11/25/2009
QA/QC													
Field Parameters													
Soil Vapours (ppm)							30	23	16	22	-	-	-
Physical Parameters													
moisture (%)							16.8	12.9	12.6	15.4	9	10.8	6.9
pH (pH units)							5.9	-	8.1	7.7	6.2	6	6.1
Halogenated Hydrocarbons													
bromodichloromethane (BDCM)	200	HH	550	HH	550	HH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
bromoform (tribromomethane)	650	HH	4,000	HH	4,000	HH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
bromomethane (methyl bromide)	45	HH	300	HH	300	HH	< 0.12	< 0.12	< 0.12	< 0.12	< 0.12	< 0.12	< 0.12
methyl ethyl ketone (2-Butanone)	20000	HH	150000	HH	150000	HH	<1.5	< 1.5	< 1.5	< 1.5	< 1.5	< 1.5	< 1.5
carbon tetrachloride	5	EH	50	EH	50	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
chlorobenzene	1	EH	10	EH	10	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
chloroethane							< 0.06	< 0.06	< 0.06	< 0.06	< 0.06	< 0.06	< 0.06
chloroform	5	EH	50	EH	50	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
chloromethane (methyl chloride)							< 0.12	< 0.12	< 0.12	< 0.12	< 0.12	< 0.12	< 0.12
dibromochloromethane (DBCM)	150	HH	400	HH	400	HH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
1,2-dibromoethane (ethylene dibromide) (EDB)	7	HH	15	HH	15	HH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
dibromomethane (methylene bromide)							< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
1,2-dichlorobenzene	1	EH	10	EH	10	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
1,3-dichlorobenzene	1	EH	10	EH	10	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
1,4-dichlorobenzene	1	EH	10	EH	10	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
dichlorodifluoromethane (freon 12)	6500	HH	45000	HH	45000	HH	< 0.06	< 0.06	< 0.06	< 0.06	< 0.06	< 0.06	< 0.06
1,1-dichloroethane	5	EH	50	EH	50	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
1,2-dichloroethane	5	EH	50	EH	50	EH	< 0.06	< 0.06	< 0.06	< 0.06	< 0.06	< 0.06	< 0.06
1,1-dichloroethene	5	EH	50	EH	50	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
1,2-dichloroethylene (cis) (1,2-dichloroethene) (cis)	5	EH	50	EH	50	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
1,2-dichloroethylene (trans) (1,2-dichloroethene) (trans)	5	EH	50	EH	50	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
1,2-dichloropropane (propylene dichloride)	5	EH	50	EH	50	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
1,3-dichloropropene (cis)	5	EH	50	EH	50	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
1,3-dichloropropene (trans)	5	EH	50	EH	50	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
2-Hexanone	150	HH	1000	HH	1000	HH	<1.5	< 1.5	<1.5	<1.5	< 1.5	< 1.5	< 1.5
methyl isobutyl ketone (4-Methyl-2-pentanone)							<0.6	< 0.6	<0.6	<0.6	< 0.6	< 0.6	< 0.6
dichloromethane (methylene chloride)	5	EH	50	EH	50	EH	< 0.1	< 0.1	<0.1	<0.1	< 0.1	< 0.1	< 0.1
1,1,2,2-tetrachloroethane	70	HH	150	HH	150	HH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
tetrachloroethylene (PERC)	2.5	AW	2.5	AW	2.5	AW	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
1,1,1-trichloroethane	5	EH	50	EH	50	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
1,1,2-trichloroethane	5	EH	50	EH	50	EH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
trichloroethylene (TCE)	0.3	AW	0.3	AW	0.3	AW	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01
trichlorofluoromethane (freon 11)	9000	HH	70000	HH	70,000	HH	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03	< 0.03
vinyl chloride (chloroethene)	2	HH	45	HH	45	HH	< 0.06	< 0.06	< 0.06	< 0.06	< 0.06	< 0.06	< 0.06

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Standards shown from the Contaminated Sites Regulation (“CSR”; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019)
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial)
MCS: most conservative standard based on applicable site-specific standards
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants;
AW = Groundwater Flow to Surface Water used by Aquatic Life, F = Fresh Water Aquatic Life and M = Marine Aquatic Life;
FDA = field duplicate available
FD = field duplicate
QA/QC = quality assurance/quality control
NS = No Standard
< = less than laboratory reporting limit; - = not analyzed

TABLE 2a
Historic Soil Sampling Results - Total Metals
Ladysmith Harbour, Ladysmith, BC

Location	CSR Standards for PL	MCS	**CSR Standards for CL**	MCS	CSR Standards for IL	MCS	BH05-1	BH05-1	MW05-2	MW05-2	MW05-3	MW05-3	MW05-4	MW05-4	MW05-5	MW05-5	MW05-6	MW05-6	MW5-6	MW05-7	MW05-7	MW05-7	MW05-7	MW05-7	MW05-8	MW05-9
SCN							0019-08	0020-03	0020-05	0020-08	0021-01	0021-03	0022-01	0022-02	0022-10	0023-06	0023-08	0023-10	0023-12	0024-04	0024-06	0024-09	0025-04	0025-09	0026-01	0026-07
Depth (m)							6.9	13.7	0.15	1.8	0.15	0.9	1.7	2.7	2.7	9.4	0.15	0.9	1.8	0.15	0.9	2.7	9.4	14.6	1	0.47
Date Sampled							28-Mar-05	28-Mar-05	28-Mar-05	28-Mar-05	28-Mar-05	28-Mar-05	29-Mar-05	29-Mar-05	29-Mar-05	29-Mar-05	29-Mar-05	29-Mar-05	29-Mar-05	29-Mar-05	29-Mar-05	29-Mar-05	29-Mar-05	29-Mar-05	30-Mar-05	30-Mar-05
Source							Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder FDA	Golder	Golder	Golder FDA	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder
QA/QC																										
Physical																										
pH							8.65	8.98	7.76	8.13	8.01	8.47	9.07	8.64	8.40	9.11	7.96	7.50	8.75	6.56	6.70	8.37	8.93	8.72	8.74	6.72
Total Metals																										
*Aluminum	55000	HH	250000	HH	250000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Antimony	20	EH	**40**	EH	40	EH	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10
Arsenic	10	AW-M/DW	**10**	AW-M	10	AW-M/DW	<5.0	<5.0	5.1	9.3	5.4	5.8	<5.0	<5.0	<5.0	<5.0	7.6	<5.0	<5.0	**16.3**	<5.0	<5.0	<5.0	**24.5**	<5.0	**14.2**
Barium	350	DW	**350**	DW	350	DW	213	175	177	226	144	198	124	192	191	190	**713**	52.2	162	79.7	52.7	131	147	213	205	61.2
Beryllium[pH]	1-150	AW-M	1-350	DW/T	1-350	DW	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50
Bismuth							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Cadmium[pH]	1-30	AW-M	**1-70**	DW/AW-M	1-70	AW-M/DW	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	1.30	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50
Calcium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
*Chromium (total)	65	AW-M/DW	**65**	DW/AW-M	65	AW-M/DW	9.1	5.4	37.6	39.9	31.5	35.9	19.1	11.1	16.5	7.0	40.7	14.2	27.1	16.0	11.1	21.8	24.4	14.2	26.1	13.0
*Cobalt	30	AW-M/DW	30	DW/AW-M	30	AW-M/DW	4.7	3.4	9.0	10.1	8.4	9.8	6.1	4.4	5.6	4.5	10.5	3.8	6.1	7.3	3.5	6.2	6.0	4.6	7.7	4.2
*Copper[pH]	100-150	AW-M/DW	**100-300**	AW-M/T	100-300	AW-M	56.4	33.4	115	79.2	84.0	80.0	76.9	53.4	74.4	45.2	226	31.9	71.3	78.4	32.6	68.8	63.1	52.6	68.8	62.7
*Iron	70000	HH	150000	HH	150000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Lead[pH]	120	I	**120-150**	AW-M/DW	120-1000	DW	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	**677**	<50	<50	<50	<50	<50	<50	<50	<50	<50
Magnesium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
*Manganese	5000	DW/T	5000	DW/T	5000	DW/T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Mercury	25	I	75	I/T	75	T	0.153	0.169	0.288	0.291	0.338	0.394	0.229	0.210	0.220	0.170	0.107	0.081	0.188	0.328	0.062	0.623	0.576	0.852	0.294	0.169
Molybdenum	15	DW	**15**	DW	15	DW	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	4.3	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0
Nickel[pH]	70-150	AW-M/DW	**70-250**	AW-M/DW	70-250	AW-M/DW	17.2	10.3	56.5	63.0	45.4	57.5	26.9	18.0	25.5	14.0	33.5	16.1	38.6	14.1	11.0	31.5	38.4	24.0	38.5	12.4
Phosphorus							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Potassium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
*Selenium	4	AW-M/DW	4	AW-M/DW	4	AW-M/DW	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0
Silver	20	EH	40	EH	40	EH	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0
Sodium[I]	200	T	**1000**	T	1000	T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Strontium	20000	HH	150000	HH	150000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Tellurium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Thallium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Tin	50	EH	300	EH	300	EH	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	23.6	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0
Titanium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
*Vanadium	200	DW	200	DW	200	DW	33.5	25.1	45.0	50.9	40.9	41.1	30.6	26.7	28.3	28.0	63.6	24.1	41.5	36.5	22.4	40.7	37.8	35.0	34.2	25.9
Zinc[pH]	150-200	AW-M	**150-200**	AW-M	150-200	AW-M	34.2	27.5	156	68.6	81.6	81.7	58.4	41.6	51.6	32.5	**1240**	79.8	77.9	94.5	94.4	51.1	47.4	34.2	66.2	102
Zirconium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Inorganic Parameters																										
Elemental Sulphur (%)							2.32	3.54	-	-	-	-	-	-	0.520	2.31	-	-	-	-	-	0.630	0.800	2.08	-	-

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019).
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial)
MCS: most conservative standard based on applicable site-specific standards
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; DW Drinking Water; AW = Groundwater Flow to Surface Water used by Aquatic Life, and M = Marine Aquatic Life.
EH = generic standard for the protection of ecological health
HH = generic standard for the protection of human health
pH = standard is pH dependant
FDA = field duplicate available, FD = field duplicate
< = less than laboratory reporting limit; - = not analyzeditalics = Laboratory Method Detection Limit above applicable standards
QA/QC = quality assurance/quality control
italics = Laboratory Method Detection Limit above applicable standards
*Background concentrations applied as detailed in Protocol 4 is applied to replace CSR Standards.
Golder = Golder Associates Ltd. 2005. Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site
Hardy = Hardy BBT Limited. 1990. Environmental Review for Town of Ladysmith Waterfront Development Slag Point Area,
EBA = EBA Environmental Ltd. 1994. Elk Falles Forest Industries Limited Phase II Environmental Assessment Ladysmith
Levelton[a] = Levelton Engineering Ltd. 2000a. Preliminary Site Investigation Stage 1 and Stage II Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District. Ladysmith, B.C. Dated August 25, 2000.

Levelton[b] = Levelton Engineering Ltd. 2000b. Detailed Site Investigation Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District. Ladysmith, B.C. Dated October 25, 2000.

TABLE 2a
Historic Soil Sampling Results - Total Metals
Ladysmith Harbour, Ladysmith, BC

Location	CSR Standards for PL	MCS	CSR Standards for CL	MCS	CSR Standards for IL	MCS	BH05-10	BH05-10	BH05-10	BH05-10	BH05-13	BH05-13	BH05-13	BH05-15	BH05-15	MW05-17	MW05-17	MW05-18	MW05-18	MW05-18	MW05-18	MW05-19	MW05-19	MW05-19	BH05-21	MW05-22
SCN							0027-04	0027-07	0028-02	0028-06	8721-06	8721-11	8715-06	33753-07	33754-08	0029-01	0029-12	0030-05	0030-07	0031-07	0031-10	0131-02	0131-09	0131-10	0130-10	0132-05
Depth (m)							0.9	3.7	9.6	14	2.8	7.1	15.5	2	13.1	1.7	12.7	0.47	1.7	8.4	12.1	0.47	6	7.5	3.1	0.15
Date Sampled							30-Mar-05	30-Mar-05	30-Mar-05	30-Mar-05	31-Mar-05	31-Mar-05	31-Mar-05	31-Mar-05	31-Mar-05	01-Apr-05	01-Apr-05	01-Apr-05	01-Apr-05	04-Apr-05	04-Apr-05	11-Apr-05	11-Apr-05	11-Apr-05	11-Apr-05	11-Apr-05
Source							Golder	Golder	Golder	Golder	Golder FDA	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder
QA/QC																										
Physical																										
pH							6.91	8.25	8.59	8.72	8.10	8.68	8.82	8.04	8.67	8.05	8.78	7.38	6.31	7.79	7.88	7.86	7.17	8.66	8.20	8.10
Total Metals																										
*Aluminum	55000	HH	250000	HH	250000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Antimony	20	EH	**40**	EH	40	EH	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10
Arsenic	10	AW-M/DW	**10**	AW-M	10	AW-M/DW	<5.0	<5.0	**21.8**	<5.0	<5.0	<5.0	<5.0	5.5	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	6.5	<5.0	<5.0	<5.0	6.6
Barium	350	DW	**350**	DW	350	DW	35.4	148	225	185	139	150	222	177	216	201	235	98.1	123	213	235	193	29.0	280	192	199
Beryllium[pH]	1-150	AW-M	1-350	DW/T	1-350	DW	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	0.51	<0.50	<0.50
Bismuth							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Cadmium[pH]	1-30	AW-M	**1-70**	DW/AW-M	1-70	AW-M/DW	**1.21**	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50
Calcium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
*Chromium (total)	65	AW-M/DW	**65**	DW/AW-M	65	AW-M/DW	16.1	29.0	12.9	7.2	30.8	11.3	6.1	30.9	7.4	33.7	7.5	29.7	22.6	8.0	11.1	33.6	2.6	27.0	18.7	21.3
*Cobalt	30	AW-M/DW	30	DW/AW-M	30	AW-M/DW	4.4	6.3	4.4	3.9	7.3	4.7	3.8	9.4	5.0	9.6	4.1	10.7	6.5	3.7	4.2	9.0	<2.0	7.9	7.3	7.8
*Copper[pH]	100-150	AW-M/DW	**100-300**	AW-M/T	100-300	AW-M	55.4	79.5	51.3	36.0	69.5	53.6	41.9	72.9	34.5	111	38.5	74.6	60.3	40.8	56.8	86.5	9.7	57.3	67.8	69.5
*Iron	70000	HH	150000	HH	150000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Lead[pH]	120	I	**120-150**	AW-M/DW	120-1000	DW	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50	<50
Magnesium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
*Manganese	5000	DW/T	5000	DW/T	5000	DW/T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Mercury	25	I	75	I/T	75	T	0.397	0.400	0.761	0.939	0.427	0.366	0.252	0.625	0.372	0.489	0.194	0.061	0.131	0.256	0.183	0.406	0.091	0.354	0.216	0.413
Molybdenum	15	DW	**15**	DW	15	DW	8.5	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	5.1	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0
Nickel[pH]	70-150	AW-M/DW	**70-250**	AW-M/DW	70-250	AW-M/DW	14.5	41.6	21.2	11.9	48.9	18.5	11.7	54.2	14.4	55.2	14.8	19.9	26.4	13.6	18.4	49.7	<5.0	27.0	28.6	31.0
Phosphorus							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Potassium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
*Selenium	4	AW-M/DW	4	AW-M/DW	4	AW-M/DW	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0
Silver	20	EH	40	EH	40	EH	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0
Sodium[i]	200	T	**1000**	T	1000	T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Strontium	20000	HH	150000	HH	150000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Tellurium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Thallium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Tin	50	EH	300	EH	300	EH	6.7	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0
Titanium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
*Vanadium	200	DW	200	DW	200	DW	37.9	40.2	35.3	35.5	38.3	40.7	30.7	40.0	35.2	39.1	30.4	65.5	33.9	28.6	29.1	44.9	5.3	43.2	31.0	43.0
Zinc[pH]	150-200	AW-M	**150-200**	AW-M	150-200	AW-M	147	69.3	33.6	42.3	61.1	80.5	24.9	73.0	35.1	78.8	26.4	68.1	**516**	29.8	37.4	78.3	10.3	46.7	59.7	65.6
Zirconium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Inorganic Parameters																										
Elemental Sulphur (%)							-	0.340	1.77	2.12	0.370	2.32	2.06	0.280	2.39	0.280	1.26	-	-	1.51	1.07	-	-	-	0.250	-

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amer
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial)
MCS: most conservative standard based on applicable site-specific standards
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; DW Drinking Wa
EH = generic standard for the protection of ecological health
HH = generic standard for the protection of human health
pH = standard is pH dependant
FDA = field duplicate available, FD = field duplicate
< = less than laboratory reporting limit; - = not analyzeditalics = Laboratory Method Detection Limit above applicable standards
QA/QC = quality assurance/quality control
italics = Laboratory Method Detection Limit above applicable standards
*Background concentrations applied as detailed in Protocol 4 is applied to replace CSR Standards.
Golder = Golder Associates Ltd. 2005. Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site
Hardy = Hardy BBT Limited. 1990. Environmental Review for Town of Ladysmith Waterfront Development Slag Point Area,
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TABLE 2a
Historic Soil Sampling Results - Total Metals
Ladysmith Harbour, Ladysmith, BC

Location	CSR Standards for PL	MCS	CSR Standards for CL	MCS	CSR Standards for IL	MCS	BH05-23	BH05-23	MW05-24	TP05-1	TP05-2	TP05-4	TP05-8	TP05-9	TP05-10	TP05-11	TP05-11	TP05-12	TP05-13	TP05-14	TP05-15	Stockpile A	Stockpile A	Stockpile A	Stockpile B	Stockpile B
SCN							0133-03	0134-06	0047-01	0032-06	0032-07	0033-04	0034-03	0034-05	0034-07	0034-09	0034-11	0035-02	0035-04	0035-06	0035-08	0094-01	0094-05	COMP 1-6	0094-09	0094-12
Depth (m)							0.47	14.9	0.15	0.67	0.9	1	0.4	0.6	0.6	0.3	1.6	2.5	0.4	0.5	1.2	n/a	n/a	n/a	n/a	n/a
Date Sampled							11-Apr-05	11-Apr-05	21-Apr-05	06-Apr-05	06-Apr-05	06-Apr-05	06-Apr-05	06-Apr-05	06-Apr-05	06-Apr-05	06-Apr-05	06-Apr-05	06-Apr-05	06-Apr-05	06-Apr-05	2005-05-18	2005-05-18	2005-05-18	2005-05-18	2005-05-18
Source							Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder FDA	Golder	Golder	Golder	Golder
QA/QC																										
Physical																										
pH							7.44	9.10	7.81	5.66	5.99	5.77	7.97	7.01	6.81	5.89	6.59	7.81	6.55	8.04	7.10	6.67	6.86	6.52	6.52	6.44
Total Metals																										
*Aluminum	55000	HH	250000	HH	250000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Antimony	20	EH	**40**	EH	40	EH	<10	<10	<10	<10	<10	<10	<10	**46**	34	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10
Arsenic	10	AW-M/DW	**10**	AW-M	10	AW-M/DW	<5.0	<5.0	<5.0	<5.0	**10.8**	<5.0	**14.2**	**15.8**	**13.5**	<5.0	<5.0	5.2	<5.0	<5.0	<5.0	6.3	5.8	5.5	6	5.9
Barium	350	DW	**350**	DW	350	DW	110	230	183	68.8	76.2	104	**2890**	**370**	**750**	61.1	32.6	156	42.6	184	296	143	89	89.4	90	86
Beryllium[pH]	1-150	AW-M	1-350	DW/T	1-350	DW	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	0.90	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50
Bismuth							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Cadmium[pH]	1-30	AW-M	**1-70**	DW/AW-M	1-70	AW-M/DW	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	1.48	2.92	**4.72**	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50
Calcium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
*Chromium (total)	65	AW-M/DW	**65**	DW/AW-M	65	AW-M/DW	29.4	8.4	34.6	16.5	26.3	21.4	38.6	**100**	**101**	14.2	11.9	29.2	15.0	25.3	29.4	22.7	21.5	22.1	28.6	29.9
*Cobalt	30	AW-M/DW	30	DW/AW-M	30	AW-M/DW	9.9	4.3	8.7	6.5	7.5	6.8	13.8	19.0	25.4	3.7	3.0	7.3	2.6	6.1	10.8	9	9.1	9	10.6	12.4
*Copper[pH]	100-150	AW-M/DW	**100-300**	AW-M/T	100-300	AW-M	59.5	43.0	75.8	57.1	64.2	75.8	**1000**	**4780**	**3540**	69.9	47.8	76.4	31.5	107	73.1	60.1	57.7	58.9	66.8	47.7
*Iron	70000	HH	150000	HH	150000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Lead[pH]	120	I	**120-150**	AW-M/DW	120-1000	DW	<50	<50	<50	<50	54	<50	**590**	**2130**	**2230**	<50	<50	<50	<50	<50	88	<50	<50	<50	<50	<50
Magnesium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
*Manganese	5000	DW/T	5000	DW/T	5000	DW/T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Mercury	25	I	75	I/T	75	T	0.065	0.283	0.301	0.144	0.123	0.147	<0.050	0.254	0.236	0.076	0.066	0.213	0.083	0.391	0.284	0.065	0.065	0.065	0.056	0.062
Molybdenum	15	DW	**15**	DW	15	DW	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	4.4	**26.0**	**26.4**	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0	<4.0
Nickel[pH]	70-150	AW-M/DW	**70-250**	AW-M/DW	70-250	AW-M/DW	21.8	14.7	51.8	16.7	22.4	19.5	59.8	**270**	**217**	14.1	10.5	45.5	8.6	36.6	25.7	17.8	17.6	18.1	21.8	20.6
Phosphorus							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Potassium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
*Selenium	4	AW-M/DW	4	AW-M/DW	4	AW-M/DW	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<3.0	<3.0	<3.0	<3.0
Silver	20	EH	40	EH	40	EH	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	2.9	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0	<2.0
Sodium[I]	200	T	**1000**	T	1000	T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Strontium	20000	HH	150000	HH	150000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Tellurium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Thallium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Tin	50	EH	300	EH	300	EH	<5.0	<5.0	<5.0	<5.0	<5.0	<5.0	44.6	172	223	5.5	<5.0	<5.0	<5.0	<5.0	13.1	<5.0	<5.0	<5.0	<5.0	<5.0
Titanium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
*Vanadium	200	DW	200	DW	200	DW	77.5	27.3	45.2	33.9	45.4	44.9	48.4	42.8	47.8	28.0	21.3	38.1	22.0	35.2	73.8	67	67.4	63.5	75.2	73.9
Zinc[pH]	150-200	AW-M	**150-200**	AW-M	150-200	AW-M	68.9	34.4	73.8	119	**551**	140	**5300**	**1820**	**2260**	143	**538**	148	81.9	62.6	131	59.3	60.2	57.5	45.1	53.8
Zirconium							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Inorganic Parameters																										
Elemental Sulphur (%)							-	1.91	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amen
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial)
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TABLE 2a
Historic Soil Sampling Results - Total Metals
Ladysmith Harbour, Ladysmith, BC

Location		CSR Standards for PL	MCS	**CSR Standards for CL**	MCS	CSR Standards for IL	MCS	Stockpile B	CPT 1	CPT 4	CPT 5	CPT 9	CPT 10	CPT 10	TP1	TP2	TP3	TP4	TP4	TP6	TP12	LOT 11G	1	2	3	5	6
SCN								COMP 7-12	-	-	-	-	-	-								009071-32					
Depth (m)								n/a	1.55	2.5	0.7	2.1	2	2	1	1	3	2	4	3	3	0.5	3	1.3	3	3	4.5
Date Sampled								2005-05-18	1990-09-18	1990-09-18	1990-09-18	1990-09-18	1990-09-18	1990-09-18	18-Jul-00	18-Jul-00	18-Jul-00	18-Jul-00	18-Jul-00	18-Jul-00	18-Jul-00	18-Jul-00	21-Sep-00	07-May-93	07-May-93	07-May-93	07-May-93	07-May-93
Source								Golder	Hardy	Hardy	Hardy	Hardy	Hardy FDA	Hardy FD	Levelton[a]	Levelton[a]	Levelton[a]	Levelton[a]	Levelton[a]	Levelton[a]	Levelton[a]	Levelton[a]	Levelton[b]	EBA	EBA	EBA	EBA	EBA
QA/QC																												
Physical																												
pH								6.53	4.87	7.2	6.52	8.37	7.77	-	6.2	6.5	-	6.8	6.3	8.0	-	6.4	-	-	-	-	-
Total Metals																												
*Aluminum		55000	HH	250000	HH	250000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	74700	65800	-	-	-
Antimony		20	EH	**40**	EH	40	EH	<10	-	-	-	-	-	-	<2	<2	-	3	<2	<2	-	<2	<1.5	<1.5	-	-	-
Arsenic		10	AW-M/DW	**10**	AW-M	10	AW-M/DW	5.5	-	*<25*	*<25*	**52**	**26**	**34.2**	2.4	2.9	-	7.4	2.8	4.1	-	5.4	6.5	**11**	-	-	-
Barium		350	DW	**350**	DW	350	DW	99.1	-	65.6	59.2	6.80	171	123	94	78	-	72	79	**508**	-	67.6	**367**	319	-	-	-
Beryllium[pH]		1-150	AW-M	1-350	DW/T	1-350	DW	<0.50	-	-	-	-	-	-	0.3	0.2	-	0.3	0.3	0.4	-	0.2	0.7	0.6	-	-	-
Bismuth								-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	<2	<2	-	-	-
Cadmium[pH]		1-30	AW-M	**1-70**	DW/AW-M	1-70	AW-M/DW	<0.50	-	<0.10	<0.10	0.17	1.06	0.6	<0.2	<0.2	-	<0.2	<0.2	0.4	-	<0.2	0.7	0.6	-	-	-
Calcium								-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	19400	10100	-	-	-
*Chromium (total)		65	AW-M/DW	**65**	DW/AW-M	65	AW-M/DW	28.7	-	26.8	28.1	49.1	**77.1**	59.8	20	16	-	34	40	24	-	**72**	31.9	25.1	-	-	-
*Cobalt		30	AW-M/DW	30	DW/AW-M	30	AW-M/DW	11.8	-	14.6	15.6	9.5	12.6	15.7	8.9	7.5	-	17	11	10	-	9.0	9.7	9.6	-	-	-
*Copper[pH]		100-150	AW-M/DW	**100-300**	AW-M/T	100-300	AW-M	84.4	-	58.9	55.9	86.4	147	**252**	51	63	-	45	40	90	-	58.4	73.9	77.9	-	-	-
*Iron		70000	HH	150000	HH	150000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	16200	17700	-	-	-
Lead[pH]		120	I	**120-150**	AW-M/DW	120-1000	DW	<50	28.0	2.8	2.7	<2.0	83.4	55.8	14	28	45	6	6	41	7	72	16	29	-	-	-
Magnesium								-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	6130	5900	-	-	-
*Manganese		5000	DW/T	5000	DW/T	5000	DW/T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	253	255	-	-	-
Mercury		25	I	75	I/T	75	T	0.054	-	0.058	0.08	0.239	0.334	0.341	<0.05	<0.05	-	0.06	0.05	0.07	-	0.08	0.44	0.7	-	-	-
Molybdenum		15	DW	**15**	DW	15	DW	<4.0	-	<5.0	<5.0	<5.0	<5.0	<4.0	<0.4	0.6	-	<0.4	0.8	1	-	3.4	<0.4	0.5	-	-	-
Nickel[pH]		70-150	AW-M/DW	**70-250**	AW-M/DW	70-250	AW-M/DW	22.6	-	20.1	25.5	44.4	113	108	19	16	-	<0.5	18	2.1	-	25.9	41.6	50.4	-	-	-
Phosphorus								-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	219	326	-	-	-
Potassium								-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	8820	7050	-	-	-
*Selenium		4	AW-M/DW	4	AW-M/DW	4	AW-M/DW	<3.0	-	0.07	0.1	0.08	0.16	0.11	<0.5	<0.5	-	<0.5	<0.5	<0.5	-	1.1	<1	<1	-	-	-
Silver		20	EH	40	EH	40	EH	<2.0	-	<2.0	<2.0	<2.0	<2.0	<2.0	<1	<1	-	<1	<1	<1	-	<1	<1	<1	-	-	-
Sodium[i]		200	T	**1000**	T	1000	T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	**3970**	822	-	-	-
Strontium		20000	HH	150000	HH	150000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	121	105	-	-	-
Tellurium								-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	3	<2	-	-	-
Thallium								-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	<2	<2	-	-	-
Tin		50	EH	300	EH	300	EH	<5.0	-	<30	<30	<30	<30	<30	<2	<2	-	<2	<2	<2	-	8	<2	<2	-	-	-
Titanium								-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1150	1030	-	-	-
*Vanadium		200	DW	200	DW	200	DW	78.6	-	-	-	-	-	-	45	41	-	82	91	70	-	41.8	101	97.5	-	-	-
Zinc[pH]		150-200	AW-M	**150-200**	AW-M	150-200	AW-M	57.6	-	46.4	40.1	91.3	**588**	**543**	59	119	-	31	28	94	-	103	65.2	78.6	-	-	-
Zirconium								-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	5.6	5.8	-	-	-
Inorganic Parameters																												
Elemental Sulphur (%)								-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2660	3030	1530	2900	2790

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amen
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial)
MCS: most conservative standard based on applicable site-specific standards
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; DW Drinking Wa
EH = generic standard for the protection of ecological health
HH = generic standard for the protection of human health
pH = standard is pH dependant
FDA = field duplicate available, FD = field duplicate
< = less than laboratory reporting limit; - = not analyzeditalics = Laboratory Method Detection Limit above applicable standards
QA/QC = quality assurance/quality control
italics = Laboratory Method Detection Limit above applicable standards
*Background concentrations applied as detailed in Protocol 4 is applied to replace CSR Standards.
Golder = Golder Associates Ltd. 2005. Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site
Hardy = Hardy BBT Limited. 1990. Environmental Review for Town of Ladysmith Waterfront Development Slag Point Area,
EBA = EBA Environmental Ltd. 1994. Elk Falles Forest Industries Limited Phase II Environmental Assessment Ladysmith
Levelton[a] = Levelton Engineering Ltd. 2000a. Preliminary Site Investigation Stage 1 and Stage II Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District. Ladysmith, B.C. Dated August 25, 2000.

Levelton[b] = Levelton Engineering Ltd. 2000b. Detailed Site Investigation Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District. Ladysmith, B.C. Dated October 25, 2000.

TABLE 2a
Historic Soil Sampling Results - Total Metals
Ladysmith Harbour, Ladysmith, BC

Location	CSR Standards for PL	MCS	**CSR Standards for CL**	MCS	CSR Standards for IL	MCS	7	10	11	12
SCN										
Depth (m)							3.5	2	0.8	0.7
Date Sampled							07-May-93	07-May-93	27-Jun-93	27-Jun-93
Source							EBA	EBA	EBA	EBA
QA/QC										
Physical										
pH							-	-	-	-
Total Metals										
*Aluminum	55000	HH	250000	HH	250000	HH	-	-	-	-
Antimony	20	EH	**40**	EH	40	EH	-	-	-	-
Arsenic	10	AW-M/DW	**10**	AW-M	10	AW-M/DW	-	-	-	-
Barium	350	DW	**350**	DW	350	DW	-	-	-	-
Beryllium[pH]	1-150	AW-M	1-350	DW/T	1-350	DW	-	-	-	-
Bismuth							-	-	-	-
Cadmium[pH]	1-30	AW-M	**1-70**	DW/AW-M	1-70	AW-M/DW	-	-	-	-
Calcium							-	-	-	-
*Chromium (total)	65	AW-M/DW	**65**	DW/AW-M	65	AW-M/DW	-	-	-	-
*Cobalt	30	AW-M/DW	30	DW/AW-M	30	AW-M/DW	-	-	-	-
*Copper[pH]	100-150	AW-M/DW	**100-300**	AW-M/T	100-300	AW-M	-	-	-	-
*Iron	70000	HH	150000	HH	150000	HH	-	-	-	-
Lead[pH]	120	I	**120-150**	AW-M/DW	120-1000	DW	-	-	-	-
Magnesium							-	-	-	-
*Manganese	5000	DW/T	5000	DW/T	5000	DW/T	-	-	-	-
Mercury	25	I	75	I/T	75	T	-	-	-	-
Molybdenum	15	DW	**15**	DW	15	DW	-	-	-	-
Nickel[pH]	70-150	AW-M/DW	**70-250**	AW-M/DW	70-250	AW-M/DW	-	-	-	-
Phosphorus							-	-	-	-
Potassium							-	-	-	-
*Selenium	4	AW-M/DW	4	AW-M/DW	4	AW-M/DW	-	-	-	-
Silver	20	EH	40	EH	40	EH	-	-	-	-
Sodium[I]	200	T	**1000**	T	1000	T	-	-	-	-
Strontium	20000	HH	150000	HH	150000	HH	-	-	-	-
Tellurium							-	-	-	-
Thallium							-	-	-	-
Tin	50	EH	300	EH	300	EH	-	-	-	-
Titanium							-	-	-	-
*Vanadium	200	DW	200	DW	200	DW	-	-	-	-
Zinc[pH]	150-200	AW-M	**150-200**	AW-M	150-200	AW-M	-	-	-	-
Zirconium							-	-	-	-
Inorganic Parameters										
Elemental Sulphur (%)							2870	2990	879	948

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amen
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial)
MCS: most conservative standard based on applicable site-specific standards
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; DW Drinking Wa
EH = generic standard for the protection of ecological health
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pH = standard is pH dependant
FDA = field duplicate available, FD = field duplicate
< = less than laboratory reporting limit; - = not analyzeditalics = Laboratory Method Detection Limit above applicable standards
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Levelton[a] = Levelton Engineering Ltd. 2000a. Preliminary Site Investigation Stage 1 and Stage II Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District. Ladysmith, B.C. Dated August 25, 2000.

Levelton[b] = Levelton Engineering Ltd. 2000b. Detailed Site Investigation Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District. Ladysmith, B.C. Dated October 25, 2000.

TABLE 2b
Historic Soil Sampling Results - Hydrocarbons
Ladysmith Harbour, Ladysmith, BC

Location	CSR Standards for PL	MCS	CSR Standards for CL	MCS	CSR Standards for IL	MCS	BH05-1	CPT 1	CPT 2	CPT 3	CPT 3	CPT 10	CPT 10	TP2	TP3	TP4	TP4	TP6	TP7	TP8	TP10	TP10	TP11	TP11	TP12	TP13
SCN							0019-08	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Depth (m)							6.9	1.55	1.8	1	1.8	2	2	1	3	2	4	3	3	-	2	2	1	1	3	1
Date Sampled							28-Mar-05	18-Sep-90	########	18-Sep-90	18-Sep-90	18-Sep-90 Hardy FDA	18-Sep-90 Hardy FD	18-Jul-00 Levelton[a]	18-Jul-00 Levelton[a]	18-Jul-00 Levelton[a]	18-Jul-00 Levelton[a]	18-Jul-00 Levelton[a]	18-Jul-00 Levelton[a]	18-Jul-00 Levelton[a]	18-Jul-00 Levelton[a] FDA	18-Jul-00 Levelton[a] FD	18-Jul-00 Levelton[a] FDA	18-Jul-00 Levelton[a] FD	18-Jul-00 Levelton[a]	18-Jul-00 Levelton[a]
Source							Golder	Hardy	Hardy	Hardy	Hardy	Hardy	Hardy													
QA/QC												FDA	FD								FDA	FD	FDA	FD		
Non-Halogenated Volatiles																										
Benzene	0.035	DW	**0.035**	DW	6.5	DW	-	**1.02**	-	-	-	-	-	-	*<0.04*	-	-	-	-	-	*<0.04*	*<0.04*	-	-	*<0.04*	-
Ethylbenzene	15	DW	15	DW	15	DW	-	11.3	-	-	-	-	-	-	<0.04	-	-	-	-	-	<0.04	<0.04	-	-	<0.04	-
Methyl t-butyl ether (MTBE)	8000	HH	20000	HH	20000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Styrene	5	EH	50	EH	50	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Toluene	6	DW	**6**	DW	6	DW	-	**24**	-	-	-	-	-	-	<0.04	-	-	-	-	-	<0.04	<0.04	-	-	<0.04	-
meta- & para-Xylene							-	74.3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
ortho-Xylene							-	30.4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Xylene	6.5	DW	**6.5**	DW	6.5	DW	-	**104.7**	-	-	-	-	-	-	<0.04	-	-	-	-	-	<0.04	<0.04	-	-	<0.04	-
Volatile Hydrocarbons (VH6-10)							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
VPH	200	HH/EH	200	EH/HH	200	EH/HH	-	-	-	-	-	-	-	-	<10	-	-	-	-	-	<10	<10	-	-	<10	-
Polycyclic Aromatic Hydrocarbons																										
Acenaphthene	2000	HH	1500	HH	15000	HH	0.682	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Acenaphthylene							0.619	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Anthracene	2.5	T	30	T	30	T	1.05	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benz(a)anthracene	1	EH	10	EH	10	EH	0.757	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benzo(a)pyrene	10	I	30	I	50	I	0.492	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benzo(b)fluoranthene							0.572	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benzo(g,h,i)perylene							0.469	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benzo(k)fluoranthene	1	EH	10	EH	10	EH	0.120	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Chrysene	400	HH	4500	HH	4500	HH	0.982	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Dibenz(a,h)anthracene	1	EH	10	EH	10	EH	0.120	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Fluoranthene	50	T	200	T	200	T	0.649	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Fluorene	1000	HH	9500	HH	9500	HH	1.65	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Indeno(1,2,3-c,d)pyrene	1	EH	10	EH	10	EH	0.106	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2-Methylnaphthalene	100	HH	950	HH	950	HH	9.95	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Naphthalene	0.6	T	20	T	20	T	4.88	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Phenanthrene	5	EH	50	EH	50	EH	9.47	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Pyrene	10	EH	100	EH	100	EH	0.813	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Extractable Hydrocarbons																										
EPH10-19[a]	1000		**2,000**		2,000		1440	-	-	-	-	-	-	<100	<100	**4,760**	**2,230**	<100	560	-	<100	<100	<100	<100	<100	<100
EPH19-32[a]	1000		5000		5,000		1710	-	-	-	-	-	-	<100	<100	550	350	<100	<100	-	<100	<100	<100	<100	<100	<100
LEPH	1000	EH/HH	**2,000**	EH/HH	2,000	EH/HH	1420	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
HEPH	1000	EH/HH	**5000**	EH/HH	5,000	EH/HH	1710	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
General																										
Light Aliphatic Hydrocarbons[b]	1000		2,000		2,000		-	475	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Mineral Oil and Grease[b]	1000		**2,000**		2,000		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Oil and Grease[b]	1000		**2,000**		2,000		-	-	317	**2,740**	**3,130**	**2,000**	**2,050**	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Extractable Hydrocarbons[c]	200		**200**		200		-	-	**439**	-	**2440**	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total PCBs	1.5	T	35	I/T	35	I/T	-	-	-	-	-	-	-	-	-	-	-	-	-	< detection	-	-	-	-	-	< detection

Notes:
Results are expressed in micrograms per gram (ug/g) unless othwise indicated.
SCN = sample control number
COC = Chain of Custody
QA/QC = quality assurance/quality control
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial).
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; AW = Groundwater Flow to Surface Water used by Aquatic Life, M = Marine; DW = Drinking Water
EH = generic standard for the protection of ecological health; HH = generic standard for the protection of human health
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019)
MCS = most conservative standard based on applicable site-specific standards
a. LEPH and HEPH criteria used as a conservative screen for EPH10-19 and EPH19-32, respectively.
b. LEPH criteria used as a conservate screen for Light Aliphatic Hydrocarbons, Mineral Oil and Grease, Oil and Grease.

c.. VPH criteria used as a conservative screen for Total Extractable Hydrocarbons.
FDA = Field duplicate available; FD = Field duplicate.
italics = Laboratory Method Detection Limit above applicable standards
< = less than laboratory reporting limit; - = not analyzed
Golder = Golder Associates Ltd. 2005. Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site Investigation Lot 16G Ladysmith Harbour Ladysmith, BC. Dated June 30, 2005.
Hardy = Hardy BBT Limited. 1990. Environmental Review for Town of Ladysmith Waterfront Development Slag Point Area, Ladysmith, B.C. Dated October 1990.
EBA = EBA Environmental Ltd. 1994. Elk Falles Forest Industries Limited Phase II Environmental Assessment Ladysmith Harbour Leases Project Ladysmith, B.C. Dated January 1994.
EBA[b] = EBA Environmental Ltd. 1994b. Elk Falles Forest Industries Limited Supplemental Information Ladysmith Harbour Leases Project Ladysmith, BC. Dated February 4, 1994.
Levelton[a] = Levelton Engineering Ltd. 2000a. Preliminary Site Investigation Stage 1 and Stage II Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, District. Ladysmith, B.C. Dated August 25, 2000.Plan VIP64405, Oyster Land
Levelton[b] = Levelton Engineering Ltd. 2000b. Detailed Site Investigation Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District., Ladysmith, BC.
EPH_{C10-19} = extractable petroleum hydrocarbons, carbon range 10-19
EPH_{C19-32} = extractable petroleum hydrocarbons, carbon range 19-32
LEPH = light extractable petroleum hydrocarbons
HEPH = heavy extractable petroleum hydrocarbons
VPH = volitile petroleum hydrocarbons
VH (C6-C10) = volitile hydrocarbons, carbon range 6-10
PAH = polycyclic aromatic hydrocarbon
ppm = parts per milion

TABLE 2b
Historic Soil Sampling Results - Hydrocarbons
Ladysmith Harbour, Ladysmith, BC

Location	CSR Standards for PL	MCS	CSR Standards for CL	MCS	CSR Standards for IL	MCS	MW00-02	BH00-03	BH00-03	MW00-04	BH00-05	MW00-06	BH00-07	BH00-07	BH00-08	BH00-08	BH00-08	MW00-09	MW00-09	BH00-10	BH00-10	MW00-11	BH00-12	BH00-13	BH00-14	1	2
SCN							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Depth (m)							1.4	0.2	0.5	1.4	2.6	1.5	1.8	2.9	0.8	1.8	2.9	2.4	2.4	1.4	3.5	4.6	2.1	1.2	1.4	3	1.3
Date Sampled							20-Sep-00	20-Sep-00	20-Sep-00	20-Sep-00	20-Sep-00	20-Sep-00	20-Sep-00	20-Sep-00	20-Sep-00	20-Sep-00	20-Sep-00	20-Sep-00	20-Sep-00	20-Sep-00	20-Sep-00	21-Sep-00	21-Sep-00	21-Sep-00	21-Sep-00	07-May-93	07-May-93
Source							Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	Levelton[b]	EBA[a]	EBA[a]
QA/QC																		FDA	FD								
Non-Halogenated Volatiles																											
Benzene	0.035	DW	**0.035**	DW	6.5	DW	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	*<0.04*	-	-	-	-	-
Ethylbenzene	15	DW	15	DW	15	DW	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	<0.04	-	-	-	-	-
Methyl t-butyl ether (MTBE)	8000	HH	20000	HH	20000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Styrene	5	EH	50	EH	50	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Toluene	6	DW	**6**	DW	6	DW	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	<0.04	-	-	-	-	-
meta- & para-Xylene							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
ortho-Xylene							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Xylene	6.5	DW	**6.5**	DW	6.5	DW	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	<0.04	-	-	-	-	-
Volatile Hydrocarbons (VH6-10)							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
VPH	200	HH/EH	200	EH/HH	200	EH/HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	<10	-	-	-	-	-
Polycyclic Aromatic Hydrocarbons																											
Acenaphthene	2000	HH	1500	HH	15000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Acenaphthylene							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Anthracene	2.5	T	30	T	30	T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benz(a)anthracene	1	EH	10	EH	10	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benzo(a)pyrene	10	I	30	I	50	I	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benzo(b)fluoranthene							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benzo(g,h,i)perylene							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benzo(k)fluoranthene	1	EH	10	EH	10	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Chrysene	400	HH	4500	HH	4500	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Dibenz(a,h)anthracene	1	EH	10	EH	10	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Fluoranthene	50	T	200	T	200	T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Fluorene	1000	HH	9500	HH	9500	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Indeno(1,2,3-c,d)pyrene	1	EH	10	EH	10	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2-Methylnaphthalene	100	HH	950	HH	950	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Naphthalene	0.6	T	20	T	20	T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Phenanthrene	5	EH	50	EH	50	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Pyrene	10	EH	100	EH	100	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Extractable Hydrocarbons																											
EPH10-19[a]	1000		**2,000**		2,000		430	1,350	875	<250	<250	<250	770	<250	**2,350**	<250	<250	<250	<250	<250	<250	-	<250	<250	<250	-	-
EPH19-32[a]	1000		5000		5,000		480	1,450	680	<250	<250	<250	270	<250	750	<250	<250	<250	<250	<250	<250	-	<250	<250	<250	-	-
LEPH	1000	EH/HH	**2,000**	EH/HH	2,000	EH/HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
HEPH	1000	EH/HH	**5000**	EH/HH	5,000	EH/HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
General																											
Light Aliphatic Hydrocarbons[b]	1000		2,000		2,000		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Mineral Oil and Grease[b]	1000		**2,000**		2,000		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	**3,400**	**5,500**
Oil and Grease[b]	1000		**2,000**		2,000		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Extractable Hydrocarbons[c]	200		**200**		200		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total PCBs	1.5	T	35	I/T	35	I/T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Notes:
Results are expressed in micrograms per gram (ug/g) unless othwise indicated.
SCN = sample control number
COC = Chain of Custody
QA/QC = quality assurance/quality control
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial).
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; AW = Groundwate
EH = generic standard for the protection of ecological health; HH = generic standard for the protection of human health
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019)
MCS = most conservative standard based on applicable site-specific standards
a. LEPH and HEPH criteria used as a conservative screen for EPH10-19 and EPH19-32, respectively.
b. LEPH criteria used as a conservate screen for Light Aliphatic Hydrocarbons, Mineral Oil and Grease, Oil and Grease.

c.. VPH criteria used as a conservative screen for Total Extractable Hydrocarbons.
FDA = Field duplicate available; FD = Field duplicate.
italics = Laboratory Method Detection Limit above applicable standards
< = less than laboratory reporting limit; - = not analyzed
Golder = Golder Associates Ltd. 2005. Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site Investigation Lot 16G Ladysmith Harbour Ladysmith, BC. Dated June 30, 2005.
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Levelton[a] = Levelton Engineering Ltd. 2000a. Preliminary Site Investigation Stage 1 and Stage II Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, District. Ladysmith, B.C. Dated August 25, 2000.Plan VIP64405, Oyster Land
Levelton[b] = Levelton Engineering Ltd. 2000b. Detailed Site Investigation Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District., Ladysmith, BC.
EPH_{C10-19} = extractable petroleum hydrocarbons, carbon range 10-19
EPH_{C19-32} = extractable petroleum hydrocarbons, carbon range 19-32
LEPH = light extractable petroleum hydrocarbons
HEPH = heavy extractable petroleum hydrocarbons
VPH = volitile petroleum hydrocarbons
VH (C6-C10) = volitile hydrocarbons, carbon range 6-10
PAH = polycyclic aromatic hydrocarbon
ppm = parts per milion

TABLE 2b
Historic Soil Sampling Results - Hydrocarbons
Ladysmith Harbour, Ladysmith, BC

Location	CSR Standards for PL	MCS	CSR Standards for CL	MCS	CSR Standards for IL	MCS	3	4	6	7	10	11	14	15	16	17	17	17	18	18	19	19
SCN							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Depth (m)							3	3	4.5	3.5	2	0.8	0.6	0.6	0.6	1	1	1.5	1	1.5	1	1.5
Date Sampled							07-May-93	07-May-93	07-May-93	07-May-93	07-May-93	27-Jun-93	16-Jul-93	10-Aug-93	10-Aug-93	28-Jan-94	28-Jan-94	28-Jan-94	28-Jan-94	28-Jan-94	28-Jan-94	28-Jan-94
Source							EBA[a]	EBA[a]	EBA[a]	EBA[a]	EBA[a]	EBA[a]	EBA[a]	EBA[a]	EBA[a]	EBA[b] FDA	EBA[b] FD	EBA[b]	EBA[b]	EBA[b]	EBA[b]	EBA[b]
QA/QC																						
Non-Halogenated Volatiles																						
Benzene	0.035	DW	**0.035**	DW	6.5	DW	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ethylbenzene	15	DW	15	DW	15	DW	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Methyl t-butyl ether (MTBE)	8000	HH	20000	HH	20000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Styrene	5	EH	50	EH	50	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Toluene	6	DW	**6**	DW	6	DW	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
meta- & para-Xylene							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
ortho-Xylene							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Xylene	6.5	DW	**6.5**	DW	6.5	DW	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Volatile Hydrocarbons (VH6-10)							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
VPH	200	HH/EH	200	EH/HH	200	EH/HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Polycyclic Aromatic Hydrocarbons																						
Acenaphthene	2000	HH	1500	HH	15000	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Acenaphthylene							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Anthracene	2.5	T	30	T	30	T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benz(a)anthracene	1	EH	10	EH	10	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benzo(a)pyrene	10	I	30	I	50	I	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benzo(b)fluoranthene							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benzo(g,h,i)perylene							-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Benzo(k)fluoranthene	1	EH	10	EH	10	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Chrysene	400	HH	4500	HH	4500	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Dibenz(a,h)anthracene	1	EH	10	EH	10	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Fluoranthene	50	T	200	T	200	T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Fluorene	1000	HH	9500	HH	9500	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Indeno(1,2,3-c,d)pyrene	1	EH	10	EH	10	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2-Methylnaphthalene	100	HH	950	HH	950	HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Naphthalene	0.6	T	20	T	20	T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Phenanthrene	5	EH	50	EH	50	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Pyrene	10	EH	100	EH	100	EH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Extractable Hydrocarbons																						
EPH10-19[a]	1000		**2,000**		2,000		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
EPH19-32[a]	1000		5000		5,000		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
LEPH	1000	EH/HH	**2,000**	EH/HH	2,000	EH/HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
HEPH	1000	EH/HH	**5000**	EH/HH	5,000	EH/HH	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
General																						
Light Aliphatic Hydrocarbons[b]	1000		2,000		2,000		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Mineral Oil and Grease[b]	1000		**2,000**		2,000		**3,200**	**2,400**	**2,500**	**2,300**	**2,400**	**16,000**	1,100	470	960	<10	<10	<10	<10	<10	<10	<10
Oil and Grease[b]	1000		**2,000**		2,000		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Extractable Hydrocarbons[c]	200		**200**		200		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total PCBs	1.5	T	35	I/T	35	I/T	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Notes:
Results are expressed in micrograms per gram (ug/g) unless othwise indicated.
SCN = sample control number
COC = Chain of Custody
QA/QC = quality assurance/quality control
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial).
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; AW = Groundwate
EH = generic standard for the protection of ecological health; HH = generic standard for the protection of human health
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019)
MCS = most conservative standard based on applicable site-specific standards
a. LEPH and HEPH criteria used as a conservative screen for EPH10-19 and EPH19-32, respectively.
b. LEPH criteria used as a conservate screen for Light Aliphatic Hydrocarbons, Mineral Oil and Grease, Oil and Grease.

c.. VPH criteria used as a conservative screen for Total Extractable Hydrocarbons.
FDA = Field duplicate available; FD = Field duplicate.
italics = Laboratory Method Detection Limit above applicable standards
< = less than laboratory reporting limit; - = not analyzed
Golder = Golder Associates Ltd. 2005. Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site Investigation Lot 16G Ladysmith Harbour Ladysmith, BC. Dated June 30, 2005.
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EBA[b] = EBA Environmental Ltd. 1994b. Elk Falles Forest Industries Limited Supplemental Information Ladysmith Harbour Leases Project Ladysmith, BC. Dated February 4, 1994.
Levelton[a] = Levelton Engineering Ltd. 2000a. Preliminary Site Investigation Stage 1 and Stage II Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, District. Ladysmith, B.C. Dated August 25, 2000.Plan VIP64405, Oyster Land
Levelton[b] = Levelton Engineering Ltd. 2000b. Detailed Site Investigation Lot 4, Plan 45800, District Lot 8G, District Lot 11G and Lot 1, Plan VIP64405, Oyster Land District., Ladysmith, BC.
EPH_{C10-19} = extractable petroleum hydrocarbons, carbon range 10-19
EPH_{C19-32} = extractable petroleum hydrocarbons, carbon range 19-32
LEPH = light extractable petroleum hydrocarbons
HEPH = heavy extractable petroleum hydrocarbons
VPH = volitile petroleum hydrocarbons
VH (C6-C10) = volitile hydrocarbons, carbon range 6-10
PAH = polycyclic aromatic hydrocarbon
ppm = parts per milion

TABLE 2c
Historic Soil Sampling Results - VOCs
Ladysmith Harbour, Ladysmith, BC

Location	CSR Standards for PL	MCS	**CSR Standards for CL**	MCS	CSR Standards for IL	MCS	MW05-2	MW05-3	MW05-4
SCN							0020-08	0021-03	0022-01
Depth (m)							1.8	0.9	1.7
Soil Type							coal fill	coal fill	coal fill
Date Sampled							28-Mar-05	28-Mar-05	29-Mar-05
Source							Golder	Golder	Golder
QA/QC									
Chlorinated Aliphatics									
Carbon Tetrachloride	**5**	EH	**50**	EH	**50**	EH	<0.10	<0.10	<0.10
Chlorobenzene	1	EH	10	EH	10		<0.10	<0.10	<0.10
Chloroethane							<0.10	<0.10	<0.10
Chloromethane							-	-	-
1,2-Dichlorobenzene	1	EH	10	EH	10	EH	<0.10	<0.10	<0.10
1,3-Dichlorobenzene	1	EH	10	EH	10	EH	<0.10	<0.10	<0.10
1,4-Dichlorobenzene	1	EH	10	EH	10	EH	<0.10	<0.10	<0.10
1,1-Dichloroethane	5	EH	50	EH	50	EH	<0.10	<0.10	<0.10
1,2-Dichloroethane	5	EH	50	EH	50	EH	<0.10	<0.10	<0.10
1,2-Dichloroethene	5	EH	50	EH	50	EH	-	-	-
cis-1,2-Dichloroethylene	5	EH	50	EH	50	EH	<0.10	<0.10	<0.10
trans-1,2-Dichloroethylene	5	EH	50	EH	50	EH	<0.10	<0.10	<0.10
1,1-Dichloroethene	5	EH	50	EH	50	EH	<0.10	<0.10	<0.10
Dichloromethane	5	EH	50	EH	50	EH	-	-	-
1,2-dichloropropane	5	EH	50	EH	50	EH	<0.10	<0.10	<0.10
cis 1,3-dichloropropene	5	EH	50	EH	50	EH	<0.10	<0.10	<0.10
trans 1,3-dichloropropene	5	EH	50	EH	50	EH	<0.10	<0.10	<0.10
1,1,1,2-tetrachloroethane	550	HH	1500	HH	1500	HH	<0.10	<0.10	<0.10
1,1,2,2-tetrachloroethane	70	HH	150	HH	150	HH	<0.10	<0.10	<0.10
tetrachloroethylene (Perc)	2.5	AW	2.5	AW	2.5	AW	<0.10	<0.10	<0.10
1,1,1-trichloroethane	5	EH	50	EH	50	EH	<0.10	<0.10	<0.10
1,1,2-trichloroethane	5	EH	50	EH	50	EH	<0.10	<0.10	<0.10
Trichloroethylene (TCE)	0.3	AW	0.3	AW	0.3	AW	<0.10	<0.10	<0.10
Trichlorofluoromethane	9000	HH	70000	HH	70,000	HH	<0.10	<0.10	<0.10
Vinyl Chloride	2	HH	45	HH	45	HH	<0.10	<0.10	<0.10
Trihalomethanes									
Bromodichloromethane	200	HH	550	HH	550	HH	<0.10	<0.10	<0.10
Bromoform	650	HH	4,000	HH	4,000	HH	<0.10	<0.10	<0.10
Chloroform	5	EH	50	EH	50	EH	<0.10	<0.10	<0.10
Dibromochloromethane	150	HH	400	HH	400	HH	<0.10	<0.10	<0.10

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019)
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial)
MCS: most conservative standard based on applicable site-specific standards
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants;
AW = Groundwater Flow to Surface Water used by Aquatic Life, F = Fresh Water Aquatic Life and M = Marine Aquatic Life;
FDA = field duplicate available
FD = field duplicate
QA/QC = quality assurance/quality control
NS = No Standard
< = less than laboratory reporting limit; - = not analyzed
italics = Laboratory Method Detection Limit above applicable standards
Golder = Golder Associates Ltd. 2005. Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site Investigation Lot 16G Ladysmith Harbour Ladysmith, BC. Dated June 30, 2005.

TABLE 2d
Historic Soil Sampling Results - Chlorinated Phenols
Ladysmith Harbour, Ladysmith, BC

Location		CSR Standards for PL	MCS	CSR Standards for CL	MCS	CSR Standards for IL	MCS	9
SCN								2
Depth (m)								
Date Sampled								07-May-93
Source								EBA
QA/QC								
Chlorinated Phenols								
Pentachlorophenol[pH]		1.5-25	DW	1.5-55	DW	1.5-55	DW	<0.005
Tetracholorophenols								<0.005
Trichlorophenols								<0.005

Notes:
Results are expressed in micrograms per gram (ug/g), unless otherwise indicated.
m bgs = metres below ground surface
SCN = sample control number
COC = Chain of Custody
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019)
Land Use abbreviations: PL (Urban Park Land); CL (Commercial); IL (Industrial)
MCS: most conservative standard based on applicable site-specific standards
Referenced site-specific factors include: I = Intake of Contaminated Soil; T = Toxicity to Invertebrates and Plants; AW = Groundwater Flow to Surface Water used by Aquatic Life, F = Fresh Water Aquatic Life and M = Marine Aqua
FDA = field duplicate available
FD = field duplicate
pH = Standard is pH dependant.
QA/QC = quality assurance/quality control
NS = No Standard
< = less than laboratory reporting limit; - = not analyzed
italics = Laboratory Method Detection Limit above applicable standards
EBA = EBA Environmental Ltd. 1994. Elk Falles Forest Industries Limited Phase II

TABLE 3a
Historic Groundwater Results - Dissolved Metals
Ladysmith Harbour, Ladysmith, BC

Location	CSR Standards for DW	CSR Standards for AW-M	MW05-2	MW05-3	MW05-4	MW05-5	MW05-6	MW05-7	MW05-8	MW05-9	MW05-11	MW05-12	MW05-16	MW05-17	MW05-18	MW05-19	MW05-20	MW05-22	MW05-24
SCN			0045-02	0045-01	0045-05	0036-02	0045-08	0036-01	0045-07	0046-10	0045-06	0046-02	0045-03	0039-01	0039-02	0046-08	0046-09	0048-01	0049-01
Date			2005-04-13	2005-04-13	2005-04-13	2005-04-14	2005-04-13	2005-04-14	2005-04-13	2005-04-14	2005-04-13	2005-04-14	2005-04-13	2005-04-14	2005-04-14	2005-04-14	2005-04-14	2005-04-21	2005-04-25
Source			Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder	Golder FDA	Golder	Golder FDA	Golder	Golder	Golder	Golder
QA/QC																			
Physical																			
Hardness (as $CaCO_3$)			345	1470	146	2510	566	3430	519	463	394	791	590	4290	4640	323	87.2	633	533
pH			7.18	6.51	7.58	7.77	7.45	7.75	7.31	7.37	7.11	6.60	6.79	7.73	7.79	7.29	8.59	6.90	6.70
Dissolved Metals																			
Aluminum	9.5		<0.020	<0.10	<0.050	<1.0	<0.10	<1.0	0.031	<0.020	<0.020	<0.10	<0.020	<0.10	<0.20	<0.010	<0.010	<0.010	<0.020
Antimony	0.006	2.5	<0.0010	<0.0050	<0.0025	*<0.050*	<0.0050	*<0.050*	<0.0010	<0.0010	<0.0010	<0.0050	<0.0010	<0.0050	*<0.010*	<0.00050	<0.00050	<0.00050	<0.0010
Arsenic	0.01	0.125	<0.0020	*<0.020*	<0.0050	*<0.10*	<0.010	*<0.10*	0.0053	<0.0060	<0.0020	<0.010	<0.0020	*<0.040*	*<0.060*	<0.0010	<0.0010	0.0032	0.0024
Barium	1	5	0.124	<0.20	0.074	<0.20	0.075	<0.20	0.132	<0.20	0.244	<0.20	0.071	0.29	0.34	<0.20	<0.20	0.100	0.228
Beryllium	0.008	1	<0.0050	*<0.050*	<0.0050	*<0.050*	*<0.010*	*<0.050*	<0.0050	*<0.050*	<0.0050	*<0.050*	<0.0050	*<0.050*	*<0.050*	*<0.050*	*<0.050*	<0.0050	<0.0050
Bismuth			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Boron	5	12	0.37	2.1	1.11	<1.0	1.23	3.0	0.51	1.5	0.21	1.1	0.29	2.2	2.0	<1.0	<1.0	0.36	0.37
Cadmium	0.005	0.015	<0.00010	<0.00050	<0.00025	<0.0050	<0.00050	<0.0050	<0.00010	<0.00010	<0.00010	<0.00050	0.00011	<0.00050	<0.0010	0.000061	<0.000050	0.000064	0.00019
Calcium			88.0	101	22.8	244	72.3	237	157	67.2	135	225	181	262	297	99.4	15.1	179	160
Chromium IV, III	0.05, 6	0.015,0.56	<0.0010	<0.0050	0.0028	<0.050	<0.0050	<0.050	<0.0050	<0.0010	<0.0010	<0.0050	<0.0050	<0.0050	<0.010	<0.00050	<0.00050	<0.00050	<0.0020
Cobalt	0.02	0.04	0.0041	*<0.0050*	0.0050	*<0.050*	*<0.0050*	*<0.050*	0.0022	0.0050	0.0089	0.0115	0.0011	*<0.0050*	*<0.010*	0.00523	0.00424	0.00322	0.0024
Copper	1.5	0.02	<0.0020	<0.010	<0.0050	<0.10	<0.010	<0.10	0.0029	0.0046	<0.0020	<0.010	<0.0020	<0.010	<0.020	<0.0010	0.0013	<0.0010	<0.0020
Iron	6.5		0.359	<0.30	<0.030	<0.30	<0.060	<0.30	1.15	<0.30	0.162	0.48	<0.030	<0.30	<0.30	<0.30	<0.30	0.615	0.085
Lead	0.01	0.02	<0.0020	*<0.010*	<0.0050	*<0.10*	<0.010	*<0.10*	<0.0020	<0.0020	<0.0020	<0.010	<0.0020	<0.010	*<0.020*	<0.0010	<0.0010	<0.0010	<0.0020
Lithium	0.033*		*<0.050*	*<0.50*	*<0.050*	*<0.50*	*<0.10*	*<0.50*	*<0.050*	*<0.50*	*<0.050*	*<0.50*	*<0.050*	*<0.50*	*<0.50*	*<0.50*	*<0.50*	*<0.050*	*<0.050*
Magnesium			30.4	295	21.5	461	93.6	690	31.0	71.7	13.5	55.7	33.4	884	946	18.3	12.0	44.9	32.4
Manganese	**1.5**		0.092	<0.10	0.021	0.33	<0.020	0.28	0.272	<0.10	**1.58**	0.73	0.053	0.77	0.86	0.13	<0.10	0.263	0.728
Mercury	0.001	0.00025	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020
Molybdenum	0.25	10	<0.0020	<0.010	0.0063	<0.10	<0.010	<0.10	<0.0020	0.0027	0.0022	<0.010	<0.0020	<0.010	<0.020	<0.0010	<0.0010	<0.0010	<0.0020
Nickel	0.08	0.083	<0.010	<0.050	<0.025	<0.50	<0.050	<0.50	<0.010	<0.010	0.012	<0.050	<0.010	<0.050	<0.10	<0.0050	<0.0050	<0.0050	<0.010
Phosphorus			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Potassium			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Selenium	0.01	0.02	<0.0020	*<0.060*	<0.0050	*<0.10*	<0.010	*<0.10*	<0.0020	*<0.014*	<0.0020	<0.010	0.0035	*<0.12*	*<0.14*	<0.0010	<0.0010	<0.0010	<0.0020
Silicon			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Silver	0.02	0.015	<0.00010	<0.00050	<0.00025	<0.0050	<0.00050	<0.0050	<0.00010	<0.00010	<0.00010	<0.00050	<0.00010	<0.00050	<0.0010	<0.000050	<0.000050	<0.000050	<0.00010
Sodium	**200**[i]		67.6	**3240**	**532**	**4000**	**1050**	**5790**	30.4	**1060**	15.1	33	10.4	**6620**	**5420**	31	50	3.4	24.4
Strontium	2.5		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Tellurium			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Thallium		0.003	<0.00040	<0.0020	<0.0010	*<0.020*	<0.0020	*<0.020*	<0.00040	<0.00040	<0.00040	<0.0020	<0.00040	<0.0020	*<0.0040*	<0.00020	<0.00020	<0.00020	*<0.00040*
Tin	2.5		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Titanium		1	<0.050	<0.50	<0.050	<0.50	<0.10	<0.50	<0.050	<0.50	<0.050	<0.50	<0.050	<0.50	<0.50	<0.50	<0.50	<0.050	<0.050
Uranium	0.02	0.085	<0.00040	<0.0020	<0.0010	<0.020	<0.0020	<0.020	<0.00040	0.00081	0.00258	<0.0020	0.00061	<0.0020	<0.0040	0.00052	<0.00020	0.00096	0.00074
Vanadium	0.02		*<0.030*	*<0.30*	*<0.030*	*<0.30*	*<0.060*	*<0.30*	*<0.030*	*<0.30*	*<0.030*	*<0.30*	*<0.030*	*<0.30*	*<0.30*	*<0.30*	*<0.30*	*<0.030*	*<0.030*
Zinc	3	0.1	<0.0050	<0.050	<0.0050	<0.050	<0.010	<0.050	<0.0050	<0.050	<0.0050	<0.050	<0.0050	<0.050	<0.050	<0.050	<0.050	<0.0050	<0.0050
Zirconium																			

Notes:
All concentrations in milligrams per litre (mg/L), unless otherwise noted.
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019).
Abbreviations: AW (Aquatic Life), M = Marine; DW (Drinking Water)
i= standard for sodium ion and chloride ion conservatively applied
V= Standard is valence dependent VI refers to chromium VI and III refers to chromium III
FDA = field duplicate available; FD = field duplicate; SCN = sample control number
italics indicate the concentration exceeds the laboratory detection limit.
*Background concentration for lithium and cobalt as detailed in Technical Bulletin 3 is applied to replace CSR DW Standard.
Golder = Golder Associates Ltd. 2005. Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site Investigation Lot 16G Ladysmith Harbour Ladysmith, BC. Dated June 30, 2005.

1	Exceeds CSR Standards for DW
1	Exceeds CSR Standards for AW-M
1	Exceeds CSR Standards for DW and AW-M

TABLE 3b
Historic Groundwater Results - Total Metals and Other Parameters
Ladysmith Harbour, Ladysmith, BC

Location	CSR Standards for DW	CSR Standards for AW-M	4	4	17	17	18	19
SCN			93011006	93015229	467-16-1	467-16-1	467-16-2	467-16-3
Date			1993-06-04	1993-07-16	1994-01-28	1994-01-28	1994-01-28	1994-01-28
Source			EBA[a]	EBA[a]	EBA[b] FDA	EBA[b] FD	EBA[b]	EBA[b]
QA/QC								
Physical								
Hardness (as $CaCO_3$)			591	-	-	-	-	-
pH			-	-	-	-	-	-
Salinity			-	-	20100	16610	4290	14890
Total Metals*								
Aluminum	9.5		9.18	-	-	-	-	-
Antimony	0.006	2.5	<0.02	-	-	-	-	-
Arsenic	**0.01**	0.125	0.004	-	**0.060**	-	**0.080**	**0.130**
Barium	1	5	0.184	-	0.715	-	0.713	2.287
Beryllium	0.008	1	<0.001	-	-	-	-	-
Bismuth			<0.02	-	-	-	-	-
Boron	5	12	0.36	-	-	-	-	-
Cadmium	**0.005**	0.015	<0.002	-	**0.006**	-	0.005	**0.042**
Calcium			175	-	-	-	-	-
Chromium (IV, III)	**0.05,6[V]**	0.015,0.56[V]	0.027	-	0.035	-	0.043	**0.159**
Cobalt	**0.02**	0.04	0.009	-	**0.062**	-	**0.100**	**0.167**
Copper	**1.5**	0.02	0.110	-	0.584	-	0.705	**1.529**
Iron	**6.5**		**10.8**	-	-	-	-	-
Lead	**0.01**	0.02	**0.03**	-	**0.21**	-	**0.170**	**2.060**
Lithium	0.033*		-	-	-	-	-	-
Magnesium			37.4	-	-	-	-	-
Manganese	1.5		0.457	-	-	-	-	-
Mercury	**0.001**	0.00025	0.00065	-	**0.002**	-	**0.003**	**0.009**
Molybdenum	0.25	10	<0.004	-	0.045	-	0.020	0.043
Nickel	**0.08**	0.083	0.04	-	**0.149**	-	**0.150**	**0.342**
Phosphorus			0.21	-	-	-	-	-
Potassium			12.4	-	-	-	-	-
Selenium	0.01	0.02	<0.005	-	<0.005	-	<0.005	<0.005
Silicon			3.8	-	-	-	-	-
Silver	0.02	0.015	<0.03	-	<0.003	-	<0.003	0.006
Sodium	200[i]		66.8	68	-	-	-	-
Strontium	2.5		1.13	-	-	-	-	-
Tellurium			<0.02	-	-	-	-	-
Thallium		0.003	*<0.03*	-	-	-	-	-
Tin	2.5		<0.02	-	<0.010	-	<0.010	<0.010
Titanium		1	0.066	-	-	-	-	-
Uranium	0.02	0.085	-	-	-	-	-	-
Vanadium	**0.02**		**0.033**	-	-	-	-	-
Zinc	3	0.1	0.08	-	0.440	-	0.400	1.360
Zirconium			<0.003	-	-	-	-	-
Other Parameters								
Sulphur			32.6	-	-	-	-	-
Sulfate	500	1280-4290[H]	-	103	-	-	-	-
Chloride	250[i]	1500[i]	-	36.9	-	-	-	-

Notes:
All concentrations in milligrams per litre (mg/L), unless otherwise noted.
Standards for dissolved metals shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019).
Abbreviations: AW (Aquatic Life), M = Marine; DW (Drinking Water)
FDA = field duplicate available; FD = field duplicate; SCN = sample control number
H = standard is Hardness dependent, i= standard for sodium ion conservatively applied
V= Standard is valence dependent VI refers to chromium VI and III refers to chromium III
italics indicate the concentration exceeds the laboratory detection limit.
*data for total metals is conservatively screened against CSR standards for dissolved metals.
*Background concentration for lithium and cobalt as detailed in Technical Bulletin 3 is applied to replace CSR DW Standard.

EBA[a] = EBA Environmental Ltd. 1994a. Elk Falles Forest Industries Limited Phase II Environmental Assessment Ladysmith Harbour Leases Project Ladysmith, B.C. Dated January 1994.

EBA[b] = EBA Environmental Ltd. 1994b. Elk Falles Forest Industries Limited Supplemental Information Ladysmith Harbour Leases Project Ladysmith, BC. Dated February 4, 1994.

1	Exceeds CSR Standards for DW
1	Exceeds CSR Standards for AW-M
1	Exceeds CSR Standards for DW and AW-M

TABLE 3c
Historic Groundwater Results - Hydrocarbons
Ladysmith Harbour, Ladysmith, BC

Location	CSR Standards for DW	CSR Standards for AW Marine	MW05-4	MW05-5	MW05-6	MW05-7	MW05-8	MW05-9	MW05-16	MW05-17	MW05-18	MW05-22	MW05-24	MW00-02	MW00-04	MW00-06	MW00-09	4	4	17	18	19
SCN			0045-05	0036-02	0045-08	0036-01	0045-07	0046-10	0045-03	0039-01	0039-02	0048-01	0049-01	009079-02	009079-01	009079-03	009079-04	93011006	93013268	467-16-1	467-16-2	467-16-3
Date			2005-04-13	2005-04-14	2005-04-13	2005-04-14	2005-04-13	2005-04-14	2005-04-13	2005-04-14	2005-04-14	2005-04-21	2005-04-25	2000-09-25	2000-09-25	2000-09-25	2000-09-25	1993-06-04	1993-06-25	1994-01-28	1994-01-28	1994-01-28
Source			Golder	Golder	Golder	Golder	Golder	Golder	Golder FDA	Golder	Golder FDA	Golder	Golder	Levelton	Levelton	Levelton	Levelton	EBA[a]	EBA[a]	EBA[b]	EBA[b]	EBA[b]
QA/QC																						
Monoaromatic Hydrocarbons																						
Benzene	0.005	1	<0.00050	-	<0.00050	-	-	<0.00050	-	-	-	<0.00050	<0.00050	-	-	-	-	-	-	-	-	-
Ethylbenzene	0.14	2.5	<0.00050	-	<0.00050	-	-	<0.00050	-	-	-	<0.00050	<0.00050	-	-	-	-	-	-	-	-	-
Styrene	0.8	0.72	<0.0010	-	<0.0010	-	-	<0.0010	-	-	-	<0.0010	<0.0010	-	-	-	-	-	-	-	-	-
Toluene	0.06	2.0	<0.0010	-	<0.0010	-	-	<0.0010	-	-	-	<0.0010	<0.0010	-	-	-	-	-	-	-	-	-
meta- & para-Xylene			<0.00050	-	<0.00050	-	-	<0.00050	-	-	-	<0.00050	<0.00050	-	-	-	-	-	-	-	-	-
ortho-Xylene			<0.00050	-	<0.00050	-	-	<0.00050	-	-	-	<0.00050	<0.00050	-	-	-	-	-	-	-	-	-
Total Xylene	0.09	0.3	<0.0010	-	<0.0010	-	-	<0.0010	-	-	-	<0.0010	<0.0010	-	-	-	-	-	-	-	-	-
VHw6-10	15	15	<0.10	-	<0.10	-	-	<0.10	-	-	-	<0.10	<0.10	-	-	-	-	-	-	-	-	-
VPHw		1.5	<0.10	-	<0.10	-	-	<0.10	-	-	-	<0.10	<0.10	-	-	-	-	-	-	-	-	-
Polycyclic Aromatic Hydrocarbons																						
Acenaphthene	0.25	0.06	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Acenaphthylene			<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Acridine		0.0005	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Anthracene	1	0.001	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Benz(a)anthracene	0.00007	0.0001	*<0.000050*	*<0.000050*	*<0.000050*	*<0.000050*	*<0.000050*	*<0.000050*	*<0.000050*	*<0.000050*	*<0.000050*	*<0.000050*	*<0.000050*	-	-	-	-	-	-	-	-	-
Benzo(a)pyrene	0.00001	0.0001	<0.000010	<0.000010	<0.000010	<0.000010	<0.000010	<0.000010	<0.000010	<0.000010	<0.000010	<0.000010	<0.000010	-	-	-	-	-	-	-	-	-
Benzo(b)fluoranthene	0.00007		<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Benzo(g,h,i)perylene			<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Benzo(k)fluoranthene			<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Chrysene	0.007	0.001	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Dibenzo(a,h)anthracene	0.00001		<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Fluoranthene	0.15	0.002	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Fluorene	0.15	0.12	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Indeno(1,2,3-c,d)pyrene			<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Naphthalene	0.08	0.01	<0.000050	<0.000050	<0.000050	0.000066	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Phenanthrene		0.003	<0.000050	<0.000050	<0.000050	0.000126	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Pyrene	0.1	0.0002	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
Quinoline	0.00005	0.034	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	<0.000050	-	-	-	-	-	-	-	-	-
EPHw10-19	5	5	<0.30	-	<0.30	-	<0.30	<0.30	<0.30	-	-	<0.30	<0.30	0.910	<0.100	0.105	0.285	-	-	-	-	-
EPHw19-32			<1.0	-	<1.0	-	<1.0	<1.0	<1.0	-	-	<1.0	<1.0	1.400	<0.100	0.260	0.590	-	-	-	-	-
LEPHw		0.5	<0.30	-	<0.30	-	<0.30	<0.30	<0.30	-	-	<0.30	<0.30	-	-	-	-	-	-	-	-	-
HEPHw			<1.0	-	<1.0	-	<1.0	<1.0	<1.0	-	-	<1.0	<1.0	-	-	-	-	-	-	-	-	-
Mineral Oil and Grease		0.5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	7	*<1*	45	*<5*	7

Notes:
All concentrations in micrograms per litre (mg/L), unless otherwise noted.
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019).
Abbreviations: AW (Aquatic Life), M = Marine; DW (Drinking Water)
EPHC10-19 = extractable petroleum hydrocarbons, carbon range 10-19
EPHC19-32 = extractable petroleum hydrocarbons, carbon range 19-32
LEPH = light extractable petroleum hydrocarbons
HEPH = heavy extractable petroleum hydrocarbons
PAH = Polycyclic Aromatic Hydrocarbons
VPH = volatile petroleum hydrocarbons
FDA = field duplicate available; FD = field duplicate; SCN = sample control number
italics indicate the concentration exceeds the laboratory detection limit.
LEPH criteria used as a conservate screen for Mineral Oil and Grease and EHw10-19.
Golder = Golder Associates Ltd. 2005. Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site Investigation Lot 16G Ladysmith Harbour Ladysmith, BC. Dated June 30, 2005.

1	Exceeds CSR Standards for DW
1	Exceeds CSR Standards for AW-M
1	Exceeds CSR Standards for DW and AW-M

TABLE 3d
Historical Groundwater Results - Phenols
Ladysmith Harbour, Ladysmith, BC

Location SCN Date Sampled Source QA/QC	CSR Standards for DW	CSR Standards for AW Marine	MW05-4 0045-05 2005-04-13 Golder	MW05-5 0036-02 2005-04-14 Golder	MW05-6 0045-08 2005-04-13 Golder	MW05-7 0036-01 2005-04-14 Golder	MW05-8 0045-07 2005-04-13 Golder	MW05-9 0046-10 2005-04-14 Golder	MW05-16 0045-03 2005-04-13 Golder FDA	MW05-17 0039-01 2005-04-14 Golder	MW05-18 0039-02 2005-04-14 Golder FDA	MW05-22 0048-01 2005-04-21 Golder	MW05-24 0049-01 2005-04-25 Golder
Physical													
pH			7.58	7.77	7.45	7.75	7.31	7.37	6.79	7.73	7.79	6.90	6.70
Phenolic Compounds													
2,3,4-trichlorophenol		0.0025-0.32*	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050
2,3,5-trichlorophenol		0.0025-0.34*	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050
2,3,6-trichlorophenol		0.008-1.08*	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050
2,4,5-trichlorophenol	0.4	0.0025-0.30*	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050
2,4,6-trichlorophenol	0.005	0.006-0.8*	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050
3,4,5-trichlorophenol		0.001-0.128*	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050	<0.00050
2,3,4,5-tetrachlorophenol		0.002-0.26*	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020
2,3,4,6-tetrachlorophenol	0.1	0.0055-0.72*	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020
2,3,5,6-tetrachlorophenol		0.0025-0.34*	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020
pentachlorophenol	0.06	0.001-0.0110*	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020

Notes:
All concentrations in micrograms per litre (mg/L), unless otherwise noted.
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019).
< = less than laboratory reporting limit; - = not analyzed
Land Use abbreviations: AW (Aquatic Life), M = marine; DW (Drinking Water)
For chlorophenol standards, refer to Technical Guidance Document #9
*The standards varies as a function of pH, Temperature and Chlorophenol Isomer
FDA = field duplicate available; FD = field duplicate; SCN = sample control number
italics indicate the concentration exceeds the laboratory detection limit.
Golder = Golder Associates Ltd. 2005. Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site Investigation Lot 16G Ladysmith Harbour Ladysmith, BC. Dated June 30, 2005.

1	Exceeds CSR Standards for DW
1	Exceeds CSR Standards for AW-M
1	Exceeds CSR Standards for DW and AW-M

TABLE 4a
Results of Groundwater Analyses
Ladysmith Harbour, Ladysmith, BC
Metals and Anions

Location			MW00-2	MW00-04	MW00-09	MW00-11	MW05-02	MW05-04	MW05-08	MW05-09	MW05-12	MW05-17	MW05-20	MW05-22	MW09-1	MW09-2	MW09-3	MW09-03	MW09-4	MW09-5
Date	CSR Standards for DW	CSR Standards for AW-M	2009-11-18	2009-12-11	2009-12-11	2009-12-11	2009-11-24	2009-11-24	2009-11-24	2009-11-24	2009-11-24	2009-11-24	2009-11-24	2009-11-24	2009-11-18	2009-11-18	2009-11-18	2009-12-11	2009-11-19	2009-11-19
SCN			21367-04	21382-03	21382-02	21382-01	21369-04	21369-05	21369-02	21369-03	21369-08	21369-01	21369-07	21369-06	21367-01	21367-02	21367-03	21382-04	21367-07	21367-08
QA/QC																				
Field Parameters																				
pH (pH units)			7.55	6.66	6.33	7.33	7.04	7.06	6.74	7.23	6.6	7.51	8.4	6.59	7.97	7.36	7.69	7.31	7.64	7.6
Temperature (°C)			10.8	8.69	9.21	10.4	9.4	10.62	10.9	13.61	12.61	11.32	9.4	11.12	10.29	10.44	10.44	11.7	12.8	10.33
Conductivity (µS/cm)			242	230	96	386	531	510	1054	9547	1331	33734	900	1032	301	370	390	355	460	410
Redox (mV)			31.6	-62.7	144.2	18.7	-15.9	-34	-39.2	17.3	110.9	-219.6	-121.6	-61.6	-130.6	-80	-121.6	-116.9	45.3	-50.4
Dissolved Oxygen (%)			11.7	2.91	4.41	2.67	10.9	9.4	13.9	18.6	21.1	14.4	60.2	9.6	36	23.6	14.3	1.56	8.9	11
Laboratory Parameters																				
pH (lab)			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Conductivity (µS/cm_)			-	-	-	-	464	639	775	12000	1100	14900	209	1060	-	-	-	-	-	-
Salinity (Low Level) (g/L)			-	-	-	-	0.22	0.31	0.38	6.89	0.55	8.7	< 0.2	0.53	-	-	-	-	-	-
Hardness $CaCO_3$			125	101	56	198	233	307	435	944	790	1570	86	688	106	151	144	-	206	175
Bicarbonate Alkalinity HCO_3			-	-	-	-	262	393	449	709	752	181	78.9	645	-	-	-	-	-	-
Carbonate Alkalinity CO_3			-	-	-	-	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	-	-	-	-	-	-
Hydroxide Alkalinity OH			-	-	-	-	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	-	-	-	-	-	-
Total Alkalinity $CaCO_3$			-	-	-	-	215	322	368	581	617	148	64.7	528	-	-	-	-	-	-
Dissolved Bromide Br			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Dissolved Chloride Cl	**250**	1500	-	-	-	-	12	14.2	21.6	**2700**	10.9	**4010**	17.4	6.1	-	-	-	-	-	-
Dissolved Fluoride F	1.5	15	-	-	-	-	< 0.25	< 0.1	< 0.25	< 0.5	< 0.1	< 0.5	< 0.05	< 0.1	-	-	-	-	-	-
Dissolved Nitrate N	10	400	-	-	-	-	< 0.25	< 0.1	< 0.25	< 0.5	1.15	< 0.5	1.91	0.15	-	-	-	-	-	-
Nitrite N	1	0.2-2[CI]	-	-	-	-	< 0.002	< 0.002	< 0.002	< 0.002	0.023	< 0.002	< 0.002	< 0.002	-	-	-	-	-	-
Nitrate and Nitrite N	10	400	-	-	-	-	< 0.25	< 0.1	< 0.25	< 0.5	1.13	< 0.5	1.91	0.15	-	-	-	-	-	-
Dissolved Sulphate SO_4	**500**	1280-4290[H]	-	-	-	-	14.6	15.1	23.8	**561**	103	466	5.72	85.2	-	-	-	-	-	-
Dissolved Metals																				
aluminum	9.5		0.017	0.086	0.071	0.005	0.015	0.013	0.093	0.023	0.014	0.019	0.014	0.01	0.079	0.013	0.036	-	0.011	0.026
antimony	0.006	2.5	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	0.0018	< 0.0005	0.0009	-	< 0.0005	< 0.0005
arsenic	**0.01**	0.125	0.009	0.001	< 0.001	< 0.001	< 0.001	0.001	< 0.001	0.003	0.002	0.004	< 0.001	< 0.001	0.005	0.003	0.004	-	< 0.001	0.003
barium	1	5	0.026	0.044	0.059	0.032	0.05	0.034	0.091	0.043	0.051	0.1	0.036	0.14	0.042	0.024	0.04	-	0.043	0.16
beryllium	0.008	1	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	-	< 0.0005	< 0.0005
bismuth			< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	0.0007	< 0.0005	< 0.0005	< 0.0005	< 0.0005	0.0005	-	< 0.0005	< 0.0005
boron	5	12	0.04	0.12	0.03	0.13	0.15	0.23	0.59	1.9	1.13	0.79	0.04	0.32	0.04	0.07	0.08	-	0.12	0.05
cadmium	0.005	0.015	< 0.00005	0.00066	0.00024	< 0.00005	< 0.00005	< 0.00005	< 0.00005	0.00012	< 0.00005	< 0.00005	< 0.00005	0.00009	< 0.00005	< 0.00005	< 0.00005	-	0.00014	< 0.00005
calcium			36.8	31.6	16.6	54.1	71.7	74.6	142	143	227	110	30.5	215	31.3	41.9	42.3	-	58.4	49.5
cesium			<0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	<0.0005	<0.0005	<0.0005	-	<0.0005	<0.0005
chromium (IV,III)	0.05,6[V]	0.015,0.56[V]	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	-	< 0.001	< 0.001
cobalt	0.02	0.04	< 0.0005	0.014	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	0.001	< 0.0005	0.0007	0.0014	0.0049	0.0029	-	0.0035	0.001
copper	1.5	0.02	0.0009	0.0036	0.0021	0.0009	0.0012	0.0047	0.016	0.013	0.0027	0.0011	0.0045	0.003	0.0038	0.0025	0.0005	-	0.0014	0.0007
iron	**6.5**		1.69	4.25	0.08	0.49	0.72	< 0.05	0.18	< 0.05	0.08	< 0.05	< 0.05	0.14	0.36	0.18	0.75	-	0.09	0.7
lanthanum			<0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	<0.0005	<0.0005	<0.0005	-	<0.0005	<0.0005
lead	0.01	0.02	< 0.00025	0.0003	< 0.00025	< 0.00025	< 0.00025	< 0.00025	< 0.00025	< 0.00025	< 0.00025	< 0.00025	< 0.00025	< 0.00025	0.0046	< 0.00025	< 0.00025	-	< 0.00025	< 0.00025
lithium	**0.033***		< 0.0005	0.0006	0.0009	0.0079	0.012	0.0099	0.016	**0.059**	0.029	**0.068**	0.0011	0.028	< 0.0005	< 0.0005	0.0017	-	< 0.0005	0.001
magnesium			8.04	5.42	3.5	15.1	13	29.3	19.7	143	53.6	314	2.26	36.5	6.61	11.2	9.38	-	14.5	12.3
manganese	**1.5**		0.689	1.31	0.353	0.934	0.059	0.127	0.01	0.0036	0.185	0.205	< 0.0005	0.092	1.05	**2.37**	1.29	-	1.11	0.572
mercury	0.001	0.00025	< 0.00002	< 0.00002	< 0.00002	< 0.00002	< 0.00002	< 0.00002	0.00005	< 0.00002	< 0.00002	< 0.00002	< 0.00002	< 0.00002	< 0.00002	< 0.00002	< 0.00002	-	< 0.00002	< 0.00002
molybdenum	0.25	10	0.0007	0.0006	< 0.0005	< 0.0005	0.0008	0.001	< 0.0005	0.0037	0.0006	0.0023	< 0.0005	< 0.0005	0.018	0.0023	0.0095	-	0.0027	0.001
nickel	0.08	0.083	0.001	0.007	< 0.001	< 0.001	< 0.001	0.001	0.003	0.002	0.002	0.003	< 0.001	< 0.001	0.003	0.003	0.004	-	0.006	0.001
phosphorus			< 0.075	< 0.075	< 0.075	< 0.075	< 0.075	< 0.075	< 0.075	< 0.075	< 0.075	0.26	< 0.075	< 0.075	< 0.075	< 0.075	< 0.075	-	< 0.075	< 0.075
potassium			1.19	1.51	0.91	1.01	6.39	14	5.49	67.2	6.65	80.5	0.72	4.35	4.42	1.22	3.54	-	1.64	0.88
rhenium			<0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	<0.0005	<0.0005	<0.0005	-	<0.0005	<0.0005
rubidium			0.0007	0.0008	0.0011	0.0005	0.0027	0.0043	0.0018	0.021	0.0026	0.0064	< 0.0005	0.0022	0.004	0.001	0.0023	-	0.0013	0.001
selenium	0.01	0.02	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	0.002	< 0.001	< 0.001	< 0.001	-	< 0.001	< 0.001
silicon			12.5	4.9	6.8	10.8	3.9	2.8	4.9	4	4.4	4	2.1	3.7	7	12	7.2	-	12.7	12.9
silver	0.02	0.015	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	-	< 0.0002	< 0.0002
sodium	**200[i]**		12.5	11	7.25	23.6	15.9	24.4	26.4	**1820**	20.7	**2170**	7.58	4.59	29.2	11.9	28.8	-	26.4	12
strontium	**2.5**		0.207	0.23	0.18	0.668	1.1	0.657	1.07	1.49	**2.68**	2.37	0.191	**3.23**	0.187	0.252	0.198	-	0.441	0.287
sulphur			<5	< 5	< 5	< 5	5	< 5	10	209	44	168	< 5	34	10	<5	13	-	<5	<5
tellurium			<0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	<0.001	<0.001	<0.001	-	<0.001	<0.001
thallium		0.003	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	-	< 0.0001	< 0.0001
thorium			0.0004	< 0.00025	< 0.00025	< 0.00025	0.0003	0.0004	0.0006	0.0008	0.0004	0.0019	0.0003	0.0004	0.0006	0.0004	0.0005	-	0.0003	0.0003
tin	2.5		<0.0006	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	0.0037	< 0.0005	< 0.0005	<0.0006	<0.0006	<0.0006	-	<0.0006	<0.0006
titanium		1	< 0.001	0.002	0.001	< 0.001	< 0.001	< 0.001	0.002	0.002	< 0.001	0.004	< 0.001	< 0.001	0.003	< 0.001	0.002	-	< 0.001	< 0.001
tungsten			0.005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	0.007	0.005	0.005	-	0.005	0.005
uranium	0.02	0.085	< 0.00025	< 0.00025	< 0.00025	< 0.00025	< 0.00025	0.0003	< 0.00025	0.001	0.0007	0.0008	< 0.00025	0.0008	< 0.00025	< 0.00025	0.0005	-	0.0004	< 0.00025
vanadium	0.02		0.0007	0.005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	0.0006	0.0008	0.0007	0.0006	< 0.0005	< 0.0005	0.0016	0.0007	0.001	-	0.001	< 0.0005
zinc	3	0.1	0.008	0.19	0.008	< 0.005	< 0.005	0.009	< 0.005	0.005	< 0.005	< 0.005	0.005	0.006	0.016	0.008	< 0.005	-	< 0.005	0.006
zirconium			< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	-	< 0.0005	< 0.0005

Notes:
All concentrations in milligrams per litre (mg/L), unless otherwise noted.
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019).
Abbreviations: AW (Aquatic Life), M = Marine; DW (Drinking Water)
H = standard is Hardness dependent, Cl = standard is chloride dependent, i= standard for sodium ion conservatively applied
V= Standard is valence dependent VI refers to chromium VI and III refers to chromium III
FDA = field duplicate available; FD = field duplicate; SCN = sample control number
italics indicate the concentration exceeds the laboratory detection limit.
*Background concentration for lithium and coblat as detailed in Technical Bulletin 3 is applied to replace CSR DW Standard.

1	Exceeds CSR Standards for DW
1	Exceeds CSR Standards for AW-M
1	Exceeds CSR Standards for DW and AW-M

TABLE 4a
Results of Groundwater Analyses
Ladysmith Harbour, Ladysmith, BC
Metals and Anions

Location	CSR Standards for DW	CSR Standards for AW-M	MW09-07	MW09-7	MW09-07	MW09-08	MW09-08	MW09-08	MW09-8	MW09-8	MW09-9	MW09-9	MW09-10	MW09-11	MW09-16	MW11-01	MW11-01	MW11-02	MW11-02	Sea Water	WHARF
Date			2011-02-16	2009-11-20	2011-02-22	2011-02-16	2011-02-16	2011-02-22	2009-11-20	2009-11-20	2009-11-25	2009-11-25	2009-11-19	2009-11-19	2009-11-20	2011-02-16	2011-02-22	2011-02-16	2011-02-22	2009-11-20	2011-02-22
SCN			0531-01	21368-07	0532-01	0531-02 FDA	0531-03 FD	0532-02	21368-05 FDA	21368-06 FD	21369-09 FDA	21369-10 FD	21367-05	21367-06	21368-02	0531-04	0532-03	0531-05	0532-04	21368-01	0532-05
QA/QC																					
Field Parameters																					
pH (pH units)			6.34	7.22	6.49	6.99	6.99	6.68	6.92	6.92	6.75	6.75	8.13	7.88	7.62	7.69	6.76	8.05	7.87	-	-
Temperature (°C)			6.6	11.33	6.2	7.77	7.77	6.84	9.66	9.66	10.3	10.3	11.49	11.31	10.51	6.6	5.1	10.2	10.03	-	-
Conductivity (µS/cm)			2960	1420	4100	8360	8360	33841	14210	14210	402	402	401	410	347	4223	41100	402	437	-	-
Redox (mV)			-	13	-	-95.9	-95.9	65.9	45.1	45.1	-54.8	-54.8	6.7	12.6	24.7	192.9	-	158.7	51.3	-	-
Dissolved Oxygen (%)			-	8.91	-	-	-	2.325	6.7	6.7	7.87	7.87	10.9	7.1	10.7	-	-	-	-	-	-
Laboratory Parameters																					
pH (lab)			7.15	-	7.11	7.50	7.55	7.53	-	-	-	-	-	-	-	7.78	7.57	7.95	8.27	-	7.70
Conductivity (µS/cm_)				-					18900	22400	-	-	-	-	436					13100	
Salinity (Low Level) (g/L)				-					11.3	13.5	-	-	-	-	0.21					7.57	
Hardness $CaCO_3$			185	102	1010	694	684	3850	1480	1890	149	149	149	145	32	425	4330	83.5	73.2	1030	4710
Bicarbonate Alkalinity HCO_3			187	-	153	133	129	119	112	123	-	-	-	-	105	106	116	107	140	43.1	123
Carbonate Alkalinity CO_3			<2.0	-	<2.0	<2.0	<2.0	<2.0	< 0.5	< 0.5	-	-	-	-	< 0.5	<2.0	<2.0	<2.0	<2.0	< 0.5	<2.0
Hydroxide Alkalinity OH			<2.0	-	<2.0	<2.0	<2.0	<2.0	< 0.5	< 0.5	-	-	-	-	< 0.5	<2.0	<2.0	<2.0	<2.0	< 0.5	<2.0
Total Alkalinity $CaCO_3$			187	-	153	133	129	119	91.7	100	-	-	-	-	86.3	106	116	107	140	35.3	123
Dissolved Bromide Br			<1.0	-	5.5	5.9	5.2	28.8	-	-	-	-	-	-	-	5.0	35.6	<0.050	<0.050	-	39.0
Dissolved Chloride Cl	**250**	1500	**384**	-	**2170**	**2340**	**2150**	**10800**	**5410**	**6480**	-	-	-	-	76.4	**2000**	**12800**	9.04	18.1	**3470**	**14500**
Dissolved Fluoride F	1.5	15	<0.40	-	<0.40	<0.40	<0.40	0.404	< 1.25	*< 2.5*	-	-	-	-	< 0.25	<0.40	0.513	0.122	0.261	< 1.25	0.714
Dissolved Nitrate N	10	400	<0.10	-	<0.10	<0.10	<0.10	<0.50	< 1.25	< 2.5	-	-	-	-	< 0.25	<0.10	<0.50	0.0871	<0.0050	< 1.25	<0.50
Nitrite N	1	0.2-2[CI]	<0.020	-	<0.020	<0.020	<0.020	*<0.10*	< 0.002	< 0.002	-	-	-	-	0.005	0.025	<0.10	0.0026	<0.0010	< 0.002	<0.10
Nitrate and Nitrite N	10	400	-	-	-	-	-	-	< 1.25	< 2.5	-	-	-	-	< 0.25	-	-	-	-	< 1.25	-
Dissolved Sulphate SO_4	**500**	1280-4290[H]	59	-	314	290	264	**1510**	**686**	**843**	-	-	-	-	20.5	279	**1790**	25.8	48.7	476	**2030**
Dissolved Metals																					
aluminum	9.5		<0.050	0.017	<0.20	<0.20	<0.20	<1.0	0.012	0.014	0.011	0.016	0.014	0.012	0.046	<0.10	<1.0	0.021	0.013	0.059	<1.0
antimony	0.006	2.5	<0.0025	< 0.0005	*<0.010*	*<0.010*	*<0.010*	*<0.050*	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	<0.0050	*<0.050*	0.00555	*0.00993*	< 0.0005	*<0.050*
arsenic	**0.01**	0.125	<0.0050	0.002	*<0.020*	*<0.020*	*<0.020*	*<0.10*	**0.011**	**0.014**	0.001	0.001	0.002	0.01	0.003	<0.010	*<0.10*	**0.0109**	**0.0304**	0.009	*<0.10*
barium	1	5	0.138	0.12	0.751	<0.040	<0.040	<0.20	0.18	0.21	0.012	0.012	0.034	0.041	0.014	0.027	<0.20	<0.020	<0.020	0.017	<0.20
beryllium	0.008	1	<0.0050	< 0.0005	*<0.010*	*<0.010*	*<0.010*	*<0.050*	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	<0.0050	*<0.050*	<0.0050	<0.0050	< 0.0005	*<0.050*
bismuth			<0.20	< 0.0005	<0.40	<0.40	<0.40	<2.0	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	<0.20	<2.0	<0.20	<0.20	< 0.0005	<2.0
boron	5	12	0.28	0.13	0.52	0.66	0.65	2.5	0.91	1.06	0.07	0.06	0.06	0.05	0.12	0.44	2.9	<0.10	<0.10	0.72	3.4
cadmium	0.005	0.015	<0.00025	0.0001	<0.0010	*<0.0010*	<0.0010	*<0.0050*	< 0.00005	< 0.00005	6.00E-05	0.00007	< 0.00005	< 0.00005	< 0.00005	<0.00050	*<0.0050*	0.000055	<0.000050	0.00018	*<0.0050*
calcium			17.8	15	90.3	50.0	49.1	262	136	169	41.4	41.5	39.2	42.6	5.25	35.6	305	26.1	20.0	78	323
cesium				< 0.0005					< 0.0005	< 0.0005	< 0.0005	< 0.0005	<0.0005	<0.0005	< 0.0005					< 0.0005	
chromium (IV,III)	0.05,6[V]	0.015,0.56[V]	<0.0025	< 0.001	<0.010	<0.010	<0.010	*<0.050*	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	<0.0050	*<0.050*	<0.00050	<0.0010	< 0.001	*<0.050*
cobalt	0.02	0.04	<0.0025	0.0035	*<0.010*	*<0.010*	*<0.010*	*<0.050*	0.0014	0.0016	0.0035	0.0035	0.0028	0.0026	0.0011	*<0.0050*	*<0.050*	<0.00050	<0.00050	< 0.0005	*<0.050*
copper	1.5	0.02	<0.0050	0.0025	<0.020	<0.020	<0.020	*<0.10*	0.0018	0.0019	0.0014	0.0012	0.0011	0.0024	0.0019	<0.010	*<0.10*	0.0047	<0.0010	0.0018	*<0.10*
iron	**6.5**		2.97	0.35	**7.68**	0.087	0.088	<0.30	0.35	0.64	< 0.05	< 0.05	1.64	0.94	0.46	<0.030	<0.30	<0.030	<0.030	0.06	<0.30
lanthanum				< 0.0005					< 0.0005	< 0.0005	< 0.0005	< 0.0005	<0.0005	<0.0005	< 0.0005					< 0.0005	
lead	0.01	0.02	<0.0050	0.0003	*<0.020*	*<0.020*	*<0.020*	*<0.10*	< 0.00025	< 0.00025	< 0.00025	< 0.00025	< 0.00025	< 0.00025	< 0.00025	<0.010	*<0.10*	<0.0010	<0.0010	< 0.00025	*<0.10*
lithium	**0.033***		*<0.050*	0.0021	<0.10	<0.10	<0.10	*<0.50*	**0.035**	**0.039**	0.0008	0.0012	< 0.0005	0.0006	0.0021	*<0.050*	*<0.50*	*<0.050*	*<0.050*	0.026	*<0.50*
magnesium			34.1	15.7	191	138	136	775	277	357	11.1	10.9	12.4	9.25	4.57	81.6	866	4.46	5.66	202	948
manganese	**1.5**		0.506	0.731	1.48	0.295	0.289	<0.10	1.04	1.4	**2.45**	**2.47**	1.32	**3.77**	0.271	0.751	1.01	0.224	0.070	0.0091	<0.10
mercury	0.001	0.00025	<0.00020	< 0.00002	<0.00020	<0.00020	<0.00020	<0.00020	< 0.00002	< 0.00002	< 0.00002	< 0.00002	< 0.00002	< 0.00002	< 0.00002	<0.00020	<0.00020	<0.00020	<0.00020	< 0.00002	<0.00020
molybdenum	0.25	10	<0.0050	0.0007	<0.020	<0.020	<0.020	<0.10	0.0054	0.0064	0.0061	0.006	0.0009	0.0033	0.0018	<0.010	<0.10	0.0031	0.0045	0.0028	<0.10
nickel	0.08	0.083	<0.025	0.005	*<0.10*	*<0.10*	*<0.10*	<0.50	0.006	0.006	0.005	0.004	0.003	0.003	0.002	<0.050	*<0.50*	<0.0050	<0.0050	0.002	*<0.50*
phosphorus			0.40	< 0.075	<0.60	<0.60	<0.60	<3.0	< 0.075	< 0.075	< 0.075	< 0.075	< 0.075	< 0.075	< 0.075	<0.30	<3.0	<0.30	<0.30	< 0.075	<3.0
potassium			15.7	8.42	42.6	54.2	53.3	221	85.7	104	0.79	0.78	0.88	1.2	3.91	35.4	254	<2.0	<2.0	66.4	301
rhenium			-	< 0.0005	-	-	-	-	< 0.0005	< 0.0005	< 0.0005	< 0.0005	<0.0005	<0.0005	< 0.0005	-	-	-	-	< 0.0005	-
rubidium			-	0.0029	-	-	-	-	0.03	0.034	0.0005	0.0005	0.0007	0.001	0.0012	-	-	-	-	0.024	-
selenium	0.01	0.02	<0.0050	< 0.001	*<0.040*	*<0.040*	*<0.040*	*<0.20*	< 0.001	0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	*<0.020*	*<0.20*	<0.0010	<0.0010	< 0.001	*<0.20*
silicon			7.72	5.2	6.60	4.88	4.77	2.71	2.6	2.4	10.9	10.9	12.5	13.7	5.6	5.20	2.23	5.04	6.62	2.3	1.82
silver	0.02	0.015	<0.00025	< 0.0002	<0.0010	<0.0010	<0.0010	<0.0050	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	< 0.0002	<0.00050	<0.0050	<0.000050	<0.000050	< 0.0002	<0.0050
sodium	**200[I]**		**287**	97.8	**1210**	**1270**	**1250**	**6310**	**2800**	**3450**	12.5	12.9	11	18.1	73.4	**869**	**7520**	33.6	70.1	**1910**	**8590**
strontium	**2.5**		0.299	0.209	1.50	0.839	0.827	**4.79**	**2.8**	**3.43**	0.248	0.256	0.304	0.35	0.051	0.524	**5.37**	0.209	0.289	1.71	**5.75**
sulphur			-	10	-	-	-	-	249	310	< 5	< 5	<5	<5	< 5	-	-	-	-	166	-
tellurium			-	< 0.001	-	-	-	-	< 0.001	< 0.001	< 0.001	< 0.001	<0.001	<0.001	< 0.001	-	-	-	-	< 0.001	-
thallium		0.003	<0.0010	< 0.0001	*<0.0040*	*<0.0040*	*<0.0040*	*<0.020*	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	< 0.0001	<0.0020	*<0.020*	<0.00020	<0.00020	< 0.0001	*<0.020*
thorium			-	0.0005	-	-	-	-	0.0012	0.0013	0.0003	0.0003	0.0004	0.0004	0.0005	-	-	-	-	0.0011	-
tin	2.5		<0.030	< 0.0005	<0.060	<0.060	<0.060	<0.30	< 0.0005	< 0.0005	< 0.0005	< 0.0005	<0.0006	<0.0006	< 0.0005	<0.030	<0.30	<0.030	<0.030	< 0.0005	<0.30
titanium		1	<0.050	0.001	<0.10	<0.10	<0.10	<0.50	0.004	0.005	< 0.001	< 0.001	< 0.001	< 0.001	0.002	<0.050	<0.50	<0.050	<0.050	0.003	<0.50
tungsten			-	< 0.0005	-	-	-	-	< 0.0005	< 0.0005	< 0.0005	< 0.0005	0.005	0.005	< 0.0005	-	-	-	-	< 0.0005	-
uranium	0.02	0.085	<0.0010	< 0.00025	<0.0040	<0.0040	<0.0040	<0.020	0.001	0.0012	0.0003	0.0003	< 0.00025	0.0003	< 0.00025	<0.0020	<0.020	0.00190	0.00252	0.0006	<0.020
vanadium	0.02		*<0.030*	< 0.0005	*<0.060*	*<0.060*	*<0.060*	*<0.30*	< 0.0005	< 0.0005	0.0009	0.0008	0.0007	0.0015	0.0015	*<0.030*	*<0.30*	*<0.030*	*<0.030*	< 0.0005	*<0.30*
zinc	3	0.1	0.0081	0.011	0.013	<0.010	<0.010	<0.050	< 0.005	< 0.005	0.005	0.005	< 0.005	0.01	0.005	<0.0050	<0.050	<0.0050	<0.0050	< 0.005	<0.050
zirconium				< 0.0005					< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005	< 0.0005					< 0.0005	

Notes:
All concentrations in milligrams per litre (mg/L), unless otherwise noted.
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375
Abbreviations: AW (Aquatic Life), M = Marine; DW (Drinking Water)
H = standard is Hardness dependent, Cl = standard is chloride dependent, i= sta
V= Standard is valence dependent VI refers to chromium VI and III refers to chrc
FDA = field duplicate available; FD = field duplicate; SCN = sample control num
italics indicate the concentration exceeds the laboratory detection limit.
*Background concentration for lithium and coblat as detailed in Technical Bulletir

1 Exceeds CSR Standards for DW
1 Exceeds CSR Standards for AW-M

1 Exceeds CSR Standards for DW and AV

TABLE4b
Results of Groundwater Monitoring Analyses
Ladysmith Harbour, Ladysmith, BC
Hydrocarbons

Location	CSR Standards for DW	CSR Standards for AW-M	MW00-2	MW00-04	MW00-09	MW00-11	MW05-02	MW05-04	MW05-08	MW05-09	MW05-12	MW05-17	MW05-20	MW05-22	MW09-1	MW09-2	MW09-3	MW09-03	MW09-4	MW09-5	MW09-6
SCN			21367-04	21382-03	21382-02	21382-01	21369-04	21369-05	21369-02	21369-03	21369-08	21369-01	21369-07	21369-06	21367-01	21367-02	21367-03	21382-04	21367-07	21367-08	21368-03
Date			18-Nov-09	10-Dec-09	10-Dec-09	10-Dec-09	25-Nov-09	25-Nov-09	25-Nov-09	25-Nov-09	25-Nov-09	25-Nov-09	25-Nov-09	25-Nov-09	18-Nov-09	18-Nov-09	18-Nov-09	10-Dec-09	19-Nov-09	19-Nov-09	20-Nov-09
QA/QC																					
Parameters																					
pH (field) (pH units)			7.55	6.66	6.33	7.33	7.04	7.06	6.74	7.23	6.6	7.51	8.4	6.59	7.97	7.36	7.69	7.31	7.64	7.6	6.76
Temperature (°C)			10.8	8.69	9.21	10.4	9.4	10.62	10.9	13.61	12.61	11.32	9.4	11.12	10.29	10.44	10.44	11.7	12.8	10.33	11.2
Conductivity (µS/cm)			242	230	96	386	531	510	1054	9547	1331	33734	900	1032	301	370	390	355	460	410	411
Redox (mV)			31.6	-62.7	144.2	18.7	-15.9	-34	-39.2	17.3	110.9	-219.6	-121.6	-61.6	-130.6	-80	-121.6	-116.9	45.3	-50.4	31
Dissolved Oxygen (%)			11.7	2.91	4.41	2.67	10.9	9.4	13.9	18.6	21.1	14.4	60.2	9.6	36	23.6	14.3	1.56	8.9	11	-
hardness (mg/L)			125	101	56	198	233	307	435	944	790	1570	86	688	106	151	144	-	206	175	-
Monoaromatic Hydrocarbons																					
benzene	**5**	1,000	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	**56**	< 0.1	< 0.1	-	< 0.1	0.2	-
ethylbenzene	140	2,500	0.2	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	87	0.3	< 0.1	-	< 0.1	< 0.1	-
styrene	800	720	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 1	< 0.1	< 0.1	-	< 0.1	< 0.1	-
toluene	**60**	2,000	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	**198**	0.2	< 0.1	-	< 0.1	0.2	-
ortho-Xylene			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
meta- & para-Xylene			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
total xylene	**90**	300	1.3	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	**1,200**	2.1	0.4	-	< 0.1	< 0.1	-
Methyl t-butyl ether (MTBE)	95	4400	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
VH_{w6-10}	15,000	15,000	< 100	< 100	< 100	< 100	< 100	< 100	< 100	< 100	< 100	< 100	< 100	< 100	5,300	< 100	< 100	-	< 100	< 100	-
VPH_w		1,500	< 100	< 100	< 100	< 100	< 100	< 100	< 100	< 100	< 100	< 100	< 100	< 100	3,800	< 100	< 100	-	< 100	< 100	-
Polycyclic Aromatic Hydrocarbons																					
acenaphthene	250	60	< 0.1	< 0.1	< 0.1	< 0.1	< 0.2	< 0.2	< 0.2	< 0.2	< 0.2	< 0.2	< 0.2	< 0.2	< 0.1	< 0.1	< 0.1	-	< 0.1	< 0.1	< 0.1
acenaphthylene			< 0.1	< 0.1	< 0.1	< 0.1	< 0.2	< 0.2	< 0.2	< 0.2	< 0.2	< 0.2	< 0.2	< 0.2	< 0.1	< 0.1	< 0.1	-	< 0.1	< 0.1	< 0.1
acridine		0.5	< 0.05	< 0.05	< 0.05	< 0.05	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.05	< 0.05	< 0.05	-	< 0.05	< 0.05	< 0.05
anthracene	1000	1	< 0.01	< 0.01	< 0.01	< 0.01	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	0.01	< 0.01	< 0.01	-	< 0.01	0.02	< 0.01
benzo(a)anthracene	0.07	1	< 0.01	< 0.01	< 0.01	< 0.01	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.01	< 0.01	< 0.01	-	< 0.01	0.01	< 0.01
benzo(a)pyrene	**0.01**	0.1	< 0.01	< 0.01	< 0.01	< 0.01	*< 0.02*	*< 0.02*	*< 0.02*	*< 0.02*	*< 0.02*	*< 0.02*	*< 0.02*	*< 0.02*	< 0.01	< 0.01	< 0.01	-	< 0.01	< 0.01	< 0.01
benzo(b)fluoranthene	**0.07**		< 0.01	< 0.01	< 0.01	< 0.01	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.01	< 0.01	< 0.01	-	< 0.01	< 0.01	< 0.01
benzo(g,h,i)perylene			< 0.01	< 0.01	< 0.01	< 0.01	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	0.01	< 0.01	< 0.01	-	< 0.01	< 0.01	< 0.01
benzo(k)fluoranthene			< 0.01	< 0.01	< 0.01	< 0.01	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.01	< 0.01	< 0.01	-	< 0.01	< 0.01	< 0.01
chrysene	7	1	< 0.01	< 0.01	< 0.01	< 0.01	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.01	< 0.01	< 0.01	-	< 0.01	0.01	< 0.01
dibenzo(a,h)anthracene	0.1		< 0.01	< 0.01	< 0.01	< 0.01	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.01	< 0.01	< 0.01	-	< 0.01	< 0.01	< 0.01
fluoranthene	150	2	< 0.04	< 0.04	< 0.04	< 0.04	< 0.08	< 0.08	< 0.08	< 0.08	< 0.08	< 0.08	< 0.08	< 0.08	< 0.04	< 0.04	< 0.04	-	< 0.04	0.05	< 0.04
fluorene	150	120	< 0.05	< 0.05	< 0.05	< 0.05	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.05	< 0.05	0.06	-	< 0.05	0.06	< 0.05
indeno(1,2,3-c,d)pyrene			< 0.01	< 0.01	< 0.01	< 0.01	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	< 0.02	0.01	< 0.01	< 0.01	-	< 0.01	< 0.01	< 0.01
naphthalene	80	10	< 0.3	< 0.3	< 0.3	< 0.3	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	8.9	0.4	0.4	-	< 0.3	0.5	< 0.3
phenanthrene		3	< 0.05	< 0.05	< 0.05	< 0.05	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	0.06	< 0.05	0.06	-	< 0.05	0.12	< 0.05
pyrene	100	0.2	< 0.02	< 0.02	< 0.02	< 0.02	< 0.04	< 0.04	< 0.04	< 0.04	< 0.04	< 0.04	< 0.04	< 0.04	< 0.02	< 0.02	< 0.02	-	< 0.02	0.04	0.03
quinoline	0.5	34	< 0.5	< 0.5	< 0.5	< 0.5	*< 1*	*< 1*	*< 1*	*< 1*	*< 1*	*< 1*	*< 1*	*< 1*	< 0.5	< 0.5	< 0.5	-	< 0.5	< 0.5	< 0.5
Other Hydrocarbons																					
$EPHw_{10-19}$	5,000	5,000	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	370	< 250	< 250	-	< 250	< 250	< 250
$EPHw_{19-32}$			600	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	-	< 250	< 250	< 250
LEPHw		500	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	360	< 250	< 250	-	< 250	< 250	< 250
HEPH			600	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	< 250	-	< 250	< 250	< 250

Notes:
All concentrations in micrograms per litre (µg/L), unless otherwise noted.
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019).
Abbreviations: AW (Aquatic Life), M = Marine; DW (Drinking Water)
EPHC10-19 = extractable petroleum hydrocarbons, carbon range 10-19
EPHC19-32 = extractable petroleum hydrocarbons, carbon range 19-32
LEPH = light extractable petroleum hydrocarbons
HEPH = heavy extractable petroleum hydrocarbons
PAH = Polycyclic Aromatic Hydrocarbons
VPH = volatile petroleum hydrocarbons
FDA = field duplicate available; FD = field duplicate; SCN = sample control number
italics indicate the concentration exceeds the laboratory detection limit.

1 Exceeds CSR Standards for DW
1 Exceeds CSR Standards for AW-M

1 Exceeds CSR Standards for DW and AW-M

TABLE4b
Results of Groundwater Monitoring Analyses
Ladysmith Harbour, Ladysmith, BC
Hydrocarbons

Location	CSR Standards for DW	CSR Standards for AW-M	MW09-07	MW09-07	MW09-7	MW09-08	MW09-08	MW09-8	MW09-8	MW09-08	MW09-9	MW09-9	MW09-10	MW09-11	MW09-16	MW11-01	MW11-01	MW11-02	MW11-02
SCN			0531-01	0532-01	21368-07	0531-02	0531-03	21368-05	21368-06	0532-02	21369-09	21369-10	21367-05	21367-06	21368-02	0531-04	0532-03	0531-05	0532-04
Date			16-Feb-11	22-Feb-11	20-Nov-09	16-Feb-11 FDA	16-Feb-11 FD	20-Nov-09 FDA	20-Nov-09 FD	22-Feb-11	25-Nov-09 FDA	25-Nov-09 FD	19-Nov-09	19-Nov-09	20-Nov-09	16-Feb-11	22-Feb-11	16-Feb-11	22-Feb-11
QA/QC																			
Parameters																			
pH (field) (pH units)			6.34	6.49	7.22	6.99	6.99	6.92	6.92	6.68	6.75	6.75	8.13	7.88	7.62	7.69	6.76	8.05	7.87
Temperature (°C)			6.6	6.2	11.33	7.77	7.77	9.66	9.66	6.84	10.3	10.3	11.49	11.31	10.51	6.6	5.1	10.2	10.03
Conductivity (µS/cm)			2960	4100	1420	8360	8360	14210	14210	33841	402	402	401	410	347	4223	41100	402	437
Redox (mV)			-	-	13	-95.9	-95.9	45.1	45.1	65.9	-54.8	-54.8	6.7	12.6	24.7	192.9	-	158.7	51.3
Dissolved Oxygen (%)			-	-	8.91	-	-	6.7	6.7	2.325	7.87	7.87	10.9	7.1	10.7	-	-	-	-
hardness (mg/L)			185	-	102	694	684	1480	1890	-	149	149	149	145	32	425	-	83.5	-
Monoaromatic Hydrocarbons																			
benzene	**5**	1,000	1.07	1.45	1	<0.50	<0.50	< 0.1	< 0.1	<0.50	< 0.1	< 0.1	< 0.1	< 0.1	-	<0.50	<0.50	<0.50	<0.50
ethylbenzene	140	2,500	0.59	<0.50	2.3	<0.50	<0.50	< 0.1	< 0.1	<0.50	< 0.1	< 0.1	< 0.1	< 0.1	-	<0.50	<0.50	<0.50	<0.50
styrene	800	720	<0.50	<0.50	< 0.1	<0.50	<0.50	< 0.1	< 0.1	<0.50	< 0.1	< 0.1	< 0.1	< 0.1	-	<0.50	<0.50	<0.50	<0.50
toluene	**60**	2,000	<1.0	<1.0	< 0.1	<1.0	<1.0	< 0.1	< 0.1	<1.0	< 0.1	< 0.1	< 0.1	< 0.1	-	<1.0	<1.0	<1.0	<1.0
ortho-Xylene			<0.50	<0.50	-	<0.50	<0.50	-	-	<0.50	-	-	-	-	-	<0.50	<0.50	<0.50	<0.50
meta- & para-Xylene			<0.50	<0.50	-	<0.50	<0.50	-	-	<0.50	-	-	-	-	-	<0.50	<0.50	<0.50	<0.50
total xylene	**90**	300	<0.71	<0.71	0.5	<0.71	<0.71	< 0.1	< 0.1	<0.71	< 0.1	< 0.1	< 0.1	< 0.1	-	<0.71	<0.71	<0.71	<0.71
Methyl t-butyl ether (MTBE)	95	4400	<1.0	<1.0	-	<1.0	<1.0	-	-	<1.0	-	-	-	-	-	<1.0	<1.0	<1.0	<1.0
VH_{w6-10}	15,000	15,000	140	240	170	<100	<100	< 100	< 100	<100	< 100	< 100	< 100	< 100	-	<100	<100	<100	<100
VPH_w		1,500	140	240	170	<100	<100	< 100	< 100	<100	< 100	< 100	< 100	< 100	-	<100	<100	<100	<100
Polycyclic Aromatic Hydrocarbons																			
acenaphthene	250	60	0.252	0.257	< 0.1	0.898	0.931	< 0.1	< 0.1	0.128	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	0.098	<0.050	<0.050	<0.050
acenaphthylene			<0.050	<0.050	< 0.1	<0.050	<0.050	< 0.1	< 0.1	<0.050	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1	0.084	<0.050	<0.050	<0.050
acridine		0.5	<0.070	<0.050	< 0.05	<0.13	<0.13	< 0.05	< 0.05	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	<0.060	<0.050	<0.050	<0.050
anthracene	1000	1	<0.050	<0.050	< 0.01	<0.10	<0.10	0.06	0.05	<0.050	0.05	0.04	< 0.01	< 0.01	< 0.01	0.145	<0.050	<0.050	<0.050
benzo(a)anthracene	0.07	1	<0.050	<0.050	< 0.01	<0.050	<0.050	0.05	0.04	<0.050	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	0.369	<0.050	<0.050	<0.050
benzo(a)pyrene	**0.01**	0.1	*<0.020*	<0.010	< 0.01	*<0.030*	<0.010	**0.03**	**0.03**	*<0.020*	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	**0.399**	<0.020	<0.010	<0.020
benzo(b)fluoranthene	**0.07**		<0.050	<0.050	< 0.01	<0.050	<0.050	**0.09**	0.07	<0.050	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	**0.467**	<0.050	<0.050	<0.050
benzo(g,h,i)perylene			<0.050	<0.050	< 0.01	<0.050	<0.050	0.03	0.04	<0.050	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	0.267	<0.050	<0.050	<0.050
benzo(k)fluoranthene			<0.050	<0.050	< 0.01	<0.050	<0.050	0.03	0.03	<0.050	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	0.184	<0.050	<0.050	<0.050
chrysene	7	1	<0.050	<0.050	< 0.01	<0.050	<0.050	0.1	0.08	<0.050	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	0.446	<0.050	<0.050	<0.050
dibenzo(a,h)anthracene	0.1		<0.050	<0.050	< 0.01	<0.050	<0.050	0.01	< 0.01	<0.050	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	0.054	<0.050	<0.050	<0.050
fluoranthene	150	2	<0.050	<0.050	< 0.04	<0.050	<0.050	0.22	0.2	<0.050	< 0.04	< 0.04	< 0.04	< 0.04	< 0.04	0.929	<0.050	<0.050	<0.050
fluorene	150	120	0.279	0.235	0.05	<0.050	<0.050	< 0.05	< 0.05	<0.050	< 0.05	< 0.05	< 0.05	< 0.05	< 0.05	0.081	<0.050	<0.050	<0.050
indeno(1,2,3-c,d)pyrene			<0.050	<0.050	< 0.01	<0.050	<0.050	0.03	0.03	<0.050	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	0.292	<0.050	<0.050	<0.050
naphthalene	80	10	<0.82	<0.37	< 0.3	<0.64	<0.29	< 0.3	< 0.3	<0.050	< 0.3	< 0.3	< 0.3	< 0.3	< 0.3	<0.96	<0.050	<0.43	<0.070
phenanthrene		3	0.193	<0.050	< 0.05	<0.050	<0.050	0.12	0.12	<0.050	0.05	< 0.05	< 0.05	< 0.05	< 0.05	0.399	<0.050	<0.050	<0.050
pyrene	100	0.2	<0.050	<0.050	0.02	0.054	0.056	**0.23**	**0.23**	<0.050	0.03	0.03	< 0.02	< 0.02	< 0.02	**0.854**	<0.050	<0.050	<0.050
quinoline	0.5	34	0.138	<0.31	< 0.5	<0.050	<0.050	< 0.5	< 0.5	<0.050	< 0.5	< 0.5	< 0.5	< 0.5	< 0.5	<0.050	<0.050	<0.050	<0.050
Other Hydrocarbons																			
$EPHw_{10-19}$	5,000	5,000	<430	<250	< 250	<340	<280	< 250	< 250	<250	< 250	< 250	< 250	< 250	< 250	<250	<250	<250	<250
$EPHw_{19-32}$			<380	<250	< 250	<510	<250	300	260	<250	< 250	< 250	< 250	< 250	280	1280	<250	<350	330
LEPHw		500	<430	<250	< 250	<340	<280	< 250	< 250	<250	< 250	< 250	< 250	< 250	< 250	<250	<250	<250	<250
HEPH			<380	<250	< 250	<510	<250	300	260	<250	< 250	< 250	< 250	< 250	280	1280	<250	<350	330

Notes:
All concentrations in micrograms per litre (µg/L), unless otherwise noted.
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1
Abbreviations: AW (Aquatic Life), M = Marine; DW (Drinking Water)
EPHC10-19 = extractable petroleum hydrocarbons, carbon range 10-19
EPHC19-32 = extractable petroleum hydrocarbons, carbon range 19-32
LEPH = light extractable petroleum hydrocarbons
HEPH = heavy extractable petroleum hydrocarbons
PAH = Polycyclic Aromatic Hydrocarbons
VPH = volatile petroleum hydrocarbons
FDA = field duplicate available; FD = field duplicate; SCN = sample control number
italics indicate the concentration exceeds the laboratory detection limit.

1 Exceeds CSR Standards for DW
1 Exceeds CSR Standards for AW-M

1 Exceeds CSR Standards for DW and

TABLE 4c
Results of Groundwater Monitoring Analyses
Ladysmith Harbour, Ladysmith, BC
VOCs

Location	CSR Stanards for DW	CSR standards for AW-M	MW09-1	MW09-2	MW09-3	MW00-2	MW09-10	MW09-11	MW09-4	MW09-5	MW09-16	MW09-6	MW09-6	MW09-07	MW09-07	MW09-08	MW09-08	MW09-8	MW09-8	MW09-7	MW05-17
SCN			21367-01	21367-02	21367-03	21367-04	21367-05	21367-06	21367-07	21367-08	21368-02	21368-03	21368-04	0531-01	0532-01	0531-02	0531-03	21368-05	21368-06	21368-07	21369-01
Date			2009-11-18	2009-11-18	2009-11-18	2009-11-18	2009-11-19	2009-11-19	2009-11-19	2009-11-19	2009-11-20	2009-11-20 FDA	2009-11-20 FD	2011-02-16	2011-02-22	2011-02-16 FDA	2011-02-16 FD	2009-11-20 FDA	2009-11-20 FD	2009-11-20	2009-11-25
QA/QC																					
Parameters																					
pH (field) (pH units)			7.97	7.36	7.69	7.55	8.13	7.88	7.64	7.6	7.62	6.76	6.76	6.34	6.49	6.99	6.99	6.92	6.92	7.22	7.51
Temperature (°C)			10.29	10.44	10.44	10.8	11.49	11.31	12.8	10.33	10.51	11.2	11.2	6.6	6.2	7.77	7.77	9.66	9.66	11.33	11.32
Conductivity (µS/cm)			301	370	390	242	401	410	460	410	347	411	411	2960	4100	8360	8360	14210	14210	1420	33734
Redox (mV)			-130.6	-80	-121.6	31.6	6.7	12.6	45.3	-50.4	24.7	31	31	-	-	-95.9	-95.9	45.1	45.1	13	-219.6
Dissolved Oxygen (%)			36	23.6	14.3	11.7	10.9	7.1	8.9	11	10.7	-	-	-	-	-	-	6.7	6.7	8.91	14.4
hardness (mg/L)			106	151	144	125	149	145	206	175	32	-	-	185	-	694	684	1480	1890	102	1570
Halogenated Hydrocarbons																					
1,1,1-Trichloroethane	8000		-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
1,1,2,2-Tetrachloroethane	0.8		-	-	-	-	-	-	-	< 0.2	-	-	-	*<1.0*	*<1.0*	*<1.0*	*<1.0*	< 0.2	-	-	-
1,1,2-Trichloroethane	3		-	-	-	-	-	-	-	< 0.1	-	-	-	<2.0	*<4.0*	<1.0	<1.0	< 0.1	-	-	-
1,1-Dichloroethane	30		-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
1,1-Dichloroethylene	14		-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
1,2-Dibromoethane	0.5		-	-	-	-	-	-	-	< 0.1	-	-	-	-	-	-	-	< 0.1	-	-	-
1,2-Dichlorobenzene	200	420	-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
1,2-Dichloroethane	5	1000	-	-	-	-	-	-	-	< 0.4	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.4	-	-	-
1,2-Dichloropropane	4.5		-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
1,3-Dichlorobenzene		1,500	-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
1,4-Dichlorobenzene	5	260	-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
2-Butanone	2500		-	-	-	-	-	-	-	< 5	-	-	-	-	-	-	-	< 5	-	-	-
2-Hexanone	20		-	-	-	-	-	-	-	< 20	-	-	-	-	-	-	-	< 20	-	-	-
4-Methyl-2-pentanone			-	-	-	-	-	-	-	< 2	-	-	-	-	-	-	-	< 2	-	-	-
Bromodichloromethane	100		-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
Bromoform	100		-	-	-	-	-	-	-	< 0.2	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.2	-	-	-
Bromomethane	5.5		-	-	-	-	-	-	-	< 0.8	-	-	-	-	-	-	-	< 0.8	-	-	-
Carbon Tetrachloride	2	130	-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
Chlorobenzene	80	250	-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
Chloroethane			-	-	-	-	-	-	-	< 0.4	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.4	-	-	-
Chloroform	100	20	-	-	-	-	-	-	-	< 0.3	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.3	-	-	-
Chloromethane			-	-	-	-	-	-	-	< 0.4	-	-	-	<5.0	<5.0	<5.0	<5.0	< 0.4	-	-	-
cis-1,2-Dichloroethylene	8		-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
cis-1,3-Dichloropropene	1.5		-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
Dibromochloromethane	100		-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
Dibromomethane			-	-	-	-	-	-	-	< 0.2	-	-	-	<5.0	<5.0	<5.0	<5.0	< 0.2	-	-	-
Dichlorodifluoromethane	800		-	-	-	-	-	-	-	< 0.2	-	-	-	-	-	-	-	< 0.2	-	-	-
Methylene Chloride	50	980	-	-	-	-	-	-	-	< 6	-	-	-	-	-	-	-	< 6	-	-	-
Tetrachloroethylene	30	1,100	-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
trans-1,2-Dichloroethylene	80		-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
trans-1,3-Dichloropropene	1.5		-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
Trichloroethylene	5	200	-	-	-	-	-	-	-	< 0.1	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.1	-	-	-
Trichlorofluoromethane	1000		-	-	-	-	-	-	-	< 0.2	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.2	-	-	-
Vinyl Chloride	2		-	-	-	-	-	-	-	< 0.2	-	-	-	<1.0	<1.0	<1.0	<1.0	< 0.2	-	-	-

Notes:
All concentrations in micrograms per litre (µg/L), unless otherwise noted.
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, updated to 16 April 2019).
Abbreviations: AW (Aquatic Life), M = Marine; DW (Drinking Water)
FDA = field duplicate available; FD = field duplicate; SCN = sample control number
italics indicate the concentration exceeds the laboratory detection limit.

1 Exceeds CSR Standards for DW
1 Exceeds CSR Standards for AW-M

1 Exceeds CSR Standards for DW and AW-M

TABLE 4c
Results of Groundwater Monitoring Analyses
Ladysmith Harbour, Ladysmith, BC
VOCs

Location			MW05-08	MW05-09	MW05-02	MW05-04	MW05-22	MW05-20	MW05-12	MW09-08	MW09-9	MW09-9	MW00-11	MW00-09	MW00-04	MW09-03	MW11-01	MW11-01	MW11-02	MW11-02
SCN	CSR Stanards for DW	CSR standards for AW-M	21369-02	21369-03	21369-04	21369-05	21369-06	21369-07	21369-08	0532-02	21369-09	21369-10	21382-01	21382-02	21382-03	21382-04	0531-04	0532-03	0531-05	0532-04
Date			2009-11-25	2009-11-25	2009-11-25	2009-11-25	2009-11-25	2009-11-25	2009-11-25	2011-02-22	2009-11-25 FDA	2009-11-25 FD	2009-12-10	2009-12-10	2009-12-10	2009-12-10	2011-02-16	2011-02-22	2011-02-16	2011-02-22
QA/QC																				
Parameters																				
pH (field) (pH units)			6.74	7.23	7.04	7.06	6.59	8.4	6.6	6.68	6.75	6.75	7.33	6.33	6.66	7.31	7.69	6.76	8.05	7.87
Temperature (°C)			10.9	13.61	9.4	10.62	11.12	9.4	12.61	6.84	10.3	10.3	10.4	9.21	8.69	11.7	6.6	5.1	10.2	10.03
Conductivity (µS/cm)			1054	9547	531	510	1032	900	1331	33841	402	402	386	96	230	355	4223	41100	402	437
Redox (mV)			-39.2	17.3	-15.9	-34	-61.6	-121.6	110.9	65.9	-54.8	-54.8	18.7	144.2	-62.7	-116.9	192.9	-	158.7	51.3
Dissolved Oxygen (%)			13.9	18.6	10.9	9.4	9.6	60.2	21.1	2.325	7.87	7.87	2.67	4.41	2.91	1.56	-	-	-	-
hardness (mg/L)			435	944	233	307	688	86	790	-	149	149	198	56	101	-	425	-	83.5	-
Halogenated Hydrocarbons																				
1,1,1-Trichloroethane	8000		-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
1,1,2,2-Tetrachloroethane	0.8		-	-	-	-	-	-	-	*<1.0*	< 0.1	< 0.1	-	-	-	-	*<1.0*	*<1.0*	*<1.0*	*<1.0*
1,1,2-Trichloroethane	3		-	-	-	-	-	-	-	<1.0	< 0.2	< 0.2	-	-	-	-	<1.0	<1.0	<1.0	<1.0
1,1-Dichloroethane	30		-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
1,1-Dichloroethylene	14		-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
1,2-Dibromoethane	0.5		-	-	-	-	-	-	-	-	< 0.1	< 0.1	-	-	-	-	-	-	-	-
1,2-Dichlorobenzene	200	420	-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
1,2-Dichloroethane	5	1000	-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
1,2-Dichloropropane	4.5		-	-	-	-	-	-	-	<1.0	< 0.4	< 0.4	-	-	-	-	<1.0	<1.0	<1.0	<1.0
1,3-Dichlorobenzene		1,500	-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
1,4-Dichlorobenzene	5	260	-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
2-Butanone	2500		-	-	-	-	-	-	-	-	< 0.1	< 0.1	-	-	-	-	-	-	-	-
2-Hexanone	20		-	-	-	-	-	-	-	-	< 5	< 5	-	-	-	-	-	-	-	-
4-Methyl-2-pentanone			-	-	-	-	-	-	-	-	< 2	< 2	-	-	-	-	-	-	-	-
Bromodichloromethane	100		-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
Bromoform	100		-	-	-	-	-	-	-	<1.0	< 0.2	< 0.2	-	-	-	-	<1.0	<1.0	<1.0	<1.0
Bromomethane	5.5		-	-	-	-	-	-	-	-	< 0.8	< 0.8	-	-	-	-	-	-	-	-
Carbon Tetrachloride	2	130	-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
Chlorobenzene	80	250	-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
Chloroethane			-	-	-	-	-	-	-	<1.0	< 0.4	< 0.4	-	-	-	-	<1.0	<1.0	<1.0	<1.0
Chloroform	100	20	-	-	-	-	-	-	-	<1.0	< 0.3	< 0.3	-	-	-	-	<1.0	<1.0	24.1	3.7
Chloromethane			-	-	-	-	-	-	-	<5.0	< 0.4	< 0.4	-	-	-	-	<5.0	<5.0	<5.0	<5.0
cis-1,2-Dichloroethylene	8		-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
cis-1,3-Dichloropropene	1.5		-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
Dibromochloromethane	100		-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
Dibromomethane			-	-	-	-	-	-	-	<5.0	< 0.2	< 0.2	-	-	-	-	<5.0	<5.0	<5.0	<5.0
Dichlorodifluoromethane	800		-	-	-	-	-	-	-	-	< 0.2	< 0.2	-	-	-	-	-	-	-	-
Methylene Chloride	50	980	-	-	-	-	-	-	-	-	< 6	< 6	-	-	-	-	-	-	-	-
Tetrachloroethylene	30	1,100	-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
trans-1,2-Dichloroethylene	80		-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
trans-1,3-Dichloropropene	1.5		-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
Trichloroethylene	5	200	-	-	-	-	-	-	-	<1.0	< 0.1	< 0.1	-	-	-	-	<1.0	<1.0	<1.0	<1.0
Trichlorofluoromethane	1000		-	-	-	-	-	-	-	<1.0	< 0.2	< 0.2	-	-	-	-	<1.0	<1.0	<1.0	<1.0
Vinyl Chloride	2		-	-	-	-	-	-	-	<1.0	< 0.2	< 0.2	-	-	-	-	<1.0	<1.0	<1.0	<1.0

Notes:
All concentrations in micrograms per litre (µg/L), unless otherwise noted.
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96,
Abbreviations: AW (Aquatic Life), M = Marine; DW (Drinking Water)
FDA = field duplicate available; FD = field duplicate; SCN = sample control number
italics indicate the concentration exceeds the laboratory detection limit.

1	Exceeds CSR Standards for DW
1	Exceeds CSR Standards for AW-M
1	Exceeds CSR Standards for DW and

TABLE 6a
Historic Seepage Water Results
- LEPH, HEPH, and PAHs
Ladysmith Harbour, Ladysmith, BC

Location SCN Date Source QA/QC	CSR Standards for DW	CSR Standards for AW-M	Seeps (M505-2) 0046-03 14-Apr-2005 Golder
Polycyclic Aromatic Hydrocarbons			
Acenaphthene	**0.25**	0.06	0.000056
Acenaphthylene			<0.000050
Acridine		0.0005	0.000069
Anthracene	**1**	0.001	0.000116
Benz(a)anthracene	**0.00007**	0.001	**0.000147**
Benzo(a)pyrene	**0.00001**	0.001	**0.000086**
Benzo(b)fluoranthene	**0.00007**		**0.000133**
Benzo(g,h,i)perylene			0.000054
Benzo(k)fluoranthene			<0.000050
Chrysene	**0.007**	0.001	0.000251
Dibenzo(a,h)anthracene	**0.0001**		<0.000050
Fluoranthene	**0.15**	0.002	0.000161
Fluorene	**0.15**	0.120	0.000093
Indeno(1,2,3-c,d)pyrene			<0.000050
Naphthalene	**0.08**	0.01	0.000251
Phenanthrene		0.003	0.00100
Pyrene	**0.1**	0.0002	0.000164
Quinoline	**0.0005**	0.034	<0.000050
EPHw10-19	**5**	5	0.38
EPHw19-32			<1.0
LEPHw		0.5	0.38
HEPHw			<1.0

Notes:
All concentrations in micrograms per litre (mg/L), unless otherwise noted.
Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, January 24, 2019).

Abbreviations: AW (Aquatic Life), M = Marine; DW (Drinking Water)
EPHC10-19 = extractable petroleum hydrocarbons, carbon range 10-19
EPHC19-32 = extractable petroleum hydrocarbons, carbon range 19-32
LEPH = light extractable petroleum hydrocarbons
HEPH = heavy extractable petroleum hydrocarbons
PAH = Polycyclic Aromatic Hydrocarbons
VPH = volatile petroleum hydrocarbons
FDA = field duplicate available; FD = field duplicate; SCN = sample control number
italics indicate the concentration exceeds the laboratory detection limit.
LEPH criteria used as a conservate screen for Mineral Oil and Grease and EHw10-19.

Golder = Golder Associates Ltd. 2005. Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site Investigation Lot 16G Ladysmith Harbour Ladysmith, BC. Dated June 30, 2005.

1	Exceeds CSR Standards for DW
1	Exceeds CSR Standards for AW-M
1	Exceeds CSR Standards for DW and AW-M

TABLE 5b
Historic Seepage Water Results - Total Metals
Ladysmith Harbour, Ladysmith, BC

Location SCN Date Source QA/QC	CSR Standards for DW	CSR Standards for AW-M	Seeps (M505-2) 0046-03 14-Apr-2005 Golder
Physical			
Hardness (as $CaCO_3$)			4210
pH			7.7
Total Metals			
Aluminum	**9.5**		<2.0
Antimony	**0.006**	2.5	*<0.10*
Arsenic	**0.01**	0.125	*<0.20*
Barium	**1**	5	<0.40
Beryllium	**0.008**	1	*<0.10*
Boron	**5**	12	3.1
Cadmium	**0.005**	0.015	*<0.010*
Calcium			277
Chromium	**0.05,6**[V]	0.015,0.56[V]	*<0.10*
Cobalt	**0.001**	0.04	*<0.10*
Copper	**1.5**	0.02	*<0.20*
Iron	**6.5**		<0.60
Lead	**0.01**	0.02	<0.20
Lithium	**0.033***		*<1.0*
Magnesium			854
Manganese	**1.5**		<0.20
Mercury	**0.001**	0.00025	<0.00020
Molybdenum	**0.25**	10	<0.20
Nickel	**0.080**	0.083	*<1.0*
Selenium	**0.01**	0.02	*<0.20*
Silver	**0.02**	0.015	<0.010
Sodium	**200**[I]		**7530**
Thallium		0.003	*<0.040*
Titanium		1	<1.0
Uranium	**0.02**	0.085	*<0.040*
Vanadium	**0.02**		*<0.60*
Zinc	**3**	0.1	<0.10

Notes:
All concentrations in milligrams per litre (mg/L), unless otherwise noted.

Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, January 24, 2019).
Abbreviations: AW (Aquatic Life), M = Marine; DW (Drinking Water)
i= standard for sodium ion conservatively applied
V= Standard is valence dependent VI refers to chromium VI and III refers to chromium III
FDA = field duplicate available; FD = field duplicate; SCN = sample control number
italics indicate the concentration exceeds the laboratory detection limit.
*Background concentration for lithium as detailed in BC CSR Technical Bulletin 3 is applied to replace CSR DW Standard.
Golder = Golder Associates Ltd. 2005. Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site Investigation Lot 16G Ladysmith Harbour Ladysmith, BC. Dated June 30, 2005.

1	Exceeds CSR Standards for DW
1	Exceeds CSR Standards for AW-M
1	Exceeds CSR Standards for DW and AW-M

TABLE 6
Historical Porewater Results - Total and Dissolved Metals
Ladysmith Harbour, Ladysmith, BC

Location Date Source	CSR Standards for DW	CSR Standards for AW-M	E6 R2 Feb-05 G3	F6 R2 Feb-05 G3	F7 R2 Feb-05 G3	I8 R2 Feb-05 G3	J8 R2 Feb-05 G3	J7 R2 Feb-05 G3	N7 R3 Feb-05 G3
Porewater Total Metals									
Arsenic			<10	<10	<10	<10	<10	<10	<10
Cadmium			<0.050	<0.050	<0.050	<0.050	<0.050	<0.050	<0.050
Copper			<10	<10	<10	<10	<10	<10	0.22
Lead			<10	<10	<10	<10	<10	<10	<10
Mercury			0.00022	<0.00020	<0.00020	<0.00020	<0.00020	0.00022	0.00061
Molybdenum			<10	<10	<10	<10	<10	<10	<10
Nickel			<0.50	<0.50	<0.50	<0.50	<0.50	<0.50	<0.50
Zinc			0.151	0.475	0.075	0.165	0.080	0.190	0.243
Porewater Dissolved Metals									
Arsenic	0.010	0.125	*<10*	*<10*	*<10*	*<10*	*<10*	*<10*	*<10*
Cadmium	0.005	0.015	<0.0050	<0.0050	<0.0050	<0.0050	<0.0050	<0.0050	<0.0050
Copper	1.5	0.020	<10	<10	<10	<10	<10	<10	<10
Lead	0.010	0.020	<10	<10	<10	<10	<10	<10	<10
Mercury	0.001	0.00025	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020	<0.00020
Molybdenum	0.250	10	<10	<10	<10	<10	<10	<10	<10
Nickel	0.080	0.083	<10	<10	<10	<10	<10	<10	<10
Zinc	3.00	0.100	<0.050	<0.050	<0.050	<0.050	<0.050	0.07	<0.050

Notes:

All concentrations in milligrams per litre (mg/L), unless otherwise noted.

Standards shown from the Contaminated Sites Regulation ("CSR"; BC Reg. 375/96, O.C. 1480/96 and M271/2004, including amendments up to BC Reg. 13/2019, January 24, 2019).

Abbreviations: AW (Aquatic Life), M = Marine; DW (Drinking Water)

FDA = field duplicate available; FD = field duplicate; SCN = sample control number

italics indicate the concentration exceeds the laboratory detection limit.

Golder = Golder Associates Ltd. 2005. Report on Supplemental Stage 1 Preliminary Site Investigation and Detailed Site Investigation Lot 16G Ladysmith Harbour Ladysmith, BC. Dated June 30, 2005.

1	Exceeds CSR Standards for DW
1	Exceeds CSR Standards for AW-M
1	Exceeds CSR Standards for DW and AW-M



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20 February 2020 Project No. 18109842-002-L-Rev0

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Ladysmith, BC V9G 1A2

REGULATORY PATH TO CLOSURE
LOTS 1, 4 AND 5, TOWN OF LADYSMITH

Mr. Belobaba:

The following outlines the possible regulatory path forward for developing the Town of Ladysmith (the Town) property adjacent to Ladysmith Harbour, including Lots 1, 4 and 5 (the Site). We have attached BC ENV Fact Sheet 9, Highlights for Developers for further information on the site development process.

It should be noted that the regulatory requirements change frequently, and we understand there are changes planned for the regulations in 2020. The information provided here is based on current requirements and the anticipated upcoming changes to the Site Profile Process.

Based on the presence of contamination, investigation and remediation activities in accordance with the BC Ministry of Environment and Climate Change Strategy (ENV) will be required if development is planned. The first section of this letter presents the current CSR requirements for the development process, followed by our opinion of probable costs for implementing the next stages of investigation and remediation for the Site.

1.0 SITE PROFILE

Applying for a development permit will include completion of a Site Profile form. The Site Profile flags potential sources of contamination at a site and results in the municipality forwarding the site profile to ENV. ENV will then suspend (place a 'freeze' on) the local governments ability to approve certain applications, including demolition permit, rezoning, subdividing or development. ENV has draft revisions to the Site Profile process which will not allow a Municipality to opt out of the regulatory process.

Triggers for completion of a Site Profile are as follows:

- Applying for a permit for:
- Subdivision
- Zoning

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- Development or development variance
- Soil removal
- Demolition
- Decommissioning a site, which includes soil removal
- You are taking over a property as a trustee, receiver or liquidator
- You are selling property that has, or has had a Schedule 2 activity on it

Completing a Site Profile for this Site would identify the known presence of contaminated media, resulting in the Site Profile form being forwarded to ENV and a subsequent freeze placed on the permits described above.

Note that the contamination identified at the Site remains a source of liability for the current and future land owners until it has been addressed via the pathway outlined in Section 3.0. Therefore, for the purposes of developing this regulatory path, we would recommend that the Town of Ladysmith obtain a Certificate of Compliance for the Site, or for individual parts of the Site as they are developed or consider obtaining a Voluntary Remediation Agreement. It is recommended that the Town of Ladysmith seek legal council for understanding of the potential future liabilities following remediation. The following section outlines the processes for obtaining an Instrument or Voluntary Remediation Agreement.

There are a variety of situations which require authorizations before redevelopment of a parcel. Typically, a Site would undergo investigation and remediation activities, and obtain a Certificate of Compliance for the entire area comprising the Site. However, it is understood that the Town of Ladysmith would remediate and develop the property in phases, potential subdividing portions for sale in advance of the remainder.

There are generally four possible paths to obtain a release of the ‘freeze’ from the Director at ENV, regulatory closure and redevelopment of the Site in phases, including

1) Obtaining a notice from the ENV Director granting a release.
2) Obtaining a Protocol 6 Pre Approval from ENV to redevelop the Site in phases
3) Obtaining an Approval in Principal.
4) Obtaining a Voluntary Remediation Agreement.

The following outlines the options for regulatory path to closure under a phased development.

1.1 ENV Director Grants a Release

Given that the Town wishes to develop the Site in Phases, site investigation and remediation activities are required to be completed in order to apply for release of development permits, including

- Preliminary Site Investigation (PSI)
- Detailed Site Investigation (DSI)
- Site Risk Classification report

  - It is assumed that the Site will not be designated as High Risk
- Either:
  - Remediation plan that concludes remediation to applicable standards is achievable and once completed, the Site will be eligible for a Certificate of Compliance
  - Remediation to numerical standards (i.e. excavation and removal of contaminated soils, with confirmatory sampling), or
  - Risk based remediation activities including a risk assessment consisting of: a) developing problem formulation, b) exposure assessment, c) effects assessment, d) risk assessment characterization, and e) uncertainty analysis.
- Written opinion from an Approved Professional Roster review (of above documents).

It is understood that ENV has drafted changes to the Site Profile process, with notable changes on how multi phased developments are completed. Once implemented, sites undergoing phased development would no longer be eligible for releases of local government applications under redevelopment activities when there is a change of proposed activity at the Site. These changes may come into effect in 2020 and as a result, the release process may not be viable option for the Town's schedule.

1.2 Protocol 6 PreApproval to Redevelop the Site in Phases

An application would be made to ENV to obtain Pre-Approval to redevelop the Site in phases. Golder recommends that dialogue with ENV about the phased development and extent of areas requiring remediation be an ongoing process. An application would include

- Preliminary Site Investigation (PSI) for the entire site
- Detailed Site Investigation (DSI) for the entire site.
- If a DSI was not yet completed, a detailed plan and schedule for completing the activities
- Site Risk Classification Report
  - It is assumed that the Site will not be designated as High Risk
  - A high-risk designation does not mean that there are unacceptable risks to human health or the environment. The high-risk designation indicates that there are concentrations present at a magnitude that requires engagement with ENV. The general expectation is that action will be taken to resolve the high-risk issues irrespective of whether a CSR regulatory instrument is being pursued. Resolution involves reclassification of the issue as either "not high-risk" (i.e., upper cap material is removed) or "high-risk, risk-managed" (i.e., risks are evaluated and appropriate management or monitoring actions taken).
- Either:
  - Remediation plan that outlines the necessary remedial activities for each of the phases of the site which would concludes remediation to applicable standards is achievable and once completed, the Site will be eligible for a Certificate of Compliance

  - Remediation to numerical standards (i.e. excavation and removal of contaminated soils, with confirmatory sampling), or
  - Risk based remediation activities including a risk assessment consisting of: a) developing problem formulation, b) exposure assessment, c) effects assessment, d) risk assessment characterization, and e) uncertainty analysis.
- Written opinion from an Approved Professional Roster review (of above documents).

As part of the application, ENV would require a commitment from the Owner of the Site that the work would be completed in accordance with the remediation plan and that once completed, the site would be eligible for a Certificate of Compliance. Note that providing financial security would provide ENV with greater certainty on the commitment to follow the remediation plan. It is also noted, whether the Town provide financial security or not, ENV may require it prior to providing a pre approval to develop the Site in phases.

1.3 Approval In Principal

An Approval in Principal (AiP) is an ENV instrument when a remediation plan has been reviewed and approved. An AiP based on the recommendation of an Approved Professional must be able to meet the requirements of the remediation plan in 5 years.

An application would be made to obtain an AiP and would be prepared in such a way to redevelop the site in phases. Golder recommends that dialogue with ENV about the phased development and extent of areas requiring remediation be an ongoing process.

An application would include

- Preliminary Site Investigation (PSI) for the entire site
- Detailed Site Investigation (DSI) for the entire site.
- If a DSI was not yet completed, a detailed plan and schedule for completing the activities
- Site risk classification report
  - It is assumed that the Site will not be designated as High Risk
- Remediation plan that outlines the necessary remedial activities for each of the phases of the site which would concludes remediation to applicable standards is achievable and once completed, the Site will be eligible for a Certificate of Compliance
- Preparation of an application including the above documents and additional documents including Contaminated Site Application form and Summary of Site Condition.
- Review and recommendation from an Approved Professional Roster review (of above documents) to ENV to issue an AiP.

Following the issuance of the AiP, the Town would follow through on the remediation plan outlined as part of the application.

If the phased development were to require a time period of greater than 5 years, a pre-approval application to ENV would be required prior to Approved Professional Roster Review or that the AiP application be submitted to ENV for their review. Once issued, if material changes were made to the phased development or remediation plan, an application for an amendment to the AiP would be required.

Following completion of remediation of the phased developments, a Confirmation of Remediation Report would be prepared to support an application for a Certificate of Compliance for each sub area of the Site.

Municipalities are typically exempt from financial security requirements, except when thy are part of a pool of responsible persons. Therefore, it is noted that ENV could require financial security prior to issuing an AiP.

1.4 Voluntary Remediation Agreement

Based on our understanding, ENV has issued one VRA in BC and the VRA provision has since some into question and ENV does not consider this as an option in recent years. ENV recommends that parties consider indemnification under the financial Administration Act for seeking a cap of liabilities on Site. In consideration of this information, Golder recommends that the Town of Ladysmith consult legal council to obtain guidance on seeking cap for environmental liabilities on Site.

Notwithstanding, a VRA would require similar scope of work as described in prior options. For your information, a VRA application would include

- Detailed Site Investigation (DSI) for the entire site.
- Site Risk Classification Report
  - It is assumed that the Site will not be designated as High Risk
- A remediation plan
- A detailed description of the responsible person's past and present activities on the Site, including the amount and characteristics of contamination at he Site that the attributable to their activities
- An estimate of the responsible person's share of the total cost of remediation and justification for the estimate
- The name and address for any other person the Town of Ladysmith believes may be a responsible person in accordance with the EMA
- A statement describing the responsible person's ability and plans to conduct and finance the remediation

2.0 OPINON OF PROBABLE COSTS

As noted above, there are various pathways to meet your business goals for the Site. Regardless of this, each path will require a completed DSI for the entire Site, development of a remedial plan and implementation of remediation. Not only is it a regulatory requirement, but from a business perspective, the DSI will characterize and delineate the nature and extent of the contamination. With that established, we will be able to determine the most cost-effective approach to remediate the delineated volume of contamination and obtain the necessary Ministry instrument(s) as per your development plans.

The following provides an outline of the opinion of probable costs to complete a preliminary supplemental investigation, complete detailed site investigation, develop a remediation plan, and implement remediation for the Site.

Please note that a DSI is usually carried out in steps due to the fact that drilling can sometimes delineate or identify new contamination, the latter necessitating additional work to complete delineation. Data for the Site was generally last collected over 10 years ago, therefore, we recommend proceeding with a preliminary supplement investigation prior to finalizing the DSI program.

Based on the historic data, soil contamination is relatively well understood, therefore we recommend completing delineation of the soil contamination as the first step. As for the groundwater, because its chemistry can vary over time and the last round of data was collected approximately ten years ago, this is not likely representative of current conditions. Therefore we recommend redeveloping and resampling the existing wells to document existing conditions and whether or not we can potentially refute some of the historic contamination, thereby limiting future drilling/sampling requirements. Following the review of the data, additional investigation requirement would be identified for completing DSI requirements.

The preliminary supplement scope of work would include the following tasks:

- Development of a health and safety plan and onsite utility locate
- Drilling of up to 30 boreholes
- Development and groundwater sampling of up to 20 existing and accessible monitoring wells
- Laboratory analysis
- Preparation of a data report consisting of a figures and tables of analytical results
- Completion of site risk classification
- Preparation of data gap and recommendations of scope of work to complete DSI

This portion of the recommended preliminary supplemental investigation work (soil delineation and groundwater resampling) noted above is estimated at approximately $95,000 before taxes.

Table 1 provides an opinion of probable cost for completing the remaining tasks to obtain ENV closure, and assumes that risk assessment is a viable option, with limited near surface soil remediation (excavation).

Table 1: Opinion of Probable Costs

Task	Estimate
Preliminary supplemental investigation	$95,000
DSI	$100,000 – $300,000
Risk Assessment Limited Soil Remediation	$50,000 - $75,000 $0 - $100,000
Develop Remedial Plan	$40,000 - $70,000
Approved Professional Review (1)	$30,000
CSAP Fees (1) AiP COC (each)	 $15,000 $2,000
ENV Fees (1) AiP and SOSC COC and SOSC (each)	 $5,000 $8,000
Range of Opinion of Probable Cost	$480,000 – $700,000

Note: Costs do not include ENV applications or approvals, with the exception of obtaining a regulatory instrument. Opinion of probable cost do not include contingency or taxes. Table does not present ENV fees for obtaining a Voluntary Remediation Agreement.
(1) not applicable for Voluntary Remediation Agreement

3.0 CLOSE

We trust that this meets your requirements at this time. Should you have any questions, please do not hesitate to contact Dawn Flotten at 604-296-4377.

Yours very truly,

Golder Associates Ltd.

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DMF/GG/lih

Attachments: Fact Sheet 9, Highlights for Developers

Highlights for Developers

During the development of the provincial contaminated sites regime, the development industry expressed concerns about uncertain legal requirements for cleaning up contaminated sites. This uncertainty increases the risks developers face, and translates into project delays, higher project costs, and fewer redevelopment projects being completed. In response, the *Environmental Management Act* and Contaminated Sites Regulation were written to set clear rules for contaminated sites management, enhancing business predictability. A number of flexible provisions help developers minimize remediation costs.

Meaning of "contaminated site" and "remediation" clarified

The Act and Regulation provide certainty by specifying standards to define "contaminated site" and acceptable "remediation." Developers may use a variety of generic and site-specific factors to decide whether a site is legally considered to be contaminated. They also have the right to choose from a variety of standards when undertaking remediation.

Screening development applications

In the absence of legislative guidance in recent years, local governments adopted a wide range of methods to screen development applications for possible contamination. The *site profile* system is designed to bring uniformity to local government reviews.

What is a site profile?

A site profile is a screening form for identifying potentially contaminated sites. It is a summary created from readily available information about a site – including its past and present uses and basic land descriptions – and should not require the assistance of a consultant to complete. Developers must provide site profiles in specified instances, most notably when applying for zoning, subdivision, demolition, and relocation of soil.

When is a site profile required?

Specific industrial or commercial land uses trigger site profile submissions. Those uses, as listed in the Regulation, are ones that tend to leave contamination (for example, service stations). The list is sometimes attached to the site profile form for those individuals needing to complete and submit a site profile.

Site profile exemptions

Under the Contaminated Sites Regulation, developers are exempt from providing site profiles for several situations, such as where:

- an accurate site profile already exists in the Site Registry (see Fact Sheet 20);
- the site has already been determined to be a contaminated site;
- the site's remediation has already been approved (for example, by a Certificate of Compliance); or
- a local government has opted out of administering the site profile system.

Site investigations

Site investigations are the key means of gathering information to determine if a site is contaminated. They must be done by experienced consultants. Developers can have them done without government involvement.

Under the contaminated sites legislation, a Director of Waste Management from the ministry may order a site investigation, prompted by a site profile or other information the Director may receive.

The Site Registry
The Site Registry was created to provide the public with easy access to reported information about sites, including their basic characteristics, legal events that have occurred, and milestones in the remediation process. The registry is, in part, a "record of decisions" about sites that have been evaluated, whether or not they are clean or contaminated. Anyone may access the registry by computer through BC OnLine.

Independent remediation

Procedures for independent remediation are in place so developers can clean up sites with minimal supervision by the ministry. Independent remediation may occur where liability is not in dispute, investigation and remediation procedures are routine, and engineering or environmental consultants have been engaged to apply the regulations and any additional guidelines or requirements. The ministry must be notified at the start and at the completion of remediation, and this information will be entered on the Site Registry. A developer who uses independent remediation may still apply for a Certificate of Compliance at a later date.

Liability for developers

A current owner or operator – including developer – of a site may be responsible for remediation at a contaminated site. However, there are several exemption situations which could be relevant to developers who may be "responsible persons":

Innocent acquisition exemption – On acquisition, a person diligently undertook inquiries, did not find contamination, and did not contribute further to the contamination.

Migration of a substance from offsite – A person is immune when a site has been contaminated, because the migration of a substance occurred from offsite.

Environmental consultant immunity – Consultants acting for a developer are immune from liability where they assist the developer in the remediation of the site, provided they are not negligent in their actions.

Immunity from future liability – A person who remediates and obtains a Certificate of Compliance can use the certificate as a defence in private cost-recovery lawsuits. This immunity applies where another person subsequently proposes to change the use of a remediated site or undertake further cleanup.

The Contaminated Sites Regulation provides further exemptions that may be relevant to developers. Of note are those exemptions pertaining to sureties, construction contractors, and transporters of contaminated soil.

Cost recovery provisions

The Act and Regulation authorize a person to recover costs from a responsible person (or persons) where those costs are associated with remediation. That is, a developer who undertakes remediation may recover costs from other responsible persons — for example, past polluters.

Minor contributor status

If a developer cannot obtain an exemption from liability, as described above, he or she may be entitled to obtain *minor contributor status*. The Province recognizes the need to treat people who only contributed a minor portion of the contamination on site in a fair and expeditious manner. Minor contributor status can also cap

and shield a person against private cost recovery lawsuits, also limiting liability.

Voluntary Remediation Agreement

This type of agreement with the ministry may be particularly attractive to a developer who wishes to settle liability expeditiously. Under such an agreement, the developer accepts the terms of remediation, including scheduling and a cap on remediation costs. A Voluntary Remediation Agreement can also form the basis for private financial transactions.

Approval in Principle

Developers urged the ministry to provide interim approval, or Approval in Principle, to enable them to seek appropriate financing and municipal development approvals. A Director may grant an Approval in Principle by approving a remediation plan which, if implemented, would lead to remediation that meets applicable standards. Investigation results, an evaluation of remediation options, public consultation input, and remediation plans would be reviewed and, if satisfactory to a Director, may lead to an Approval in Principle being issued.

Certificate of Compliance

A developer can seek approval of remediation in the form of a Certificate of Compliance. Such a certificate may be issued by a Director when a site has been cleaned up according to a remediation plan and meets numerical remediation standards in the Contaminated Sites Regulation.

The ministry recognizes that, at some sites or for some types of contamination, remediation that meets numerical standards may not be technically feasible or financially justified. The legislation allows onsite management of contaminants, provided these contaminants are managed according to prescribed risk standards. A Director may also issue a Certificate of Compliance where the remediation meets risk-based standards in the Regulation.

Security

The *Environmental Management Act* and Contaminated Sites Regulation establish a hierarchy of security: restrictive covenants under the *Land Titles Act* can be imposed only where an entry on the Site Registry is not adequate to achieve effective remediation; and financial security can only be imposed where restrictive covenants are not adequate. The Regulation specifies circumstances under which a Director must discharge the relevant security.

Contaminated Soil Relocation Agreement

Such an agreement regulates the movement of soils from contaminated sites, taking into account the soil quality and environmental conditions at the deposit site. While local bylaws to regulate soil deposit are allowed, they must not conflict with provincial standards if immunity provisions under this legislation are to apply to those local governments.

Public review

Not all contaminated sites raise significant public concerns. However, a Director may require a responsible person to carry out public consultation and review of remediation in cases where there is significant public interest. For those sites, public consultations could achieve better public understanding of proposed remediation, as well as provide useful commentary with which to evaluate remediation alternatives.

Addressing industry concerns, the Regulation bars a Director from ordering public consultation for remediation for which an Approval in Principle or Certificate of Compliance have already been issued.

Contaminated sites fees

The contaminated site legislation authorizes the provincial and local governments to collect fees to offset their costs of contaminated sites regulation and administration. See Fact Sheet 25, "Fees for Contaminated Sites Services," for more information on fees.

Note: This summary is solely for the convenience of the reader. The current legislation and regulations should be consulted for complete information.

For more information, contact the Environmental Management Branch at site@gov.bc.ca